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GCSE AQA

Chemistry

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GCSE AQA

Chemistry

There's no need to worry about GCSE Chemistry with this brilliant CGP book — it's got it all covered, from covalent bonding to fractional distillation.

We've included exam-style practice for every topic, *and* a full set of practice papers too. Plus, there are fully-worked answers at the back, so it's easy to mark your work!

There's also lots of matching online revision in CGP RevisionHub. It's full of quick quizzes, summary tests, videos and more to use alongside your book! Grab your lab coat and get stuck in.

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Throughout this book you'll see grade stamps like these:

These grade stamps help to show how difficult the questions are.
Remember — to get a top grade you need to be able to answer **all** the questions, not just the hardest ones.

In the real exams, some questions test how well you can write (as well as your scientific knowledge).
In this book, we've marked these questions with an asterisk (*).

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Contributors: Michael Bossart, Paddy Gannon and Mike Thompson

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CGP, Broughton House, Griffin Street, Broughton-in-Furness, Cumbria, LA20 6HH. EU Rep: International Associates Auditing & Certification Limited, The Black Church, St Mary's Place, Dublin 7, D07 P4AX, Ireland. EUAR@ie.ia-net.com

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How to Revise with *CGP RevisionHub*

*There's more to **revision** than just reading this book — you need to **understand** and **remember** it too. Read on for how to get the most out of your **book** and **RevisionHub**...*

Here's a handy guide to using RevisionHub.

Video

1) You Need to **Learn Your Stuff**

This book covers everything you need to know for GCSE AQA Chemistry.

1) Read it — but not all at once... break it into manageable chunks.
2) Once you've read over a chunk, try making your own mind maps and flash cards — writing down and thinking about the content will help it to really stick.
3) We've also included some videos to help with the trickier bits — watch out for the QR codes.

2) Check That You Can **Remember** What You've Learnt

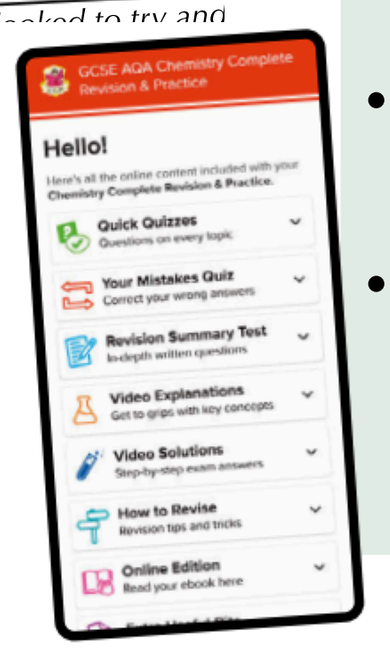
Ta-dah!

Revision Hub

If you don't test yourself, you won't know what you don't know. So try covering up a page and seeing if you can write it all out. Then try quizzing yourself using RevisionHub:

Quick Quizzes

- The QR code at the top of each main page takes you to a quick quiz of the stuff on that page.
- These quizzes are a quick check of the key facts and your maths skills.
- Revisit questions that you get wrong by doing the 'Your Mistakes' quiz, and there are Paper 1 and Paper 2 quizzes too.



Revision Summary Tests

- At the end of each topic in the book is a fiendish memory test on as much stuff as we could cram in.
- You can do these tests on paper or you can complete them on the RevisionHub — follow the QR code.
- Online, you can also find sample answers, assess your progress, and look back at your results for previous topics to see the areas you need to work on.

3) Make Sure That You're **Exam Ready**

To see if you're exam ready, try some exam-style questions. Luckily for you, there are loads in this book.

- You'll find one or more exam-style questions at the bottom of some pages.
- Every few pages, you'll find some warm-up and exam questions.
- There are also two practice papers at the back of the book.

Use these resources to check that you can apply what you've learnt to exams.

There are answers in the back of the book, and most questions at the bottom of the main pages also have a video solution for you to watch — just use the QR code next to the question.

You probably won't stick strictly to this order of steps — for example, you might want to try some exam-style questions as you go, and you'll need to look back at bits of this book if you get a question wrong or are finding a topic hard.

Revision — there's no better path to exam success...

Yep, revision can seem quite daunting. But it's a key part of preparing for your exams. If it's making you feel like you don't know where to start, then fear not — follow this QR code to see what we recommend.

Start Revising

What to Expect in the Exams

*Before you get cracking with your **revision** and **exam practice**, here's a **handy guide** to what you'll have to face in the exams. You're welcome.*

Topics are Covered in **Different Papers**

You're expected to know the basic concepts of chemistry in both papers.

For GCSE AQA Higher Chemistry, you'll sit two exam papers at the end of your course.

Paper	Time	No. of Marks	Topics Assessed	Maths Skills	Practical Skills
1	1 hr 45 mins	100	1, 2, 3, 4 and 5	At least 20% of total marks	At least 15% of total marks
2	1 hr 45 mins	100	6, 7, 8, 9 and 10		

There are **Different Question Types**

In each exam, you'll be expected to answer a mixture of multiple choice questions, structured questions, questions that have short, closed answers, as well as open response questions.

For some open response questions, you'll be marked on the overall quality of your answer, not just its scientific content. So...

Always make sure:

- You answer the question fully.
- You include detailed, relevant information.
- Your answer is clear and has a logical structure.

You'll be Tested on Your **Maths**...

We've indicated Maths Skills throughout this book with a handy stamp that looks like this:

At least 20% of the total marks for GCSE Chemistry will come from questions that test your maths skills. For these questions, always remember to:

- Show your working — you could get marks for this, even if your final answer's wrong.
- Check that the units of your answer are the same as the ones they asked for in the question.
- Make sure your answer is given to an appropriate number of significant figures.

...and on Your **Practical Skills**

Required practical activities are marked up with a stamp like this.

GCSE Chemistry contains 8 required practical activities that you'll do during the course. You can be asked about these, and the practical skills involved in them, in the exams.

E.g. you might have to comment on experiment design (the apparatus and method), make predictions, analyse or interpret results... Pretty much anything to do with planning and carrying out investigations.

At least 15% of the total marks will be for questions that test your understanding of the practical activities and practical skills.

You'll Need to Know About **Working Scientifically**

Working Scientifically is all about how science is applied in the outside world by real scientists.

For example, you might be asked about ways that scientists communicate an idea to get their point across without being biased, or about the limitations of a scientific theory.

You need to think about the situation that you've been given and use all your scientific savvy. Always read the question and any data you've been given really carefully before you start writing.

Atoms

Quick Quiz

*All substances are made of **atoms**. Atoms are really **tiny** — too small to see, even with your microscope. They're so tiny that a **50p piece** contains about 77 400 000 000 000 000 000 000 of them. Quite a lot then...*

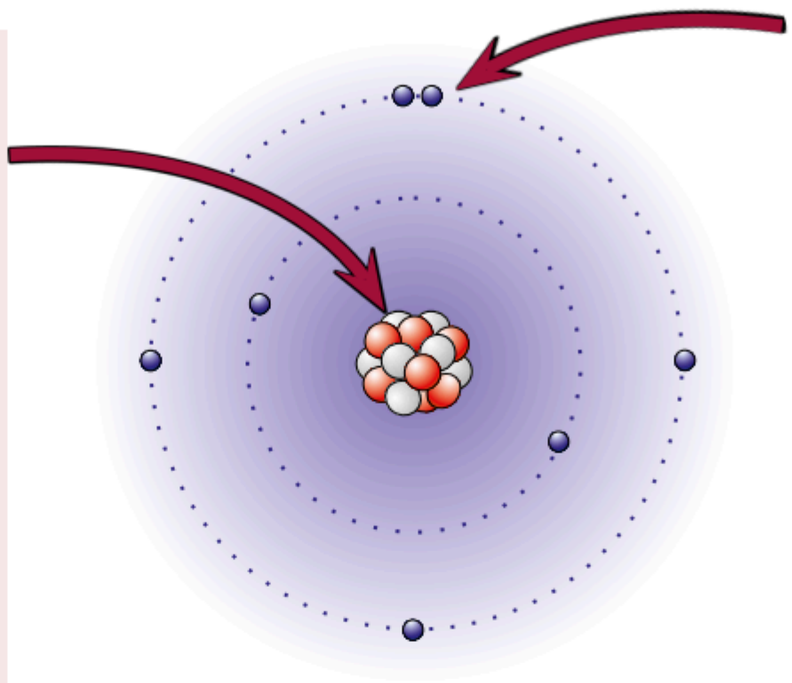
Atoms Contain **Protons**, **Neutrons** and **Electrons**

Atoms have a radius of about 0.1 nanometres (that's 1×10^{-10} m). There are a few different (and equally useful) models of the atom — but chemists tend to like the model below best.

A nanometre (nm) is one billionth of a metre. Shown in standard form, that's 1×10^{-9} m. You can read more about standard form on page 148.

The Nucleus

- Contains protons and neutrons and is in the centre of the atom.
- It has a radius of around 1×10^{-14} m (that's around 1/10 000 of the atom's radius).
- It has a positive charge because of the protons.
- Almost the whole mass of the atom is concentrated in the nucleus.



The Electrons

- Move around the nucleus in electron shells.
- They're negatively charged and tiny, but they cover a lot of space.
- The volume of their orbits determines the size of the atom.
- Electrons have virtually no mass.

You need to know the relative mass and charge for each type of particle in an atom:

Particle	Relative Mass	Relative Charge
Proton	1	+1
Neutron	1	0
Electron	Very small	–1

Video

Number of Protons **Equals** Number of Electrons

An ion is an atom or group of atoms that has lost or gained electrons.

1) Atoms are neutral — they have no charge overall (unlike ions).
2) This is because they have the same number of protons as electrons.
3) The charges on the electrons and protons are the same size but opposite — so they cancel out.

In an ion, the number of protons doesn't equal the number of electrons. This means it has an overall charge. For example, an ion with a 2– charge has two more electrons than protons.

Atomic Number and **Mass Number** Describe an Atom

The nuclear symbol of an atom tells you its atomic (proton) number and mass number:

Atomic number — tells you how many protons there are.

$^{23}_{11}\mathrm{Na}$

Mass number — tells you the total number of protons and neutrons in the atom.

This is the nuclear symbol for sodium. See the next page for more on symbols.

To get the number of neutrons, just subtract the atomic number from the mass number, e.g. this atom of sodium has 23 – 11 = 12 neutrons.

Atoms make up pretty much everything

Ahh, the very beginning of GCSE Chemistry... Make sure you've got these basics down, then let's go.

Q1 An atom of gallium has an atomic number of 31 and a mass number of 70. Give the number of electrons, protons and neutrons in the atom. [3 marks]

Q1 Video Solution

Quick Quiz

Elements

*An **element** is a substance made up of atoms that all have the **same** number of **protons** in their nucleus.*

Elements Consist of Atoms with the Same Atomic Number

1) Atoms can have different numbers of protons, neutrons and electrons.
2) It's the number of protons in the nucleus that decides what type of atom it is — a substance that only contains atoms with the same number of protons is called an element. There are about 100 different elements.
3) Each element has a different number of protons.

E.g. an atom with one proton is hydrogen and an atom with two protons is helium.

Atoms can be Represented by Symbols

Atoms of each element can be represented by a one or two letter symbol — it's a type of shorthand that saves you the bother of having to write the full name of the element.

Some make perfect sense, e.g. O = oxygen. Others less so, e.g. Na = sodium.

You'll see these symbols on the periodic table (see page 17).

Most odd symbols actually come from the Latin names of the elements.

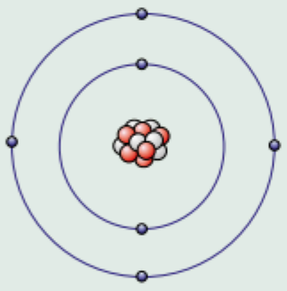
Isotopes are the Same Except for the Number of Neutrons

- Isotopes are different forms of the same element, which have the same number of protons but a different number of neutrons.
- So isotopes have the same atomic number but different mass numbers.
- A very popular example of a pair of isotopes are carbon-12 and carbon-13.

Carbon-12

$^{12}_{6}C$

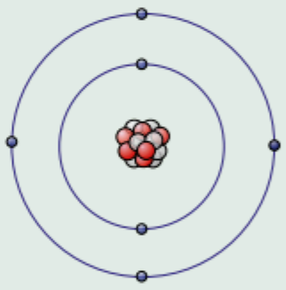
6 Protons
6 Electrons
6 Neutrons



Carbon-13

$^{13}_{6}C$

6 Protons
6 Electrons
7 Neutrons



- Because many elements can exist as a number of different isotopes, relative atomic mass (A_r) is used instead of mass number when referring to the element as a whole. This is an average mass taking into account the different masses and abundances (amounts) of all the isotopes that make up the element.
- You can use this formula to work out the relative atomic mass of an element:

$$\text{relative atomic mass } (A_r) = \frac{\text{sum of (isotope abundance} \times \text{isotope mass number)}}{\text{sum of abundances of all the isotopes}}$$

EXAMPLE

Copper has two stable isotopes. Cu-63 has an abundance of 69.2% and Cu-65 has an abundance of 30.8%. Calculate the relative atomic mass of copper to 1 decimal place.

MATHS SKILLS

$$\text{Relative atomic mass} = \frac{(69.2 \times 63) + (30.8 \times 65)}{69.2 + 30.8} = \frac{4359.6 + 2002}{100} = \frac{6361.6}{100} = 63.616 = 63.6 \text{ (1 d.p.)}$$

Isotopes of an element have different mass numbers

Atoms, elements and isotopes — make sure you know what they are and the differences between them.

Q2 Video Solution

Q1 A substance consists of atoms which all have the same number of protons and electrons but different numbers of neutrons. Explain why this substance is an element. [1 mark]

Q2 Silicon, Si, has three stable isotopes. Si-28 has an abundance of 92.2%, Si-29 has an abundance of 4.7% and Si-30 has an abundance of 3.1%. Calculate silicon's relative atomic mass to 1 decimal place. [2 marks]

Compounds

Quick Quiz

*It would be great if we only had to deal with elements. But, unluckily for you, elements can mix and match to make lots of new substances called **compounds**. And this makes things a little bit more complicated...*

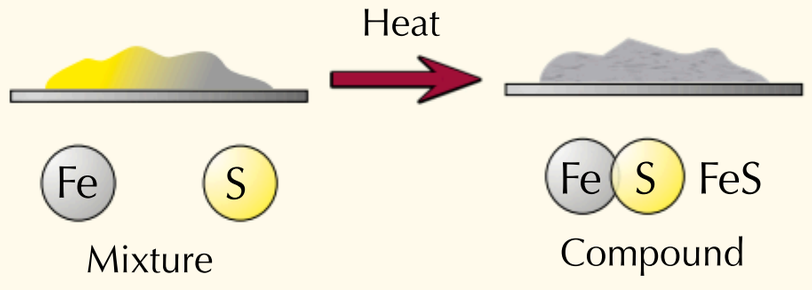
Atoms **Join Together** to Make **Compounds**

You can usually measure a change in energy during a chemical reaction, such as a temperature change.

1) When elements react, atoms join with other atoms to form at least one new substance.
2) Compounds are substances formed from two or more elements. The atoms of each element are in fixed proportions throughout the compound and they're held together by chemical bonds.
3) Making bonds involves atoms giving away, taking or sharing electrons. Only the electrons are involved — the nuclei of the atoms aren't affected at all when a bond is made.
4) It's usually difficult to separate the original elements of a compound out again — a chemical reaction is needed to do this.

The properties of a compound are usually totally different from the properties of the original elements.

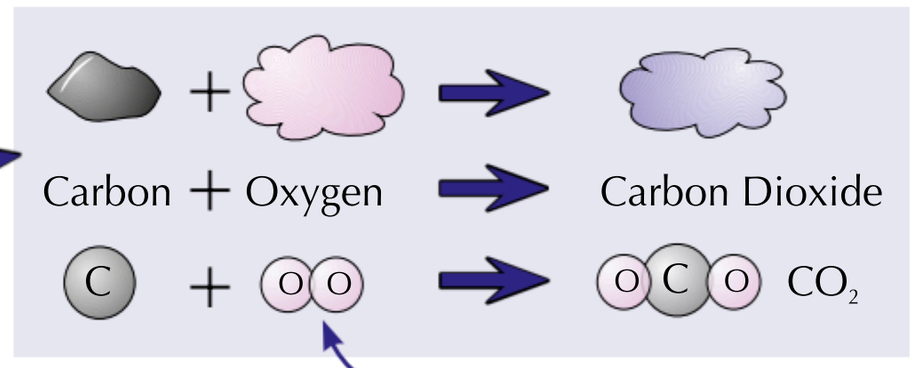
For example, if iron (a lustrous magnetic metal) and sulfur (a nice yellow powder) react, the compound formed (iron sulfide) is a dull grey solid lump, and doesn't behave anything like either iron or sulfur.



A **Formula** Shows What **Atoms** are in a **Compound**

Just as elements can be represented by symbols, compounds can be represented by formulas. The formulas are made up of elemental symbols in the same proportions that the elements can be found in the compound.

- For example, carbon dioxide, CO_2, is a compound formed from a chemical reaction between carbon and oxygen. It contains 1 carbon atom and 2 oxygen atoms.
- Here's another example: the formula of sulfuric acid is H_2SO_4. So, each molecule contains 2 hydrogen atoms, 1 sulfur atom and 4 oxygen atoms.
- There might be brackets in a formula, e.g. calcium hydroxide is $Ca(OH)_2$. The little number outside the bracket applies to everything inside the brackets. So in $Ca(OH)_2$ there's 1 calcium atom, 2 oxygen atoms and 2 hydrogen atoms.



Here are some more examples of formulas which might come in handy:

Ammonia — NH_3	Sodium chloride — $NaCl$	Sodium hydroxide — $NaOH$
Water — H_2O	Carbon monoxide — CO	Calcium chloride — $CaCl_2$
Nitric acid — HNO_3	Hydrochloric acid — HCl	Sodium carbonate — Na_2CO_3

Every chemical compound has its own formula

Q2 Video Solution

Being able to work out the type and number of atoms in a compound from its formula is a pretty important skill in Chemistry, so make sure you're feeling confident with this stuff before moving on.

Q1 How many atoms are represented in the formula Na_2CO_3? [1 mark]

Q2 A compound has the formula $Al_2(SO_4)_3$. Name the elements and state how many atoms of each element are represented in its formula. [1 mark]

Quick Quiz

Chemical Equations

Formulas *and* ***equations*** *are used to show what is happening to substances involved in* ***chemical reactions****. They tell you what* ***atoms*** *are involved and how the substances* ***change*** *during a reaction.*

Chemical Changes are Shown Using Chemical Equations

1) One way to show a chemical reaction is to write a word equation. It's not as quick as using chemical symbols and you can't tell straight away what's happened to each of the atoms, but it's dead easy.

Here's an example — you're told that methane burns in oxygen giving carbon dioxide and water:

The molecules on the left-hand side of the equation are called the reactants (because they react with each other).

methane + oxygen → carbon dioxide + water

The molecules on the right-hand side are called the products (because they've been produced from the reactants).

2) Another way to show a chemical reaction is to write the symbol equation. This is a kind of shorthand that shows the symbols or formulas of the reactants and products. Here's the symbol equation for the reaction above:

$$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$$

You'll have spotted that there's a '2' in front of the O_2 and the H_2O. The reason for this is explained below...

Symbol Equations Need to be Balanced

- There must always be the same number of atoms on both sides — they can't just disappear.
- You balance the equation by putting numbers in front of the formulas where needed. Take this equation for reacting sulfuric acid with sodium hydroxide:

$$H_2SO_4 + NaOH \rightarrow Na_2SO_4 + H_2O$$

- The formulas are all correct but the numbers of some atoms don't match up on both sides.
- You can't change formulas like H_2SO_4 to H_2SO_5. You can only put numbers in front of them.

The more you practise, the quicker you get, but all you do is this:

1) Find an element that doesn't balance and pencil in a number to try to sort it out.
2) See where it gets you. It may create another imbalance, but if so, pencil in another number and see where that gets you.
3) Carry on chasing unbalanced elements and it'll sort itself out pretty quickly.

EXAMPLE

In the equation above, we're short of H atoms on the RHS (Right-Hand Side).

MATHS SKILLS

1) The only thing you can do about that is make it $2H_2O$ instead of just H_2O:

$$H_2SO_4 + NaOH \rightarrow Na_2SO_4 + 2H_2O$$

2) But that now gives too many H atoms and O atoms on the RHS, so to balance that up you could try putting 2NaOH on the LHS (Left-Hand Side):

$$H_2SO_4 + 2NaOH \rightarrow Na_2SO_4 + 2H_2O$$

3) And suddenly there it is! Everything balances. And you'll notice the Na just sorted itself out.

Getting good at balancing equations is all about practice

Q2 Video Solution

There's a lot of numbers floating about on this page. Remember, a number in front of a formula applies to the whole thing — $3Na_2SO_4$ means three lots of Na_2SO_4. The little numbers within or at the end of a formula only apply to the atom or brackets right before. So the 4 in Na_2SO_4 means there are 4 Os, but there's just 1 S not 4.

Q1 Balance the equation: $Fe + Cl_2 \rightarrow FeCl_3$ [1 mark]

Q2 Hydrogen and oxygen molecules are formed in a reaction where water splits apart.
For this reaction: a) State the word equation. b) Give a balanced symbol equation. [3 marks]

Warm-Up & Exam Questions

So, you reckon you know your elements from your compounds?... Have a go at these questions and see how you do. If you get stuck on something just flick back and give it another read through.

Warm-Up Questions

1) What does the mass number tell you about an atom?
2) What is the definition of an element?
3) What is a compound?
4) Balance this equation for the reaction of glucose ($C_6H_{12}O_6$) and oxygen:
   $C_6H_{12}O_6 + O_2 \rightarrow CO_2 + H_2O$

Exam Questions

1 This question is about atomic structure. GRADE 4-6 GRADE

1.1 Use your knowledge to complete **Table 1**.

Table 1

Name of particle	Relative charge
Proton	.....
Neutron	.....

[2 marks]

1.2 Where are protons and neutrons found in an atom?

[1 mark]

1.3 An atom has 8 electrons. How many protons does the atom have?

[1 mark]

1.4 What is the relative mass of a proton?

[1 mark]

2 **Table 2** gives some information about nitrogen. GRADE 4-6 GRADE

Table 2

Element	Number of protons	Mass number
nitrogen	7	14

2.1 How many neutrons does nitrogen have?

[1 mark]

2.2 Describe how the information in **Table 2** can be used to work out the atomic number of nitrogen.

[1 mark]

2.3 What is the chemical symbol for nitrogen?

[1 mark]

Exam Questions

3 Methane (CH_4) burns in oxygen to make carbon dioxide and water. GRADE 6-7

3.1 State the names of the reactants in this reaction.

[1 mark]

3.2 State the names of the products in this reaction.

[1 mark]

3.3 Which molecule involved in the reaction is composed of only one element?

[1 mark]

3.4 Complete and balance the symbol equation for the reaction below.

$$\text{.....}CH_4 + \text{.....} \rightarrow \text{.....}CO_2 + \text{.....}H_2O$$

[2 marks]

4 Sulfuric acid reacts with ammonia to form ammonium sulfate, $(NH_4)_2SO_4$. GRADE 6-7

4.1 Complete and balance the symbol equation below.

$$\text{.....} + \text{.....} \rightarrow (NH_4)_2SO_4$$

[3 marks]

4.2 In the balanced equation, how many atoms are there in the reactants?

[1 mark]

4.3 How many hydrogen atoms are present in ammonium sulfate?

[1 mark]

5 Carbon has several isotopes. These include carbon-12 and carbon-13. Details about the carbon-13 isotope are shown below. GRADE 7-9

$$\boxed{{}^{13}_{6}C}$$

5.1 Explain what an isotope is.

[3 marks]

5.2 Give the number of protons, neutrons and electrons that carbon-13 contains.

[3 marks]

5.3 Details of element **X** are shown below.

$$\boxed{{}^{13}_{7}X}$$

Explain how you can tell that element **X** is not an isotope of carbon.

[1 mark]

5.4 There are two isotopes of element **X**. One isotope has a mass number of 13 and a percentage abundance of 79%. The other isotope has a mass number of 14 and a percentage abundance of 21%.

Use this information to calculate the relative atomic mass of element **X**. Give your answer to 1 d.p.

[4 marks]

Mixtures and Chromatography

Quick Quiz

*A **mixture** is exactly what it says on the tin — two or more **elements** or **compounds** mixed together.*

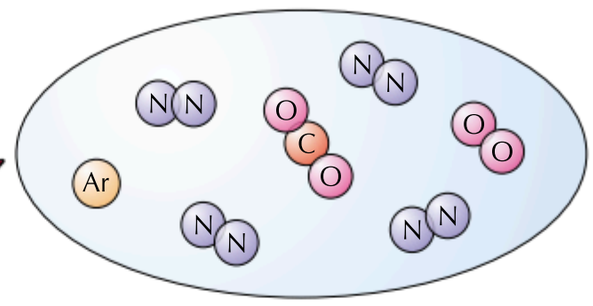
Mixtures are Easily Separated

Unlike in a compound, there's no chemical bond between the different parts of a mixture.

The parts of a mixture can be either elements or compounds, and they can be separated out by physical methods such as filtration (p. 10), crystallisation (p.10), simple distillation (p.11), fractional distillation (p.11) and chromatography (see below).

A physical method is one that doesn't involve a chemical reaction, so doesn't form any new substances.

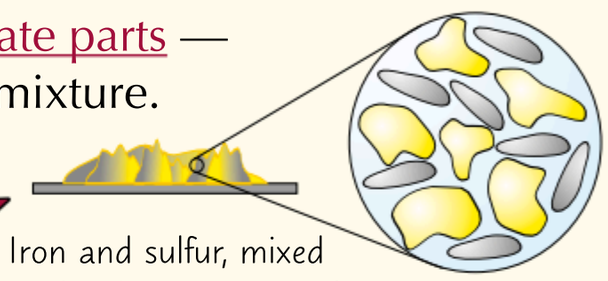
- For example, air is a mixture of gases, mainly nitrogen, oxygen, carbon dioxide and argon. The gases can all be separated out.



- Crude oil is a mixture of different length hydrocarbon molecules.

The properties of a mixture are just a mixture of the properties of the separate parts — the chemical properties of a substance aren't affected by it being part of a mixture.

For example, a mixture of iron powder and sulfur powder will show the properties of both iron and sulfur. It will contain grey magnetic bits of iron and bright yellow bits of sulfur.



Iron and sulfur, mixed together but not reacted.

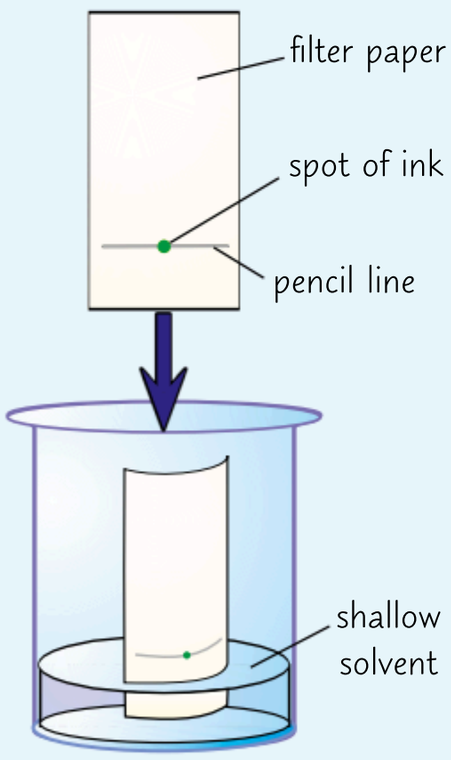
You Need to Know How to Do Paper Chromatography

PRACTICAL

One method of separating substances in a mixture is through chromatography.

Method

This technique can be used to separate different dyes in an ink. Here's how you can do it:



1) Draw a line near the bottom of a sheet of filter paper. (Use a pencil to do this — pencil marks are insoluble and won't dissolve in the solvent.)
2) Add a spot of the ink to the line and place the sheet in a beaker of solvent, *e.g* water. The solvent used depends on what's being tested. Some compounds dissolve well in water, but sometimes other solvents, like ethanol, are needed.
3) Make sure the ink isn't touching the solvent — you don't want it to dissolve into it.
4) Place a lid on top of the container to stop the solvent evaporating.
5) The solvent seeps up the paper, carrying the ink with it.
6) When the solvent has nearly reached the top of the paper, take the paper out of the beaker and leave it to dry.

The point the solvent has reached as it moves up the paper is called the solvent front.

7) The end result is a pattern of spots called a chromatogram.

- Each spot on the chromatogram is a different dye in the ink.
- The dyes separate out because each one moves up the paper at a different rate, so they form spots in different places.
- If any of the dyes in the ink are insoluble (won't dissolve) in the solvent you've used, they'll stay on the baseline.

Video

Chromatography separates the different dyes in inks

Always use a pencil to draw the baseline on your chromatogram. If you use a pen, all the components of the ink in the pen will get separated, along with the substance you're analysing. And that'd make your results pretty confusing.

More Separation Techniques

PRACTICAL

Quick Quiz

*Chemists use these techniques loads to separate **solids** from **liquids**, so make sure you know how to do them.*

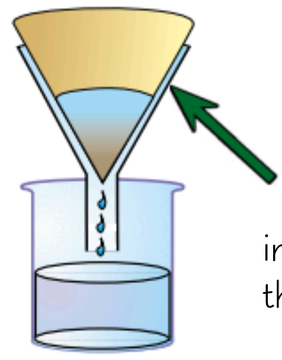
Filtration Separates Insoluble Solids from Liquids

Video

If a solid is insoluble, it won't dissolve in a liquid.

Filtration can be used to separate an insoluble solid product from a liquid reaction mixture.

It can be used in purification as well. For example, solid impurities in a reaction mixture can be separated out using filtration.



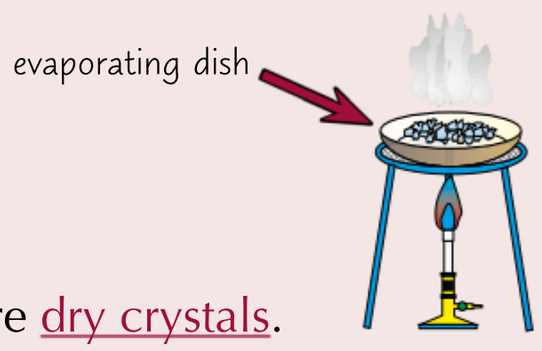
Filter paper folded into a cone shape — the solid is left in the filter paper.

Two Ways to Separate Soluble Solids from Solutions

If a solid can be dissolved it's described as being soluble. You can separate a soluble salt from a solution by either evaporation or crystallisation. Here's how you could go about these methods:

EVAPORATION

1) Pour the solution into an evaporating dish.
2) Slowly heat the solution. The solvent will evaporate and the solution will get more concentrated. Eventually, crystals will start to form.
3) Keep gently heating the dish until all you have left are dry crystals.



You don't have to use a Bunsen burner — you could use a water bath, or an electric heater. See p.154 for more about them.

Evaporation is a really quick way of separating a soluble salt from a solution, but you can only use it if the salt doesn't decompose (break down) when it's heated. Otherwise, you'll have to use crystallisation.

CRYSTALLISATION

1) Pour the solution into an evaporating dish and gently heat the solution. Some of the solvent will evaporate and the solution will get more concentrated.
2) Once some of the solvent has evaporated, or when you see crystals start to form (the point of crystallisation), remove the dish from the heat and leave the solution to cool.
3) The salt should start to form crystals as it becomes insoluble in the cold, highly concentrated solution.
4) Filter the crystals out of the solution, and leave them in a warm place to dry. You could also use a drying oven or a desiccator.

You should also use crystallisation if you want to make nice big crystals of your salt.

Salt crystallising out of solution.

Filtration and Evaporation can be Used to Separate Rock Salt

- Rock salt is simply a mixture of salt and sand (they spread it on the roads in winter).
- The salt and sand are both solids — but salt dissolves in water and sand doesn't.

This vital difference in their physical properties gives a great way to separate them. Here's what to do...

1) Grind the mixture to make sure the salt crystals are small, so will dissolve easily.
2) Put the mixture in water and stir. The salt will dissolve, but the sand won't.
3) Filter the mixture. The grains of sand won't fit through the tiny holes in the filter paper, so they collect on the paper instead. The salt passes through the filter paper as it's part of the solution.
4) Evaporate the water from the salt so that it forms dry crystals.

You can heat the mixture to help dissolve the salt.

You could also use crystallisation here.

Separating rock salt requires filtration and evaporation

You might get asked how to separate another type of mixture containing insoluble and soluble solids. It's the same principles as for rock salt, so just apply the same method and think carefully about what's happening in each stage.

Distillation

Quick Quiz

*Distillation is used to separate mixtures of **liquids**. This page covers two types — **simple** and **fractional**.*

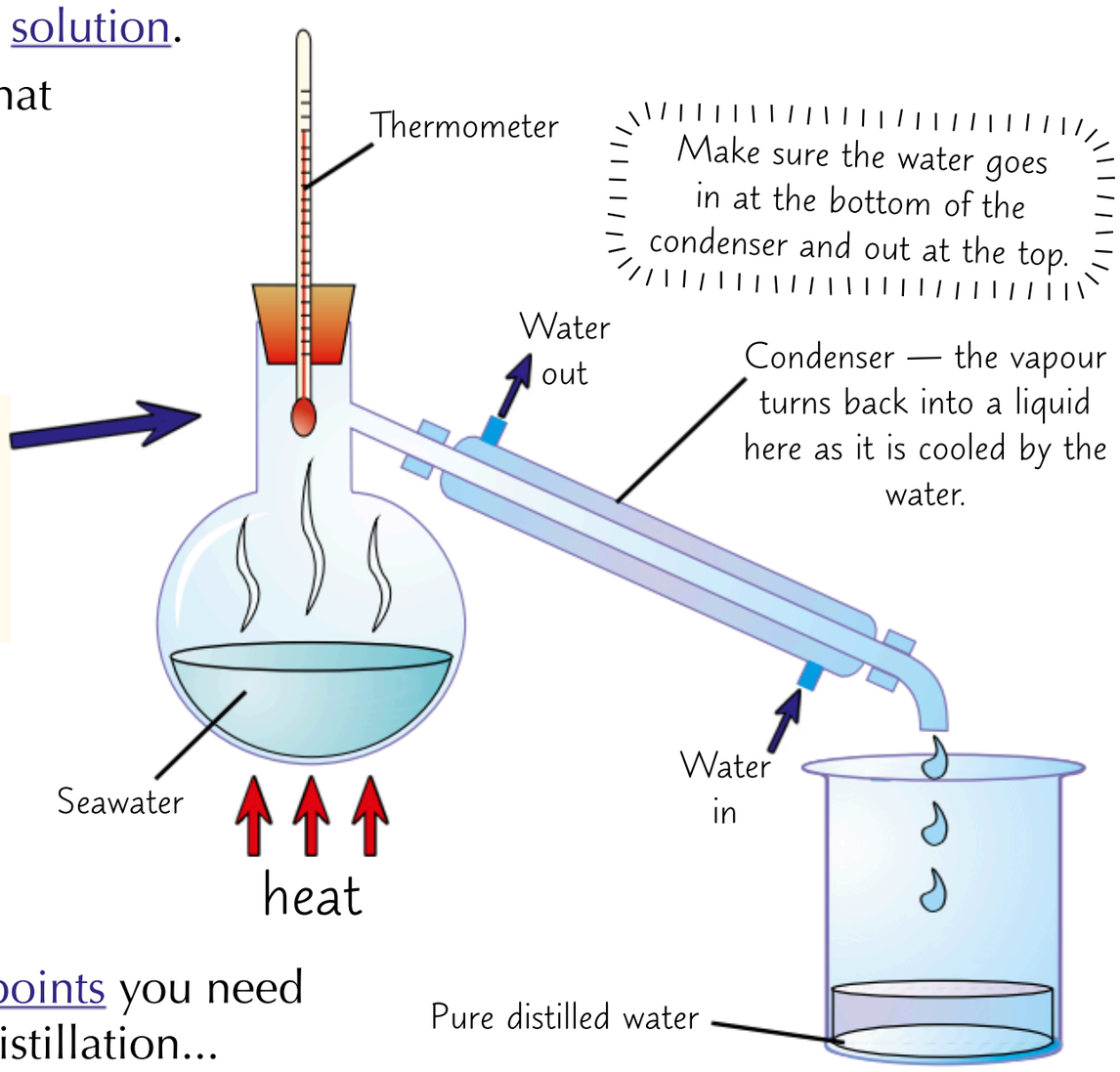
Simple Distillation is Used to Separate Out Solutions

PRACTICAL

Simple distillation is used to separate a liquid from a solution.

- The solution is heated. The part of the solution that has the lowest boiling point evaporates first.
- The vapour is then cooled, condenses (turns back into a liquid) and is collected.
- The rest of the solution is left behind in the flask.

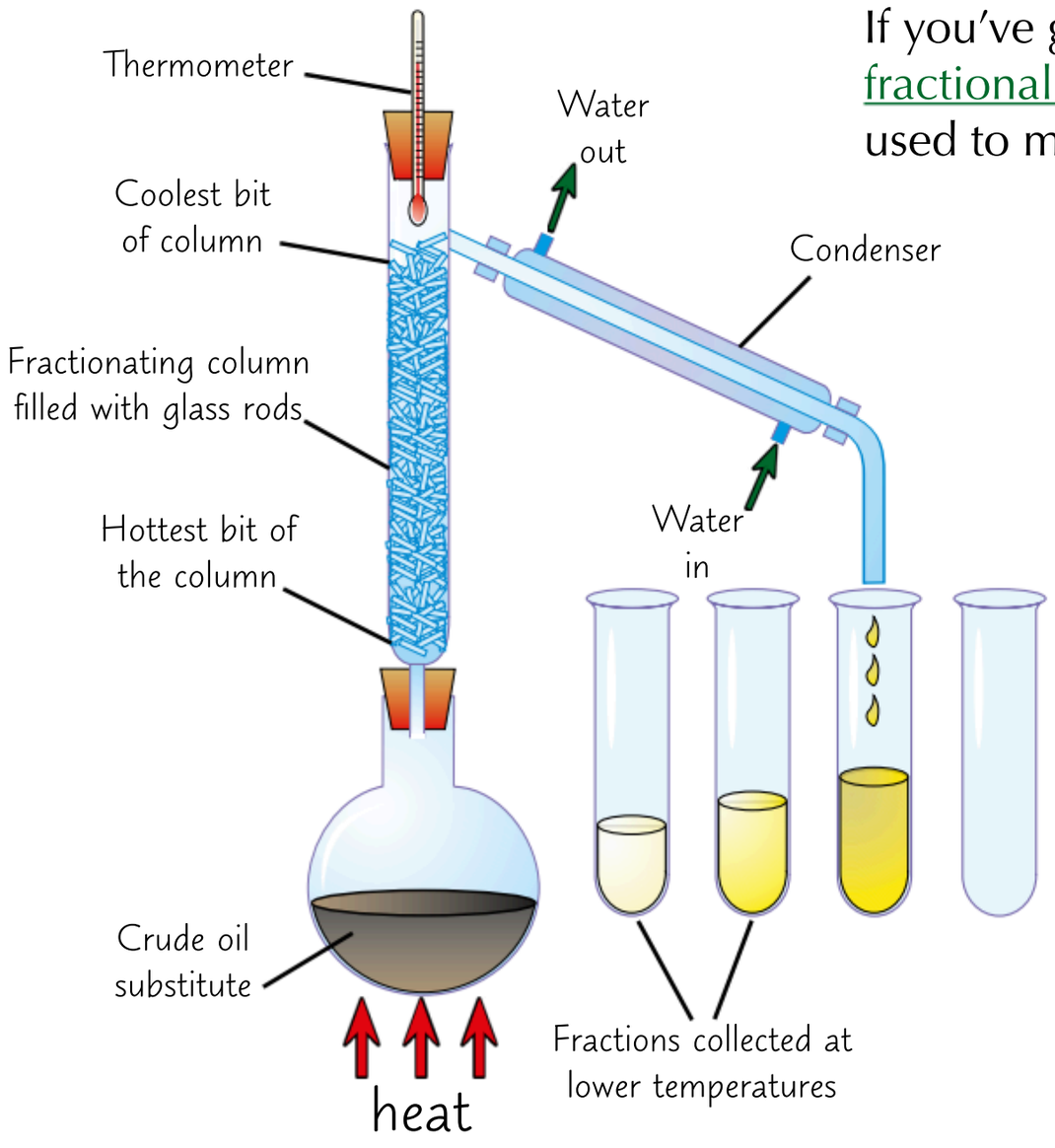
You can use simple distillation to get pure water from seawater. The water evaporates and is condensed and collected. Eventually you'll end up with just salts left in the flask.



The problem with simple distillation is that you can only use it to separate things with very different boiling points — if the temperature goes higher than the boiling point of the substance with the highest boiling point, they will mix again.

If you have a mixture of liquids with similar boiling points you need another method to separate them — like fractional distillation...

Fractional Distillation is Used to Separate a Mixture of Liquids



If you've got a mixture of liquids you can separate it using fractional distillation. Here is a lab demonstration that can be used to model fractional distillation of crude oil at a refinery.

1) You put your mixture in a flask and stick a fractionating column on top. Then you heat it.
2) The different liquids will all have different boiling points — so they will evaporate at different temperatures.
3) The liquid with the lowest boiling point evaporates first. When the temperature on the thermometer matches the boiling point of this liquid, it will reach the top of the column.
4) Liquids with higher boiling points might also start to evaporate. But the column is cooler towards the top. So they will only get part of the way up before condensing and running back down towards the flask.
5) When the first liquid has been collected, you raise the temperature until the next one reaches the top.

Fractional distillation is used in the lab and industry

You made it to the end of separation techniques, so make sure you understand what each method can be used to separate and the apparatus set up for each technique. Then have a crack at this question.

Q1 Propan-1-ol, methanol and ethanol have boiling points of 97 °C, 65 °C and 78 °C respectively. A student uses fractional distillation to separate a mixture of these compounds. State which liquid will be collected in the second fraction and explain why. [2 marks]

Q1 Video Solution

Warm-Up & Exam Questions

PRACTICAL

So the last few pages have all been about mixtures and how to separate them. Here are some questions to get stuck into and make sure you know your filtration from your distillation...

Warm-Up Questions

1) What is the definition of a mixture?
2) Which technique could you use to separate a mixture of liquids with similar boiling points?

Exam Questions

1 A forensic scientist is using paper chromatography to compare different inks which contain a mixture of dyes. He draws a pencil line near the bottom of a sheet of filter paper and adds a spot of each of the different inks to the line with a gap between each spot.

GRADE 4-6

1.1 Describe the next step that's required to carry out paper chromatography.

[1 mark]

1.2 Why is pencil used to make the line on the filter paper?

[1 mark]

2 Lawn sand is a mixture of insoluble sharp sand and soluble ammonium sulfate fertiliser.

GRADE 6-7

2.1 Describe how you would obtain pure, dry samples of the two components of lawn sand in the lab.

[3 marks]

2.2 A student separated 51.4 g of lawn sand into sharp sand and ammonium sulfate.
After separation, the total mass of the two products was 52.6 g.
Suggest a reason for the difference in mass.

[1 mark]

3 **Table 1** gives the boiling points of three liquids.

GRADE 6-7

3.1 State why simple distillation cannot be used to separate water from a solution of water and methanoic acid.

[1 mark]

Table 1

Liquid	Boiling point (°C)
Methanoic acid	101
Propanone	56
Water	100

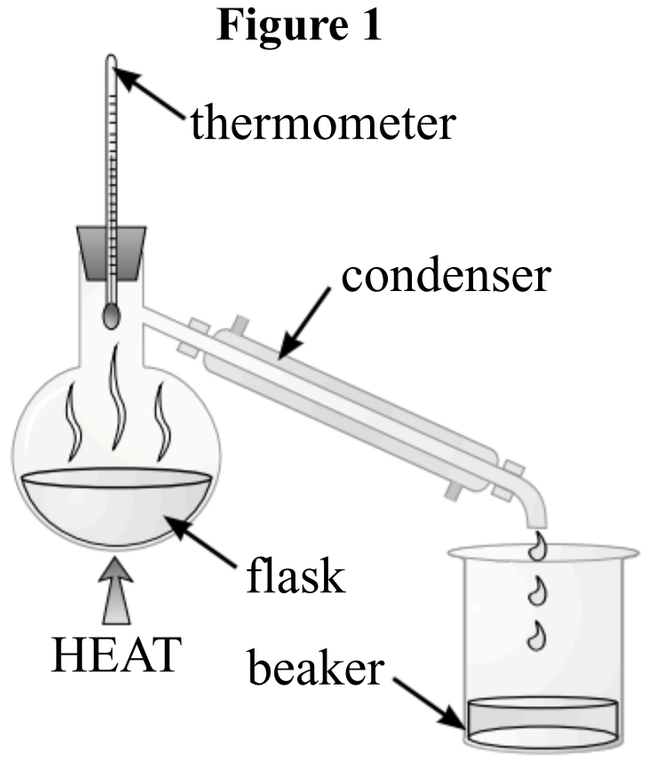
3.2 The apparatus in **Figure 1** was used to separate a mixture of propanone and water.

Complete the table using the options below.

no liquid **water** **propanone** **both liquids**

Temperature on thermometer	Contents of the flask	Contents of the beaker
30 °C	.....	.....
65 °C	.....	.....
110 °C	.....	.....

Figure 1



[3 marks]

Revision Summary Test for Topic 1a

Well, that's the first section over and done with. To check how well you remember it, you can:

- Tackle the revision summary test below, or scan the QR code to do it online.
  The questions are hard, but they'll show you how well you really know your stuff.
- Track your progress online and see which areas need more work.
- Compare your answers with sample answers for the test here: cgpbooks.co.uk/Base

Summary Test

Atoms and Elements (p.3-4) ☑

1) What is the approximate radius of an atom?
2) Where is most of the mass of an atom found?
3) Where in an atom would you find:
   a) electrons, b) protons, c) neutrons?
4) Give the relative charges and relative masses of electrons, protons and neutrons.
5) Why do atoms have no overall charge?
6) How can you use the atomic number and mass number of an atom to find out how many protons, electrons and neutrons it has?
7) Define an element, in terms of protons.
8) What are isotopes?
9) What is relative atomic mass (A_r)? Give the formula used to calculate it.

Compounds and Equations (p.5-6) ☑

10) Describe what a compound is and how one is formed.
11) How many atoms are represented in each of these formulas?
    a) NH_3
    b) $CaCl_2$
    c) H_2SO_4
    d) $Mg(OH)_2$
12) Balance these equations:
    a) $Mg + O_2 \rightarrow MgO$
    b) $CuO + HCl \rightarrow CuCl_2 + H_2O$

Mixtures and Separation (p.9-11) ☑

13) Give three differences between a mixture and a compound.
14) Outline a method for separating dyes in an ink using paper chromatography.
15) What causes the dyes in an ink to separate during chromatography?
16) What type of mixture can be separated by filtration?
17) Give two ways of separating a soluble salt from a solution.
    What apparatus would you need for both techniques?
18) What steps would you need to take to separate the salt and sand in rock salt?
19) What is simple distillation used to separate? How does it work?
20) How is a mixture of liquids separated using fractional distillation?

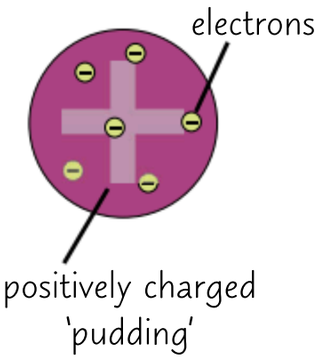
Quick Quiz

The History of The Atom

*Our picture of **the atom** kept changing due to new discoveries. Hold on — you're going on a trip through **time**...*

The Theory of **Atomic Structure** Has **Changed** Over Time

- At the start of the 19th century John Dalton described atoms as solid spheres which couldn't be divided. He thought that different spheres made up the different elements.
- In 1897 J J Thomson showed that atoms contain smaller, negatively charged particles — electrons. The 'solid sphere' idea of atomic structure had to be changed. Thomson's theory was known as the 'plum pudding model' — it showed the atom as a ball of positive charge with electrons stuck in it.

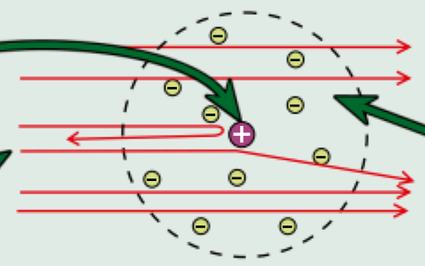


Rutherford Showed that the **Plum Pudding Model** was Wrong

1) In 1909 Ernest Rutherford and his student Ernest Marsden carried out their alpha particle scattering experiments. They fired positively charged alpha particles at a thin sheet of gold.
2) From the plum pudding model, they were expecting the particles to pass straight through the sheet or be slightly deflected at most. This was because the positive charge of each atom was thought to be very spread out through the 'pudding' of the atom. Though most of the particles did go straight through, some were deflected more than expected and a small number were deflected backwards. So the plum pudding model couldn't be right.
3) Rutherford came up with a new idea to explain the results — a nuclear model of the atom.

All the positive charge and most of the mass is concentrated in a tiny nucleus.

Alpha particles that pass near the positive charge of the nucleus are deflected by large amounts — some are deflected straight back.

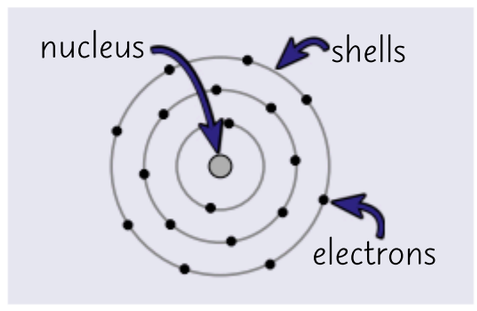


A cloud of negative electrons surrounds the nucleus, but most of the atom is empty space, so most alpha particles pass straight through.

Video

Bohr's Nuclear Model Explains a Lot

1) Scientists realised that electrons in a 'cloud' around the nucleus of an atom, as Rutherford described, would be attracted to the nucleus and cause the atom to collapse.
2) Niels Bohr came up with a new nuclear model of the atom. He suggested that electrons orbit the nucleus in fixed shells and aren't found anywhere in between. Each shell is a fixed distance from the nucleus.
3) Bohr's theory of atomic structure was supported by many experiments and it helped to explain lots of other scientists' observations at the time.



Further Experiments Showed the **Existence** of **Protons**

Further experiments by Rutherford and others showed that the nucleus can be divided into smaller particles, which each have the same charge as a hydrogen nucleus. These particles were named protons.

About 20 years after scientists accepted that atoms have nuclei, James Chadwick carried out an experiment which provided evidence for neutral particles in the nucleus. His discovery of neutrons resulted in a model of the atom pretty close to the modern day accepted version (see page 3).

Our understanding of what an atom looks like has changed

Our understanding of the atom has gone through many stages thanks to other people's work being built upon with new evidence and new predictions made. A fine example of the scientific method — there's more about that on page 139.

Electronic Structure

Quick Quiz

*The fact that electrons occupy '**shells**' around the nucleus is what causes the whole of chemistry. It's ace.*

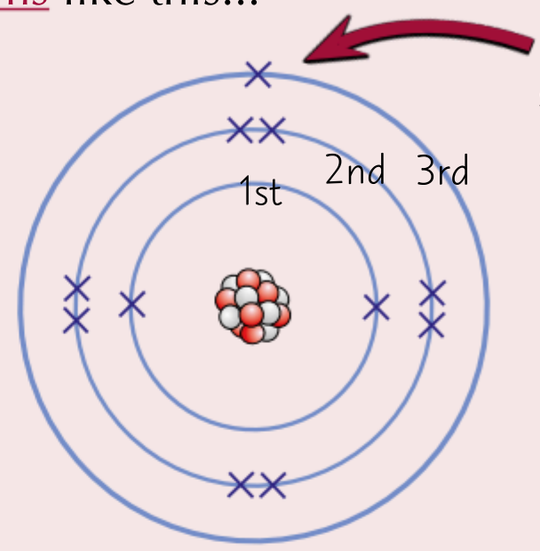
Electron Shell **Rules**:

1) Electrons always occupy shells (sometimes called energy levels).
2) The lowest energy levels are always filled first — these are the ones closest to the nucleus.
3) Only a certain number of electrons are allowed in each shell:

1st shell	2nd shell	3rd shell
2	8	8

4) Atoms are happier when they have full electron shells — like the noble gases in Group 0.
5) In most atoms, the outer shell is not full and this makes the atom want to react to fill it.

Electronic structures can be shown as diagrams like this...



...or as numbers like this: 2, 8, 1

Both of these show the element sodium.

Follow the Rules to **Work Out** Electronic Structures

You need to be able to work out the electronic structures for the first 20 elements of the periodic table.

EXAMPLE

What is the electronic structure of nitrogen?

1) Nitrogen's atomic number is 7. This means it has 7 protons, so it must have 7 electrons.
2) Look at the Electron Shell Rules: the first shell can only take 2 electrons and the second shell can take a maximum of 8 electrons.
   So the electronic structure for nitrogen must be 2, 5.

EXAMPLE

What is the electronic structure of magnesium?

1) Magnesium's atomic number is 12. This means it has 12 protons, so it must have 12 electrons.
2) Look at the Electron Shell Rules: the first shell can only take 2 electrons and the second shell can take a maximum of 8 electrons, so the third shell must also be partially filled.
   So the electronic structure for magnesium must be 2, 8, 2.

Here are some more examples of electronic structures:

H Hydrogen	He Helium	Li Lithium	C Carbon	Ne Neon	Ca Calcium
1 Proton no. = 1	2 Proton no. = 2	2,1 Proton no. = 3	2,4 Proton no. = 6	2,8 Proton no. = 10	2,8,8,2 Proton no. = 20

Electron shells — one of the most important ideas in chemistry

Q2 Video Solution

Electronic structures may seem a bit complicated at first but once you learn the rules, it's a piece of cake. And just like cake, you'll never regret going back for some more. Better get practising.

Q1 Give the electronic structure of aluminium (atomic number = 13). [1 mark]

Q2 Give the electronic structure of argon (atomic number = 18). [1 mark]

Development of the Periodic Table

Quick Quiz

*It's shocking I know, but we haven't always known as much about chemistry as we do nowadays. Early chemists looked to try and explain **patterns** in the properties of elements to get a bit of understanding.*

In the **Early 1800s** Elements were **Arranged** by **Atomic Weight**

Until quite recently, there were two obvious ways to categorise elements:

Their physical and chemical properties.

Their atomic weight.

Atomic weight is equivalent to what we would call "relative atomic mass" today.

1) Back in the day, nobody knew that protons, neutrons or electrons existed. (It was only after protons were discovered in the 20th century that scientists realised the best way of arranging the elements was to put them in order of their atomic number.)
2) The only thing scientists could measure was atomic weight, so they tried arranging the known elements in order of atomic weight. When this was done, a periodic pattern was noticed in the properties of the elements. This is where the name 'periodic table' comes from — ta da...
3) However, early periodic tables were not complete and putting the elements in the exact order of their atomic weights meant that some elements were placed in the wrong group.

Dmitri Mendeleev Left Gaps and **Predicted** New Elements

In 1869, Dmitri Mendeleev took 50 known elements and arranged them into his Table of Elements.
He overcame some of the problems of earlier tables by leaving gaps and swapping a few elements around:

H																	
Li	Be											B	C	N	O	F	
Na	Mg											Al	Si	P	S	Cl	
K	Ca	*	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	*	*	As	Se	Br	
Rb	Sr	Y	Zr	Nb	Mo	*	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	
Cs	Ba	*	*	Ta	W	*	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi			

Leaving gaps

Mendeleev left some gaps in his table.
He did this to make sure that elements with similar properties stayed in the same groups. These gaps represented undiscovered elements. Mendeleev was able to predict what their properties might be.

When new elements were found, their properties fitted Mendeleev's predictions, helping to confirm his ideas. For example, he made really good predictions about the chemical and physical properties of an element he called ekasilicon, which we know today as germanium.

Swapping elements

Mendeleev put the elements mainly in order of atomic weight — but he switched the order if the properties suggested it should be changed. For example, iodine (I) has a smaller atomic weight than tellurium (Te), but Mendeleev swapped them over to keep iodine in a group of elements with similar properties.

The discovery of isotopes (see p.4) in the early 20th century confirmed that Mendeleev was right to not place elements in a strict order of atomic weight but to also take account of their properties. Isotopes of the same element have different masses but the same atomic number and the same chemical properties — so they occupy the same position on the periodic table.

By leaving gaps in the table Mendeleev had the right idea

Mendeleev's table was an important step in getting to the periodic table we use today. But before you flick over for more on that, make sure you can describe what Mendeleev did to overcome some of the problems with earlier periodic tables.

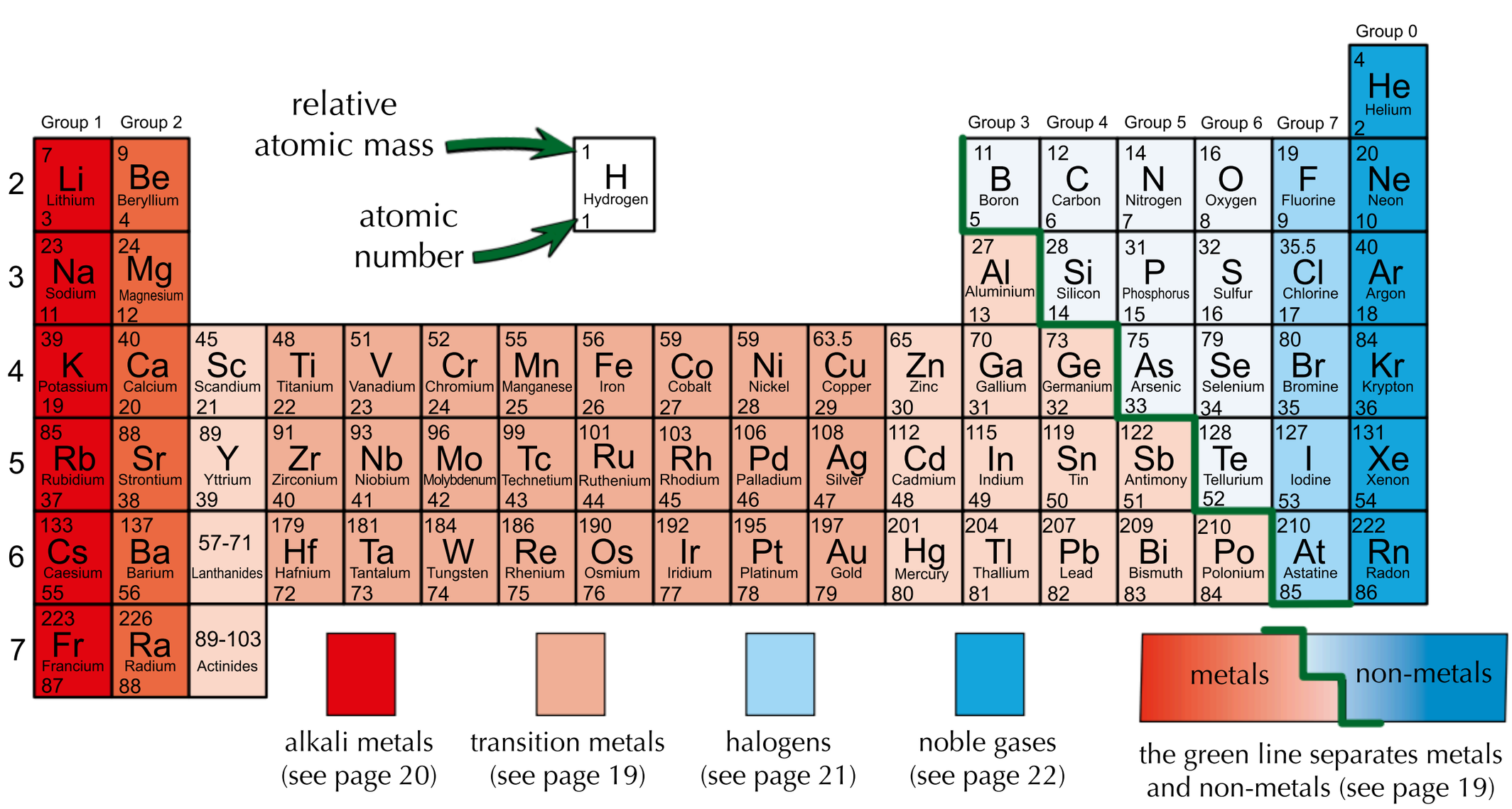
The Modern Periodic Table

Quick Quiz

*Chemists were getting pretty close to producing something that you might **recognise** as a periodic table. The big breakthrough came when the **structure** of the **atom** was understood a bit better.*

The Periodic Table Helps you to See Patterns in Properties

1) There are about 100 elements in total. All materials are made up of elements.
2) The periodic table is a useful way of organising all these elements.
In the periodic table the elements are laid out in order of increasing atomic (proton) number:



3) Arranging the elements like this means there are repeating patterns in the properties of the elements.
4) Elements with similar properties form columns. These vertical columns are called groups.

- The group number tells you how many electrons there are in the outer shell.
- For example, Group 1 elements all have one electron in their outer shell and Group 7 all have seven electrons in their outer shell. This is useful as the way atoms react depends on the number of electrons in their outer shell. So elements in the same group react in similar ways.
- If you know the properties of one element, you can predict properties of other elements in that group. For example the Group 1 elements are Li, Na, K, Rb, Cs and Fr — they're all metals and they react in a similar way (see page 20).
- You can use trends in the groups to make predictions about the reactivity of elements too. E.g. in Group 7 reactivity decreases as you go down the group (see page 21).

The exception to this is Group 0. Helium has two electrons in its outer shell. All the other Group 0 elements have eight.

5) The rows are called periods. Each new period represents another full shell of electrons.

The modern periodic table is vital for understanding chemistry

The Periodic Table is a good example of how science progresses. A scientist has a basically good (though incomplete) hypothesis (see p.139). Other scientists question it and come up with new evidence. The hypothesis may be modified, or even scrapped, to take account of this evidence. Only when all of the evidence supports a hypothesis will it be accepted.

Warm-Up & Exam Questions

The last few pages have been pretty tricky — there's a lot of new information to learn.
Luckily, here are some questions to get your head around, to help you test your understanding.

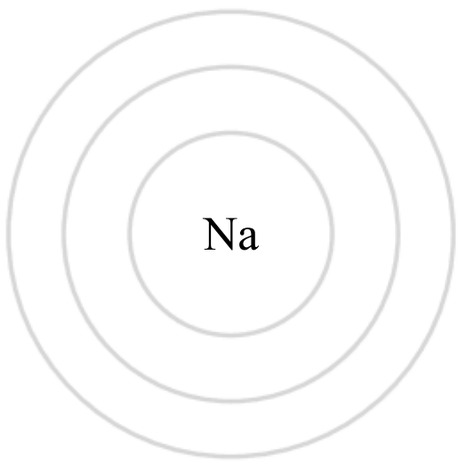
Warm-Up Questions

1) Describe J J Thomson's model of the atom.
2) How many electrons can be held in the following:
   a) the first shell of an atom
   b) the second shell of an atom
3) What discovery supported Mendeleev's decision not to place elements in order of atomic weight?

Exam Questions

1 Sodium has an atomic number of 11. GRADE 4-6 GRADE

1.1 Complete the dot and cross diagram to show the electron structure of sodium.



[1 mark]

1.2 Sodium is in Group 1.
Name another element that would have the same number of outer shell electrons.

[1 mark]

1.3 How many electrons does sodium need to lose so that it has a full outer shell?

[1 mark]

2 The periodic table contains all the known elements arranged in order. GRADE 6-7 GRADE

2.1 How were elements generally ordered in early periodic tables?

[1 mark]

2.2 How are the elements arranged in the modern periodic table?

[1 mark]

2.3 Calcium and strontium are both found in Group 2 of the periodic table.
Explain why these two elements react in similar ways.

[1 mark]

3* Describe how the theory of atomic structure has changed over time. GRADE 7-9 GRADE

[6 marks]

Metals and Non-Metals

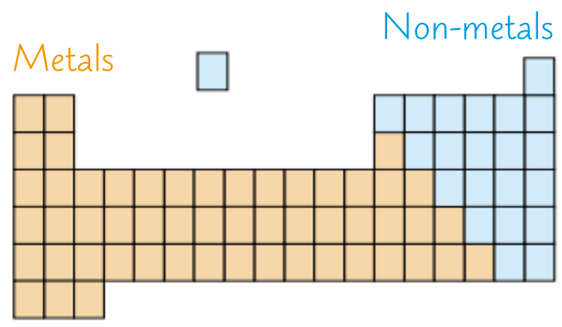
Quick Quiz

*I can almost guarantee you'll touch something **metallic** today — that's how important metals are to modern life.*

Most Elements in the Periodic Table are **Metals**

Metals are elements which form positive ions when they react.
They're towards the bottom and to the left of the periodic table.

Non-metals don't generally form positive ions when they react.
Non-metals are at the far right and top of the periodic table.



The **Electronic Structure** of Atoms Affects How They Will React

Atoms generally react to form a full outer shell. They do this by losing, gaining or sharing electrons.

- Metals at the left of the periodic table don't have many outer electrons to remove.
- Metals at the bottom of the periodic table have outer electrons which are a long way from the nucleus — so they only feel a weak attraction.
- These things mean it generally takes less energy for metals to lose their outer electrons than to gain more. So metals tend to react by forming positive ions.

- Non-metals at the right of the periodic table have lots of outer electrons.
- Non-metals at the top of the periodic table have outer electrons that are close to the nucleus — so they feel a strong attraction.
- These things mean it can take lots of energy to remove the outer electrons from a non-metal. They tend to react by either sharing or gaining electrons to get a full outer shell.

Metals and **Non-Metals** Have **Different Physical Properties**

All metals have metallic bonding, so they have similar basic physical properties.

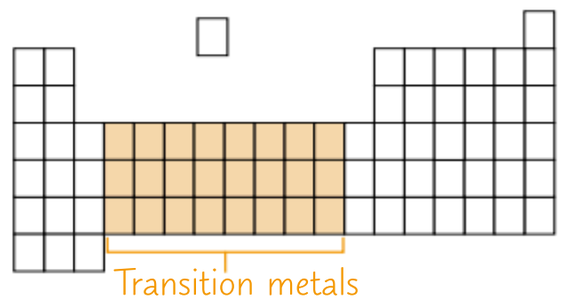
- They're strong (hard to break).
- They can be bent or hammered into different shapes (malleable).
- They're great at conducting heat and electricity.
- They generally have high boiling and melting points.

Non-metals don't have metallic bonding, so they have different properties to metals. They tend to be dull looking, brittle, aren't always solids at room temperature, don't generally conduct electricity and often have low densities.

Non-metals form a variety of different structures, so they have a wide range of chemical properties.

Transition Metals are Found Between **Group 2** and **Group 3**

1) Transition metals are in the centre of the periodic table.
2) Transition metals are typical metals. They have the properties you would expect a metal to have — they're good conductors of heat and electricity, and they're dense, strong and shiny.
3) Transition metals also have some pretty special properties...



- They can form ions with different charges.
  For example copper can form Cu^{+} and Cu^{2+} ions and cobalt can form Co^{2+} and Co^{3+} ions.
- Transition metal ions are often coloured, so compounds that contain them are colourful.
  For example, potassium chromate(VI) is yellow, and potassium manganate(VII) is purple.
- Many transition metals make good catalysts (things that speed up the rate of a reaction, see p.81).
  For example, an iron catalyst is used in the Haber process for making ammonia (p.134).

Transition metals are pretty useful elements...

Transition metals have similar properties to 'typical metals' but they also have a few more unusual characteristics. These include having multiple ions with different charges, coloured compounds, and the ability to act as catalysts.

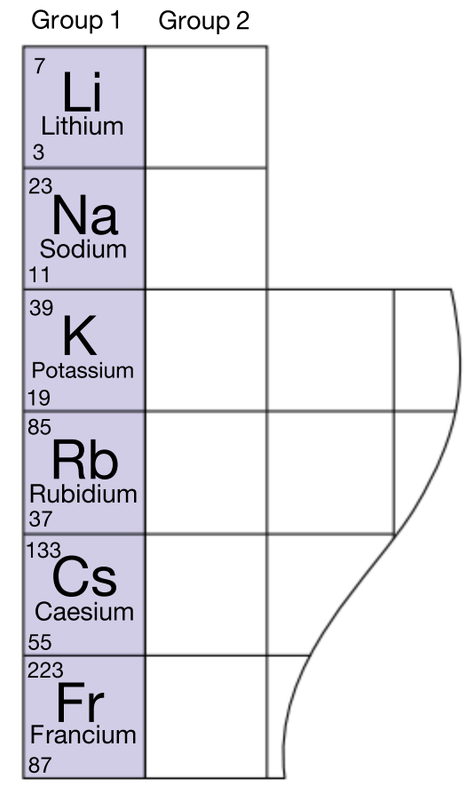
Group 1 Elements

Quick Quiz

*Group 1 elements are known as the **alkali metals**. You'll see below that as metals go, they're pretty **reactive**.*

The **Group 1** Elements are **Reactive, Soft** Metals

1) The alkali metals are lithium, sodium, potassium, rubidium, caesium and francium.
2) They all have one electron in their outer shell, which gives them similar properties — they are very reactive, soft metals that have a low density.
3) As you go DOWN Group 1, the alkali metals:
   - have HIGHER RELATIVE ATOMIC MASSES.
   - have LOWER MELTING AND BOILING POINTS.
   - become MORE REACTIVE — as you go down the group, the outer electron gets further away from the nucleus. So the attraction between the nucleus and electron decreases and it gets easier to remove.



Alkali Metals Form **Ionic Compounds** with **Non-Metals**

There's more about ionic compounds on page 28.

The Group 1 elements don't need much energy to lose their one outer electron, so they form 1+ ions very easily. They react to form ionic compounds — these compounds are generally white solids that dissolve in water to form colourless solutions.

Here's a bit more about the reactions of Group 1 metals with water, chlorine and oxygen:

Water

- When Group 1 metals are put in water, they react vigorously to produce hydrogen gas and metal hydroxides — compounds that dissolve in water to produce alkaline solutions.

$$2Na_{(s)} + 2H_2O_{(l)} \rightarrow 2NaOH_{(aq)} + H_{2\,(g)}$$
sodium + water → sodium hydroxide + hydrogen

- The more reactive (lower down in the group) an alkali metal is, the more violent the reaction — the reaction with potassium releases enough energy to ignite the hydrogen.

Chlorine

- Group 1 metals react vigorously when heated in chlorine gas to form metal chloride salts.
- As you go down the group, reactivity increases so the reaction with chlorine gets more vigorous.

$$2Na_{(s)} + Cl_{2(g)} \rightarrow 2NaCl_{(s)}$$
sodium + chlorine → sodium chloride

Oxygen

Group 1 metals can react with oxygen to form a metal oxide. Different types of oxide will form depending on the metal:

- Lithium reacts to form lithium oxide (Li_2O).
- Sodium reacts to form a mixture of sodium oxide (Na_2O) and sodium peroxide (Na_2O_2).
- Potassium reacts to form a mixture of potassium peroxide (K_2O_2) and potassium superoxide (KO_2).

These reactions are why Group 1 metals tarnish — they react with oxygen in the air to form a dull, metal oxide layer.

Group 1 Metals Have **Different Properties** to Transition Metals

1) Group 1 metals are much more reactive than transition metals — for example, they react more vigorously with water, oxygen and Group 7 elements.
2) They're also less dense, strong and hard than the transition metals, and have much lower melting points. E.g. manganese melts at 2000 °C, but sodium melts at 98 °C.

The properties of Group 1 metals change down the group

Reactions of alkali metals need safety precautions — some of them catch fire in water and can even explode.

Q1 Write a word equation for the reaction between lithium and water. [1 mark]

Q1 Video Solution

Group 7 Elements

Quick Quiz

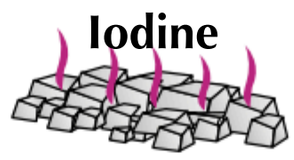
*The Group 7 elements are known as the **halogens**. Halogen means **salt maker** — you find them in lots of salts...*

The **Halogens** are All **Non-Metals** with **Coloured Vapours**:

very reactive
yellow gas

fairly reactive
dense green gas

dense red-brown
volatile liquid



dark grey crystalline
solid or a purple vapour

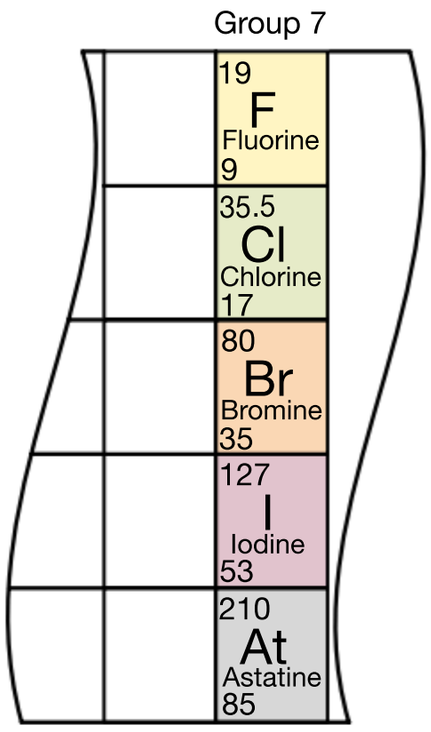
Halogens all exist as molecules which are pairs of atoms, i.e. F_2, Cl_2, Br_2, I_2.

Learn These **Trends**:

As you go DOWN Group 7, the halogens:

- have HIGHER RELATIVE ATOMIC MASSES.
- have HIGHER MELTING AND BOILING POINTS.
- become LESS REACTIVE — as you go down the group, the outer electrons get further from the nucleus. So the attraction between the electrons and the nucleus decreases and it gets harder to attract an extra electron.

This trend also occurs in Group 0 elements — see the next page for more on why this happens.



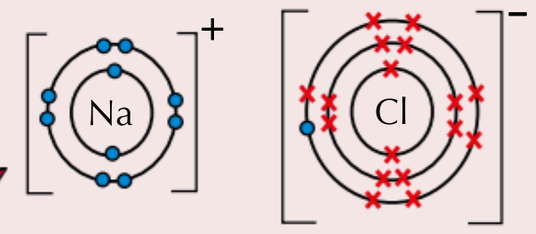
You can use these trends to predict properties of the halogens — e.g. iodine must have a higher boiling point than chlorine as it's further down Group 7.

Halogens **React** with **Metals** and **Non-Metals**

All the Group 7 elements react in similar ways, because they all have seven electrons in their outer shell:

Halogen atoms can share electrons with other non-metals to get a full outer shell — this is called covalent bonding (see page 30). For example HCl, Br_2 and CCl_4 all contain covalent bonds. Compounds that form when halogens react with non-metals have simple molecular structures (see p.31).

The halogens form 1– ions called halides (F^-, Cl^-, Br^- and I^-) when they bond with metals, for example Na^+Cl^- or $Mg^{2+}Br^-_2$.
The compounds that form have ionic bonds (see p.27).
This diagram shows the bonding in sodium chloride, NaCl.

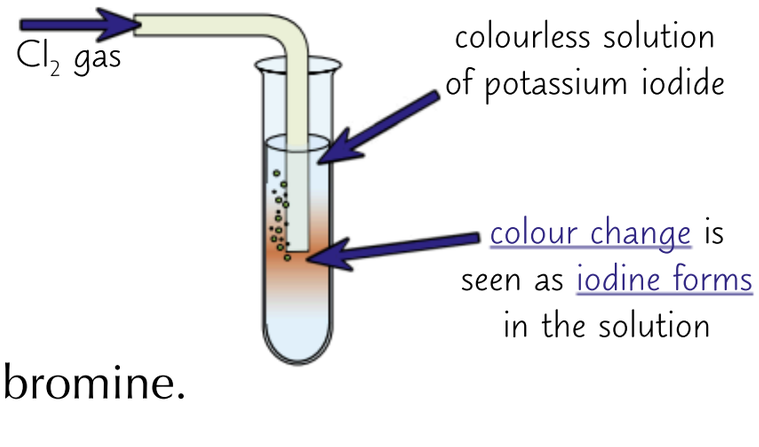


More Reactive Halogens Will **Displace** Less Reactive Ones

Chlorine can displace bromine or iodine from an aqueous solution of its salt (a bromide or iodide):

$$Cl_{2\,(g)} + 2KI_{(aq)} \rightarrow I_{2\,(aq)} + 2KCl_{(aq)}$$
pale green / colourless → brown / colourless

$$Cl_{2\,(g)} + 2KBr_{(aq)} \rightarrow Br_{2\,(aq)} + 2KCl_{(aq)}$$
pale green / colourless → orange / colourless



Bromine can also displace iodine because of the trend in reactivity, but it can't displace chlorine. Iodine can't displace either chlorine or bromine.

Halogens all have different reactivities

You don't have to be a mind-reader to be able to guess what they could ask you about halogens in the exam. Something about displacement reactions seems possible, and make sure you learn those trends too.

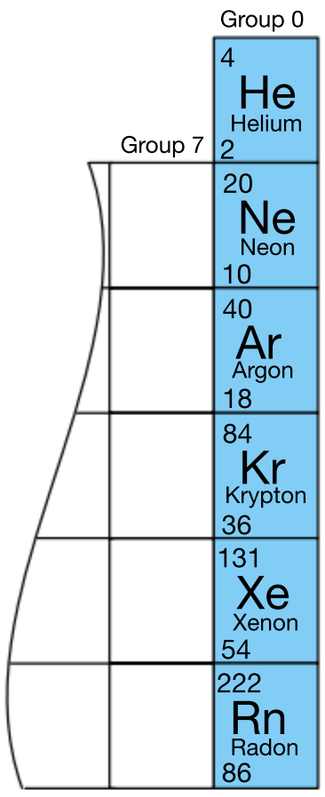
Q1 Give the balanced symbol equation for the displacement reaction between bromine and sodium iodide. [1 mark]

Q1 Video Solution

Group 0 Elements

Quick Quiz

*The **noble gases don't react** with very much and you can't even see them — making them, well, a bit **dull** really.*



Group 0 Elements are All Inert, Colourless Gases

1) Group 0 elements are called the noble gases and include the elements helium, neon and argon (plus a few others).
2) They all have eight electrons in their outer energy level, (apart from helium which has two), giving them a full outer shell.
3) As their outer shell is energetically stable they don't need to give up or gain electrons to become more stable. This means they are more or less inert — they don't react with much at all.
4) They exist as monatomic gases — single atoms not bonded to each other.
5) All the Group 0 elements are colourless gases at room temperature.
6) As they are inert they're also non-flammable — they won't set on fire.

Helium only has electrons in the first shell, which can only fit 2 electrons.

There are Patterns in the Properties of the Noble Gases

1) The boiling points of the noble gases increase as you go down the group, along with increasing relative atomic mass.
2) The increase in boiling point is due to an increase in the number of electrons in each atom leading to greater intermolecular forces between them which need to be overcome. There's more on intermolecular forces for small molecules on page 31.
3) In the exam you may be given the boiling point of one noble gas and asked to estimate the value for another one. So make sure you know the pattern.

Noble Gas
helium
neon
argon
krypton
xenon
radon

Increasing boiling point

EXAMPLE

Neon is a gas at 25 °C. Predict what state helium is in at this temperature.

Helium has a lower boiling point than neon, because it is further up the group.

So, helium must also be a gas at 25 °C.

EXAMPLE

Radon has a boiling point of –62 °C. Krypton has a boiling point of –153 °C. Predict the boiling point of xenon.

MATHS SKILLS

Xenon comes between radon and krypton in the group. So you can predict that its boiling point will be about halfway between their boiling points.

Add the boiling points of radon and krypton: (–153) + (–62) = –215

Then divide this by two: –215 ÷ 2 = –107.5 = –108 °C to the nearest whole number

So, xenon should have a boiling point of about –108 °C.

The actual boiling point of xenon is –108 °C — just as predicted. Neat!

Just like other groups of elements, the noble gases follow patterns...

Although they're unreactive and colourless, the noble gases have some unique properties that actually make them pretty useful. It took a while to discover them — but now we know all about them, including the trend in their boiling points.

Warm-Up & Exam Questions

These questions are all about the groups of the periodic table that you need to know about. Treat the exam questions like the real thing — don't look back through the book until you've finished.

Warm-Up Questions

1) What kind of ions do metals form?
2) Where are the transition metals found in the periodic table?
3) During a reaction, an atom gains electrons to end up with a full outer shell. Is the atom likely to be a metal or a non-metal?
4) Which product is formed when potassium reacts with chlorine?
5) Do halide ions have a positive or a negative charge?
6) In which group of the periodic table are the noble gases?

Exam Questions

1 **Table 1** shows some of the physical properties of four of the halogens. GRADE 4-6 GRADE

Table 1

Halogen	Properties			
	Atomic number	Colour	Physical state at room temperature	Reactivity
Fluorine	9	yellow	gas	.....
Chlorine	17	green	gas	.....
Bromine	35	red-brown	liquid	.....
Iodine	53	dark grey	solid	.....

1.1 **Table 1** has a column for reactivity. Write an **X** in the row of the halogen with the **highest** reactivity and a **Y** in the row of the halogen with the **lowest** reactivity.

[2 marks]

1.2 Which halogen in **Table 1** has the highest melting point?

[1 mark]

2 **Figure 1** shows the periodic table. GRADE 4-6 GRADE

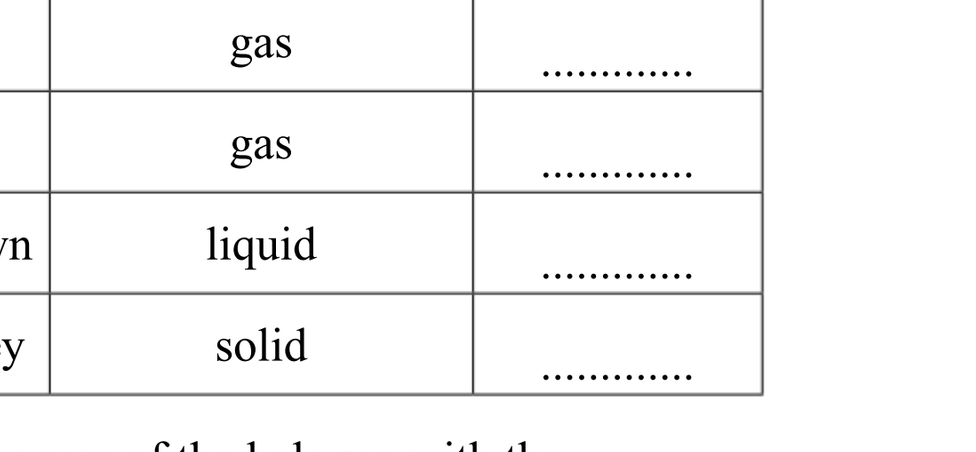
2.1 Element **X** is found in the centre of the periodic table. What name is given to the elements found in this part of the table?

[1 mark]

2.2 Element **Y** does not conduct electricity. Predict whether element **Y** will be found to the left or the right of line **A** in **Figure 1**. Explain your answer.

[2 marks]

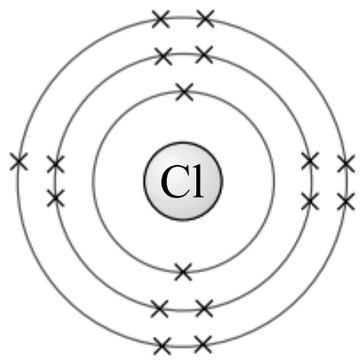
Figure 1



Exam Questions

3 Chlorine is a Group 7 element.
Its electron arrangement is shown in the diagram below.

GRADE 6-7 GRADE



3.1 Chlorine is very reactive and forms compounds with metals.
What type of bonds form between chlorine and metals?

[1 mark]

When chlorine is bubbled through potassium iodide solution a reaction occurs.
The equation below shows the reaction.

$$Cl_{2(g)} + 2KI_{(aq)} \rightarrow I_{2(aq)} + 2KCl_{(aq)}$$

3.2 State what type of reaction this is.

[1 mark]

3.3 Which is the less reactive halogen in this reaction? Explain your answer.

[2 marks]

3.4 Chlorine gas can also react with potassium bromide.
Using your knowledge of Group 7 elements, predict the products of this reaction.
Give an explanation for your answer.

[3 marks]

3.5 None of the elements in Group 0 will react with potassium iodide or potassium bromide.
Using your knowledge of the electronic structure of the Group 0 elements,
explain why no reaction occurs.

[2 marks]

4 Group 1 elements are metals.
They include lithium, sodium and potassium.

GRADE 6-7 GRADE

4.1 Explain why the Group 1 elements react vigorously with water.

[1 mark]

4.2 The Group 1 elements all react with water at a different rate. Explain why this is.

[3 marks]

4.3 Give the two products formed when potassium reacts with water.

[2 marks]

4.4 During these reactions a solution is formed.
Is this solution acidic, neutral or alkaline?

[1 mark]

Revision Summary Test for Topic 1b

That's the Periodic Table done and dusted. To make sure you've got your head round it, you can:

- Tackle the revision summary test below, or scan the QR code to do it online.
  The questions are hard, but they'll show you how well you really know your stuff.
- Track your progress online and see which areas need more work.
- Compare your answers with sample answers for the test here: cgpbooks.co.uk/Base

Summary Test

The History of the Atom (p.14)

1) Explain how Thomson's discovery of the electron changed the model of the atom.
2) In Rutherford's alpha particle scattering experiments, what did he expect to see if the plum pudding was correct? What did he actually see? How did it support the idea that the atom had a nucleus?
3) Explain how Bohr's nuclear model of the atom was different to Rutherford's nuclear model.
4) Rutherford showed that the nucleus contained positively charged particles. What are they called?
5) Who first carried out an experiment that proved the existence of the neutron?

Electronic Structure (p.15)

6) When electrons are filling energy levels (shells) in an atom, which ones are filled first?
7) How many electrons can you fit in: a) the 1st shell, b) the 2nd shell, c) the 3rd shell?
8) Work out the electronic structure of silicon (atomic number = 14).

The Periodic Table (p.16-17)

9) How were the elements arranged in early periodic tables?
10) When he was making his periodic table, Mendeleev left gaps and swapped the order of some elements.
    a) Why did he do these things?
    b) How did later discoveries help to prove that his ideas were right?
11) How are the elements in the modern periodic table arranged? Why is it called the 'periodic' table?
12) Why do elements in the same group of the periodic table have similar properties?
13) What are the rows in the periodic table called? What does each row represent, in terms of electrons?

Metals and Non-Metals (p.19)

14) Where do you find metals in the periodic table? How about non-metals?
15) Why do metals tend to react by losing electrons?
16) List some of the typical properties of metals. Do the same for non-metals.
17) Give three special properties of transition metals, which are not generally shared by other metals.

Groups of the Periodic Table (p.20-22)

18) What are the Group 1 elements also known as? How many electrons are in their outer shells?
19) What happens to the reactivity of the elements as you go down Group 1? Explain why.
20) Name the products made when lithium reacts with: a) water, b) chlorine, c) oxygen.
21) Give three ways that the properties of Group 1 metals are different to transition metals.
22) Which group of the periodic table is called the halogens?
    How many electrons do they have in their outer shells?
23) Describe three trends in the properties of the halogens as you go down the group.
24) What type of compound is formed when a halogen reacts with: a) a metal, b) a non-metal?
25) What happens when you mix: a) chlorine and potassium bromide, b) iodine and potassium chloride?
26) What are the Group 0 elements also known as? Why are they usually pretty inert?
27) What is the trend in the boiling point of the elements as you go down Group 0? Explain why.

Quick Quiz

Formation of Ions

*Ions crop up all over the place in chemistry. You're gonna have to be able to explain **how** they form and predict the **charges** of simple ions formed by elements in Groups 1, 2, 6 and 7. You'd better get on...*

Ions are Made When Electrons are Transferred

- Ions are charged particles — they can be single atoms (e.g. Cl^-) or groups of atoms (e.g. NO_3^-).
- When metals form ions, they lose electrons from their outer shell to form positive ions.
- When non-metals form ions, they gain electrons into their outer shell to form negative ions.
- The number of electrons lost or gained is the same as the charge on the ion. E.g. if 2 electrons are lost the charge is 2+. If 3 electrons are gained the charge is 3–.

When atoms lose or gain electrons to form ions, all they're trying to do is get a full outer shell like a noble gas (also called a "stable electronic structure"). Atoms with full outer shells are very stable.

Remember that the noble gases are in Group 0 of the periodic table.

Groups 1 & 2 and 6 & 7 are the Most Likely to Form Ions

1) The elements that most readily form ions are those in Groups 1, 2, 6 and 7.
2) Group 1 and 2 elements are metals and they lose electrons to form positive ions (cations).
3) Group 6 and 7 elements are non-metals. They gain electrons to form negative ions (anions).
4) You don't have to remember what ions most elements form — nope, you just look at the periodic table.
5) Elements in the same group all have the same number of outer electrons. So they have to lose or gain the same number to get a full outer shell. And this means that they form ions with the same charges.

Group 1 elements form 1+ ions. Group 2 elements form 2+ ions. Group 6 elements form 2– ions. Group 7 elements form 1– ions

H																	He
Li	Be											B	C	N	O	F	Ne
Na	Mg											Al	Si	P	S	Cl	Ar
K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr
Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe
Cs	Ba	La	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn
Fr	Ra	Ac	Rf	Db	Sg	Bh	Hs	Mt	Ds	Rg							

- A sodium atom (Na) is in Group 1 so it loses 1 electron to form a sodium ion (Na^+) with the same electronic structure as neon: $Na \rightarrow Na^+ + e^-$.
- A magnesium atom (Mg) is in Group 2 so it loses 2 electrons to form a magnesium ion (Mg^{2+}) with the same electronic structure as neon: $Mg \rightarrow Mg^{2+} + 2e^-$.
- A chlorine atom (Cl) is in Group 7 so it gains 1 electron to form a chloride ion (Cl^-) with the same electronic structure as argon: $Cl + e^- \rightarrow Cl^-$.
- An oxygen atom (O) is in Group 6 so it gains 2 electrons to form an oxide ion (O^{2-}) with the same electronic structure as neon: $O + 2e^- \rightarrow O^{2-}$.

Have a look back at page 15 for how to work out electronic structures.

Metals and non-metals can both form ions

Metal atoms lose electrons and non-metal atoms gain electrons when they form ions. Either way, you need to remember that they will end up with a full outer shell of electrons and a stable electronic configuration.

Q1 Explain why simple ions often have noble gas electronic structures. [2 marks]

Q1 Video Solution

Ionic Bonding

Quick Quiz

Time to find out how particles bond together to form compounds (bet you can't wait). First up, it's ***ionic bonds****.*

Ionic Bonding — **Transfer** of Electrons

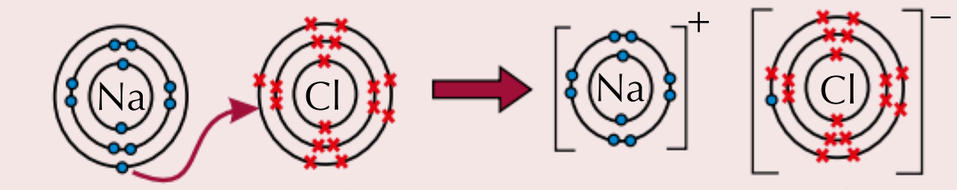
When a metal and a non-metal react together, the metal atom loses electrons to form a positively charged ion and the non-metal gains these electrons to form a negatively charged ion. These oppositely charged ions are strongly attracted to one another by electrostatic forces. This attraction is called an ionic bond.

Dot and Cross Diagrams Show How **Ionic Compounds** are **Formed**

Dot and cross diagrams show the arrangement of electrons in an atom or ion. Each electron is represented by a dot or a cross. So these diagrams can show which atom the electrons in an ion originally came from.

Sodium Chloride ($NaCl$)

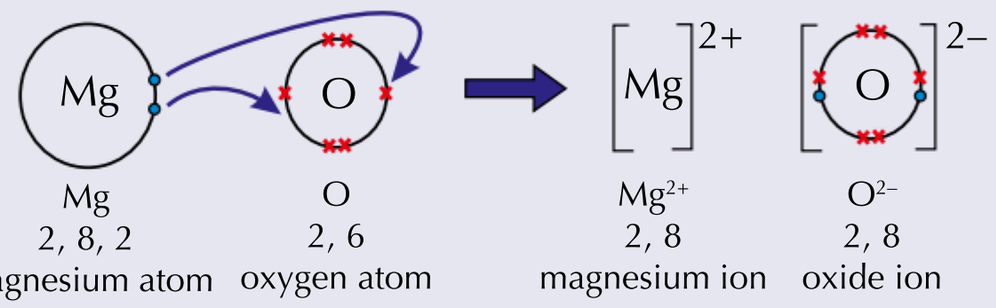
The sodium atom gives up its outer electron, becoming an Na^+ ion (see the previous page). The chlorine atom picks up the electron, becoming a Cl^- (chloride) ion.



Here, the dots represent the Na electrons and the crosses represent the Cl electrons (all electrons are really identical, but this is a good way of following their movement).

Magnesium Oxide (MgO)

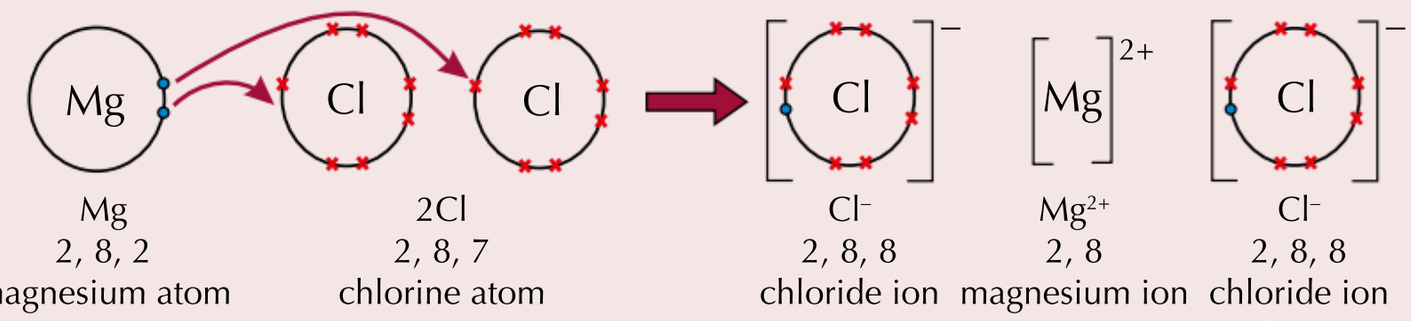
The magnesium atom gives up its two outer electrons, becoming an Mg^{2+} ion. The oxygen atom picks up the electrons, becoming an O^{2-} (oxide) ion.



Here we've only shown the outer shells of electrons on the dot and cross diagram — it makes it much simpler to see what's going on.

Magnesium Chloride ($MgCl_2$)

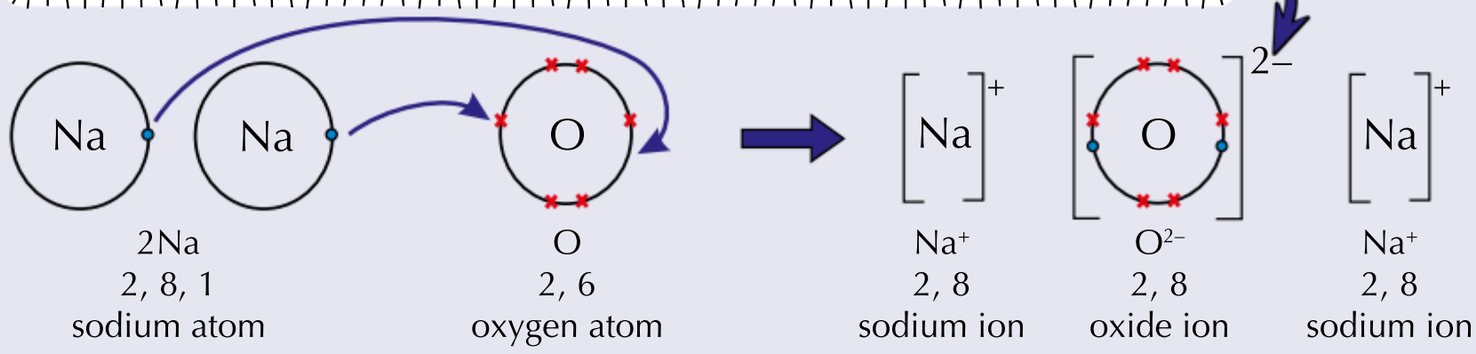
The magnesium atom gives up its two outer electrons, becoming an Mg^{2+} ion. The two chlorine atoms pick up one electron each, becoming two Cl^- ions.



All the ions in the formula are shown in the diagram, not just one of each type.

Sodium Oxide (Na_2O)

Two sodium atoms each give up their single outer electron, becoming two Na^+ ions. The oxygen atom picks up the two electrons, becoming an O^{2-} ion.



Dot and cross diagrams are useful for showing how ionic compounds are formed, but they don't show the structure of the compound, the size of the ions or how they're arranged. But hey-ho — nothing's perfect.

Show the electronic structure of ions with square brackets

Q2 Video Solution

Figuring out how ionic compounds are formed gets easier with practice, so have at go at these questions.

Q1 Describe what is represented on a dot and cross diagram by an arrow pointing from an electron shell of one atom to an electron shell of another atom. [1 mark]

Q2 Draw a dot and cross diagram to show how potassium (a Group 1 metal) and bromine (a Group 7 non-metal) form potassium bromide (KBr). [3 marks]

Quick Quiz

Ionic Compounds

*An **ionic compound** is a compound that contains **ionic bonds**. Sounds obvious, but there's a bit to come yet...*

Ionic Compounds Have A Regular Lattice Structure

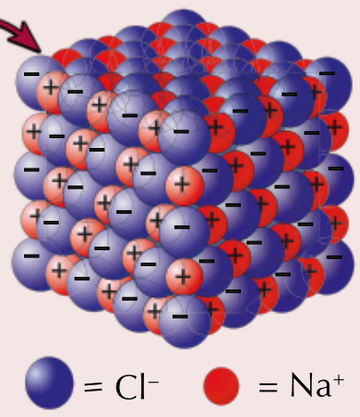
The electrostatic attraction between the oppositely charged ions is ionic bonding.

- Ionic compounds have a structure called a giant ionic lattice.
- The ions form a closely packed regular lattice arrangement and there are very strong electrostatic forces of attraction between oppositely charged ions, in all directions in the lattice.

A single crystal of sodium chloride (table salt) is one giant ionic lattice of Na^+ and Cl^- ions.

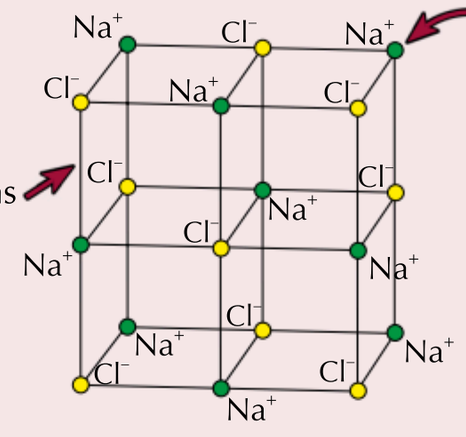
The lattice can be represented in different ways...

This model shows the relative sizes of the ions, as well as the regular pattern of an ionic crystal, but it only lets you see the outer layer of the compound.



Make sure you learn what the structure of sodium chloride looks like.

The Na^+ and Cl^- ions alternate.



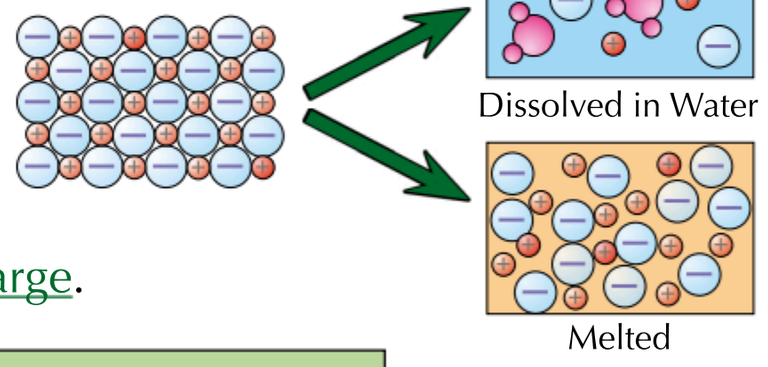
This is a ball and stick model. It shows the regular pattern of an ionic crystal and shows how all the ions are arranged. The model isn't to scale though, so the relative sizes of the ions may not be shown. Also, in reality, there aren't gaps between the ions.

E.g. the ball and stick diagram above shows 9 Na^+ and 9 Cl^-. 9:9 cancels down to 1:1, which makes the formula NaCl.

The formula for ionic compounds is an empirical formula — it tells you the simplest ratio of ions in the lattice. To find the empirical formula of an ionic compound from a diagram of its structure, count the number of each of the ions in the diagram then cancel them down to the lowest whole numbers ratio of ions.

Ionic Compounds All Have Similar Properties

- They all have high melting points and high boiling points due to the many strong bonds between the ions. It takes lots of energy to overcome this attraction.
- When they're solid, the ions are held in place, so the compounds can't conduct electricity. When ionic compounds melt, the ions are free to move and they'll carry electric charge.
- Some ionic compounds also dissolve in water. The ions separate and are all free to move in the solution, so they'll carry electric charge.



Look at Charges to Find the Formula of an Ionic Compound

You might have to work out the empirical formula of an ionic compound from its ions.
To do that you need to balance their charges so that the overall charge on the compound is zero.

EXAMPLE

Potassium oxide contains potassium and oxygen ions. What is its empirical formula?

1) Work out what charges the ions will form.
   Potassium (K) is in Group 1 so forms 1+ ions.
   Oxygen (O) is in Group 6 so forms 2– ions.
2) Balance the charges so the charge of the empirical formula is zero.
   A potassium ion has a 1+ charge, so you'll need two of them to balance out the 2– charge of an oxide ion.
   The empirical formula is K_2O.

Ionic compounds have regular lattice structures

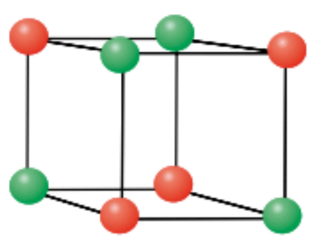
As long as you can find the charge of the ions in an ionic compound, you can work out the empirical formula. Try practising with different ionic compounds.

Q1 The structure of an ionic compound is shown on the right.

a) Predict, with reasoning, whether the compound has a high or a low melting point. [2 marks]

b) Explain why the compound can conduct electricity when molten. [1 mark]

c) Use the diagram to find the empirical formula of the compound. [1 mark]



Q1 Video Solution

Warm-Up & Exam Questions

Congratulations, you got to the end of the pages on ions — time to make sure you know them back to front...

Warm-Up Questions

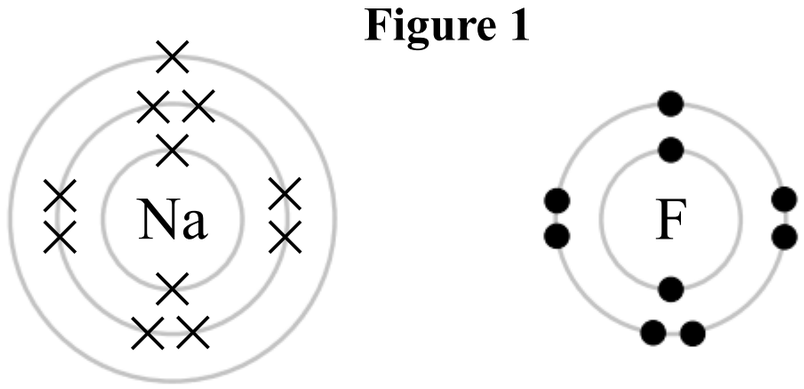
1) What is an ion?
2) What is the charge on an ion formed from a Group 2 element?
3) What is the charge on an ion formed from a Group 7 element?
4) Sodium chloride has a giant ionic structure. Does it have a high or a low boiling point?
5) Why do some ionic compounds conduct electricity when dissolved in water?
6) What is the formula of the compound containing Al^{3+} and OH^- ions only?

Exam Questions

1 **Figure 1** shows the electronic structures of sodium and fluorine.

GRADE 4-6 GRADE

Figure 1



Sodium Fluorine

1.1 Describe what will happen when sodium and fluorine react, in terms of electrons.

[2 marks]

1.2 When sodium and fluorine react they form an ionic compound.
Describe the structure of an ionic compound

[3 marks]

2 When lithium reacts with oxygen it forms the ionic compound Li_2O.

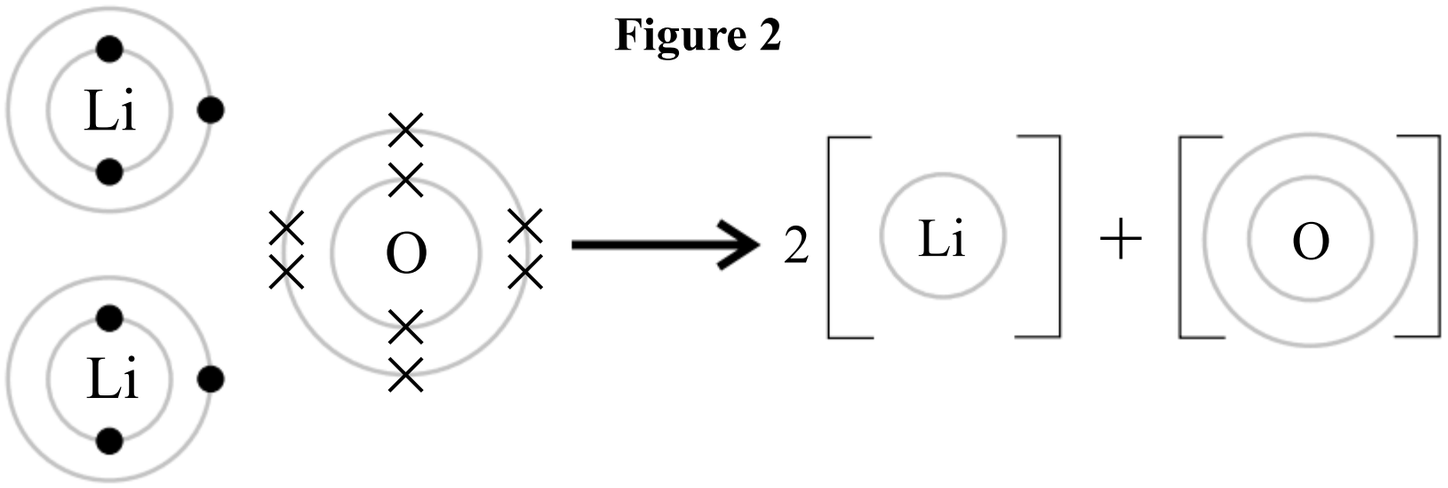
GRADE 6-7 GRADE

2.1 Name the compound formed.

[1 mark]

2.2 Complete **Figure 2** below using arrows to show how the electrons are transferred when Li_2O is formed.
Show the electron arrangements and the charges on the ions formed.

Figure 2



[3 marks]

2.3 Explain why Li_2O conducts electricity when molten.

[2 marks]

2.4 Lithium forms an ionic compound with chlorine.
What is the formula of this compound? Explain why this is.

[2 marks]

Quick Quiz

Covalent Bonding

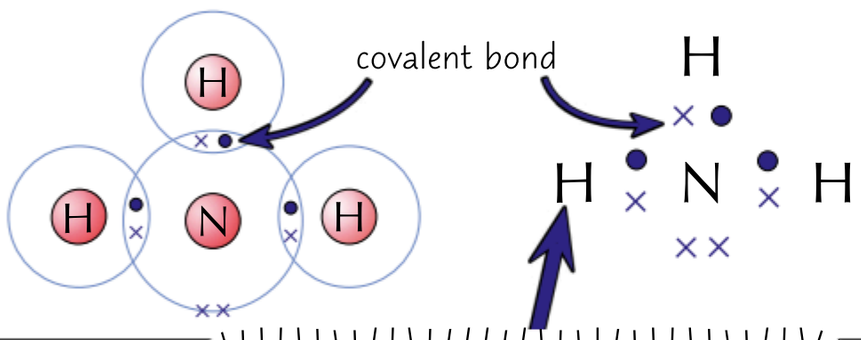
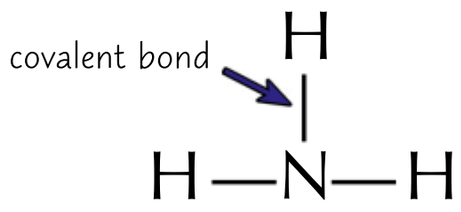
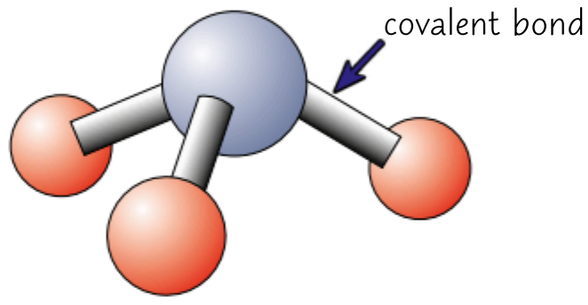
*Ionic bonding has its attractions, but enough about that for now. Up next, it's **covalent** bonds...*

Covalent Bonds — **Sharing** Electrons

1) When non-metal atoms bond together, they share pairs of electrons to make covalent bonds.
2) The positively charged nuclei of the bonded atoms are attracted to the shared pair of electrons by electrostatic forces, making covalent bonds very strong.
3) Atoms only share electrons in their outer shells (highest energy levels).
4) Each single covalent bond provides one extra shared electron for each atom.
5) Each atom involved generally makes enough covalent bonds to fill up its outer shell. Having a full outer shell gives them the electronic structure of a noble gas, which is very stable.
6) Covalent bonding happens in compounds of non-metals (e.g. H_2O) and in non-metal elements (e.g. Cl_2).

There are **Different Ways** of **Drawing** Covalent Bonds

Here are three ways of drawing covalent compounds, using ammonia (NH_3) as an example:

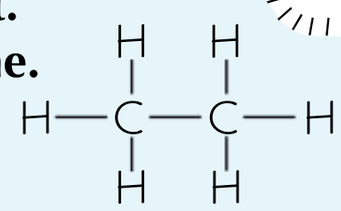
Dot and Cross Diagrams	Displayed Formula	3D Model
Nitrogen has five outer electrons. So it needs to form three covalent bonds to make up the extra three electrons needed to fill its outer shell:  You don't have to draw the orbitals. The important thing is that you get all the dots and crosses in the right places. • Electrons drawn in the overlap between the outer orbitals of two atoms are shared between those atoms. • Useful for showing which atoms the electrons in a covalent bond come from. • Doesn't show the relative sizes of the atoms, or how the atoms are arranged in space.	 • Shows covalent bonds as single lines between atoms. • Great way of showing how atoms are connected in large molecules. • Doesn't show the 3D structure of the molecule, or which atoms the electrons in the covalent bond have come from.	 • Shows the atoms, the covalent bonds and their arrangement in space next to each other. • Can get confusing for large molecules where there are lots of atoms to include. • Doesn't show where the electrons in the bonds have come from either.

You can find the molecular formula of a simple molecular compound from any of these diagrams by counting up how many atoms of each element there are.

A molecular formula shows you how many atoms of each element are in a molecule.

EXAMPLE

A diagram of the molecule ethane is shown on the right. Use the diagram to find the molecular formula of ethane.



In the diagram, there are two carbon atoms and six hydrogen atoms.
So the molecular formula is C_2H_6.

Dot and cross diagrams can be used to show covalent bonds

There are dot and cross diagrams for more covalent molecules on the next page, but make sure you can draw the different diagrams that can be used to show the bonding in ammonia first.

Q1 Draw a dot and cross diagram to show the bonding in a molecule of ammonia (NH_3). [2 marks]

Q1 Video Solution

Simple Molecular Substances

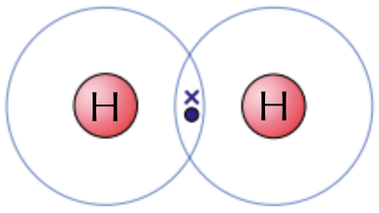
Quick Quiz

*These molecules might be **simple**, but you've still gotta know about them. I know, the world is a cruel place.*

Learn These Examples of Simple Molecular Substances

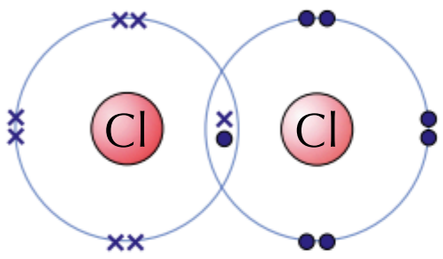
Simple molecular substances are made up of molecules containing a few atoms joined together by covalent bonds. Here are some common examples that you should know...

Hydrogen, H_2



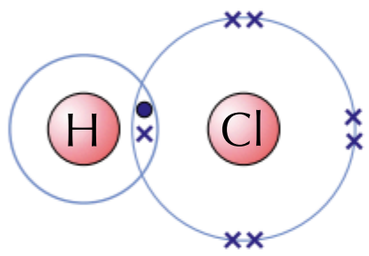
- H atoms have just one electron.
- They need one more to complete the first shell, so form single covalent bonds.

Chlorine, Cl_2



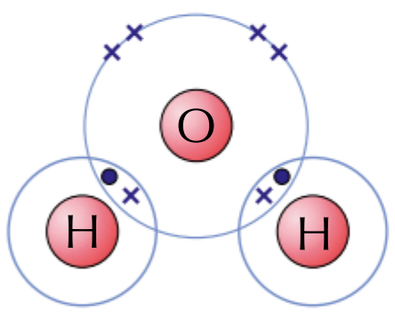
- Cl needs one more electron to complete its outer shell.
- Two Cl atoms share one pair of electrons.
- They form a single bond.

Hydrogen Chloride, HCl



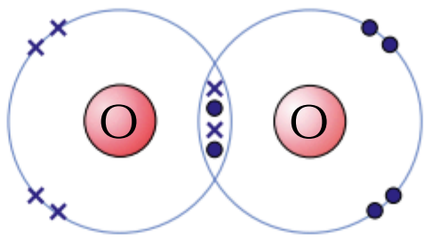
- Like in H_2 and Cl_2, each atom needs one electron to complete its outer shell.
- They form a single bond.

Water, H_2O



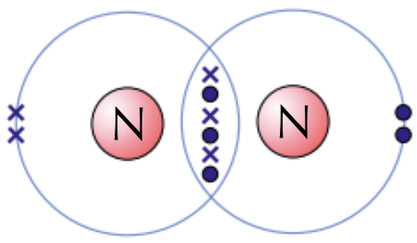
- The oxygen shares a pair of electrons with each H atom.
- Forms two single covalent bonds — one with each H atom.

Oxygen, O_2



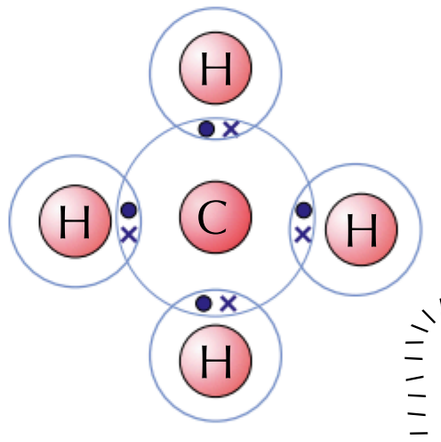
- Each O atom needs two more electrons to complete its outer shell.
- The two O atoms share two pairs of electrons.
- This is a double covalent bond.

Nitrogen, N_2



- N atoms need three more electrons to complete their outer shells.
- The two N atoms share three pairs of electrons between them.
- This is a triple bond.

Methane, CH_4

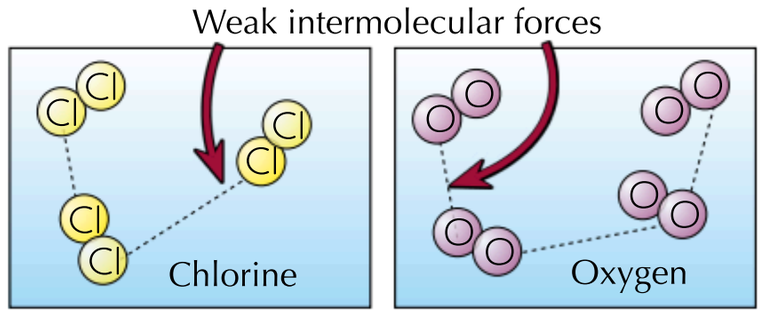


- The C atom has four outer electrons (half a full shell).
- Forms four single covalent bonds with H atoms to fill its outer shell.

Make sure you can also draw the dot and cross diagram of ammonia, NH_3, which is on the previous page.

Properties of Simple Molecular Substances

- Substances containing covalent bonds usually have simple molecular structures, like the examples above.
- The atoms within the molecules are held together by very strong covalent bonds. By contrast, the forces of attraction between these molecules are very weak.



- This means most molecular substances are gases or liquids at room temperature — to melt or boil a simple molecular compound, you only need to break the feeble intermolecular forces and not the covalent bonds. So the melting and boiling points are very low, because the molecules are easily parted from each other.
- As molecules get bigger, the strength of the intermolecular forces increases, so more energy is needed to break them, and the melting and boiling points increase.
- They don't conduct electricity — they aren't charged, so there are no free electrons or ions.

Covalent bonding involves sharing electrons

You might be asked to draw a dot and cross diagram for a simple molecule in the exam. The molecules shown on this page are good examples to learn, so make sure you're familiar with them (oh, and ammonia is handy too).

Warm-Up & Exam Questions

The questions on this page are all about covalent bonding. Go through them and if you have any problems, make sure you look back at the relevant pages again until you've got to grips with it.

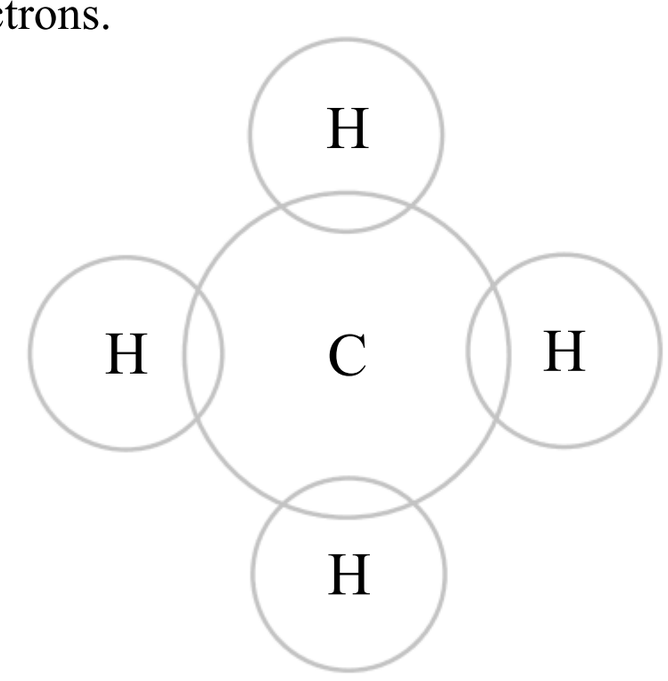
Warm-Up Questions

1) What is a covalent bond?
2) How many covalent bonds does a molecule of nitrogen have?
3) In which states are most simple molecular substances at room temperature?
4) Which forces are stronger in simple molecular substances — covalent bonds or intermolecular forces?
5) What forces need to be overcome to boil a simple molecular compound?

Exam Questions

1 Methane is a covalently bonded molecule with the formula CH_4. GRADE 4-6 GRADE

Complete the dot and cross diagram for the methane molecule.
Show only the outer electrons.



[2 marks]

2 Dot and cross diagrams can be used to show the position of electrons in covalent molecules. GRADE 4-6 GRADE

2.1 Draw a dot and cross diagram for oxygen (O_2). Only show the outer electrons.

[2 marks]

2.2 Nitrogen is in Group 5 of the periodic table.
How many bonds does it need to make to gain a full outer shell?

[1 mark]

3 Hydrogen chloride is a simple molecular substance. GRADE 6-7 GRADE

3.1 Explain why hydrogen chloride has poor electrical conductivity.

[1 mark]

3.2 Explain how the atoms are held together in a molecule of hydrogen chloride.

[2 marks]

3.3 A molecule of hydrogen chloride has a stronger bond than a molecule of chlorine (Cl_2).
However, hydrogen chloride boils at –85 °C, whereas chlorine boils at –34 °C.
Suggest and explain why chlorine has a higher boiling point than hydrogen chloride.

[3 marks]

Polymers and Giant Covalent Structures

Quick Quiz

Simple molecular substances aren't the only compounds held together by covalent bonds.
***Polymers** and **giant covalent structures** also contain covalent bonds — read on to find out more about them.*

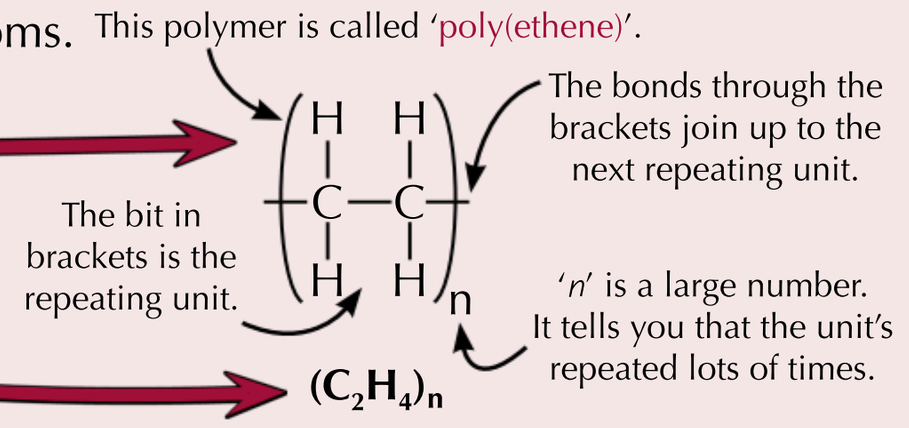
Polymers Are **Long Chains** of **Repeating** Units

1) In a polymer, lots of small units are linked together to form a long molecule that has repeating sections.
2) All the atoms in a polymer are joined by strong covalent bonds.
3) The intermolecular forces between polymer molecules are larger than between simple covalent molecules, so more energy is needed to break them. This means most polymers are solid at room temperature.
4) The intermolecular forces are still weaker than ionic or covalent bonds, so they generally have lower boiling points than ionic or giant covalent compounds.

There's more about how polymers are made on p.98 and p.102 and more about their properties and uses on p.121-122.

A polymer can contain thousands or even millions of atoms. But to draw one, you can just draw the shortest repeating section, called the repeating unit, like this:

To find the molecular formula of a polymer, write down the molecular formula of the repeating unit in brackets, and put an 'n' outside.

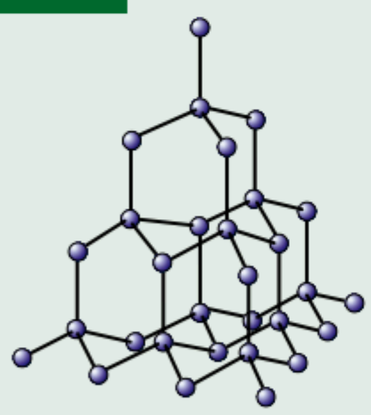


Giant Covalent Structures Are **Macromolecules**

- In giant covalent structures, all the atoms are bonded to each other by strong covalent bonds.
- They have very high melting and boiling points as lots of energy is needed to break the covalent bonds between the atoms.
- They don't contain charged particles, so they don't conduct electricity — not even when molten. (There are a few weird exceptions like graphite.)
- The main examples that you need to know about are diamond and graphite, which are both made from carbon atoms only, and silicon dioxide (silica).

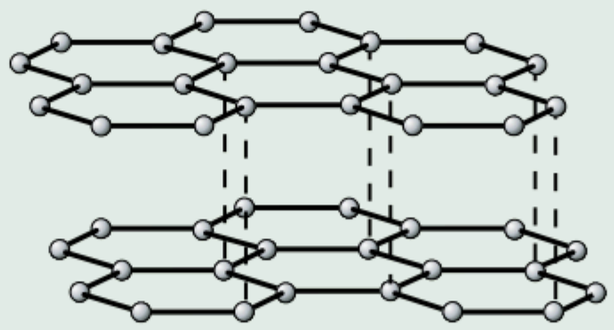
There's more about diamond and graphite (and other types of carbon structure) on the next page.

Diamond



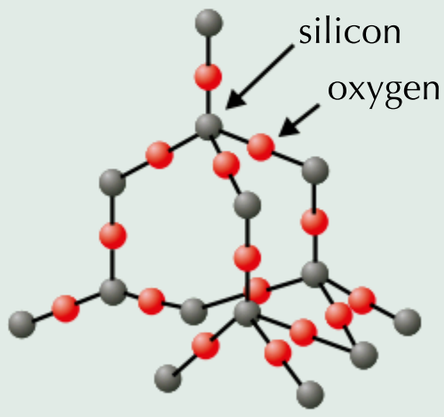
Each carbon atom forms four covalent bonds in a very rigid giant covalent structure.

Graphite



Each carbon atom forms three covalent bonds to create layers of hexagons. Each carbon atom also has one delocalised (free) electron.

Silicon dioxide

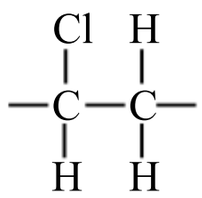


This is what sand is made of. Each grain of sand is one giant structure of silicon and oxygen.

Polymers are very large molecules

Make sure you can work out the molecular formula of a polymer. To do this you need to be comfortable with what the repeating unit of a polymer is.

Q1 The repeating unit of poly(chloroethene) is shown on the right. What's the molecular formula of poly(chloroethene)?



[1 mark]

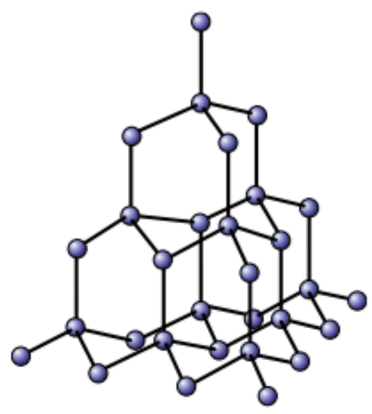
Q1 Video Solution

Allotropes of Carbon

Quick Quiz

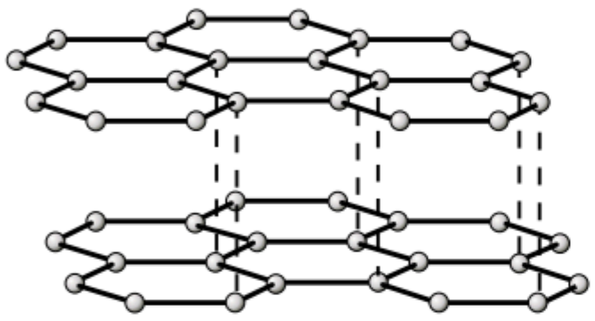
*Allotropes are **different structural forms** of the **same element** in the **same physical state**. Carbon's got lots...*

Diamond is Very **Hard**



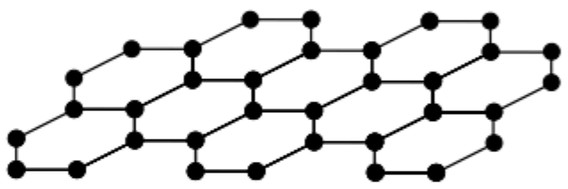
- Diamond has a giant covalent structure, made up of carbon atoms that each form four covalent bonds. This makes diamond really hard.
- Those strong covalent bonds take a lot of energy to break and give diamond a very high melting point.
- It doesn't conduct electricity because it has no free electrons or ions.

Graphite Contains Sheets of **Hexagons**



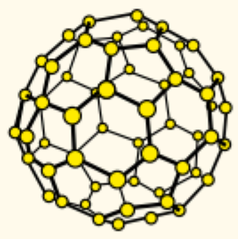
- In graphite, each carbon atom only forms three covalent bonds, creating sheets of carbon atoms arranged in hexagons.
- There aren't any covalent bonds between the layers — they're only held together weakly, so they're free to move over each other. This makes graphite soft and slippery, so it's ideal as a lubricating material.
- Graphite's got a high melting point — the covalent bonds in the layers need loads of energy to break.
- Only three out of each carbon's four outer electrons are used in bonds, so each carbon atom has one electron that's delocalised (free) and can move. So graphite conducts electricity and thermal energy.

Graphene is **One Layer** of Graphite



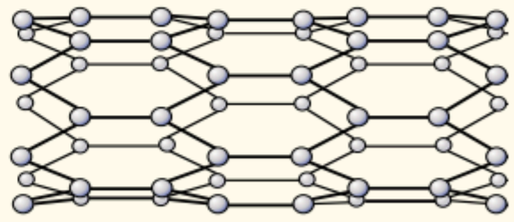
Graphene is a sheet of carbon atoms joined together in hexagons.
The sheet is just one atom thick, making it a two-dimensional substance.
The network of covalent bonds makes it very strong. It's also incredibly light, so can be added to composite materials to improve their strength without adding much weight.
Like graphite, it contains delocalised electrons so can conduct electricity through the whole structure. This means it has the potential to be used in electronics.

Fullerenes Form **Spheres** and **Tubes**



Buckminsterfullerene was the first fullerene to be discovered. It's got the molecular formula C_{60} and forms a hollow sphere.

1) Fullerenes are molecules of carbon, shaped like closed tubes or hollow balls.
2) They're mainly made up of carbon atoms arranged in hexagons, but can also contain pentagons (rings of five carbon atoms) or heptagons (rings of seven carbon atoms).
3) Fullerenes can be used to 'cage' other molecules. The fullerene structure forms around another atom or molecule, which is then trapped inside. This could be used to deliver a drug into the body.
4) Fullerenes have a huge surface area, so they could help make great industrial catalysts — individual catalyst molecules could be attached to the fullerenes. Fullerenes also make great lubricants.



Fullerenes can form nanotubes — tiny carbon cylinders.
The ratio between the length and the diameter of nanotubes is very high.
Nanotubes can conduct both electricity and thermal energy (heat).
They also have a high tensile strength (they don't break when they're stretched).
Technology that uses very small particles such as nanotubes is called nanotechnology. Nanotubes can be used in electronics or to strengthen materials without adding much weight, such as in tennis racket frames.

Diamond, graphite and graphene contain exactly the same atoms

... and fullerenes too. The different substances on this page are made purely from carbon — there's no difference at all in their atoms. The difference in properties is all down to the way the atoms are held together.

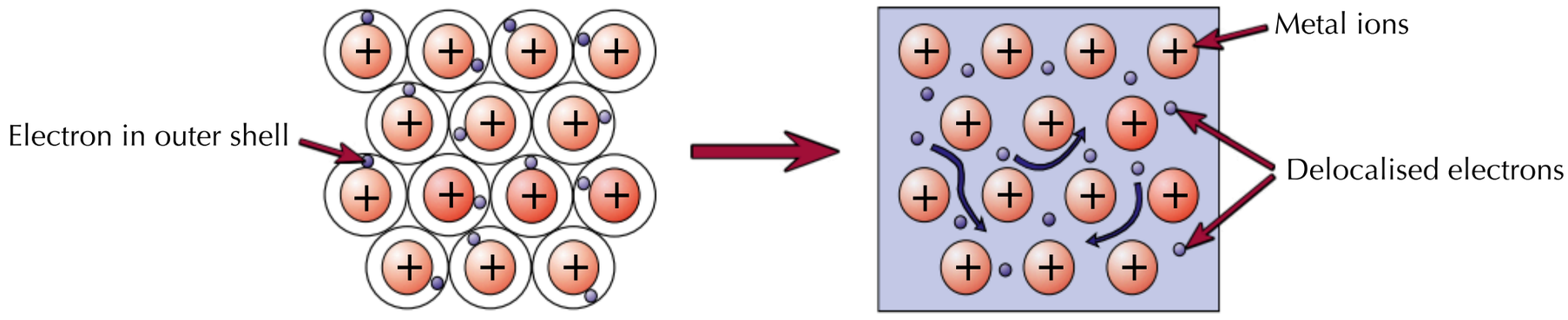
Metallic Bonding

Quick Quiz

*Ever wondered what gives a metal its **properties**? Most of it comes down to bonding...*

Metallic Bonding Involves Delocalised Electrons

- Metals also consist of a giant structure.
- The electrons in the outer shell of the metal atoms are delocalised (free to move around).



- There are strong forces of electrostatic attraction between the positive metal ions and the shared negative electrons.
- The forces of attraction hold the atoms together in a regular structure and are known as metallic bonding. Metallic bonding is very strong.
- Substances that are held together by metallic bonding include metallic elements and alloys (see below).
- It's the delocalised electrons in the metallic bonds which produce many of the properties of metals.

MOST METALS ARE SOLID AT ROOM TEMPERATURE

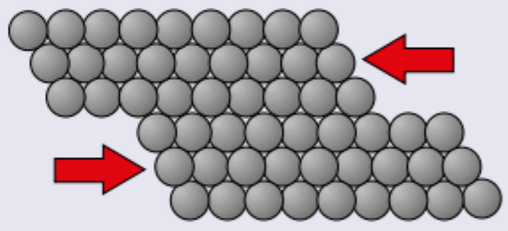
The electrostatic forces between the metal atoms and the delocalised sea of electrons are very strong, so need lots of energy to be broken. This means that most compounds with metallic bonds have very high melting and boiling points, so they're generally solid at room temperature.

METALS ARE GOOD CONDUCTORS OF ELECTRICITY AND HEAT

The delocalised electrons can move through the whole structure, carrying electric charge and thermal (heat) energy, so metals are good conductors of electricity and heat.

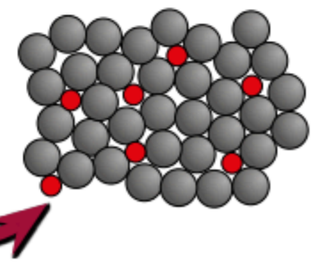
MOST METALS ARE MALLEABLE

The layers of atoms in a metal can slide over each other, making metals malleable — this means that they can be bent or hammered or rolled into flat sheets.



Alloys are Harder Than Pure Metals

- Pure metals often aren't quite right for certain jobs — they're often too soft when they're pure so are mixed with other metals to make them harder. Most of the metals we use everyday are alloys — a mixture of two or more metals or a metal and another element.
- Different elements have different sized atoms. So when another element is mixed with a pure metal, the new metal atoms will distort the layers of metal atoms, making it more difficult for them to slide over each other. This makes alloys harder than pure metals.



Metallic bonding is what makes metals, well... metals

You should understand what metallic bonding is, and be able to relate how it causes metals to have the properties they do. Also, don't forget to learn the differences between alloys and pure metals.

Warm-Up & Exam Questions

Lots of information to learn on the previous few pages — here are some questions to test yourself on.

Warm-Up Questions

1) How are the repeating units in a polymer bonded together?
2) At room temperature, what state are most polymers in?
3) Describe the differences in the hardness and electrical conductivity of diamond and graphite.
4) Which fullerene was the first to be discovered?
5) Explain why most metals are malleable.

Exam Questions

1 Silicon carbide has a giant covalent structure and is a solid at room temperature. GRADE 4-6 GRADE

1.1 Explain, in terms of its bonding and structure, why silicon carbide has a high melting point.

[2 marks]

1.2 Give **one** other example of a substance with a giant covalent structure.

[1 mark]

2 Graphite, diamond and fullerenes are entirely made from carbon but have different properties. GRADE 6-7 GRADE

2.1 Why does the structure of graphite make it a useful lubricant?

[2 marks]

2.2 Using your knowledge of the structure of diamond, suggest why it is useful as a cutting tool.

[2 marks]

2.3 Suggest **one** possible use for fullerenes.

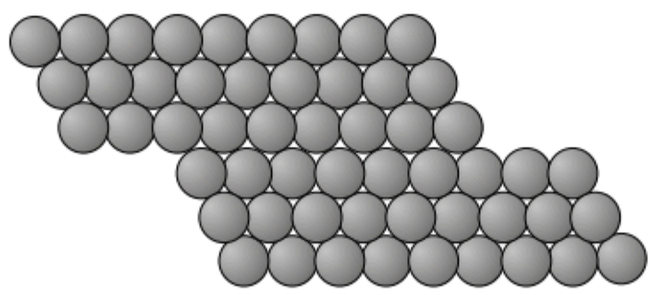
[1 mark]

2.4 Explain why graphite is able to conduct electricity.

[1 mark]

3 **Figure 1** shows the arrangement of atoms in pure iron. GRADE 6-7 GRADE

Figure 1



3.1 Steel is an alloy of iron and carbon.
Sketch a similar diagram to show the arrangement of atoms in steel.

[2 marks]

3.2 Steel is harder than iron. Explain why.

[3 marks]

Revision Summary Test for Topic 2a

So that's three types of bonding you need to know about. To test your memory, you can:

- Tackle the revision summary test below, or scan the QR code to do it online.
  The questions are hard, but they'll show you how well you really know your stuff.
- Track your progress online and see which areas need more work.
- Compare your answers with sample answers for the test here: cgpbooks.co.uk/Base

Summary Test

Ions and Ionic Compounds (p.26-28) ☑

1) How do ions form from atoms of metals? What about from atoms of non-metals?
2) For each of these groups in the periodic table, state whether the elements in them form anions or cations, and give the charge of the ions formed:
   a) Group 1, b) Group 2, c) Group 6, d) Group 7.
3) Describe how an ionic bond forms between a metal and a non-metal.
4) Sketch dot and cross diagrams to show the formation of:
   a) sodium chloride, b) magnesium oxide, c) magnesium chloride, d) sodium oxide.
5) Describe the structure of sodium chloride and name two models that can be used to represent it in a diagram. What are the downsides to each model?
6) How would you work out the empirical formula of an ionic compound from a diagram?
7) List the main properties of ionic compounds.
8) How would you work out the formula of an ionic compound from its ions?

Covalent Substances (p.30-34) ☑

9) Describe how covalent bonds form.
10) Name three types of diagram used to draw covalent compounds.
    What are the advantages and disadvantages of each one?
11) Sketch dot and cross diagrams showing the bonding in a molecule of:
    a) hydrogen, b) water, c) hydrogen chloride.
12) Why do simple molecular compounds typically have low melting and boiling points?
13) Describe the structure of a polymer.
14) How do the melting and boiling points of polymers compare to those of simple molecular structures and why?
15) What is a 'repeating unit'?
16) Why do giant covalent substances have high melting and boiling points?
17) Give three examples of giant covalent substances.
18) Describe the structure and bonding for each of these allotropes of carbon:
    a) diamond, b) graphite, c) graphene.
19) Explain why graphite can conduct electricity and diamond can't.
20) What are fullerenes? Describe their typical structure.
21) Give three properties of carbon nanotubes.

Metallic Bonding (p.35) ☑

22) What is metallic bonding?
23) List three properties of metals and explain how the structure and bonding in metals causes each property.
24) Explain why alloys are harder than pure metals.

Quick Quiz

States of Matter

*Solids, liquids and gases... You can explain quite a bit of stuff in **chemistry** if you get your head round this lot.*

Materials come in **Three States of Matter** — Solid, Liquid and Gas

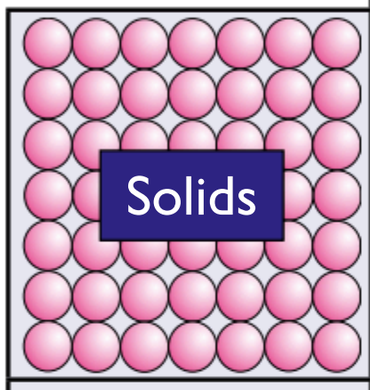
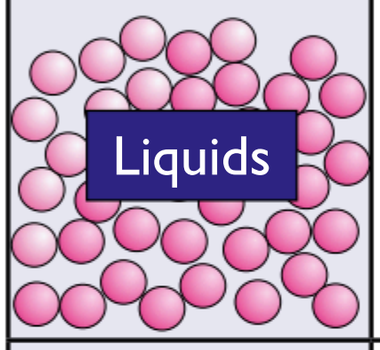
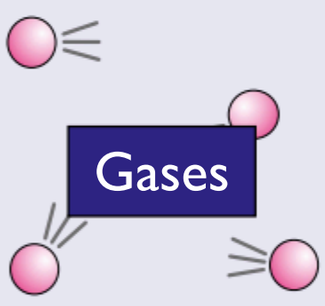
Video

The physical state something is in at a certain temperature depends on how strong the forces of attraction are between its particles. How strong the forces are depends on THREE THINGS:

1) the material (i.e. its structure and bonding) | 2) the temperature | 3) the pressure

You can use a model called particle theory to explain how the particles in a material behave in each of the three states of matter. Particle theory considers each particle as a small, solid, inelastic sphere.

The particles could be atoms, ions or molecules.

	Forces	Shape and Volume	Movement
	There are strong forces of attraction between particles, which hold them close together in fixed positions to form a regular lattice arrangement.	The particles don't move from their positions, so all solids keep a definite shape and volume, and don't flow like liquids.	The particles vibrate about their positions — the hotter the solid becomes, the more they vibrate (causing solids to expand slightly when heated).
	There are weak forces of attraction between the particles. They're randomly arranged and free to move past each other, but they tend to stick closely together.	Liquids have a definite volume but don't keep a definite shape, and will flow to fill the bottom of a container.	The particles are constantly moving with random motion. The hotter the liquid gets, the faster they move. This causes liquids to expand slightly when heated.
	The forces of attraction between the particles are very weak — so the particles are free to move and are far apart.	Gases don't keep a definite shape or volume and will always fill any container.	The particles move constantly and randomly in straight lines. The hotter the gas gets, the faster they move. Gases either expand when heated, or their pressure increases.

Particle theory is a great model for explaining the three states of matter, but it isn't perfect. In reality, the particles aren't solid or inelastic and they aren't spheres — they're atoms, ions or molecules. Also, the model doesn't show the forces between the particles, so there's no way of knowing how strong they are.

State Symbols Tell You the **State** of a Substance in an **Equation**

Symbol equations can include state symbols next to each substance — they tell you what physical state the reactants and products are in:

There's more about chemical equations on page 6.

(s) — solid (l) — liquid (g) — gas (aq) — aqueous

'Aqueous' means 'dissolved in water'.

For example, aqueous hydrochloric acid reacts with solid calcium carbonate to form aqueous calcium chloride, liquid water and carbon dioxide gas:

$$2HCl_{(aq)} + CaCO_{3(s)} \rightarrow CaCl_{2(aq)} + H_2O_{(l)} + CO_{2(g)}$$

Gases and liquids do not have a definite shape

Solids keep the same shape and volume because their particles are in fixed positions. Liquids and gases are a bit more adaptable because their particles are free to move — their shape will change depending on the container they are in.

Changing State

Quick Quiz

*When you change the **temperature**, particles start doing all kinds of funky things...*

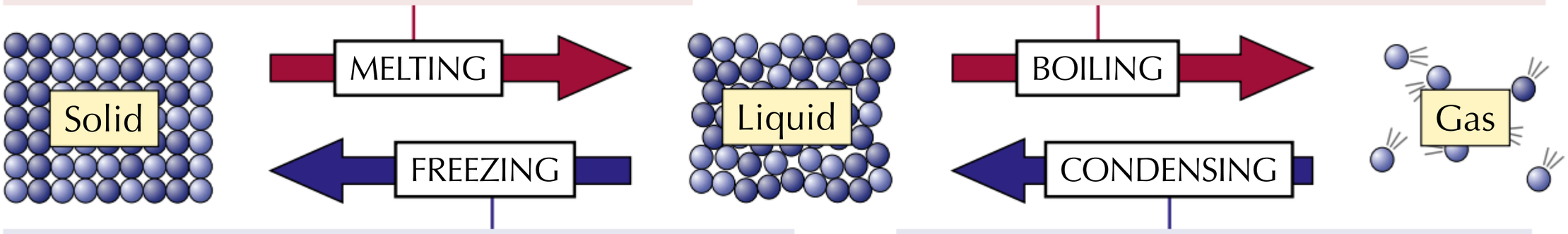
Video

Substances Can **Change** from **One State** to **Another**

Changes of state are physical changes. Physical changes don't change the particles — just their arrangement or their energy.

- When a solid is heated, its particles gain energy and vibrate more, weakening the forces that hold the solid together.
- At the melting point, the particles have enough energy to break free from their positions, so the solid turns into a liquid.

- When a liquid is heated, the particles get even more energy and move faster, which weakens the forces between them.
- At the boiling point, the particles have enough energy to overcome these forces, so the liquid becomes a gas.



- When a liquid cools, the particles have even less energy, so move slower. There's not enough energy to overcome the attraction between them, so even stronger attractions form.
- At the melting point, the forces between the particles are so strong that they're held in place. The liquid becomes a solid.

- As a gas cools, the particles no longer have enough energy to overcome the forces of attraction between them. These attractions get stronger.
- At the boiling point, the forces are so strong that the gas becomes a liquid.

The amount of energy needed for a substance to change state depends on how strong the forces between particles are. The stronger the forces, the more energy is needed to overcome them, and so the higher the melting and boiling points of the substance are.

You can **Predict** the **State** of a Substance at a Certain **Temperature**

- If a substance's temperature is below the melting point of the substance, it'll be a solid.
- If it's above the boiling point, it'll be a gas.
- If it's in between the two points, then it's a liquid.

Melting point and boiling point are bulk properties of materials (they depend on how lots of atoms interact). An atom on its own doesn't have bulk properties.

EXAMPLE

Which of the molecular substances in the table is a liquid at room temperature (25 °C)?

	melting point	boiling point
oxygen	–219 °C	–183 °C
nitrogen	–210 °C	–196 °C
bromine	–7 °C	59 °C

Oxygen and nitrogen have boiling points below 25 °C, so will both be gases at room temperature.

So the answer's bromine. It melts at –7 °C and boils at 59 °C. So, it'll be a liquid at room temperature.

Physical changes are reversible

Q1 Video Solution

Make sure you can describe what happens to particles, and the forces between them, as a substance is heated and cooled. Don't forget to learn the technical terms for each state change.

Q1 Ethanol melts at –114 °C and boils at 78 °C. Predict the state that ethanol is in at:
a) –150 °C b) 0 °C c) 25 °C d) 100 °C [4 marks]

Quick Quiz

Nanoparticles

*Using nanoparticles is known as **nanoscience**. Many **new uses** of nanoparticles are being developed.*

Nanoparticles Are Really Really Really Really Tiny

1 nm = 0.000 000 001 m
(Like I said — really tiny.)

Particles are put into categories depending on their diameter. For example:

Coarse particles (PM_{10})	Have a diameter between 2500 nm (2.5×10^{-6} m) and 10 000 nm (1×10^{-5} m). They're also called dust.
Fine particles ($PM_{2.5}$)	Have a diameter between 100 nm (1×10^{-7} m) and 2500 nm (2.5×10^{-6} m).
Nanoparticles	Have a diameter between 1 nm (1×10^{-9} m) and 100 nm (1×10^{-7} m). These are particles that contain only a few hundred atoms.

A typical atom has a diameter of about 0.1 nm (1×10^{-10} m).

The area of science that investigates the uses and properties of nanoparticles is called nanoscience.

Nanoparticles Have a Large Surface Area to Volume Ratio

1) The surface area to volume ratio is an important factor as it can affect the way that a particle behaves.

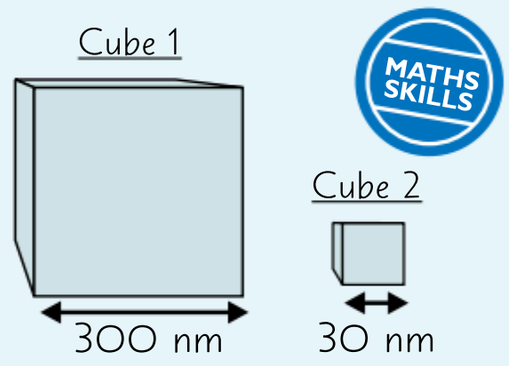
surface area to volume ratio = surface area ÷ volume

2) As particles decrease in size, the size of their surface area increases in relation to their volume. This means the surface area to volume ratio increases.

3) You can see this happening by using two cubes as an example:

EXAMPLE

Find the surface area to volume ratio for each of these cubes.



MATHS SKILLS

Cube 1 (300 nm sides):

Area of each face = $(300 \text{ nm})^2 = 90\,000 \text{ nm}^2$

There are six faces, so total surface area = $6 \times 90\,000 \text{ nm}^2 = 540\,000 \text{ nm}^2$

Volume = $(300 \text{ nm})^3 = 27\,000\,000 \text{ nm}^3$

Surface area to volume ratio = surface area ÷ volume = $540\,000 \text{ nm}^2 \div 27\,000\,000 \text{ nm}^3 = 0.02 \text{ nm}^{-1}$

You can also write this as a ratio: $540\,000 \text{ nm}^2 : 27\,000\,000 \text{ nm}^3 = 1 \text{ nm}^2 : 50 \text{ nm}^3$

Divide both sides by the surface area value (540 000) to cancel the ratio down to 1 : n.

Cube 2 (30 nm sides):

Area of each face = $(30 \text{ nm})^2 = 900 \text{ nm}^2$. Total surface area = $6 \times 900 \text{ nm}^2 = 5400 \text{ nm}^2$.

Volume = $(30 \text{ nm})^3 = 27\,000 \text{ nm}^3$

Surface area to volume ratio = $5400 \text{ nm}^2 \div 27\,000 \text{ nm}^3 = 0.2 \text{ nm}^{-1}$

As a ratio: $5400 \text{ nm}^2 : 27\,000 \text{ nm}^3 = 1 \text{ nm}^2 : 5 \text{ nm}^3$

Divide both sides by 5400

As you decrease the side of any cube by a factor of ten, the surface area to volume ratio will always increase by a factor of ten.

4) Nanoparticles have a very high surface area to volume ratio — this means their surface area is very large compared to their volume.

5) This can cause the properties of a material to be different depending on whether it's a nanoparticle or whether it's in bulk. For example, you'll often need a smaller amount of a material that's made up of nanoparticles (each containing a few hundred atoms) to work as an effective catalyst compared to a material made up of 'normal' sized particles (each containing billions of atoms).

Smaller particles have a larger surface area to volume ratio

Nanoparticles are between ten and one hundred times larger than atoms and molecules.

Q1 a) The sides of a cube are 50 nm long. What's its surface area to volume ratio in nm^{-1}? [3 marks]

b) Predict how the surface area to volume ratio would change if the size of the cube was increased so its sides had a length of 500 nm. [2 marks]

Q1 Video Solution

Uses of Nanoparticles

Quick Quiz

'What's the use of something so small that not even a gnat can see it?' I hear you cry. Well, as you're about to find out, ***scientists*** *have developed some pretty swanky* ***uses*** *for nanoparticles.*

Nanoparticles Can Be **Used** in **Lots** of Things

Finding new ways to use nanoparticles is a really important area of scientific research. Here are some of the uses that have already been developed...

COSMETICS — For example, they're used to improve moisturisers without making them really oily.

CATALYSTS — Nanoparticles have a huge surface area to volume ratio, so they could help make new catalysts.

ANTIBACTERIALS — Silver nanoparticles have antibacterial properties. They can be added to polymer fibres that are then used to make surgical masks and wound dressings and can also be added to deodorants.

COMPUTING — Some nanoparticles conduct electricity, so they can be used in tiny electric circuits for computer chips.

NANOMEDICINE — Tiny particles (such as fullerenes from page 34) are absorbed more easily by the body than most particles. This means they can be used to deliver drugs right into the cells where they're needed.

The **Effects** of Nanoparticles on **Health** Aren't Fully **Understood**

1) Although nanoparticles are useful, the way they affect the body isn't fully understood, so it's important that any new products are tested thoroughly to minimise the risks.

2) Some people are worried that products containing nanoparticles have been made available before the effects on human health have been investigated properly, and that we don't know what the long-term impacts on health will be.

3) As the long-term impacts aren't known, many people believe that products containing nanoparticles should be clearly labelled, so that consumers can choose whether or not to use them.

- Nanoparticles are being used in sun creams as they have been shown to be better than the materials in traditional sun creams at protecting skin from harmful UV rays.
- They also give better skin coverage than traditional sun creams.
- But it's not yet clear whether the nanoparticles can get into your body, and, if they do, whether they might damage cells.
- It's also possible that when they are washed away they might damage the environment.

There are so many possible uses for nanotechnology

In your exam, you might be asked to explain why nanoparticles are used for certain applications, given their properties. Nanoparticles are pretty amazing, but just remember — they may have a currently unknown dark side too...

Warm-Up & Exam Questions

Reckon you know all there is to know about states of matter and nanoparticles? Have a go at these questions and see how you get on. If you get stuck on something — just flick back and give it another read through.

Warm-Up Questions

1) What state of matter has a fixed lattice arrangement?
2) What happens to the particles in a gas when it is heated?
3) Write down how you would show that sodium ions (Na^+) are in solution in a symbol equation.
4) Name three categories that particles can be put into based on their diameter.
5) How does the surface area to volume ratio of a particle change it gets smaller?

Exam Questions

1.1 **Figure 1** shows a vessel in a distillery.
The walls of the vessel are solid copper.

GRADE 4-6 GRADE

Figure 1

Complete the sentences about solids using words from the box.
Each word may be used once, more than once or not at all.

weak	move	colder	hotter	random
strong	expand	heavier	dissolve	regular

In solids, there are ..... forces of attraction between particles, which hold them in fixed positions in a ..... arrangement.

The particles don't ..... from their positions, so solids keep their shape.

The ..... the solid becomes, the more the particles in the solid vibrate.

[4 marks]

1.2 Inside the vessel, liquid ethanol is turned into ethanol gas.
Describe the changes in arrangement, movement and energy of the particles when the liquid ethanol is heated to become a gas.

[2 marks]

2 Some people are concerned about the use of nanoparticles in medicines and cosmetics.
Suggest one reason why some people are concerned about the use of nanoparticles and suggest what could be done to minimise the risks.

GRADE 6-7 GRADE

[2 marks]

3 This question is on states of matter.

GRADE 6-7 GRADE

3.1 Using particle theory, explain why gases fill their containers.

[2 marks]

3.2 Chlorine has a melting point of –101.5 °C and a boiling point of –34 °C.
Predict what state chlorine will be in at –29 °C.

[1 mark]

Revision Summary Test for Topic 2b

That was some pretty interesting stuff — but how much do you remember? To find out, you can:

- Tackle the revision summary test below, or scan the QR code to do it online.
  The questions are hard, but they'll show you how well you really know your stuff.
- Track your progress online and see which areas need more work.
- Compare your answers with sample answers for the test here: cgpbooks.co.uk/Base

Summary Test

States of Matter (p.38-39) ☐

1) What three things affect the strength of the forces of attraction between particles in a material? ☐
2) For each of the three states of matter, describe the forces of attraction between the particles in that state and how this affects their arrangement. ☐
3) Which states of matter have: a) a definite shape, b) a definite volume? ☐
4) For each of the three states of matter, describe the movement of the particles in that state and how this is affected by heat. ☐
5) Give two reasons why the particle theory model is not perfect. ☐
6) Give the state symbols for: a) solids, b) liquids, c) gases, d) aqueous substances. ☐
7) What happens to the particles of a substance during a physical change? ☐
8) What happens to the arrangement and energy of particles when:
   a) a solid turns into a liquid? b) a liquid turns into a gas?
   c) a gas turns into a liquid? d) a liquid turns into a solid? ☐
9) How does the strength of the forces between particles influence the temperature at which a substance changes state? ☐
10) Give the state of a substance when its temperature is: a) below the substance's melting point, b) above the substance's boiling point, c) between the substance's melting and boiling points. ☐

Nanoparticles and their Uses (p.40-41) ☐

11) Give the diameters of: a) coarse particles, b) fine particles, c) nanoparticles. ☐
12) Roughly how many atoms are there in a nanoparticle? ☐
13) What is nanoscience? ☐
14) What formula can you use to calculate the surface area to volume ratio of a cube? ☐
15) What would happen to the surface area to volume ratio of a cube if the length of its side decreased by a factor of ten? ☐
16) Give an example of how the properties of a material can be different depending on whether it is made of 'normal' sized particles or nanoparticles. ☐
17) Describe how nanoparticles can be used in computing and as antibacterials. ☐
18) Now briefly describe three more uses of nanoparticles. ☐
19) Explain why people may be wary of using products that contain nanoparticles. ☐
20) Give two advantages and two possible disadvantages of using nanoparticles in sun creams. ☐

Quick Quiz

Relative Formula Mass

*Calculating **relative formula mass** is straightforward enough, but things can get a bit more confusing when you start working out the **percentage compositions** of compounds. Best get cracking, I suppose...*

Compounds Have a Relative Formula Mass, M_r

The relative formula mass (M_r) of a compound is just the relative atomic masses (A_r) of all the atoms in the compound's formula added together.

EXAMPLE

Find the relative formula mass of $MgCl_2$.

1) Look up the A_r for each element in the compound.
A_r of Mg = 24 and the A_r of Cl = 35.5.

2) Add up all the relative atomic masses of the atoms in the compound.
Mg + (2 × Cl) = 24 + (2 × 35.5) = 95 So M_r of $MgCl_2$ = 95

There are 2 Cl atoms in the formula for $MgCl_2$ (see p.5), so the relative atomic mass of chlorine needs to be multiplied by 2.

You might be given relative atomic masses in the question. If not, you can find them using the periodic table — the A_r is the bigger of the two numbers by each element — see p.17.

You Can Calculate the % Mass of an Element in a Compound

This is actually dead easy — so long as you've learnt this formula:

$$\text{Percentage mass of an element in a compound} = \frac{A_r \times \text{number of atoms of that element}}{M_r \text{ of the compound}} \times 100$$

EXAMPLE

Find the percentage mass of sodium in sodium carbonate, Na_2CO_3.

MATHS SKILLS

A_r of sodium = 23, A_r of carbon = 12, A_r of oxygen = 16

M_r of Na_2CO_3 = (2 × 23) + 12 + (3 × 16) = 106

$$\text{Percentage mass} = \frac{A_r \times \text{number of Na atoms}}{M_r \text{ of } Na_2CO_3} \times 100 = \frac{23 \times 2}{106} \times 100 = 43.39...\% = 43\% \text{ (to the nearest \%)}$$

You might also come across more complicated questions where you need to work out the percentage mass.

EXAMPLE

A mixture contains 20% iron ions by mass. What mass of iron chloride ($FeCl_2$) is needed to provide the iron ions in 50 g of the mixture? A_r of Fe = 56, A_r of Cl = 35.5.

MATHS SKILLS

1) Find the mass of iron in the mixture.
The mixture contains 20% iron by mass, so 50 g contains 20% of 50 = $\frac{20}{100} \times 50$ = 10 g of iron.

2) Calculate the percentage mass of iron in iron chloride.
$$\text{Percentage mass} = \frac{A_r \times \text{number of iron atoms}}{M_r \text{ of iron chloride}} \times 100 = \frac{56 \times 1}{56 + (2 \times 35.5)} \times 100 = 44.09...\%$$

3) Calculate the mass of iron chloride that contains 10 g of iron.
If iron chloride is 44.09...% iron by mass, then 100 g of iron chloride contains 44.09... g of iron.
So to get 1 g of iron you need (100 ÷ 44.09...) g of iron chloride,
and to get 10 g of iron you need 10 × (100 ÷ 44.09...) = 22.67... = 23 g of iron chloride (2 s.f.).

Relative formula mass — add up all the relative atomic masses

The best way to get to grips with all this stuff is by practising. Start by having a go at these questions...

Q1 Calculate the relative formula mass (M_r) of: a) H_2O b) LiOH c) H_2SO_4 [3 marks]

Q2 Calculate the percentage composition by mass of potassium in potassium hydroxide (KOH). [2 marks]

Q2 Video Solution

The Mole

Quick Quiz

The mole *can be pretty confusing. I think it's the word that puts people off. It's difficult to see the relevance of the word "mole" to anything but a small burrowing animal, but you're going to have to ignore them for now.*

"The Mole" is Simply the Name Given to an **Amount** of a Substance

- Just like "a million" is 1 000 000 and "a billion" is 1 000 000 000, "the Avogadro constant" is this many: 602 000 000 000 000 000 000 000 or 6.02×10^{23}. That's all it is — just a number.
- One mole of any substance is the amount of that substance that contains an Avogadro number of particles (so 6.02×10^{23} particles). The particles could be atoms, molecules, ions or electrons.
- The burning question, of course, is why is it such a strange long number like that?
- The answer is that the mass of that number of atoms or molecules of any substance is exactly the same number of grams as the relative atomic mass (if it's an element) or the relative formula mass (if it's a compound).
- In other words, one mole of a substance has a mass in grams equal to the relative formula mass (A_r or M_r) for that substance. Here are some examples:

Carbon has an A_r of 12.	So one mole of carbon weighs 12 g.
Nitrogen gas, N_2, has an M_r of 28 (2 × 14).	So one mole of N_2 weighs 28 g.
Carbon dioxide, CO_2, has an M_r of 44 (12 + [2 × 16]).	So one mole of CO_2 weighs 44 g.

12 g of carbon, or 28 g of N_2, or 44 g of CO_2, all contain the same number of particles, i.e. one mole or 6.023×10^{23} atoms or molecules.

Nice Formula to Find the **Number of Moles** in a **Given Mass**:

$$\text{Number of moles} = \frac{\text{mass in g (of an element or compound)}}{M_r \text{ (of the element or compound)}}$$

EXAMPLE

How many moles are there in 66 g of carbon dioxide (CO_2)?

1) Calculate the M_r of carbon dioxide.
2) Use the formula above to find out how many moles there are.

M_r of CO_2 = 12 + (16 × 2) = 44
no. of moles = mass (g) ÷ M_r
= 66 ÷ 44 = 1.5 mol (2 s.f.)

MATHS SKILLS

'mol' is the symbol for the unit 'moles'.

You can rearrange the equation above to find the mass of a known number of moles of a substance, or to find the M_r of a substance from a known mass and number of moles.

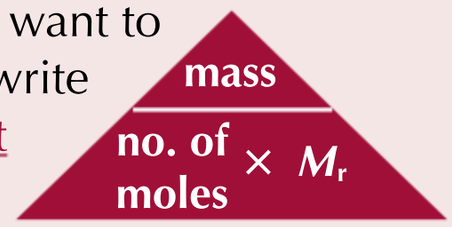
EXAMPLE

What mass of carbon is there in 4 moles of carbon dioxide?

MATHS SKILLS

There are 4 moles of carbon in 4 moles of CO_2.
Number of moles = mass ÷ M_r, so mass = number of moles × M_r.
So the mass of 4 moles of carbon = 4 × A_r(C) = 4 × 12 = 48 g

You can also use this handy formula triangle to help you with rearranging. Cover up the thing you want to find and write what's left showing.



Putting an equation in a formula triangle makes rearranging easy

Calculations involving moles can send some people into a spin. Don't be one of those people — there's really no need to freak out about moles. Go back over this page until you've got your head round it all.

Q2 Video Solution

Q1 Calculate the number of moles in 90 g of water (H_2O). A_r(O) = 16, A_r(H) = 1. [2 marks]

Q2 Calculate the mass of 0.20 mol of potassium bromide (KBr). A_r(K) = 39, A_r(Br) = 80. [2 marks]

Quick Quiz

Conservation of Mass

You've probably realised by now that you can't magic stuff out of thin air, and you can't make it magically disappear, either. This fact is pretty useful for working out the ***amounts of substances*** *in chemical reactions.*

In a Chemical Reaction, **Mass** is **Always Conserved**

1) During a chemical reaction no atoms are destroyed and no atoms are created.
2) Because of this, no mass is lost or gained — we say that mass is conserved during a reaction.

When you have a balanced symbol equation, the left-hand side and right-hand side should have:

- the same number of each type of atom — because no atoms are created or destroyed.
- the same total M_r — because mass is conserved (and the atoms are the same).

For example:

Left-hand side has 2 Li atoms and 2 F atoms.

Right-hand side has 2 Li atoms and 2 F atoms.

$$2Li + F_2 \rightarrow 2LiF$$

Total M_r on left-hand side
$= [2 \times A_r(Li)] + M_r(F_2)$
$= (2 \times 7) + (2 \times 19) = 14 + 38 = 52$

Total M_r on right-hand side
$= 2 \times M_r(LiF) = 2 \times (7 + 19)$
$= 2 \times 26 = 52$

If the Mass **Seems to Change**, There's Usually a **Gas** Involved

In some experiments, you might observe a change of mass of an unsealed reaction vessel during a reaction. This doesn't mean the mass of the products is different to the mass of reactants — it happens because your measurement doesn't account for all the reactants or products. A change in mass can usually be explained by one of these two possibilities:

Video

Mass increases

If the mass increases, it's probably because one of the reactants is a gas from the air (e.g. oxygen) and all the products are solids, liquids or aqueous.

- Before the reaction, the gas is floating around in the air. It's not contained in the reaction vessel, so you can't account for its mass.
- When the gas reacts, it becomes part of the contents of the reaction vessel — so the total mass of the stuff inside the vessel increases.

E.g. the reaction of metals with oxygen:

$$metal_{(s)} + oxygen_{(g)} \rightarrow metal\ oxide_{(s)}$$

If the container isn't sealed, its mass will increase by the same amount as the mass of oxygen that reacts.

Mass decreases

If the mass decreases, it's probably because one of the products is a gas and all the reactants are solids, liquids or aqueous.

- Before the reaction, all the reactants are contained in the reaction vessel.
- When a gas is produced, its particles are free to move (see p.38). If the container isn't sealed, then the gas can escape from it. It's no longer contained in the reaction vessel, so you can't account for its mass — the total mass of the stuff inside the reaction vessel decreases.

E.g. the thermal decomposition of metal carbonates:

$$metal\ carbonate_{(s)} \rightarrow metal\ oxide_{(s)} + carbon\ dioxide_{(g)}$$

If the container isn't sealed, its mass will decrease by the same amount as the mass of carbon dioxide produced.

Mass is ALWAYS conserved

There's quite a bit to take in on these pages, but it's all important in everything you do in chemistry. Make sure you learn the equation on the previous page — it'll crop up again in the rest of the topic.

Q1 Show that mass is conserved in the reaction: $H_2SO_{4(aq)} + 2NaOH_{(aq)} \rightarrow Na_2SO_{4(aq)} + 2H_2O_{(l)}$

$A_r(H) = 1$, $A_r(O) = 16$, $A_r(Na) = 23$, $A_r(S) = 32$ [5 marks]

Q1 Video Solution

Warm-Up & Exam Questions

If you feel like you need some practice with relative masses and mole calculations, then you're in luck. Here's a nice bunch of questions for you to have a go at.

Warm-Up Questions

1) What name is given to the sum of the relative atomic masses of the atoms in a molecule?
2) What is the name given to the particular number of particles equal to one mole of a substance?
3) Write down the definition of a mole.
4) What is the mass of one mole of iron?
5) What is the mass of one mole of oxygen gas?

Exam Questions

1 Boron can form a number of covalent compounds.
Use the A_r values B = 11, O = 16, F = 19 and H = 1 to calculate the relative formula masses of these boron compounds:

GRADE 4-6

1.1 BF_3

[1 mark]

1.2 $B(OH)_3$

[1 mark]

2 A student was asked to calculate the number of moles and the masses of different compounds for use in a lab practical.
State the formula used to work out the number of moles from the mass of a compound.

GRADE 4-6

[1 mark]

3 A teacher has a 140 g sample of potassium hydroxide (KOH).
Calculate, in grams, how much more KOH the teacher needs to have a 4 mole sample.

GRADE 6-7

[3 marks]

4 A scientist burnt 300 g of metal **X** in an unsealed beaker. The scientist then weighed the contents of the beaker and found that it now weighed 500 g.
All of the metal **X** reacted to form a single product, a metal oxide.

GRADE 6-7

4.1 Explain why the mass appeared to increase following the reaction.

[1 mark]

4.2 Analysis of the metal oxide shows that it contains 40% oxygen by mass.
What is the percentage mass of metal **X** in the oxide?

[1 mark]

4.3 The scientist plans to use the metal oxide as part of a mixture.
The final mixture contains 24% metal **X** by mass, which all comes from the metal oxide.
What mass of the metal oxide was used to make 8.0 g of the mixture?

[2 marks]

Quick Quiz

The Mole and Equations

*This is the moment where the 'number of **moles** = **mass** ÷ M_r' equation from page 45 comes into its own.*

You Can Use **Moles** to **Calculate Masses** in Reactions

Remember those balanced equations from page 6? Well, the big numbers in front of the chemical formulas of the reactants and products tell you the relative amount of each substance in the reaction. You can treat these amounts as a number of moles — for example:

The little numbers tell you how many atoms of each element there are in the formula — look back at p.5 for more on formulas.

$$Mg_{(s)} + 2HCl_{(aq)} \rightarrow MgCl_{2(aq)} + H_{2(g)}$$

In this reaction, every 1 mole of magnesium reacts with 2 moles of hydrochloric acid to form 1 mole of magnesium chloride and 1 mole of hydrogen gas.

You Can **Balance Equations** Using **Reacting Masses**

If you know the masses of the reactants and products that took part in a reaction, you can work out the balanced symbol equation for the reaction. Here are the steps you should take:

1) Divide the mass of each substance by its relative formula mass to find the number of moles.
2) Divide the number of moles of each substance by the smallest number of moles in the reaction.
3) If any of the numbers aren't whole numbers, multiply all the numbers by the same amount so that they all become whole numbers.
4) Write the balanced symbol equation by putting these numbers in front of the chemical formulas.

EXAMPLE

8.1 g of zinc oxide (ZnO) reacts completely with 0.6 g of carbon to form 2.2 g of carbon dioxide and 6.5 g of zinc. Write a balanced symbol equation for this reaction.
A_r(C) = 12, A_r(O) = 16, A_r(Zn) = 65.

MATHS SKILLS

1) Work out M_r for each of the substances in the reaction:

ZnO: 65 + 16 = 81 C: 12 CO_2: 12 + (2 × 16) = 44 Zn: 65

2) Use the equation number of moles = mass ÷ M_r to calculate how many moles of each substance reacted or were produced:

ZnO: $\frac{8.1}{81}$ = 0.10 mol C: $\frac{0.6}{12}$ = 0.05 mol

CO_2: $\frac{2.2}{44}$ = 0.05 mol Zn: $\frac{6.5}{65}$ = 0.10 mol

3) Divide by the smallest number of moles, which is 0.05:

ZnO: $\frac{0.10}{0.05}$ = 2 C: $\frac{0.05}{0.05}$ = 1

CO_2: $\frac{0.05}{0.05}$ = 1 Zn: $\frac{0.10}{0.05}$ = 2

These numbers give the ratio of the amounts of each substance in the reaction equation.

4) The numbers are all whole numbers, so you can write the balanced symbol equation now:

$$2ZnO + C \rightarrow CO_2 + 2Zn$$

Practise, Practise, Practise

Balanced equations always have the same number of atoms of each element on each side of the equation.

Q1 Video Solution

Q1 84 g of N_2 reacts completely with 18 g of H_2 to produce 102 g of NH_3.
$M_r(N_2) = 28$, $M_r(H_2) = 2$, $M_r(NH_3) = 17$.
a) Calculate how many moles of each substance reacted or was produced. [3 marks]
b) Use your answer to part a) to write a balanced symbol equation for this reaction. [2 marks]

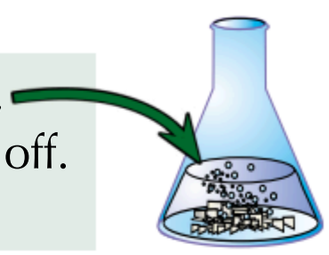
Limiting Reactants

Quick Quiz

*Reactions don't go on forever — the stuff in the reaction flask that reacts can get **used up**...*

Reactions **Stop** When **One** Reactant is **Used Up**

When some magnesium carbonate ($MgCO_3$) is placed into a beaker of hydrochloric acid, you can tell a reaction is taking place because you see lots of bubbles of gas being given off. After a while, the amount of fizzing slows down and the reaction eventually stops...



1) A reaction stops when all of one of the reactants is used up. This reactant is called the limiting reactant (because it limits the amount of product that's formed).
2) The amount of product formed is directly proportional to the amount of limiting reactant. If you halve the amount of limiting reactant, the amount of product formed will also halve. If you double the amount of limiting reactant, the amount of product will double (as long as it is still the limiting reactant).
3) This is because if you add more reactant there will be more reactant particles to take part in the reaction, which means more product particles.

Any other reactants left over when the reaction stops are in excess. Often reactants are added in excess to ensure that the limiting reactant gets completely used up.

The Amount of **Product** Depends on the **Limiting Reactant**

You can calculate the mass of a product formed in a reaction by using the mass of the limiting reactant and the balanced reaction equation:

1) Write out the balanced equation.
2) Work out relative formula masses (M_r) of the limiting reactant and the product you want.
3) Find the number of moles of the substance you know the mass of.
4) Use the balanced equation to work out how many moles there'll be of the other substance. In this case, that's how many moles of product will be made from this many moles of reactant.
5) Use the number of moles to calculate the mass.

You could also use this method to find the mass of a reactant needed to produce a known mass of a product.

EXAMPLE

Calculate the mass of aluminium oxide formed when 135 g of aluminium is burned in air.

MATHS SKILLS

1) Write out the balanced equation: $4Al + 3O_2 \rightarrow 2Al_2O_3$
2) Calculate the relative formula masses: Al: 27 Al_2O_3: $(2 \times 27) + (3 \times 16) = 102$ (You don't have to find the M_r of oxygen because it's in excess.)
3) Calculate the number of moles of Al in 135 g: Moles of Al $= \frac{\text{mass}}{M_r} = \frac{135}{27} = 5$
4) Look at the ratio of moles in the equation: 4 moles of Al react to make half as many moles (2) of Al_2O_3, so 5 moles of Al will make $5 \div 2 = 2.5$ moles of Al_2O_3
5) Calculate the mass of 2.5 moles of Al_2O_3: moles = mass ÷ M_r, so mass = moles × M_r = $2.5 \times 102 = 255$ g (3 s.f.)

The mass of product is the yield of a reaction. Masses you calculate this way are called theoretical yields. In practice you never get 100% of the yield — the amount of product you get will be less than you calculated.

The limiting reactant determines the amount of product made

Q1 Video Solution

The important thing to remember here is that reactions are limited by the reactant which gets used up first.

Q1 The balanced equation for the reaction between chlorine and potassium bromide is:

$$Cl_2 + 2KBr \rightarrow Br_2 + 2KCl$$

Calculate the mass of potassium chloride produced when 23.8 g of potassium bromide reacts in an excess of chlorine. $A_r(K) = 39$, $A_r(Br) = 80$, $A_r(Cl) = 35.5$. [4 marks]

Gases and Solutions

Quick Quiz

*This page deals with calculating the **concentrations of solutions**. But first, here's a bit about gases...*

One Mole of Any Gas Occupies 24 dm³ at 20 °C

- At the same temperature and pressure, equal numbers of moles of any gas occupy the same volume.
- At room temperature and pressure (r.t.p. = 20 °C and 1 atm) one mole of any gas occupies 24 dm^3.

You can use this formula to find the volume of a known mass of any gas at r.t.p.:

$$\text{Volume of gas } (dm^3) = \frac{\text{Mass of gas (g)}}{M_r \text{ of gas}} \times 24$$

EXAMPLE

What's the volume of 319.5 g of chlorine at r.t.p.? $\text{Volume} = \frac{\text{Mass}}{M_r} \times 24 = \frac{319.5}{71} \times 24 = 108\ dm^3$

MATHS SKILLS

For reactions between gases, you can use the volume of one gas to find the volume of another.

EXAMPLE

How much carbon dioxide is formed when 30 dm^3 of oxygen reacts with an excess of carbon monoxide? $2CO_{(g)} + O_{2(g)} \rightarrow 2CO_{2(g)}$

MATHS SKILLS

1 mole of O_2 reacts to form 2 moles of CO_2, and the volume of gas is proportional to the number of moles, so every dm^3 of O_2 that reacts will form 2 dm^3 of CO_2.
So 30 dm^3 of O_2 reacts to form 2 × 30 dm^3 = 60 dm^3 of CO_2 (1 s.f.).

Concentration is a Measure of How Crowded Things Are

- Lots of reactions in chemistry involve substances that are dissolved in a solution. The amount of a substance in a certain volume of solvent is called its concentration.
- The more solute (dissolved substance) there is in a certain volume of solvent, the more concentrated the solution is.
- You can calculate concentration as the mass of substance in a given volume — e.g. giving it in units of g/dm^3. Or it can be the number of moles of substance in a given volume — e.g. in mol/dm^3.

$$\text{concentration } (g/dm^3) = \frac{\text{mass of solute (g)}}{\text{volume of solvent } (dm^3)}$$

$$\text{concentration } (mol/dm^3) = \frac{\text{number of moles of solute (mol)}}{\text{volume of solvent } (dm^3)}$$

- Don't forget you can rearrange formulas (see p.148). E.g. to work out the mass of solute in a given volume of solution, rearrange the first formula above to give mass = concentration × volume.

EXAMPLES

1) What's the concentration, in mol/dm^3, of a solution with 1.6 moles of salt in 2 dm^3 of solvent?

MATHS SKILLS

$$\text{concentration} = \frac{\text{number of moles of solute}}{\text{volume of solvent}} = \frac{1.6}{2} = 0.8\ mol/dm^3 \text{ (1 s.f.)}$$

2) What mass of NaCl is dissolved in 500 cm^3 of a 200 g/dm^3 solution of NaCl?

1 dm^3 = 1000 cm^3

Convert the volume to dm^3 by dividing by 1000: 500 cm^3 = 500 ÷ 1000 = 0.5 dm^3

Rearrange the formula for concentration in g/dm^3 to make mass the subject, then substitute in the numbers:

mass = concentration × volume
= 200 g/dm^3 × 0.5 dm^3 = 100 g (1 s.f.)

Concentration can be expressed in two ways

You'll be using the calculations above on the next page to work out concentrations of solutions from reacting volumes. So it's well worth that extra bit of practice to make sure you're on top of them before you move on.

Concentration Calculations

Quick Quiz

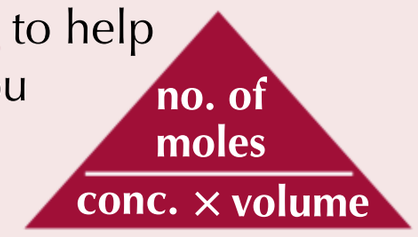
*If you've ever found yourself wondering how to find the **concentration** of a solution from the concentration of a **different** solution and a couple of **volume** values, then here's how to do it.*

You Might Be Asked to **Calculate Concentration** from Some **Results**

Titrations are experiments that let you find the volumes needed for two solutions to react together completely. If you know the concentration of one of the solutions, you can use the volumes from the titration experiment, along with the reaction equation, to find the concentration of the other solution.

There's more about titrations on page 59.

You'll need to rearrange the formula for concentration in mol/dm^3 from the previous page for these calculations. You could also use this formula triangle to help — cover up the thing you want to find and write what's left showing.



EXAMPLE

A student started with 30.0 cm^3 of sulfuric acid (H_2SO_4) of unknown concentration in a flask. She found that it took 25.0 cm^3 of 0.100 mol/dm^3 sodium hydroxide (NaOH) solution to neutralise the sulfuric acid. Find the concentration of the acid in mol/dm^3. The balanced equation is: $2NaOH + H_2SO_4 \rightarrow Na_2SO_4 + 2H_2O$

MATHS SKILLS

1) Work out the number of moles of the "known" substance you have.
Rearrange concentration = number of moles ÷ volume to give no. of moles = conc. × volume.

25.0 cm^3 = 25.0 ÷ 1000 = 0.0250 dm^3 — The volume is in cm^3, so you need to convert it to dm^3 to match the units of concentration. 1 dm^3 = 1000 cm^3.

no. of moles = 0.100 mol/dm^3 × 0.0250 dm^3
= 0.00250 moles of NaOH

2) Use the reaction equation to work out how many moles of the "unknown" stuff you must have had.

Using the equation, you can see that two moles of sodium hydroxide react with one mole of sulfuric acid.
So 0.00250 moles of NaOH must have reacted with 0.00250 ÷ 2 = 0.00125 moles of H_2SO_4.

3) Work out the concentration of the "unknown" stuff from the number of moles and the volume.

Volume of H_2SO_4 = 30.0 cm^3 = 30.0 ÷ 1000 = 0.0300 dm^3
Concentration = number of moles ÷ volume
= 0.00125 mol ÷ 0.0300 dm^3 = 0.041666... mol/dm^3 = 0.0417 mol/dm^3 (3 s.f.)

Converting **mol/dm^3** to **g/dm^3**

If you're asked a question like the one above and you have to give the concentration in g/dm^3, it's best to work it out in mol/dm^3 first and use this method to convert your answer.

Don't worry if you need to convert between the different units for concentration — it's just like converting mass to moles. That handy equation from p.45 will be useful to know yet again.

- To go from mol/dm^3 to g/dm^3, multiply by the M_r (because mass = moles × M_r).
- To go from g/dm^3 to mol/dm^3, divide by the M_r (because moles = mass ÷ M_r).

EXAMPLE

What's the concentration, in g/dm^3, of the sulfuric acid solution in the example above?

MATHS SKILLS

1) Work out the relative formula mass for the acid. $M_r(H_2SO_4) = (2 \times 1) + 32 + (4 \times 16) = 98$

2) Multiply the number of moles per dm^3 by the M_r to get the mass per dm^3.

concentration in g/dm^3 = conc. in mol/dm^3 × M_r
= 0.041666... × 98
= 4.08333... = 4.08 g/dm^3 (3 s.f.)

So many calculations...

There's a lot to process on this page, so give it another read and have a go at one of these calculations yourself:

Q1 A student found that 22.5 cm^3 of 0.150 mol/dm^3 potassium hydroxide (KOH) was needed to neutralise 25.0 cm^3 of nitric acid (HNO_3). Calculate the concentration of nitric acid in mol/dm^3.
The balanced equation for the reaction is: $HNO_3 + KOH \rightarrow KNO_3 + H_2O$ [4 marks]

Q1 Video Solution

Warm-Up & Exam Questions

Think you know everything there is about moles, limiting reactants, volumes of gases and concentration calculations? Time to put that to the test with some questions...

Warm-Up Questions

1) What do the numbers in front of the reactants and products in a balanced chemical equation stand for?
2) State why you might want to make sure one reactant is in excess when carrying out a chemical reaction.
3) In a reaction, what happens to the amount of product formed if the limiting reactant is halved?
4) How many moles of fluorine gas are in 2.4 dm^3 at r.t.p?
5) What is the concentration, in g/dm^3, of a solution that contains 20.0 g of solute in 160 cm^3 of solvent?
6) How many moles of hydrochloric acid are there in 25 cm^3 of a 0.10 mol/dm^3 solution?
7) A solution of sodium carbonate, Na_2CO_3, has a concentration of 0.0250 mol/dm^3. What is the concentration of this solution in g/dm^3?

Exam Questions

1 3.5 g of Li reacts completely with 4 g of O_2 to produce 7.5 g of Li_2O. GRADE 6-7 GRADE
$A_r(Li) = 7$, $M_r(O_2) = 32$, $M_r(Li_2O) = 30$.

1.1 Calculate how many moles of each substance reacted or was produced.

[2 marks]

1.2 Use your answer to part **1.1** to write a balanced symbol equation for this reaction.

[2 marks]

2 Sulfuric acid reacts with sodium hydrogen carbonate to produce aqueous sodium sulfate, water and carbon dioxide. The balanced equation for this reaction is: GRADE 6-7 GRADE

$$H_2SO_{4(aq)} + 2NaHCO_{3(s)} \rightarrow Na_2SO_{4(aq)} + 2H_2O_{(l)} + 2CO_{2(g)}$$

2.1 A student reacted 6.0 g of solid $NaHCO_3$ with an excess of sulfuric acid. Calculate the theoretical yield of Na_2SO_4 for this reaction.

[5 marks]

2.2 Sodium hydrogen carbonate is the limiting reactant in the example in **2.1**. Describe what is meant by the limiting reactant.

[1 mark]

Exam Questions

3 Calcium carbonate decomposes under heating to form calcium oxide and carbon dioxide. The balanced equation for this reaction is shown below:

GRADE 6-7 GRADE

$$CaCO_{3(s)} \rightarrow CaO_{(s)} + CO_{2(g)}$$

25 g of $CaCO_3$ were heated by a student.
Assuming that all the calcium carbonate decomposed, calculate the volume occupied at r.t.p. by the quantity of CO_2 produced by this reaction.
1 mol of gas occupies 24 dm^3 at r.t.p.

[4 marks]

4 In a titration, 30.4 cm^3 of a solution of 1.00 mol/dm^3 sodium hydroxide was required to neutralise 25.0 cm^3 of a solution of sulfuric acid.

GRADE 6-7 GRADE

4.1 Calculate the number of moles of sodium hydroxide used in the titration.

[1 mark]

4.2 Work out the number of moles of sulfuric acid used in the titration.

The balanced equation for this reaction is:

$$2NaOH_{(aq)} + H_2SO_{4(aq)} \rightarrow Na_2SO_{4(aq)} + 2H_2O_{(l)}$$

[1 mark]

4.3 Calculate the concentration of the sulfuric acid solution in mol/dm^3.

[2 marks]

4.4 Give your answer for **4.3** in g/dm^3.

[2 marks]

5 A student is concerned about the amount of acid in soft drinks. They decide to use a titration method to find the acid content of their favourite lemonade. The student uses a solution of 0.10 mol/dm^3 sodium hydroxide in titrations with 25 cm^3 samples of the lemonade. The results are shown in the table.

GRADE 7-9 GRADE

Repeat	Initial burette reading (cm^3)	Final burette reading (cm^3)	Vol. of NaOH needed (cm^3)
1	0.0	9.4	9.4
2	9.4	18.4	9.0
3	18.4	27.4	9.0

The student calculates that the average volume of 0.10 mol/dm^3 NaOH needed is 9.0 cm^3.

The equation for the reaction in the titration can be written:
$HA + NaOH \rightarrow NaA + H_2O$, where HA is the acid present in the lemonade.
Calculate the concentration of acid HA present in the lemonade.

[4 marks]

Atom Economy

Quick Quiz

*It's important in industrial reactions that as much of the **reactants** as possible get turned into **useful products**. This depends on the **atom economy** and the **percentage yield** (see next page) of the reaction.*

"Atom Economy" — % of Reactants Forming Useful Products

- A lot of reactions make more than one product. Some of them will be useful, but others will be waste.
- The atom economy (or atom utilisation) of a reaction tells you how much of the mass of the reactants is wasted when manufacturing a chemical and how much ends up as useful products. Learn the equation:

$$\text{atom economy} = \frac{\text{relative formula mass of desired products}}{\text{relative formula mass of all reactants}} \times 100$$

- 100% atom economy means that all the atoms in the reactants have been turned into useful (desired) products. The higher the atom economy the 'greener' the process.

EXAMPLE

Calculate the atom economy of the following reaction to produce hydrogen gas:

MATHS SKILLS

$$CH_{4(g)} + H_2O_{(g)} \rightarrow CO_{(g)} + 3H_{2(g)}$$

1) Identify the desired product. H_2 is the desired product
2) Work out the M_r of all the reactants: $M_r(CH_4) = 12 + (4 \times 1) = 16$, $M_r(H_2O) = (2 \times 1) + 16 = 18$
   Total = 16 + 18 = 34
3) Work out the total M_r of the desired product: Total = $3 \times M_r(H_2) = 3 \times (2 \times 1) = 6$
4) Use the formula to calculate the atom economy: $\frac{6}{34} \times 100 = 17.647... = 18\%$ (to the nearest %)

So in this reaction, over 80% of the mass of the starting materials is wasted.

High Atom Economy is Better for Profits and the Environment

- Pretty obviously, if you're making lots of waste, that's a problem. Reactions with a low atom economy:
  1) Use up resources very quickly.
  2) Make lots of waste materials that have to be disposed of.
  3) Tend to be unsustainable (the raw materials may run out and the waste must go somewhere).
  4) Tend to be unprofitable (it's expensive to deal with waste responsibly and buy lots of raw materials).
- The best way around the problem is to find a use for the waste products rather than just throwing them away. There's often more than one way to make the product you want, so the trick is to come up with a reaction that gives useful "by-products" rather than useless ones.
- The reactions with the highest atom economy are the ones that only have one product, as they have an atom economy of 100%. The more products there are, the lower the atom economy is likely to be.

Atom economy isn't the only factor to be considered when choosing which reaction to use to make a certain product. Things such as the yield (see next page), the rate of the reaction (see page 80) and the position of equilibrium for reversible reactions (see page 87) also need to be thought about. A reaction with a low atom economy, that produces useful by-products, might also be used.

Atom economy is really important in industry

Q1 Video Solution

You may get an unfamiliar industrial reaction in the exam — whatever they give you, the same stuff applies.

Q1 The following two reactions, A and B, both produce NH_3:

A. $2NH_4Cl + 2CaO \rightarrow CaCl_2 + Ca(OH)_2 + 2NH_3$ B. $N_2 + 3H_2 \rightarrow 2NH_3$

$A_r(N) = 14$, $A_r(H) = 1$, $A_r(Cl) = 35.5$, $A_r(Ca) = 40$, $A_r(O) = 16$

a) Which reaction, A or B, has the higher atom economy? Explain your answer. [1 mark]

b) Calculate the atom economy for the production of NH_3 by reaction A. [5 marks]

Percentage Yield

Quick Quiz

*Percentage yield tells you about the **overall success** of an experiment. It **compares** what you think **you should get** (theoretical yield) with what **you get in practice** (actual yield).*

Percentage Yield Compares **Actual** and **Theoretical** Yield

The amount of product you get is known as the yield. The more reactants you start with, the higher the actual yield will be — that's not surprising. But the percentage yield doesn't depend on the amount of reactants you started with — it's a percentage. The equation is:

$$\text{percentage yield} = \frac{\text{mass of product actually made (g)}}{\text{maximum theoretical mass of product (g)}} \times 100$$

This is just the maximum theoretical yield. You need to know how to calculate it from the mass of limiting reactant and the balanced reaction equation (see p.49).

- Percentage yield is always somewhere between 0 and 100%. 100% yield means that you got all the product that you could possibly get. 0% yield means that no reactants were converted into product, i.e. no product at all was made.
- Industrial processes should have as high a percentage yield as possible to reduce waste and reduce costs.

Yields are Always **Less Than 100%**

In real life, you never get a 100% yield — sad, I know. Some product or reactant will always get lost along the way — this is just as true for big, fancy industrial processes as it is for your school lab experiments.

How this happens depends on what sort of reaction it is and what apparatus is being used.
Lots of things can go wrong, but three common problems are:

1 Reactants aren't fully used up

In reversible reactions (see page 87), the products can turn back into reactants, so the yield will never be 100%.

For example, in the Haber process, at the same time as the reaction $N_2 + 3H_2 \rightarrow 2NH_3$ is taking place, the reverse reaction $2NH_3 \rightarrow N_2 + 3H_2$ is also happening.

So the reaction $N_2 + 3H_2 \rightarrow 2NH_3$ never goes to completion (the reactants don't all get used up).

2 Side reactions

The reactants sometimes react differently to how you expect. They might react with gases in the air, or impurities in the reaction mixture, so they end up forming extra products other than the ones you want.

3 Product is lost during separation

When you filter a liquid to remove solid particles, you nearly always lose a bit of liquid or a bit of solid.

- If you want to keep the liquid, you'll lose the bit that remains with the solid and filter paper (as they always stay a bit wet).
- If you want to keep the solid, some of it'll get left behind when you scrape it off the filter paper.

You'll also lose a bit of material when you transfer it from one container to another — even if you're careful not to spill it. Some of it always gets left behind on the inside surface of the old container.

A yield of 100% in a reaction is pretty much impossible

Q1 Video Solution

You can do yield calculations for the practicals you do in the lab as well as for industrial processes.
Try this question about the yield of a student's lab experiment:

Q1 A student carries out the following reaction: $Zn + 2HCl \rightarrow ZnCl_2 + H_2$. She calculates that her maximum theoretical yield of zinc chloride is 2.72 g. The actual yield after purification is 2.31 g. Calculate the percentage yield of zinc chloride. [1 mark]

Warm-Up & Exam Questions

Atom economy and percentage yield aren't too hard to calculate once you've had some practice.
So here are some questions to get you started...

Warm-Up Questions

1) What is the atom economy of the reaction shown? $2SO_2 + O_2 \rightarrow 2SO_3$
2) Why might a reaction with a low atom economy be bad for sustainable development?
3) What is the percentage yield of a reaction which produced 4 g of product if the predicted yield was 5 g?

Exam Questions

1 Which one of the following statements about atom economy is **not** true? Tick **one** box.

GRADE 4-6

- ☐ Reactions that only have one product have the highest atom economy.
- ☐ Reactions with a low atom economy tend to be less sustainable than reactions with high atom economies.
- ☐ Reactions with low atom economies will generally produce more waste than reactions with high atom economies.
- ☐ To calculate the atom economy of a reaction, you need to know the mass of products you expected to form, and the mass of products that actually formed.

[1 mark]

2 Ethanol produced by the fermentation of sugar can be converted into ethene, as shown below. The ethene can then be used to make polythene.

GRADE 4-6

$$C_2H_6O_{(g)} \rightarrow C_2H_{4(g)} + H_2O_{(g)}$$

Calculate the atom economy of this reaction. (A_r values: C = 12, O = 16, H = 1)

[3 marks]

3* Discuss the reasons why yields from chemical reactions are always less than 100%.

GRADE 6-7

[6 marks]

4 A sample of copper was made by reducing 4.0 g of copper oxide with methane gas.
When the black copper oxide turned orange-red, the sample was scraped out into a beaker.
Sulfuric acid was added to dissolve any copper oxide that remained.
The sample was then filtered, washed and dried. 2.8 g of copper was obtained.
(A_r values: H = 1, Cu = 63.5, O = 16.) The equation for this reaction is:

GRADE 7-9

$$CH_4 + 4CuO \rightarrow 4Cu + 2H_2O + CO_2$$

4.1 Use the equation to calculate the maximum mass of copper which could be obtained from the reaction (the theoretical yield).

[3 marks]

4.2 Calculate the percentage yield of the reaction.

[2 marks]

Revision Summary Test for Topic 3

Holy mole-y, that was a long topic. Have a well-earned breather, then try these things:

- Tackle the revision summary test below, or scan the QR code to do it online.
  The questions are hard, but they'll show you how well you really know your stuff.
- Track your progress online and see which areas need more work.
- Compare your answers with sample answers for the test here: cgpbooks.co.uk/Base

Summary Test

Moles and Equations (p.44-49) ☑

1) How do you calculate the relative formula mass (M_r) of a substance? ☑
2) What is the formula for calculating the percentage mass of an element in a compound? ☑
3) What is the value of the Avogadro constant?
   Explain how the Avogadro constant links moles, mass and relative atomic mass for an element. ☑
4) Write down the formula for the number of moles of a compound in terms of its mass and M_r.
   Now rearrange the formula to make mass the subject. ☑
5) What does conservation of mass mean? What two things should be the same
   on both sides of a balanced equation due to mass being conserved? ☑
6) For each of the observations below, explain how it is possible to observe
   a change in mass when a reaction takes place in an unsealed reaction vessel,
   and give an example of a reaction that could produce the observation:
   a) the mass of the reaction vessel increases, b) the mass of the reaction vessel decreases. ☑
7) How can you use the masses of reactants and products taking part
   in a reaction to work out a balanced symbol equation? ☑
8) What is a limiting reactant? How does the amount of limiting reactant affect the amount of product? ☑
9) Explain what it means if a reactant is in excess, and why reactants are often added in excess. ☑
10) How would you work out the theoretical yield of a product of a reaction
    from the mass of the limiting reactant and the balanced equation? ☑

Volumes of Gases and Concentrations of Solutions (p.50-51) ☑

11) What volume does one mole of any gas occupy at room temperature and pressure?
    And what are the conditions for room temperature and pressure? ☑
12) Give the formula that links the volume of a gas at r.t.p. to its mass. ☑
13) What does the concentration of a solution tell you? ☑
14) Write a formula to work out concentration in g/dm^3. Then write one for mol/dm^3. ☑
15) Describe how to work out the concentration of an acid solution. Assume that you know:
    the exact volumes of the acid solution and an alkali solution that react completely,
    the balanced equation for the reaction and the concentration of the alkali solution. ☑
16) How would you convert a concentration in mol/dm^3 to g/dm^3? What about the other way? ☑

Atom Economy and Percentage Yield (p.54-55) ☑

17) What does the atom economy of a reaction tell you? Write a formula for calculating it. ☑
18) How can atom economy affect how sustainable and profitable a reaction is? ☑
19) What's the atom economy of a reaction with only one product? ☑
20) Give four factors that are considered, along with atom economy,
    when choosing which reaction to use to make a certain product. ☑
21) Explain what these terms mean: a) yield, b) theoretical yield, c) percentage yield. ☑
22) Give the formula for percentage yield. ☑
23) Give three reasons why percentage yield is always less than 100%. ☑

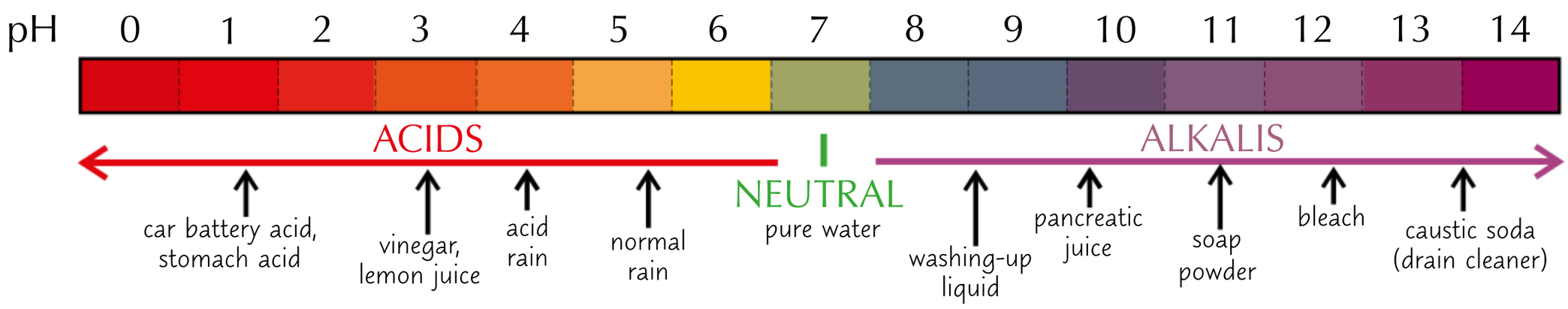
Quick Quiz

Acids and Bases

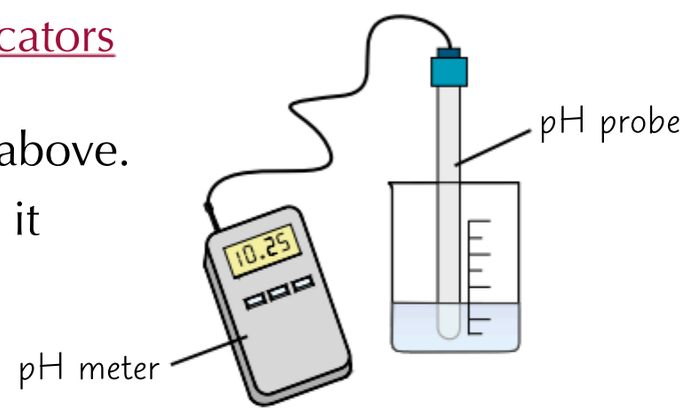
__Acids__ and __bases__ crop up everywhere in chemistry — so here's the lowdown on the basics of pH...

The **pH Scale** Goes From **0 to 14**

1) The pH scale is a measure of how acidic or alkaline a solution is.
2) The lower the pH of a solution, the more acidic it is. The higher the pH, the more alkaline the solution is.
3) A neutral substance (e.g. pure water) has pH 7.



4) Indicators can be used to measure pH approximately. They are dyes that change colour depending on whether they are above or below a certain pH.
5) Some indicators contain a mixture of dyes that means they gradually change colour over a range of pH values. These are wide range indicators and they're useful for estimating the pH of a solution. For example, universal indicator gives the red-to-green-to-purple pH scale shown above.
6) You can also estimate pH using indicators in paper form, or measure it electronically using a pH probe attached to a pH meter. There's more about measuring pH on p.152.



Acids and **Bases** Neutralise Each Other

Acids	Form aqueous solutions containing H^+ ions. The solutions have a pH less than 7.
Bases	Substances that react with acids to neutralise them.
Alkalis	Bases that dissolve in water to form solutions containing OH^- ions. The solutions have a pH greater than 7.

Reactions between acids and bases are called neutralisation reactions:

acid + base → salt + water

The salt and water products of neutralisation have a pH of 7 — i.e. they are neutral.

When acid and alkali solutions neutralise each other, the neutralisation reaction is basically the reaction between the H^+ ions from the acid and the OH^- ions from the alkali, making water:

$$H^+_{(aq)} + OH^-_{(aq)} \rightarrow H_2O_{(l)}$$

You can use an indicator to show when a neutralisation reaction reaches pH 7 and is complete. Neutralisation reactions can also be used to find concentrations by titration. There's more about all of this on the next page.

Interesting fact — your skin is slightly acidic (pH 5.5)

Neutralisation reactions are an important bit of chemistry. Remember, when you mix an acid and an alkali, hydrogen ions from the acid react with hydroxide ions from the alkali to make water. The leftover bits of the acid and alkali make a salt.

PRACTICAL

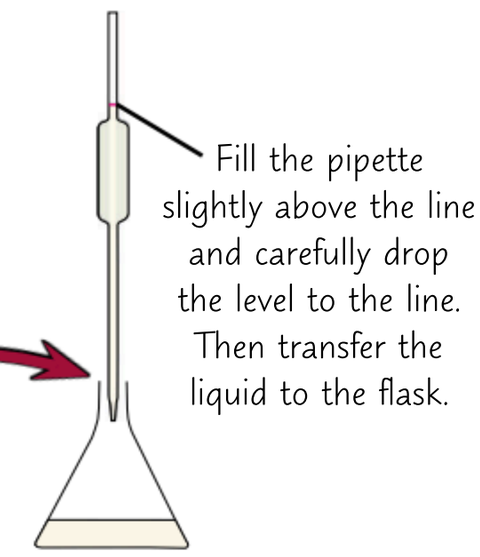
Titrations

Quick Quiz

*Titrations are a method of analysing the **concentrations** of solutions. They're pretty important.*

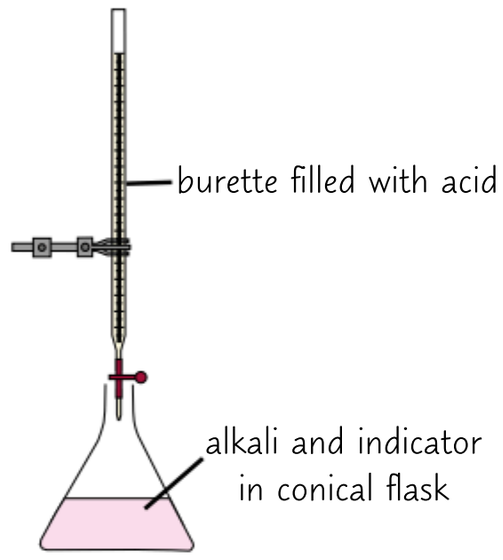
Titrations are Used to Find **Concentrations**

Titrations allow you to find the exact quantity of acid needed to neutralise a quantity of alkali (or vice versa). You can use this to work out the concentration of the acid/alkali — see p.51 for how to do these calculations. Here's how you could carry out a titration where you want to find the concentration of the alkali:



1) Using a pipette and pipette filler, add a set volume of the alkali to a conical flask.
2) Add two or three drops of a suitable indicator too.
3) Use a funnel to fill a burette with some acid of known concentration. Do this below eye level — you don't want to be looking up if some acid spills over. (You should wear safety glasses too.)
4) Record the initial volume of the acid in the burette.
5) Do a rough titration:

There's more about using burettes and pipettes on page 151.



- Quickly add the acid from the burette to the alkali, swirling the flask as you go.
- Stop adding the acid when the indicator changes colour. This happens once all the alkali has been neutralised — it's called the end-point of the titration.
- Work out how much acid you added — it's the difference between your initial and final readings on the burette.
- This won't be a very accurate measurement — the point is to show you roughly where the end-point is, so you know when you need to be really careful.

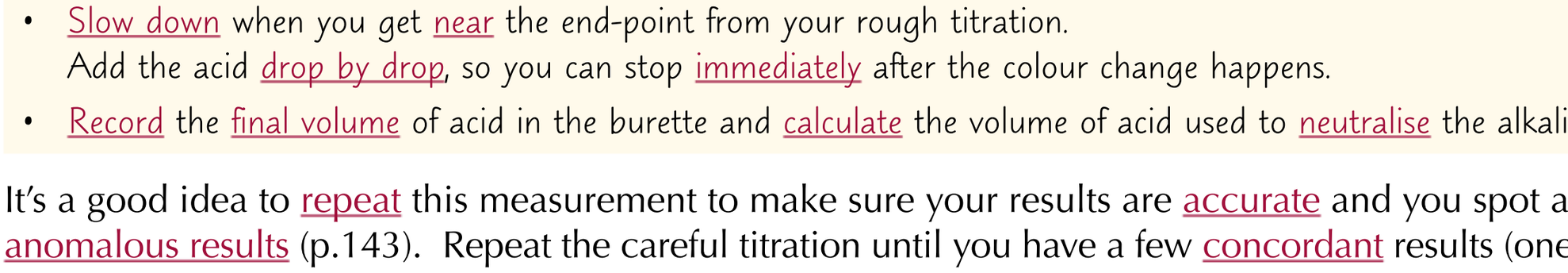
6) Now do a more careful titration:

- Set up the titration again, then slowly add the acid to the alkali again while swirling the flask.
- Slow down when you get near the end-point from your rough titration. Add the acid drop by drop, so you can stop immediately after the colour change happens.
- Record the final volume of acid in the burette and calculate the volume of acid used to neutralise the alkali.

7) It's a good idea to repeat this measurement to make sure your results are accurate and you spot any anomalous results (p.143). Repeat the careful titration until you have a few concordant results (ones which are very close to each other, e.g. within 0.1 cm^3). You can then calculate a mean volume from these results.

Use **Single Indicators** for Titrations

1) As you saw on the previous page, wide range indicators can be used to estimate the pH of a solution because they give you a variety of colours — the classic example is universal indicator.
2) This isn't so useful during a titration — instead of a gradual colour change, you want there to be a sudden colour change as soon as the reaction reaches the end-point.
3) Using a single indicator will give you this sudden colour change. Some examples are:



Accuracy is everything...

When you're doing a titration you need to make sure that your results are as accurate as possible. To do this you need to a) go very slowly as you reach the end-point of your careful titrations, and b) do a few repeats and find a mean.

Quick Quiz

Strong Acids and Weak Acids

*Right, it's time for **strong acids** versus **weak acids**. Brace yourself, it's not the simplest bit of chemistry ever...*

Acids can be **Strong** or **Weak**

As you saw on p.58, acids ionise when they dissolve in water — they dissociate (break apart) into ions. This produces hydrogen ions, H^+.

An H^+ ion is just a proton.

	Dissociation in water	Examples
Strong acids	Completely ionise in water — all acid particles dissociate to release H^+ ions. $HCl \longrightarrow H^+ + Cl^-$	hydrochloric acid, HCl nitric acid, HNO_3 sulfuric acid, H_2SO_4
Weak acids	Partially ionise in water — only a small proportion of acid particles dissociate to release H^+ ions. $CH_3COOH \rightleftharpoons H^+ + CH_3COO^-$	ethanoic acid, CH_3COOH (p.101) citric acid carbonic acid

You don't need to know the formulas for these last two.

- The ionisation of a weak acid is a reversible reaction, so there is an equilibrium (p.87) between the undissociated and dissociated acid. Since only a few of the acid particles release H^+ ions, the position of equilibrium lies well to the left.
- Reactions of acids involve the H^+ ions reacting with other substances. The higher the concentration of H^+ ions, the faster the rate of reaction (p.81), so strong acids are more reactive than weak acids of the same concentration.

pH is a Measure of the **Concentration** of **Hydrogen Ions**

1) You can use pH values to compare the concentrations of H^+ ions in solutions. For every decrease of 1 on the pH scale, the concentration of H^+ ions increases by a factor of 10. A pH 1 solution has 10 times the H^+ ion concentration of a pH 2 solution, and 100 times the H^+ ion concentration of a pH 3 solution.

2) The general rule for this is:

Factor H^+ ion concentration changes by = 10^{-X}

X is the difference in pH.

EXAMPLE

A solution's pH falls from 7 to 5.
By what factor did the H^+ ion concentration change?

MATHS SKILLS

X is the difference in pH, so *X* = final pH − initial pH = 5 − 7 = −2
Factor H^+ ion concentration changes by = $10^{-X} = 10^{-(-2)} = 10^2 = 100$

This means the concentration of H^+ ions got 100 times stronger.

3) If they have the same concentration, the pH of a strong acid is always less than the pH of a weaker acid.

Don't Confuse **Strong** Acids with **Concentrated** Acids

1) Concentration is how much acid there is in a certain volume of solvent (e.g. water). The more acid there is in a certain volume of solvent, the more concentrated (or less dilute) it is.
2) Concentration measures how many dissolved acid particles there are, whether they have ionised in the solvent or not, so it's not the same as acid strength (whether the acid is strong or weak).
3) It's possible to have a dilute solution of a strong acid, or a concentrated solution of a weak acid.
4) If you increase the concentration of an acid, its pH will decrease regardless of if it's a strong or weak acid.

Strong acids ionise completely, but weak acids don't

Q2 Video Solution

Calculating the factor that hydrogen ion concentration changes by as pH changes can be a bit tricky. The best way to get to grips with it is just to get lots of practice at using the formula.

Q1 Name a strong acid. [1 mark]

Q2 A student added strong acid to a weakly acidic solution of pH 6. The pH of the new solution was found to be pH 3. By how many times did the concentration of H^+ increase? [1 mark]

Reactions of Acids

Quick Quiz

Remember neutralisation from page 58? Well, there's more stuff on ***neutralisation reactions*** *coming up...*

Metal **Oxides** and Metal **Hydroxides** are **Bases**

All metal oxides and metal hydroxides are bases, so they react with acids in neutralisation reactions:

Acid + Metal Oxide → Salt + Water

Acid + Metal Hydroxide → Salt + Water

Most metal oxides and metal hydroxides are insoluble bases, but some are alkalis (i.e. they are soluble bases — see p.58). They react with acids in the same way whether or not they are soluble.

The salt that's produced depends on which acid is used and what the metal ion in the base is:

hydrochloric acid	+	copper oxide	→	copper chloride	+	water
$2HCl$	+	CuO	→	$CuCl_2$	+	H_2O
sulfuric acid	+	potassium hydroxide	→	potassium sulfate	+	water
H_2SO_4	+	$2KOH$	→	K_2SO_4	+	$2H_2O$
nitric acid	+	sodium hydroxide	→	sodium nitrate	+	water
HNO_3	+	$NaOH$	→	$NaNO_3$	+	H_2O

To work out the formula of an ionic compound, you need to balance the charges of the positive and negative ions so the overall charge is neutral — see p.28.

Acids and Metal **Carbonates** Produce **Carbon Dioxide**

Metal carbonates are also bases. They react with acids to produce a salt, water and carbon dioxide.

Acid + Metal Carbonate → Salt + Water + Carbon Dioxide

hydrochloric acid	+	sodium carbonate	→	sodium chloride	+	water	+	carbon dioxide
$2HCl$	+	Na_2CO_3	→	$2NaCl$	+	H_2O	+	CO_2
sulfuric acid	+	calcium carbonate	→	calcium sulfate	+	water	+	carbon dioxide
H_2SO_4	+	$CaCO_3$	→	$CaSO_4$	+	H_2O	+	CO_2

You can Make a **Soluble Salt** Using an **Insoluble Base**

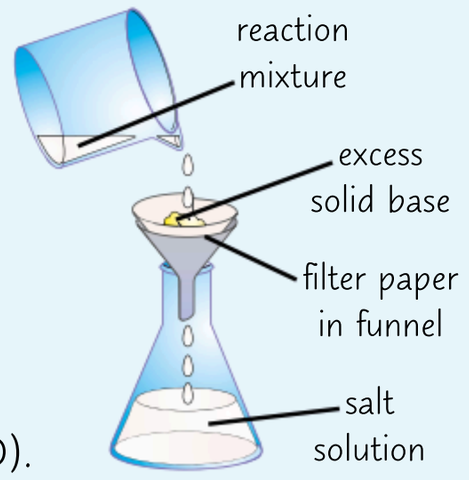
PRACTICAL

Here's one way you could make a pure, dry sample of a soluble salt from an acid and an insoluble base:

Method

1) Choose the right acid and insoluble base, such as an insoluble metal oxide, hydroxide or carbonate. E.g. to make copper chloride, you could use hydrochloric acid and copper oxide (see above).
2) Gently warm a dilute solution of your acid using a Bunsen burner, then turn off the Bunsen burner.
3) Add the insoluble base to the acid a bit at a time, until no more reacts (i.e. the base is in excess). You'll know when all the acid has been neutralised because, even after stirring, the excess solid will just sink to the bottom of the flask.
4) Then filter out the excess solid to get the salt solution (see p.10).
5) To get pure, dry crystals of the salt, gently heat the solution using a water bath or an electric heater to evaporate some of the water, making the solution more concentrated. Then stop heating the solution and leave it to cool. Crystals of the salt should form, which can be filtered out of the solution and then dried. This is called crystallisation (p.10).

It's also possible to use an acid-metal reaction — see the page 63.



Quite a few reactions to learn here...

You could be asked to describe how to make a soluble salt, so make sure you understand the method.

Q1 Calcium carbonate is added to hydrochloric acid.
Write the word equation and the balanced symbol equation for the reaction that occurs. [3 marks]

Q1 Video Solution

Warm-Up & Exam Questions

So you think you know everything there is to know about acids? Time to put yourself to the test.

Warm-Up Questions

1) What range of values can pH take?
2) What term is used to describe a solution with a pH of 7?
3) What's the difference between a strong acid and a weak acid?
4) If the pH of a solution rises from 2 to 5, what factor has the hydrogen ion concentration of the solution changed by?
5) Name the two substances formed when nitric acid reacts with copper hydroxide.

Exam Questions

1 A student had a sample of acid in a test tube. He gradually added some alkali to the acid. GRADE 4-6

1.1 Name the type of ion that acids produce in aqueous solutions.

[1 mark]

1.2 What type of reaction took place in the student's experiment? Tick **one** box.

☐ thermal decomposition ☐ neutralisation ☐ redox ☐ combustion

[1 mark]

2 All metal hydroxides are bases. They can react with acids to form a salt and water. GRADE 6-7

2.1 Sodium hydroxide is a soluble base. What name is given to bases that dissolve in water?

[1 mark]

2.2 Complete and balance the symbol equation given below for the reaction of nitric acid with magnesium hydroxide.

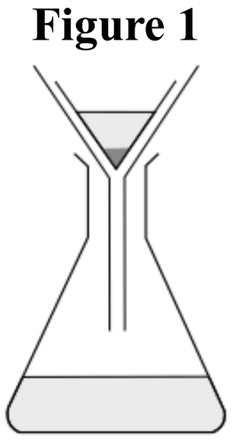
$$2HNO_3 + Mg(OH)_2 \rightarrow \text{.....} + \text{.....}$$

[2 marks]

PRACTICAL

3 Silver nitrate is a soluble salt. It can be made by adding an excess of insoluble silver carbonate to nitric acid until no further reaction occurs. GRADE 6-7

Figure 1



3.1 Give **one** observation that would indicate that the reaction is complete.

[1 mark]

3.2 Once the reaction is complete, the excess silver carbonate can be separated from the silver nitrate solution using the apparatus shown in **Figure 1**. What is this method of separation called?

[1 mark]

3.3 Describe how you could produce solid silver nitrate from silver nitrate solution.

[3 marks]

PRACTICAL

4* Describe the method you would use to carry out an accurate titration to find the volume of acid needed to neutralise 25.0 cm^3 of an alkali. Assume you have already done a rough titration. GRADE 6-7

[6 marks]

The Reactivity Series

Quick Quiz

*Metals can be placed in order of **reactivity**. This can be really handy for **predicting** their **reactions**.*

The **Reactivity Series** — How Well a **Metal** Reacts

1) The reactivity series lists metals in order of their reactivity towards other substances. Make sure you learn this list:

2) Metals react by losing electrons to form positive ions — see p.20. The more easily a metal forms positive ions, the higher up it goes in the reactivity series.

3) You can find the order of the reactivity series by observing reactions:

- E.g. you'll see below that some metals react with water or acid by losing electrons and forming positive ions.
- You can put metals in order from most to least reactive by comparing whether they react at all with acid or water, and how violent the reaction is if they do react.

The Reactivity Series

Potassium	K	Very reactive
Sodium	Na	
Lithium	Li	
Calcium	Ca	
Magnesium	Mg	Fairly reactive
Carbon	C	
Zinc	Zn	
Iron	Fe	
Hydrogen	H	Not very reactive
Copper	Cu	

Carbon and hydrogen are non-metals but are often included in the reactivity series.

How **Metals** React with **Acids** Tells You About Their **Reactivity**

Some metals react with acids: **Acid + Metal → Salt + Hydrogen**

The more reactive the metal, the faster the reaction will go. The speed of reaction is indicated by the rate at which the bubbles of hydrogen are given off.

MORE REACTIVE ← → LESS REACTIVE

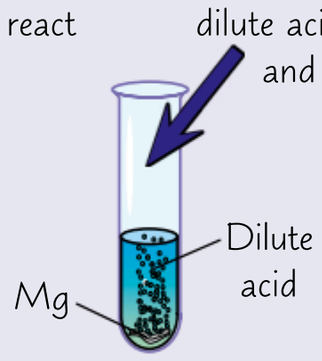
Very reactive metals like potassium, sodium, lithium and calcium react explosively with acids.

Magnesium reacts vigorously with cold, dilute acids such as $HCl_{(aq)}$ or $H_2SO_{4(aq)}$ and produces loads of bubbles.

Both zinc and iron react slowly with cold, dilute acids but more strongly if you heat the reactants.

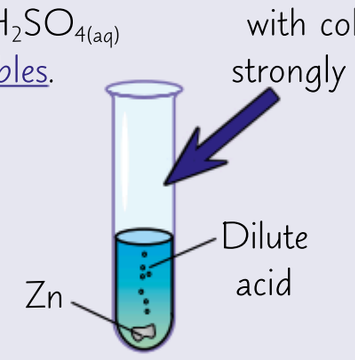
In general, copper won't react with cold, dilute acids.

HCl reacts to form chloride salts, H_2SO_4 reacts to form sulfate salts.



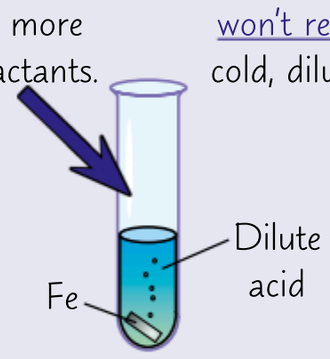
$Mg_{(s)} + 2HCl_{(aq)} \rightarrow MgCl_{2(aq)} + H_{2(g)}$

$Mg_{(s)} + H_2SO_{4(aq)} \rightarrow MgSO_{4(aq)} + H_{2(g)}$



$Zn_{(s)} + 2HCl_{(aq)} \rightarrow ZnCl_{2(aq)} + H_{2(g)}$

$Zn_{(s)} + H_2SO_{4(aq)} \rightarrow ZnSO_{4(aq)} + H_{2(g)}$



$Fe_{(s)} + 2HCl_{(aq)} \rightarrow FeCl_{2(aq)} + H_{2(g)}$

$Fe_{(s)} + H_2SO_{4(aq)} \rightarrow FeSO_{4(aq)} + H_{2(g)}$

Metals Also React with **Water**

You can see more on the reactions of Group 1 metals with water on p.20.

- Some metals react with water at room temperature: **Metal + Water → Metal Hydroxide + Hydrogen**

  E.g. when calcium reacts with water, the equation is: $Ca_{(s)} + 2H_2O_{(l)} \rightarrow Ca(OH)_{2\,(aq)} + H_{2\,(g)}$.

- The reactions of metals with water also demonstrate differences in reactivity. More reactive metals (e.g. K, Na, Li, Ca) will react with water, but less reactive metals (e.g. Zn, Fe, Cu) won't react.

You can also investigate the reactivity of metals by measuring the temperature change of the reaction with an acid or water over a set time period. If you use the same mass and surface area of metal each time, and the same amount of acid or water, then the more reactive the metal, the greater the temperature change should be.

Metals at the top of the reactivity series are highly reactive

You can use the results of experiments like these to work out an order of reactivity for the metals that you used. The more vigorous the reaction with acid or water, the more reactive the metal.

Q1 Give the balanced equation, including state symbols, for the reaction of sodium and water. [3 marks]

Q1 Video Solution

Quick Quiz

Separating Metals From Metal Oxides

*Most metals aren't just found as pure lumps — you have to **extract** them from their ores. How inconvenient...*

Metals Often Have to be Separated from their **Oxides**

1) Lots of common metals, like iron and aluminium, react with oxygen to form oxides.
These oxides are often the ores found in the ground that metals need to be extracted from.

2) The reactions of metals with oxygen to make oxides are oxidation reactions.
A reaction that separates a metal from its oxide is called a reduction reaction.

An ore is a type of rock that contains metal compounds.

OXIDATION	REDUCTION
Gain of oxygen	Loss of oxygen
E.g. formation of metal ore — metal is oxidised to form metal oxide.	E.g. extraction of metal — metal oxide is reduced to form metal.
E.g. $2Mg + O_2 \rightarrow 2MgO$	E.g. $2CuO + C \rightarrow 2Cu + CO_2$

Magnesium gains oxygen in the oxidation reaction.

Copper oxide loses oxygen in the reduction reaction.

Some Metals can be **Extracted** by **Reduction** with **Carbon**

1) One method of extracting a pure metal from a metal oxide involves reacting the metal oxide with carbon.

2) The metal ore is reduced as oxygen is removed from it. The carbon gains oxygen so is oxidised.

For example:

iron(III) oxide	+	carbon	→	iron	+	carbon dioxide
$2Fe_2O_3$	+	$3C$	→	$4Fe$	+	$3CO_2$

3) The position of the metal in the reactivity series determines whether it can be extracted by reduction with carbon:

- Only metals below carbon in the reactivity series can be extracted by reduction using carbon. (E.g. iron is extracted from iron oxide in a blast furnace.)
- One way of thinking about this is that carbon can only take oxygen away from the metal oxide if carbon is more reactive than the metal.
- Metals higher than carbon in the reactivity series can't be extracted by reduction using carbon. They have to be extracted using electrolysis (p.68) instead, which is much more expensive.

More reactive than carbon — extracted using electrolysis.

Less reactive than carbon — extracted by reduction using carbon.

The Reactivity Series

Potassium	K
Sodium	Na
Lithium	Li
Calcium	Ca
Magnesium	Mg
Carbon	C
Zinc	Zn
Iron	Fe
Copper	Cu

Make sure you can explain how different metals are extracted, and why different metals are extracted in different ways.

Some metals are so unreactive they are found in the earth as the metal itself. For example, gold is mined as its elemental form.

Carbon can't reduce things above it in the reactivity series

Metals are incredibly useful. It's a shame that extracting them isn't always cheap. Make sure you know the difference between reduction and oxidation and why carbon can only be used to extract some metals.

Q1 Write a balanced equation for the reduction of lead oxide, PbO, by carbon, C. [2 marks]

Q2 Explain why calcium can't be extracted from calcium oxide by reduction using carbon. [1 mark]

Q1 Video Solution

Redox Reactions and Half Equations

Quick Quiz

*In chemistry, oxidation **doesn't** just mean gain of oxygen. No, that would be far too easy.*

If **Electrons** are **Transferred**, It's a **Redox Reaction**

1) As you saw on page 64, oxidation is the gain of oxygen and reduction is the loss of oxygen. But you can also define oxidation and reduction in terms of electrons.
2) Oxidation is a loss of electrons. Reduction is a gain of electrons.
3) REDuction and OXidation happen at the same time — for one reactant to gain electrons, another reactant has to lose them. That's why the term "REDOX" is used for this type of reaction.

A handy way to remember this is the mnemonic OIL RIG — Oxidation Is Loss, Reduction Is Gain.

E.g. the formation of magnesium oxide (see previous page) is a redox reaction.

- Magnesium is a metal, so it reacts by losing electrons to form Mg^{2+} ions — it is oxidised.
- Oxygen is a non-metal, so it reacts by gaining electrons to form O^{2-} ions — it is reduced.

Half Equations Show Which Substances are **Oxidised** and **Reduced**

Half equations show one half of a redox reaction — they either show just electrons being gained or just electrons being lost.

Oxidation of magnesium: $Mg \rightarrow Mg^{2+} + 2e^-$

E.g. here's how to write balanced half equations for the reaction $Ca + Cl_2 \rightarrow CaCl_2$:

1) Write out the full balanced equation again, but break up any ionic compounds into ions.

$Ca + Cl_2 \rightarrow Ca^{2+} + 2Cl^-$

Use the periodic table to work out the charges on the ions (see p.26 for more).

2) Split it into half equations. Each half equation shows the change in one species (usually one element).

$Ca \rightarrow Ca^{2+}$ $\quad$ $Cl_2 \rightarrow 2Cl^-$

3) Balance the charges in each half equation by adding electrons (written e^-) to the side with the more positive total charge.

Ca has a charge of 0.
Ca^{2+} has a charge of 2+. Add $2e^-$ to the more positive side (Ca^{2+}):
$Ca \rightarrow Ca^{2+} + 2e^-$

Cl_2 has a charge of 0.
$2Cl^-$ has a charge of 2–. Add $2e^-$ to the more positive side (Cl_2):
$Cl_2 + 2e^- \rightarrow 2Cl^-$

Remember to check that both the atoms and the charges are balanced in any half equations you write.

You can **Combine** Half Equations to get **Redox Equations**

You can combine half equations to get an overall equation for a redox reaction. The key thing to remember is that each half equation must have the same number of electrons when you combine them.

E.g. the half-equations when sodium reacts with an acid are:

$Na \rightarrow Na^+ + e^-$ $\qquad$ $2H^+ + 2e^- \rightarrow H_2$

1) Double the first half equation so both have $2e^-$: $2Na \rightarrow 2Na^+ + 2e^-$
2) Combine them and cancel out the electrons:
$2Na + 2H^+ + \cancel{2e^-} \rightarrow 2Na^+ + H_2 + \cancel{2e^-}$
3) This gives you the overall equation:
$2Na + 2H^+ \rightarrow 2Na^+ + H_2$

This is an ionic equation — see the next page.

All reactions of acids with metals (see p.63) are redox reactions — they involve electrons being transferred from the metal to the hydrogen.

Remember OIL RIG — 'Oxidation Is Loss, Reduction Is Gain'

OIL RIG is a pretty handy way of reminding yourself about what goes on during a redox reaction. Make sure you don't forget that redox reactions are all about the transfer of electrons — one thing loses them and something else gains them.

Displacement Reactions & Ionic Equations

Quick Quiz

Next up, ***displacement reactions*** *and* ***ionic equations*** *— it's more things to do with* ***redox reactions****, really...*

Displacement Reactions are also **Redox Reactions**

In a displacement reaction, one metal kicks another metal out of a compound and replaces it. The important rule you need to learn is:

See the reactivity series on p.63 for which metals are more reactive than other metals.

A MORE REACTIVE metal will displace a LESS REACTIVE metal from its compound.

E.g. you can see from the reactivity series that iron is more reactive than copper. So if you put iron in a solution of copper sulfate the more reactive iron will 'kick out' the less reactive copper from the solution. You end up with iron sulfate solution and copper metal.

iron + copper sulfate → iron sulfate + copper

$Fe_{(s)}$ + $CuSO_{4(aq)}$ → $FeSO_{4(aq)}$ + $Cu_{(s)}$

In displacement reactions, it's always the metal ion (i.e. the one in the compound) that gains electrons and is reduced. The metal atom always loses electrons and is oxidised.

E.g. in the reaction between iron and copper sulfate:

The metal ion is reduced — copper ions gain electrons to form copper atoms:

$Cu^{2+} + 2e^- \rightarrow Cu$

The metal is oxidised — iron atoms lose electrons to form iron ions:

$Fe \rightarrow Fe^{2+} + 2e^-$

Ionic Equations Show Just the **Useful Bits** of Reactions

In an ionic equation only the particles that change during a reaction are shown — i.e. just the things that react and the products they form. Here's how to write one, using the reaction above as an example:

1) Start by writing out the full equation:

$Fe_{(s)} + CuSO_{4(aq)} \rightarrow FeSO_{4(aq)} + Cu_{(s)}$

2) Now write it out again including all the ions for dissolved (aqueous) compounds:

$Fe_{(s)} + Cu^{2+}_{(aq)} + SO_4^{2-}{}_{(aq)} \rightarrow Fe^{2+}_{(aq)} + SO_4^{2-}{}_{(aq)} + Cu_{(s)}$

3) The sulfate ions don't change in the reaction (they're spectator ions), so cross them out:

$Fe_{(s)} + Cu^{2+}_{(aq)} + \cancel{SO_4^{2-}{}_{(aq)}} \rightarrow Fe^{2+}_{(aq)} + \cancel{SO_4^{2-}{}_{(aq)}} + Cu_{(s)}$

4) Crossing out all the spectator ions leaves you with the ionic equation:

$Fe_{(s)} + Cu^{2+}_{(aq)} \rightarrow Fe^{2+}_{(aq)} + Cu_{(s)}$

Ionic equations are helpful for redox reactions like this displacement reaction. Because the ionic equation only shows the things that react, it only contains the substances that are oxidised and reduced. You can also get the ionic equation by combining the half equations using the method from the previous page.

Ionic equations can be used for any reaction involving ions, not just displacement reactions. E.g. the equation for neutralisation on p.58 is an ionic equation.

A displacement reaction is a type of redox reaction

The best way to get the hang of ionic equations is to try writing some. Start with this question...

Q1 The equation for the reaction of zinc and iron sulfate is: $Zn_{(s)} + FeSO_{4(aq)} \rightarrow ZnSO_{4(aq)} + Fe_{(s)}$

a) Write an ionic equation for the reaction. [1 mark]

b) State which species is being reduced and which is being oxidised. [2 marks]

Q1 Video Solution

Warm-Up & Exam Questions

Hoping to test your knowledge with some tricky chemistry questions? You're in luck...

Warm-Up Questions

1) What determines the reactivity of a metal?
2) Magnesium is reacted with dilute hydrochloric acid. Give the name of the salt formed.
3) Which two substances are formed when a metal reacts with water?
4) Here are the half-equations for the redox reaction of lithium metal with nitric acid:
$Li \rightarrow Li^+ + e^-$ and $2H^+ + 2e^- \rightarrow H_2$
Which substance is being reduced and which substance is being oxidised in this reaction?
5) This equation shows how zinc reacts with copper chloride: $Zn_{(s)} + CuCl_{2(aq)} \rightarrow ZnCl_{2(aq)} + Cu_{(s)}$
   a) Write an ionic equation for this reaction.
   b) State which species in this reaction is being oxidised.

Exam Questions

1 **Figure 1** shows part of the reactivity series of metals. Hydrogen has also been included in this reactivity series.

GRADE 4-6

Figure 1

Potassium	K
Sodium	Na
Calcium	Ca
Magnesium	Mg
Zinc	Zn
Iron	Fe
HYDROGEN	H
Copper	Cu

1.1 Name **one** metal from **Figure 1** that is more reactive than magnesium.

[1 mark]

1.2 Name **one** metal from **Figure 1** which would not react with acids at all.

[1 mark]

1.3 A student places a small piece of zinc into dilute acid. The mixture produces bubbles of hydrogen gas fairly slowly. Use this information to predict what would happen if the student repeated the experiment using iron.

[2 marks]

2 Iron can be extracted by the reduction of iron(III) oxide (Fe_2O_3) with carbon (C), to produce iron and carbon dioxide.

GRADE 6-7

2.1 Write a balanced symbol equation for this reaction.

[2 marks]

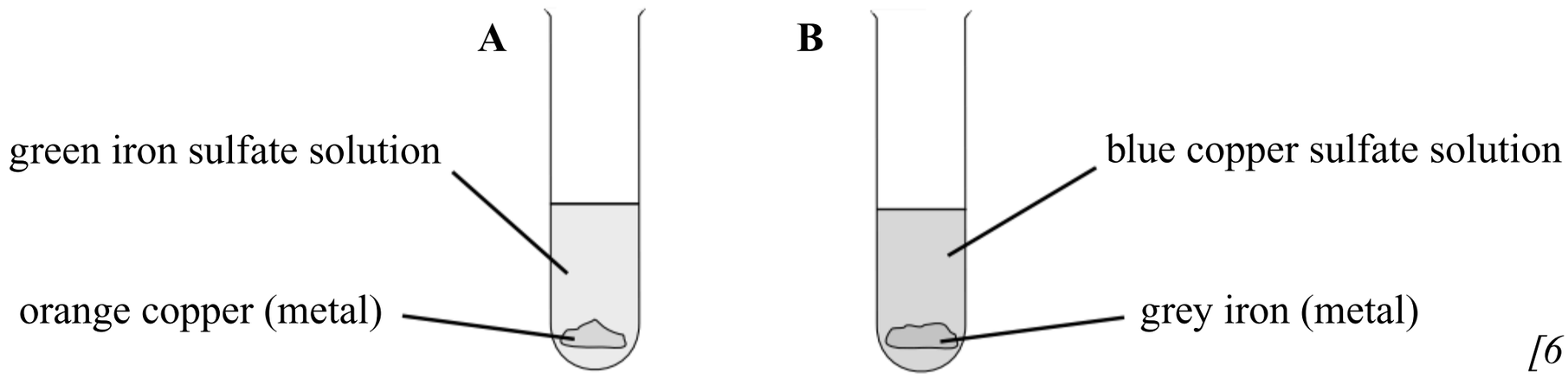
2.2 Explain why the iron(III) oxide is described as being reduced during this reaction.

[1 mark]

3 A student adds a piece of copper to some iron sulfate solution and a piece of iron to some copper sulfate solution. **Figure 2** shows what the test tubes looked like just after he added the metals. Predict what he will observe in both tubes, **A** and **B**, after 2 hours. Explain your answer.

GRADE 7-9

Figure 2



[6 marks]

Quick Quiz

Electrolysis

*You need to know the ins and outs of how **electrolysis** works. So buckle up, here we go...*

Electrolysis Means 'Splitting Up with Electricity'

1) During electrolysis, an electric current is passed through an electrolyte — a liquid or solution that conducts electricity (i.e. a molten or dissolved ionic compound). The ions move towards the electrodes, where they react, and the compound decomposes.

Electrodes are solid rods that conduct electricity and are submerged in the electrolyte. They should be inert so they don't react with the electrolyte.

2) Positive ions are attracted to the cathode (negative electrode) and gain electrons (are reduced). Negative ions are attracted to the anode (positive electrode) and lose electrons (are oxidised).
3) The movement of ions creates a flow of charge through the electrolyte.
4) As ions gain or lose electrons, they form the uncharged element and are discharged from the electrolyte.

Electrolysis of Molten Ionic Compounds Forms Elements

- An ionic solid can't be electrolysed because the ions are in fixed positions and can't move.
- Molten ionic compounds can be electrolysed because the ions can move freely and conduct electricity.
- Molten ionic compounds are always broken up into their elements. Positive metal ions are reduced to the element at the cathode and negative non-metal ions are oxidised to the element at the anode.

E.g. in the electrolysis of lead bromide, lead ions are reduced to lead at the cathode ($Pb^{2+} + 2e^- \rightarrow Pb$) and bromide ions are oxidised to bromine at the anode ($2Br^- \rightarrow Br_2 + 2e^-$).

Metals can be Extracted from Their Ores Using Electrolysis

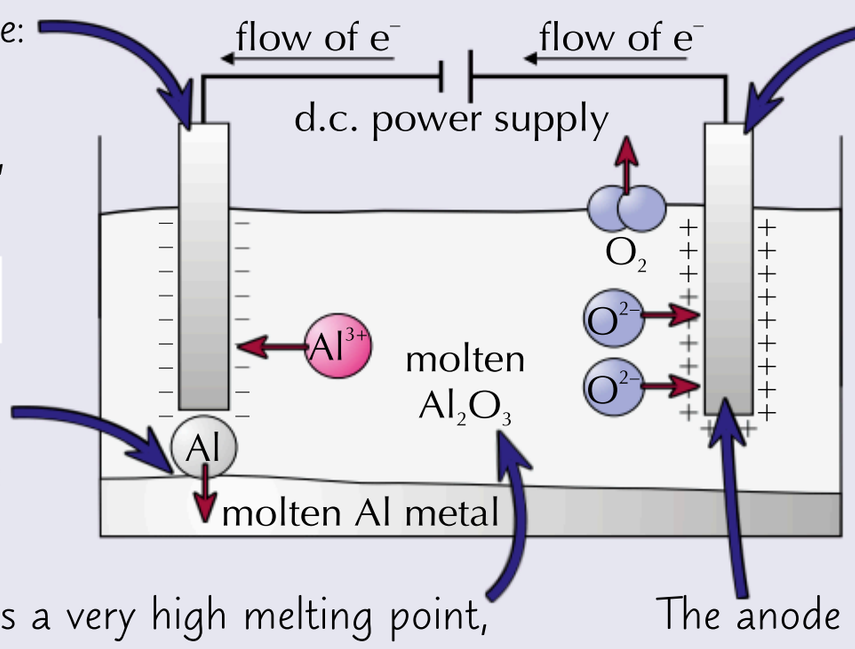
If a metal is too reactive to be reduced by carbon (p.64) or would react with carbon, electrolysis can be used to extract it from its metal oxide. This is very expensive — lots of energy is required to melt the ore and provide the necessary current.

E.g. electrolysis is used to extract aluminium from bauxite ore, which contains aluminium oxide, Al_2O_3.

At the negative electrode: Positive Al^{3+} ions gain electrons (are reduced), forming Al metal.

$Al^{3+} + 3e^- \rightarrow Al$

The aluminium metal formed is a liquid at this temperature — it sinks to the bottom.



At the positive electrode: Negative O^{2-} ions lose electrons (are oxidised), forming O_2 molecules.

$2O^{2-} \rightarrow O_2 + 4e^-$

Each half equation shows the reaction at one electrode. There's more on redox reactions and half equations on p.65.

Aluminium oxide has a very high melting point, so it's mixed with cryolite (another aluminium compound) to lower the melting point.

The anode is made of carbon. It needs replacing regularly as it reacts with the oxygen to form carbon dioxide.

The overall equation for the redox reaction is:

aluminium oxide	→	**aluminium**	+	**oxygen**
$2Al_2O_{3(l)}$	→	**$4Al_{(l)}$**	+	**$3O_{2(g)}$**

Extracting metals by electrolysis is a very expensive process

Make sure you understand why aluminium oxide is mixed with cryolite before being electrolysed.

Q1 A student carries out electrolysis on molten calcium chloride. What is produced at:
a) the anode? b) the cathode? [2 marks]

Q1 Video Solution

Electrolysis of Aqueous Solutions

Quick Quiz

*When you electrolyse an aqueous solution, you also have to factor in the ions in the **water**.*

It May be Easier to **Discharge Ions** from **Water** than the **Solute**

Video

Aqueous solutions of ionic compounds can be electrolysed, as they contain ions that can move. These solutions contain the ions from the compound, and will also contain hydrogen ions (H^+) and hydroxide ions (OH^-) from the water: $H_2O_{(l)} \rightleftharpoons H^+_{(aq)} + OH^-_{(aq)}$.

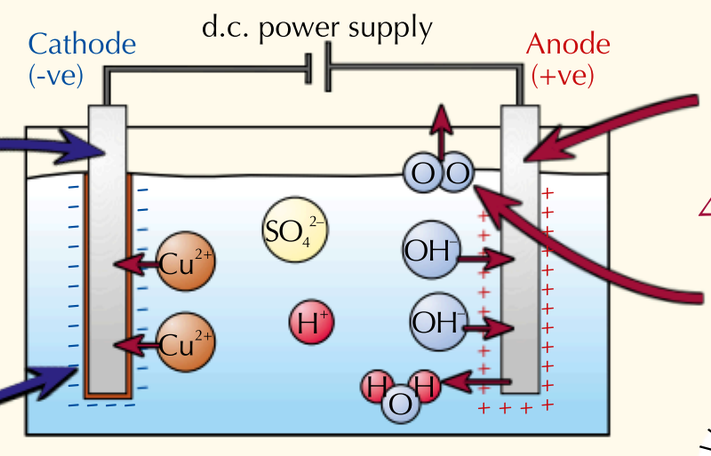
You need to know some rules to be able to predict which ions get discharged during electrolysis:

Positive ions react at the cathode:	Negative ions react at the anode:
If the metal in the compound is more reactive than hydrogen, the H^+ ions react to form hydrogen gas (H_2).	If halide ions are present, the halide ions react to form halogen molecules.
If the metal is less reactive than hydrogen, the metal ions react to form a solid layer of metal.	If halide ions are not present, the OH^- ions react to form oxygen (O_2) and water (H_2O).

E.g. copper(II) sulfate ($CuSO_4$) solution contains Cu^{2+}, SO_4^{2-}, H^+ and OH^- ions.

Copper metal is less reactive than hydrogen, so the copper ions react at the cathode:
$Cu^{2+} + 2e^- \rightarrow Cu$

Solid copper metal will coat the cathode.



There are no halide ions present, so the hydroxide ions react at the anode:
$4OH^- \rightarrow O_2 + 2H_2O + 4e^-$

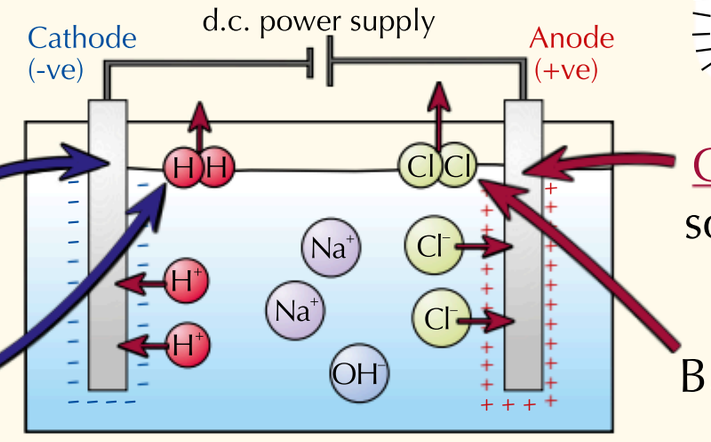
The oxygen will be seen as bubbles.

You need to be able to remember the rules for what reacts at each electrode and the half equations for the reactions.

E.g. sodium chloride (NaCl) solution contains Na^+, Cl^-, H^+ and OH^- ions.

Sodium is more reactive than hydrogen, so the H^+ ions react at the cathode:
$2H^+ + 2e^- \rightarrow H_2$

Bubbles of hydrogen gas will form.



Chloride ions are halides, so they react at the anode:
$2Cl^- \rightarrow Cl_2 + 2e^-$

Bubbles of Cl_2 gas will form.

You Need to **Test** the **Gases** Formed to **Identify** Them

PRACTICAL

Here's how you could perform an electrolysis experiment in the lab:

Method

1) Set up the solution, power supply and electrodes as shown on p.153. The best way to collect the gases produced at the electrodes is to place upside-down test tubes over the electrodes.
2) Make sure you know which electrode is which — the anode is the electrode on the same side as the positive terminal of the power supply (and the cathode is the other one).
3) Switch on the power supply and let the electrolysis run for a few minutes.
4) Write down what you observe — e.g. whether a metal forms, or bubbles of gas are produced at the electrodes. If gases are collected, you could do tests to identify them — e.g. see p.110 for the tests for H_2, O_2 and Cl_2 gas.

You should make a hypothesis about what you expect to be produced at each electrode, based on the electrolyte you're using.

Remember — all aqueous solutions contain OH^- and H^+ ions

Electrolysis is important, but it's tricky — make sure you've got the hang of the rules about what's produced.

Q1 An aqueous solution of copper bromide, $CuBr_2$, is electrolysed using inert electrodes.
State what is produced at: a) the anode, b) the cathode. [2 marks]

Q1 Video Solution

Warm-Up & Exam Questions

Time to test your mettle. Try and get through the following questions. If there's anything you're not quite sure about, have a look at the pages again until you can answer the questions without batting an eyelid.

Warm-Up Questions

1) At which electrode can metals be deposited during electrolysis?
2) What products are made when molten zinc chloride is electrolysed?
3) During the electrolysis of molten aluminium oxide are the aluminium ions being reduced or oxidised?
4) What is formed at the cathode during the electrolysis of an aqueous solution of potassium hydroxide?

Exam Questions

1 Aluminium is extracted by electrolysis of molten aluminium oxide (Al_2O_3). GRADE 6-7 GRADE

1.1 The aluminium oxide is dissolved in molten cryolite. State why.

[1 mark]

1.2 Complete the half equation below for the reaction that occurs at the negative electrode.

..... + $3e^-$ → .....

[1 mark]

1.3 Complete the half equation below for the reaction that occurs at the positive electrode.

..... → ..... + $4e^-$

[1 mark]

1.4 The positive electrode is made of carbon. Explain why it will need to be replaced over time.

[2 marks]

2 When sodium chloride solution is electrolysed, a gas is produced at each electrode. GRADE 6-7 GRADE

2.1 Name the gas produced at the negative electrode.

[1 mark]

2.2 Give the half equation for the reaction at the negative electrode.

[1 mark]

2.3 Name the gas produced at the positive electrode.

[1 mark]

2.4 Give the half equation for the reaction at the positive electrode.

[1 mark]

2.5 Explain why sodium hydroxide is left in solution at the end of the reaction.

[2 marks]

2.6 If copper chloride solution is electrolysed, copper metal is produced at the negative electrode, instead of the gas named in **2.1**. Explain why.

[1 mark]

Revision Summary Test for Topic 4

Here's what you can do to check that you've got the hang of the information in this topic:

- Tackle the revision summary test below, or scan the QR code to do it online.
  The questions are hard, but they'll show you how well you really know your stuff.
- Track your progress online and see which areas need more work.
- Compare your answers with sample answers for the test here: cgpbooks.co.uk/Base

Summary Test

Acids and their Reactions (p.58-61) ☐

1) State whether the following pH values are acidic, alkaline or neutral: a) 9, b) 2, c) 7, d) 6. ☐
2) What is a wide range indicator? Give an example of one. Now suggest another way to measure pH. ☐
3) What is: a) an acid, b) a base, c) an alkali? ☐
4) Write the general word equation for a neutralisation reaction. Now write the symbol equation for the reaction between H^+ and OH^- ions that happens when an acid reacts with an alkali. ☐
5) Describe how to: a) set up the equipment for an acid-alkali titration, b) carry out a rough titration, c) carry out a more careful titration. ☐
6) Why is it a good idea to repeat a titration until you have a few results that are very close together? ☐
7) Why should you use a single indicator in a titration, rather than a wide range indicator? ☐
8) What is the difference between strong and weak acids? ☐
9) What happens to the concentration of H^+ ions of a solution if its pH decreases by 1? ☐
10) Explain the difference between an acid's strength and its concentration. ☐
11) Give the symbol equation for the reaction of HCl with: a) CuO, b) NaOH, c) $CaCO_3$. ☐
12) Describe one way to make a pure, dry sample of a soluble salt from an acid and an insoluble base. ☐

The Reactivity Series and Extracting Metals (p.63-64) ☐

13) How does the position of a metal in the reactivity series relate to how easily it forms positive ions? ☐
14) Describe how you could determine the order of reactivity of some metals by observing reactions. ☐
15) Will iron react with dilute HCl? What about water at room temperature? Write a word equation for any reaction that would happen. ☐
16) What do oxidation and reduction mean in terms of oxygen? ☐
17) Why are some metals extracted by reduction with carbon, but others are extracted by electrolysis? ☐

Redox Reactions (p.65-66) ☐

18) What do oxidation and reduction mean in terms of electrons? What is a redox reaction? ☐
19) What are half equations? Write half equations for this reaction: $Mg_{(s)} + 2H^+_{(aq)} \rightarrow Mg^{2+}_{(aq)} + H_{2(g)}$. Which one shows oxidation and which one shows reduction? ☐
20) Give the rule for whether a metal will displace another metal from a compound. ☐
21) Which metal is oxidised and which is reduced when iron metal displaces copper ions from a compound? ☐
22) What is an ionic equation? Describe how to write an ionic equation from a full equation. ☐

Electrolysis (p.68-69) ☐

23) What is electrolysis? Outline how it works, including what can be used as an electrolyte. ☐
24) Give the half equations for the reactions at the anode and cathode in the electrolysis of Al_2O_3. ☐
25) In the extraction of aluminium by electrolysis, explain why: a) the electrolyte contains cryolite as well as bauxite, b) the carbon anode needs replacing regularly. ☐
26) Describe the rules for predicting the products formed during electrolysis of an aqueous solution. ☐
27) Give the half equation for the reaction of H^+ ions at the cathode during electrolysis of an aqueous solution. Now give the half equation for the reaction of OH^- ions at the anode. ☐
28) How could you collect and identify any gases formed during electrolysis of an aqueous solution? ☐

Quick Quiz

Exothermic & Endothermic Reactions

*Whenever chemical reactions occur, there are changes in **energy**. This means that when chemicals get together, things either hot up or cool right off. I'll give you a heads up — this page is a good 'un.*

Energy is **Moved Around** in **Chemical Reactions**

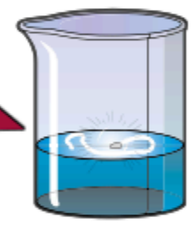
- Chemicals store a certain amount of energy — and different chemicals store different amounts.
- If the products of a reaction store more energy than the original reactants, then they must have taken in the difference in energy between the products and reactants from the surroundings during the reaction.
- But if they store less, then the excess energy was transferred to the surroundings during the reaction.
- The overall amount of energy doesn't change. This is because energy is conserved in reactions — it can't be created or destroyed, only moved around. This means the amount of energy in the universe always stays the same.

Reactions are **Exothermic** or **Endothermic**

	Exothermic reaction	Endothermic reaction
Energy is...	...transferred to the surroundings.	...taken in from the surroundings.
It is usually shown by a...	...rise in temperature.	...fall in temperature.

Many Types of Reaction are **Exothermic**

1) Exothermic reactions include:
   - Combustion reactions (burning fuels) — these give out a lot of energy, so are very exothermic.
   - Neutralisation reactions (acid + alkali).
   - Many oxidation reactions — e.g. adding sodium to water (see p.20). This reaction releases energy and the sodium moves about on the surface of the water as it is oxidised.



2) Exothermic reactions have lots of everyday uses. For example:

Some hand warmers use the exothermic oxidation of iron in air (with a catalyst) to release energy. Self-heating cans of hot drinks rely on exothermic reactions between chemicals in their bases.

Endothermic Reactions are Less Common

1) Endothermic reactions include:
   - The reaction between citric acid and sodium hydrogencarbonate.
   - Thermal decomposition — e.g. heating calcium carbonate causes it to decompose into calcium oxide and carbon dioxide.
2) Endothermic reactions also have everyday uses. For example:

Endothermic reactions are used in some sports injury packs — the chemical reaction allows the pack to become instantly cooler without having to put it in the freezer.

Energy is always conserved in a chemical reaction

Don't get exothermic and endothermic mixed up. Remember, "exo-" = exit, "-thermic" = heat, so an exothermic reaction is one that gives out heat. And endothermic means the opposite — the particles take in energy from their surroundings.

More Exothermic & Endothermic Reactions

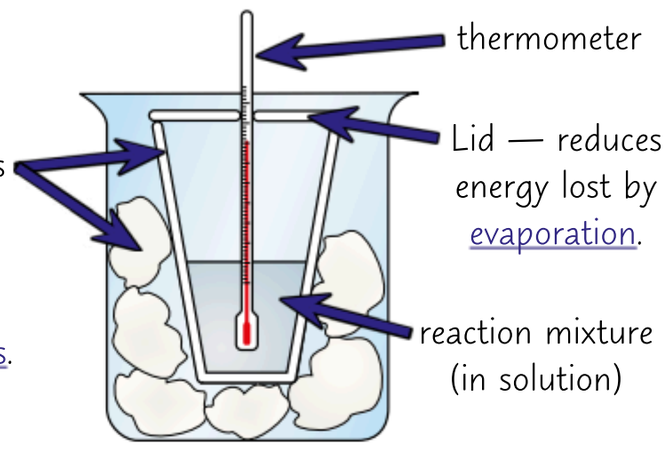
Quick Quiz

*Sometimes it's not enough to just know if a reaction is endothermic or exothermic. You may also need to know **how much** energy is absorbed or released — you can do experiments to find this out. Fun, fun, fun...*

Energy Transfer can be Measured

PRACTICAL

1) To measure the energy released by a reaction in solution, take the temperature of the reagents, mix them in a polystyrene cup and measure the temperature of the solution at the end of the reaction.
2) This works for neutralisation reactions or reactions between metals and acids, or carbonates and acids.
3) You can also use it to investigate how different variables (e.g. the mass or concentration of the reactants) affect the amount of energy transferred.



Method

Here's one way you can test the effect of acid concentration on the energy released in a neutralisation reaction between hydrochloric acid (HCl) and sodium hydroxide (NaOH):

1) Put 25 cm^3 of 0.25 mol/dm^3 hydrochloric acid and sodium hydroxide in separate beakers.
2) Place the beakers in a water bath set to 25 °C until they are both at the same temperature (25 °C).
3) Add the HCl followed by the NaOH to a polystyrene cup with a lid — as in the diagram above.
4) Take the temperature of the mixture every 30 seconds, and record the highest temperature.
5) Repeat steps 1-4 using 0.5 mol/dm^3 and then 1 mol/dm^3 of hydrochloric acid.

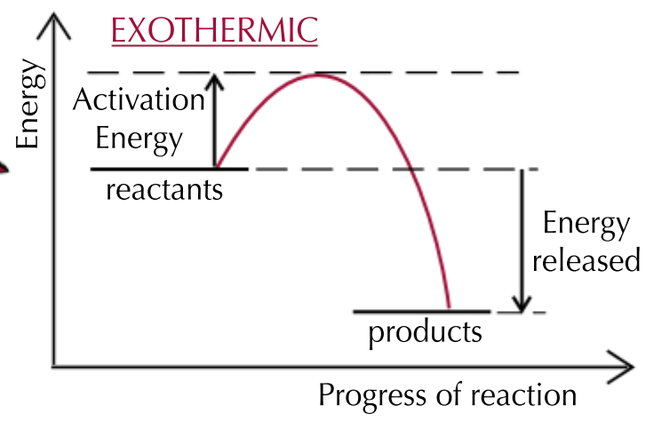
A higher maximum temperature means more energy has been released.

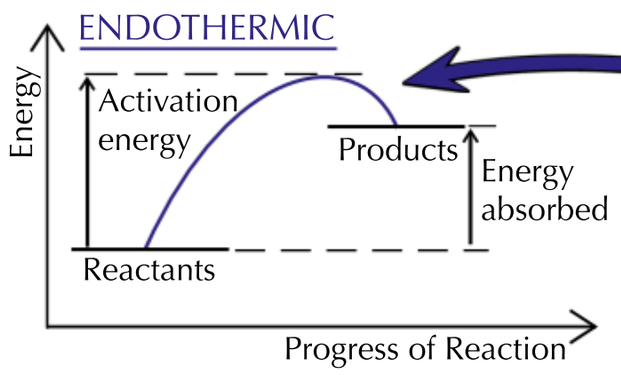
Reaction Profiles Show Energy Changes

Reaction profiles (or energy level diagrams) are diagrams that show the relative energies of the reactants and products in a reaction, and how the energy changes over the course of the reaction.

Exothermic reactions:

- The products are at a lower energy than the reactants.
- The difference in height represents the overall energy change in the reaction (the energy given out) per mole.





Endothermic reactions:

- The products are at a higher energy than the reactants.
- The difference in height represents the overall energy change during the reaction (the energy taken in) per mole.

- The activation energy (E_a) is the minimum amount of energy the reactants need to react when they collide with each other — it's needed to start the reaction.
- The greater the activation energy, the more energy needed to start the reaction — this has to be supplied, e.g. by heating the reaction mixture.

There's more on activation energy and collision theory on pages 80-81.

To get a reasonably accurate reading, insulate your reaction

In an exothermic reaction the particles transfer energy to their surroundings. So the particles lose energy, but the reaction mixture gets warmer. And surprise, surprise, it's the opposite for an endothermic reaction.

Q1 Here is the equation for the combustion of methane in air: $CH_{4(g)} + 2O_{2(g)} \rightarrow CO_{2(g)} + 2H_2O_{(g)}$
Draw a reaction profile for this reaction. [3 marks]

Q1 Video Solution

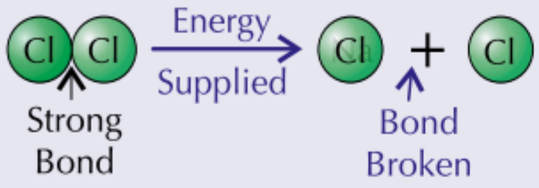
Bond Energies

Quick Quiz

*This page is all about what **causes** the energy changes that you've seen on the last couple of pages.*

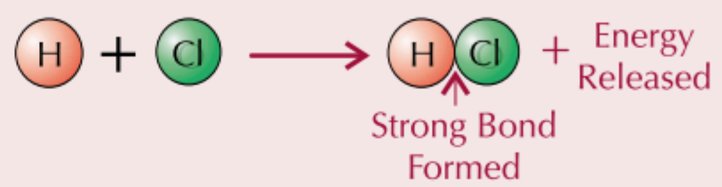
In a Chemical Reaction, **Old Bonds Break** and **New Bonds Form**

BOND BREAKING — ENDOTHERMIC PROCESS



Energy must be supplied to break existing bonds.

BOND FORMING — EXOTHERMIC PROCESS



Energy is released when new bonds are formed.

In endothermic reactions, the energy used to break bonds is greater than the energy released by forming them. In exothermic reactions, the energy released by forming bonds is greater than the energy used to break them.

There's more on energy transfer on page 72.

Bond Energy Calculations — Need to be Practised

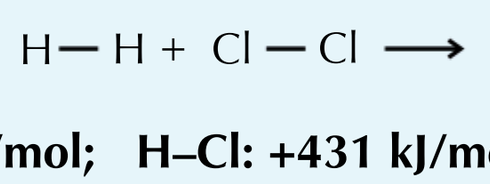
Every chemical bond has a particular bond energy associated with it that varies slightly depending on the compound the bond occurs in.

You can use known bond energies to calculate the overall energy change for a reaction:

Overall energy change = sum of energy required to break bonds in reactants − sum of energy released by forming bonds in products

EXAMPLE

Using the bond energies given below, calculate the energy change for the reaction between H_2 and Cl_2 forming HCl:



The bond energies you need are: H–H: +436 kJ/mol; Cl–Cl: +242 kJ/mol; H–Cl: +431 kJ/mol.

MATHS SKILLS

1) Find the energy required to break the original bonds in the reactants by adding together the bond energies of the bonds that are broken:

   (1 × H–H) + (1 × Cl–Cl) = 436 kJ/mol + 242 kJ/mol = 678 kJ/mol

2) Find the energy released by forming the new bond in the products by adding together the bond energies of the bonds that are formed:

   2 × H–Cl = 2 × 431 kJ/mol = 862 kJ/mol

3) Find the overall energy change for the reaction:

   Overall energy change = energy required to break bonds − energy released by forming bonds

   = 678 kJ/mol − 862 kJ/mol = −184 kJ/mol

To compare the overall energy changes of reactions, you need to know the numerical difference in bond energies. E.g. chlorine and bromine react with hydrogen in a similar way. Less energy is needed to break the bonds in the bromine reaction, but less energy is also released when the new bonds form. So you need to know the exact difference to say which reaction releases more energy overall.

You need a balanced chemical equation for these calculations

Q1 Video Solution

These calculations with bond energies might look hard at the moment, but with a bit of practice it's dead easy and it'll win you easy marks if you understand all the theory behind it. See how you get on with this question:

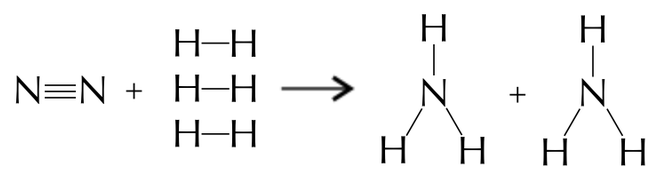
Q1 N_2 reacts with H_2 in the following reaction:

$N_2 + 3H_2 \rightarrow 2NH_3$

The bond energies for these molecules are:

N≡N: 941 kJ/mol; H–H: 436 kJ/mol; N–H: 391 kJ/mol.

Calculate the overall energy change for this reaction.



[3 marks]

Warm-Up & Exam Questions

Funny diagrams, bond energy calculations, a practical — there's a lot to get your head around in just a few pages here. Here are some questions so that you can check how you're getting on.

Warm-Up Questions

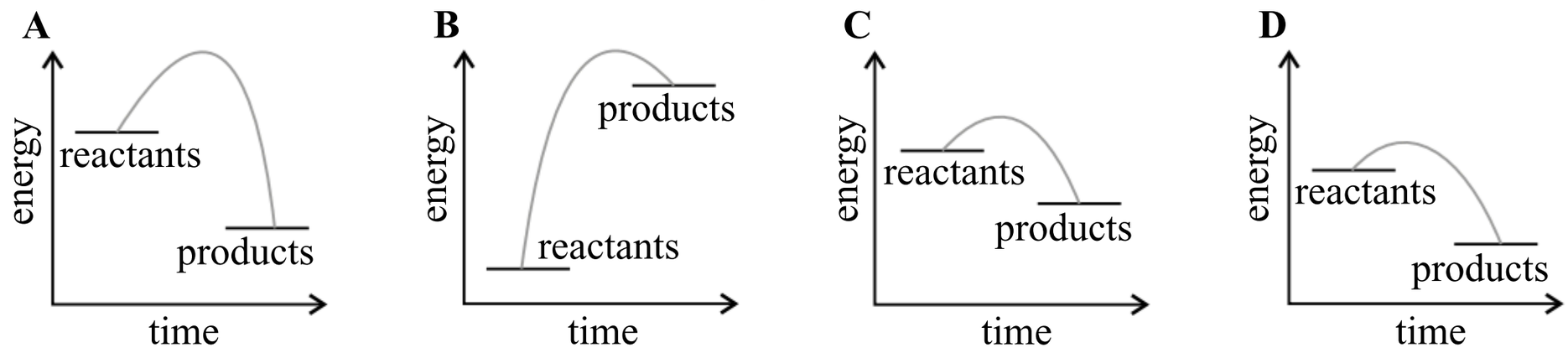
1) What is an exothermic reaction?
2) What is an endothermic reaction?
3) Is energy released when bonds are formed or when bonds are broken?
4) What is meant by the activation energy of a reaction?

Exam Questions

1 The diagrams in **Figure 1** represent the energy changes in four different chemical reactions.

GRADE 4-6

Figure 1



Write the letter of **one** diagram, **A**, **B**, **C** or **D**, which illustrates an endothermic reaction.

[1 mark]

PRACTICAL

2 A student places two beakers of ethanoic acid and dilute potassium hydroxide into a water bath until they are both 25 °C. The student adds the ethanoic acid and then the potassium hydroxide to a polystyrene cup. After 1 minute the temperature of the mixture is 28.5 °C.

GRADE 6-7

2.1 Is this reaction endothermic or exothermic? Explain your answer.

[2 marks]

2.2 The student put a lid on the polystyrene cup during the experiment. Suggest why this was done.

[1 mark]

3 When methane burns in air it produces carbon dioxide and water, as shown in **Figure 2**. The bond energies for each bond in the molecules involved are shown in **Figure 3**.

GRADE 7-9

Figure 2

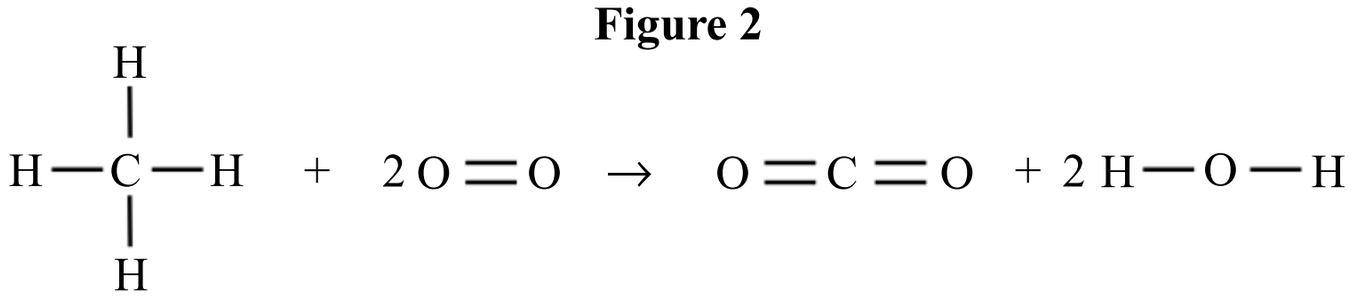


Figure 3

	Bond energies (kJ/mol):
C – H	414
O = O	494
C = O	800
O – H	459

3.1 Which **two** types of bond are broken during the reaction shown in **Figure 2**?

[1 mark]

3.2 Calculate the overall energy change for the reaction shown in **Figure 2**.

[3 marks]

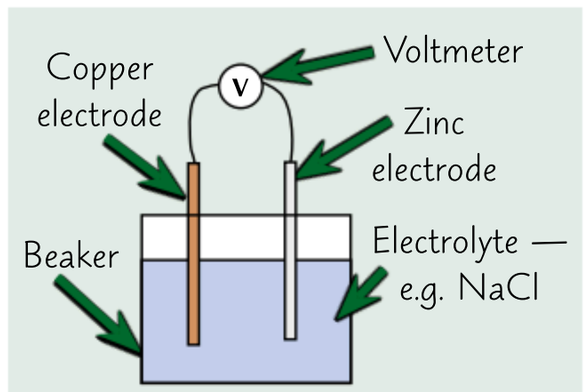
Quick Quiz

Cells and Batteries

__Batteries__ are dead useful for all sorts of things and, great news — now you can find out __how they work__.

__Chemical Reactions__ in a __Cell__ Produce __Electricity__

1) An electrochemical cell is a basic system made up of:
   - Two different electrodes that conduct electricity (usually metals).
   - An electrolyte (a liquid containing ions that react with the electrodes).
2) The chemical reactions between the electrodes and the electrolyte set up a charge difference between the electrodes.
3) Connecting the electrodes by a wire allows charge to flow, producing electricity. The cell voltage can be measured by connecting a voltmeter.



The __Voltage__ of a Cell Depends on Many __Factors__

1) Since different metals will react differently with the same electrolyte, the type of electrodes used will affect the voltage of the cell.
2) The bigger the difference in reactivity of the electrodes, the bigger the voltage of the cell.
3) You can predict the voltage of a cell if you know the voltages of other cells with those electrodes.

EXAMPLE

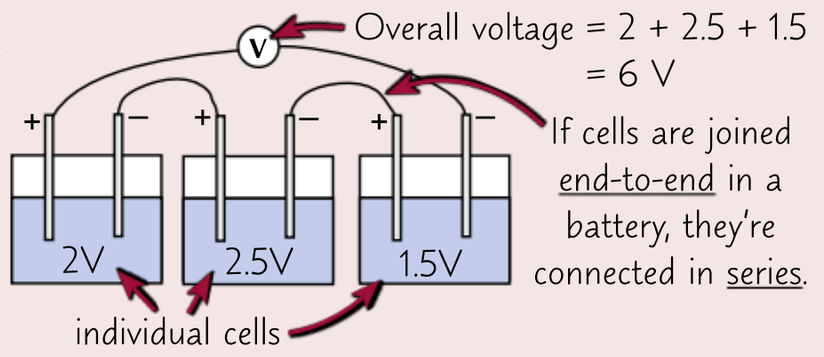
The voltages of two simple cells with the same electrolyte and different electrodes are shown in the table. If the metal used for Electrode A is less reactive than the other metal electrode, the voltage is positive and vice versa. Calculate the voltage of a cell where Electrode A is lead and Electrode B is tin.

Electrode A	Electrode B	Voltage (V)
Iron	Tin	–0.30
Iron	Lead	–0.31

The voltages of both cells are negative, so iron is more reactive than tin and lead. The voltage for the iron/lead cell is more negative than the iron/tin cell, so lead is less reactive than tin. The difference in voltage of these two cells is 0.01 V. So, for a cell where Electrode A is lead and Electrode B is tin, the voltage is +0.01 V.

4) The electrolyte used in a cell will also affect the size of the voltage since different ions in solution will react differently with the metal electrodes used.

A battery is made by connecting two or more cells together in series. The voltages of the cells in the battery are combined so there's a bigger voltage overall.



The __Reactants__ in __Non-Rechargeable__ Batteries Get __Used Up__

- In some cells, the reactions occurring are irreversible.
- Over time, the reacting particles get used up as they're turned into products. Once any one of the reactants is used up, the reaction stops and no electricity is produced.
- The products can't be turned back into the reactants, so the cell can't be recharged.

  The cells in non-rechargeable batteries, e.g. alkaline batteries, use irreversible reactions. Once one reactant is used up, they stop producing charge and need replacing.
- In a rechargeable cell, the reaction can be reversed by connecting it to an external electric current.

Remember, 'voltage' just means the 'charge difference'

Don't forget that the voltage of a battery is the total voltage of its individual cells.

Q1 Video Solution

Q1 The voltages of three cells, where one electrode is iron and the other electrode changes each time are shown in the table. Iron is less reactive than the other three metals. Write the metals in order of reactivity from lowest to highest. Explain your answer. [3 marks]

Metal	1	2	3
Voltage (V)	4.1	2.2	3.7

Fuel Cells

Quick Quiz

*We need **fuel** to move around. Unfortunately, most fuels we use are very **polluting** — **fuel cells** might help...*

Fuel Cells Use Fuel and Oxygen to Produce Electrical Energy

1) A fuel cell is an electrical cell that's supplied with a fuel and oxygen (or air) and uses energy from the reaction between them to produce electrical energy efficiently.
2) When the fuel enters the cell it becomes oxidised, which sets up a potential difference within the cell.
3) There are a few different types of fuel cells, using different fuels and different electrolytes.
4) A hydrogen-oxygen fuel cell combines hydrogen and oxygen to make nice clean water and release energy.

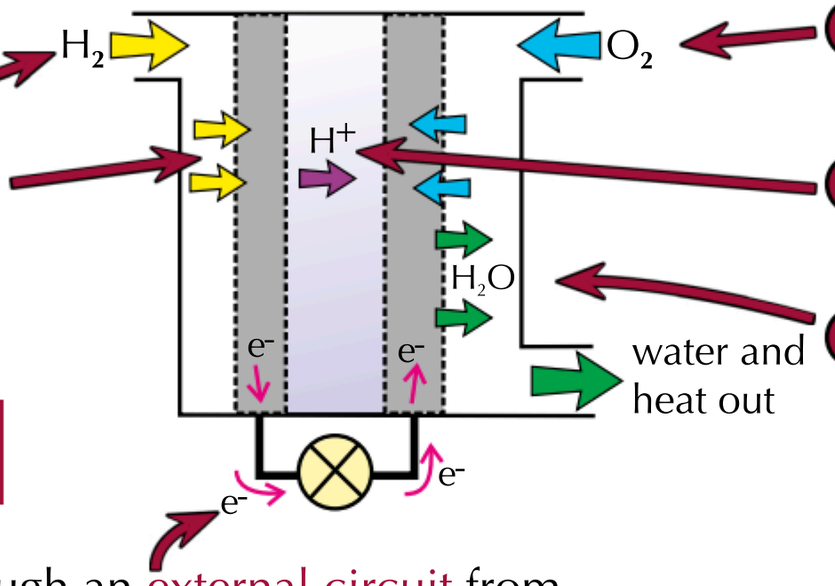
Hydrogen-Oxygen Fuel Cells Involve a Redox Reaction

The electrolyte is often an acid, such as phosphoric acid. The electrodes are often porous carbon with a catalyst. The positive electrode is the cathode and the negative electrode is the anode.
(Yep, it's the opposite to electrolysis — sorry...)

1 Hydrogen goes into the anode compartment.

3 At the anode, hydrogen loses electrons to produce H^+ ions — this is oxidation:

$$H_2 \rightarrow 2H^+ + 2e^-$$



2 Oxygen goes into the cathode compartment.

4 H^+ ions in the electrolyte move to the cathode.

5 At the cathode, oxygen gains electrons (from the cathode) and reacts with H^+ ions (from the acidic electrolyte) to make water — this is reduction:

$$O_2 + 4H^+ + 4e^- \rightarrow 2H_2O$$

6 The electrons flow through an external circuit from the anode to the cathode — this is the electric current.

7 The overall redox reaction is hydrogen plus oxygen, which gives water: $2H_2 + O_2 \rightarrow 2H_2O$

Video

There's more on half equations and redox reactions on p.65.

Hydrogen-Oxygen Fuel Cells Could be Used in Vehicles

Conventional fuels for vehicles (such as petrol) are very polluting and there's a finite supply of them — they won't last forever. So vehicles that use cleaner electrical energy, such as batteries and hydrogen fuel cells, are becoming more and more popular.

Advantages of using hydrogen fuel cells over batteries:	Disadvantages of using hydrogen fuel cells over batteries:
The only by-products of fuel cell vehicles are water and heat. Electric vehicles don't produce many pollutants either, but their batteries are more polluting to dispose of (they're made from highly toxic metal compounds).	Hydrogen gas needs more space for storage than a rechargeable battery.
Batteries in electric vehicles are rechargeable but there's a limit to how many times they can be recharged before they need replacing.	Hydrogen is explosive when mixed with air so it's hard to store safely.
Fuel cells are less expensive to make than batteries.	Hydrogen fuel is often made from hydrocarbons (from fossil fuels), or by electrolysis of water, which uses electricity (that's usually generated from fossil fuels).
Batteries store less energy than fuel cells and recharging them can take a long time compared to refuelling with hydrogen.	

Hydrogen-oxygen fuel cells don't produce nasty pollutants

These fuel cells sound great, but don't go thinking they're perfect. Once you've got the hydrogen, they're pretty eco-friendly to use. But producing hydrogen takes a lot of energy (usually from burning fossil fuels). That doesn't mean they won't be more important in the future — but there are some issues to sort out first.

Warm-Up & Exam Questions

These questions should help you get all of that electrical knowledge into your head.
These Warm-Up questions should get you started and then you can crack on with the Exam questions.

Warm-Up Questions

1) In an electrochemical cell, what is an electrolyte?
2) How do you form a battery from two or more cells?
3) What is the product formed by a hydrogen-oxygen fuel cell?
4) In a hydrogen-oxygen fuel cell, what solution is often used as the electrolyte?

Exam Questions

1 Which of the following can be used to measure the charge difference across a cell? GRADE 4-6 GRADE
Tick **one** box.

☐ cathode ☐ electrolyte ☐ anode ☐ voltmeter

[1 mark]

2 Hydrogen-oxygen fuel cells involve redox reactions. GRADE 6-7 GRADE
One of the half-equations for the hydrogen-oxygen fuel cell is:

$$O_2 + 4H^+ + 4e^- \rightarrow 2H_2O$$

2.1 Explain why the reaction in a hydrogen-oxygen fuel cell is a redox reaction.

[2 marks]

2.2 Is the half equation shown above an oxidation or a reduction reaction? Explain your choice.

[1 mark]

2.3 What is the overall chemical reaction for a hydrogen-oxygen fuel cell?

[2 marks]

2.4* Explain the advantages of using hydrogen-oxygen fuel cells rather than rechargeable batteries to power cars.

[4 marks]

3 A student investigated the effect of changing one electrode on the voltage produced by a cell. GRADE 7-9 GRADE
The other electrode was made of zinc, which is less reactive than the metals tested.
The student's results are shown in **Table 1**.

Table 1

Metal used (alongside zinc)	1	2	3
Voltage produced	4 V	2.5 V	3 V

3.1 Write the order of reactivity of the metals used from **lowest** to **highest**.
Explain your answer.

[3 marks]

3.2 Explain how you could construct a 9 V battery using these cells.

[2 marks]

Revision Summary Test for Topic 5

I hope you've still got some energy left after all that. Here's what you can do to test yourself:

- Tackle the revision summary test below, or scan the QR code to do it online. The questions are hard, but they'll show you how well you really know your stuff.
- Track your progress online and see which areas need more work.
- Compare your answers with sample answers for the test here: cgpbooks.co.uk/Base

Summary Test

Exothermic and Endothermic Reactions (p.72-73) ☑

1) Explain how the products of a reaction can store more or less energy than the reactants, if energy can't be created or destroyed. ☑
2) What are exothermic reactions? What about endothermic reactions? ☑
3) Name three different types of reaction that are exothermic. Now give one type of endothermic reaction. ☑
4) Give an everyday use of an exothermic reaction and an endothermic reaction. ☑
5) What is the purpose of putting the reaction container into a beaker containing cotton wool during an experiment to investigate the temperature change of an exothermic reaction? ☑
6) Outline how you could investigate the effect of acid concentration on the energy released in a neutralisation reaction between hydrochloric acid (HCl) and sodium hydroxide (NaOH). ☑
7) What do reaction profiles show? How do they differ between exothermic and endothermic reactions? ☑
8) What is activation energy? What does it mean if one reaction has a greater activation energy than another? ☑

Bond Energies (p.74) ☑

9) Explain whether breaking bonds is an exothermic process or an endothermic process. How about forming new bonds? ☑
10) How does the amount of energy used to break bonds compare to the energy used to form new bonds in: a) an endothermic reaction, b) an exothermic reaction? ☑
11) Give a formula that you can use to work out the energy change of a reaction from bond energies. ☑
12) Explain why you need to know the numerical differences in bond energies if you want to compare the overall energy change of two different reactions. ☑

Cells and Batteries (p.76) ☑

13) Describe how to set up a chemical cell where the voltage can be measured. ☑
14) What causes the charge difference in an electrochemical cell? ☑
15) How does the choice of electrodes used in a cell affect the voltage of the cell? Give another factor that affects the voltage and explain why. ☑
16) What is a battery? What determines the voltage of a battery? ☑
17) Explain why a non-rechargeable battery will need replacing after a while. ☑

Fuel Cells (p.77) ☑

18) What is a fuel cell? How is a potential difference set up within a fuel cell? ☑
19) Outline how a hydrogen-oxygen fuel cell works. Which molecule is oxidised and which is reduced? ☑
20) Give the overall equation for the reaction in hydrogen-oxygen fuel cells. ☑
21) Give three advantages and three disadvantages of using hydrogen-oxygen fuel cells to power cars instead of rechargeable batteries. ☑

Quick Quiz

Rates of Reaction

*Rates of reaction are pretty **important**. In **industry**, the **faster** you make **chemicals**, the **faster** you make **money**.*

Reactions Can Go at All Sorts of **Different Rates**

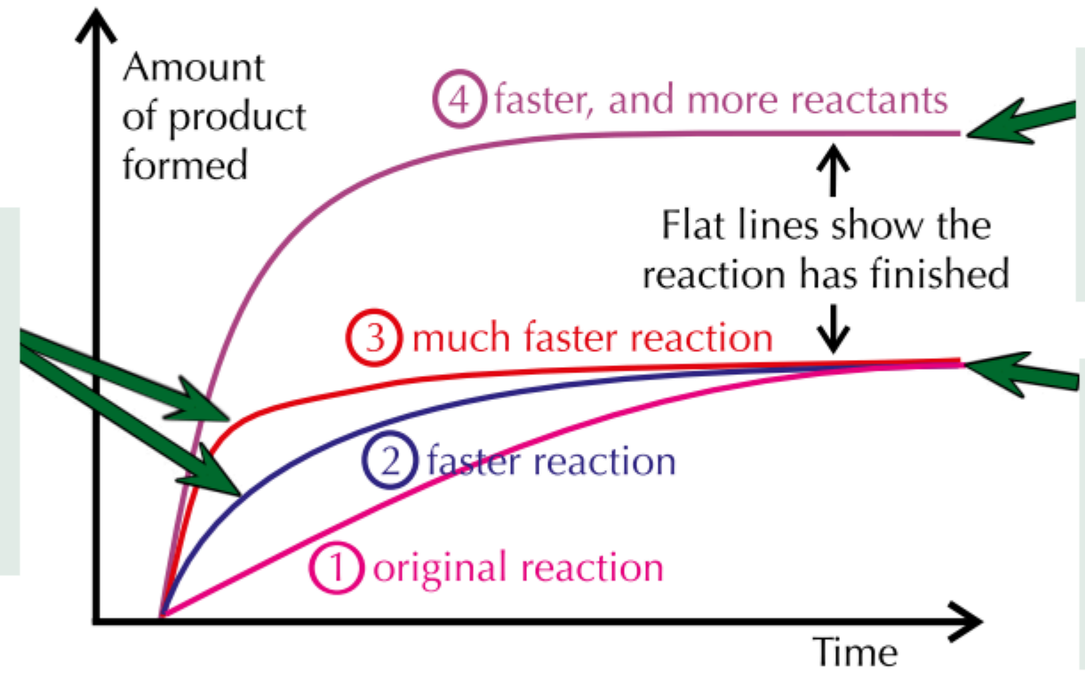
The rate of a chemical reaction is how fast the reactants are changed into products.

- An example of a slow reaction is the rusting of iron.
  (It's not slow enough though — what about my little Mini...)
- Burning is a fast reaction, but explosions are even faster and release a lot of gas. Explosive reactions are all over in a fraction of a second.

You Need to Understand **Graphs** for the **Rate of Reaction**

- You can find the speed of a reaction by recording the amount of product formed, or the amount of reactant used up over time (see page 82).
- The steeper the line on the graph, the faster the rate of reaction.
  Over time the line becomes less steep as the reactants are used up.
- The quickest reactions have the steepest lines and become flat in the least time.
- The plot below uses the amount of product formed over time to show how the speed of a particular reaction varies under different conditions.

For more on the conditions that affect the rate of reaction — see next page.



These reactions are quicker than reaction 1, but with the same initial amounts of reactants. The slopes of the graphs are steeper.

Graph 4 shows more product. This can only happen if more of the limiting reactant is added at the start.

Graphs 1, 2 and 3 all converge at the same level, showing that they all produce the same amount of product although they take different times to produce it.

Particles Must **Collide** with **Enough Energy** in Order to **React**

Reaction rates are explained perfectly by collision theory. It's simple really.
The rate of a chemical reaction depends on:

1) The collision frequency of reacting particles (how often they collide). The more collisions per second there are, the faster the reaction is. E.g. doubling the frequency of collisions doubles the rate.
2) The energy transferred during a collision. Particles have to collide with enough energy for the collision to be successful.

The minimum amount of energy needed to start the reaction is called the activation energy — you might remember it from page 73.

A successful collision is a collision that ends in the particles reacting to form products.

Factors that increase the number of collisions (so that a greater proportion of reacting particles collide) or the amount of energy particles collide with will increase the rate of the reaction (see next page for more).

Particles won't react unless they have enough energy

A lot of this topic boils down to this collision theory stuff, so make sure you have a solid grasp of it before moving on.

Factors Affecting Rates of Reaction

Quick Quiz

*I'd ask you to **guess** what this page is about, but the **title** pretty much says it all really. Read on...*

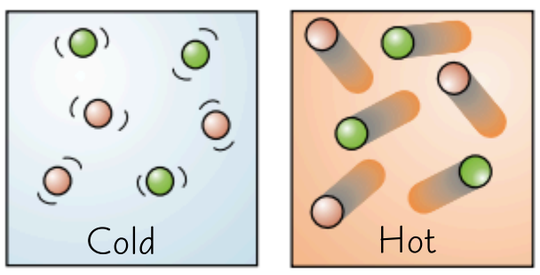
Video

The **Rate of Reaction** Depends on **Four Things**

1) Temperature.
2) The concentration of a solution or the pressure of gas.
3) Surface area — this changes depending on the size of the lumps of a solid.
4) The presence of a catalyst.

All four methods of increasing the rate of a reaction can be explained in terms of increasing the number of successful collisions between the reacting particles:

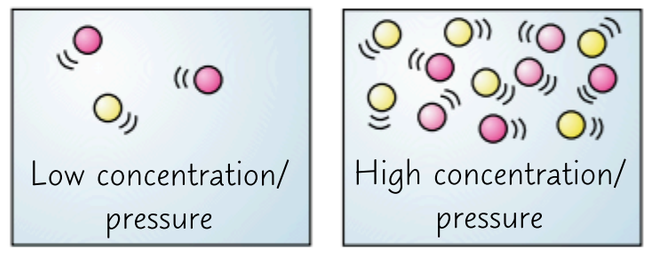
Increasing the **Temperature** Increases the Rate



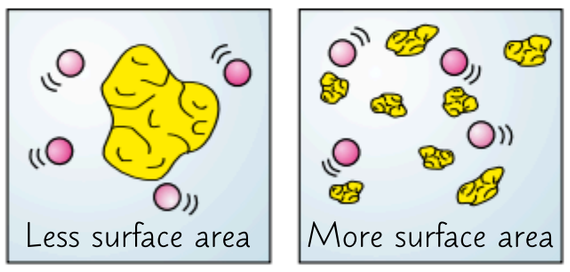
- When the temperature is increased, the particles all move faster.
- If they're moving faster, they're going to collide more frequently.
- Also the faster they move the more energy they have, so more of the collisions will have enough energy to make the reaction happen.

Increasing the **Concentration** or **Pressure** Increases the Rate

- If a solution is made more concentrated, it means there are more particles knocking about in the same volume of water (or other solvent).
- Similarly, when the pressure of a gas is increased, it means that the same number of particles occupies a smaller space.
- This makes collisions between the reactant particles more frequent.



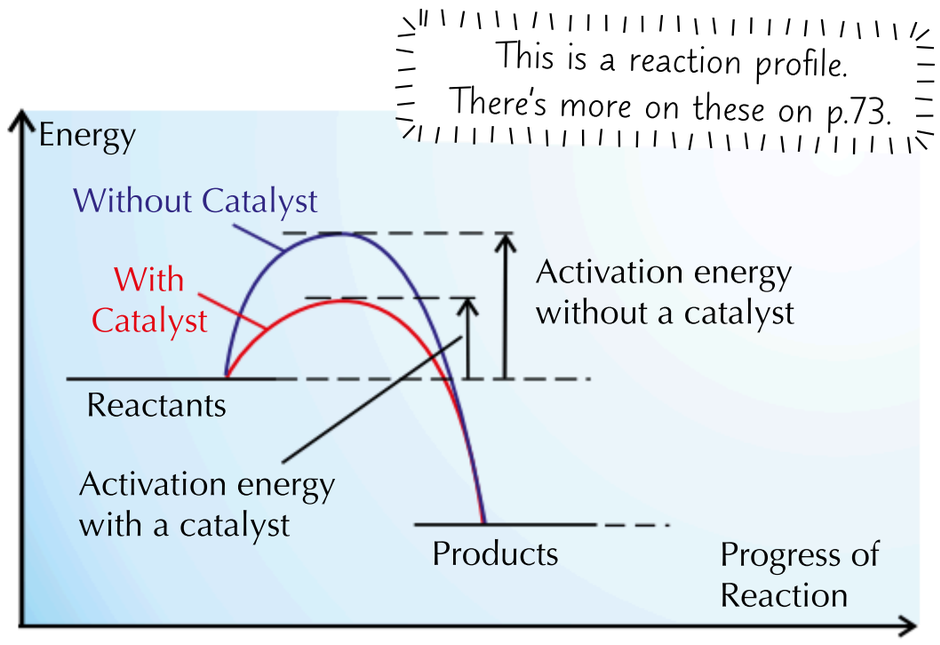
Increasing the **Surface Area** Increases the Rate



- If one of the reactants is a solid, then breaking it up into smaller pieces will increase its surface area to volume ratio.
- This means that for the same volume of the solid, the particles around it will have more area to work on — so there will be collisions more frequently.

Using a **Catalyst** Increases the Rate

- A catalyst is a substance that speeds up a reaction, without being used up in the reaction. This means it's not part of the overall reaction equation.
- Different catalysts are needed for different reactions, but they all work by decreasing the activation energy needed for the reaction to occur. They do this by providing an alternative reaction pathway with a lower activation energy.
- Enzymes are biological catalysts — they catalyse reactions in living things.



It's easier to learn stuff when you know the reasons for it

Once you've had a good read of this page, rates of reaction should start to make more sense. Just remember that the more often particles bump into each other, and the harder they hit when they do, the faster the reaction happens.

Quick Quiz

Measuring Rates of Reaction

PRACTICAL

All this talk about rates of reactions is fine and dandy, but it's no good if you can't ***measure*** *them.*

Here Are **Three Ways** to Measure the **Rate** of a Reaction

The rate of a reaction can be observed either by how quickly the reactants are used up or how quickly the products are formed:

$$\textbf{Rate of Reaction} = \frac{\textbf{Amount of reactant used or amount of product formed}}{\textbf{Time}}$$

This is the mean rate of reaction. To find the rate of a reaction at a particular time, you'll need to plot a graph and find the gradient at that time (see page 84).

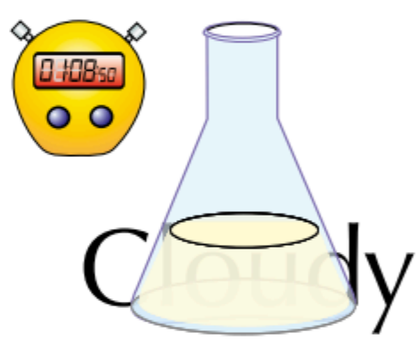
When the product or reactant is a gas you usually measure the amount in cm^3. If it's a solid, then you use grams (g). You can also measure the amount of product or reactant in moles. Time is often measured in seconds (s). This means that the units for rate may be in cm^3/s, g/s or mol/s.

Here are three different ways of measuring the rate of a reaction:

1) **Precipitation** and **Colour Change**

A posh way of saying this is that its 'turbidity' changes.

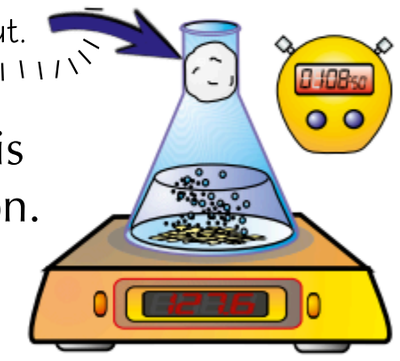
- You can record the visual change in a reaction if the initial solution is transparent and the product is a precipitate which clouds the solution (it becomes opaque).
- You can observe a mark through the solution and measure how long it takes for it to disappear — the faster it disappears, the quicker the products have formed, so the quicker the reaction.
- If the reactants are coloured and the products are colourless (or vice versa), you can time how long it takes for the solution to lose (or gain) its colour.
- The results are very subjective — different people might not agree over the exact point when the mark 'disappears' or the solution changes colour. Also, if you use this method, you can't plot a rate of reaction graph from the results as easily.



2) **Change in Mass** (Usually Gas Given Off)

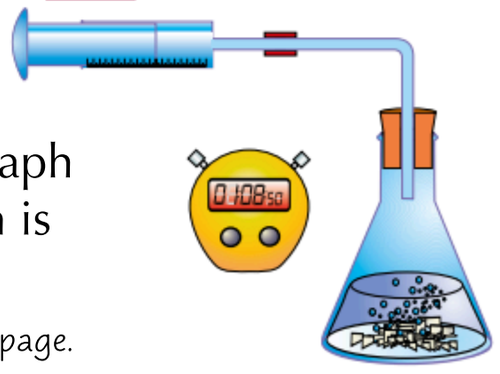
Putting cotton wool in the top of the flask lets the gas escape but stops any liquid spitting out.

- If a reaction produces a gas, you can measure how quickly the gas forms by letting it escape.
- The reaction vessel is placed on a mass balance so you can read how much mass is lost over time. The quicker the reading on the balance drops, the faster the reaction.
- If you take measurements at regular intervals, you can plot a graph of mass lost over time and find the rate from that (see page 84 for more).
- This is the most accurate of the three methods described on this page because the mass balance is very accurate. But it has the disadvantage of releasing the gas straight into the room.



3) The **Volume** of Gas Given Off

- You can also see how quickly a gas is produced by measuring its volume over time.
- Gas syringes usually give volumes accurate to the nearest cm^3, so they're a quite accurate way to measure the amount of gas given off. You can take measurements at regular intervals and plot a rate of reaction graph using this method too. You have to be quite careful though — if the reaction is too vigorous, you can easily blow the plunger out of the end of the syringe.



You can also measure the volume of gas released by collecting it in an upturned measuring cylinder — see next page.

Each of these three methods has pros and cons

For example, the mass balance method is only accurate as long as the flask isn't too hot, otherwise some of the loss in mass you see might be due to evaporation of liquid. And the precipitation method is very subjective, so it's not very accurate.

PRACTICAL

Two Rate Experiments

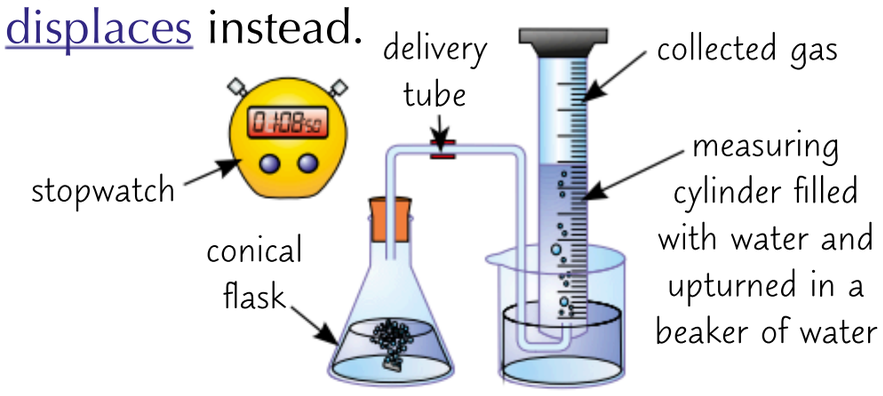
Quick Quiz

*Here's a lovely page on **practical investigations** into the effect of **concentration** on the rate of a reaction.*

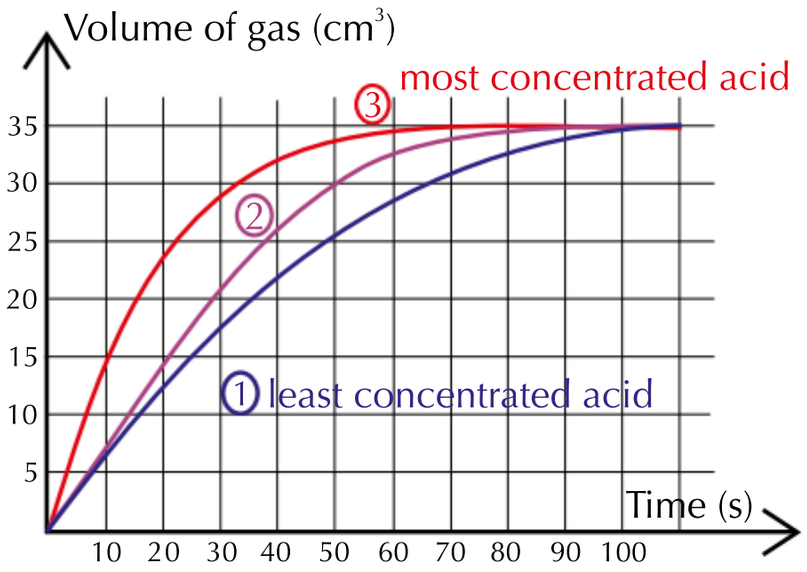
Magnesium and HCl React to Produce H_2 Gas

If you want to measure the volume of gas but don't have a gas syringe to hand, don't panic — you can measure the amount of water that the gas displaces instead.

You can read all about how to set this up on page 153.



1) Set up the apparatus as shown on the right.
2) Record the initial volume of gas in the measuring cylinder.
3) Add some magnesium ribbon to the acid in the conical flask, then quickly put the bung in and start the stopwatch.
4) Take readings of the volume of gas at regular intervals.



5) Work out the volume of gas produced at each point by subtracting the initial volume of gas from each reading. Now you can plot a graph with time on the *x*-axis and volume of gas on the *y*-axis.
6) Repeat with more concentrated acid solutions. Variables such as the amount of magnesium ribbon and the volume of acid used should be kept the same each time — only change the acid's concentration. This is to make your experiment a fair test — see p.142.
7) The three graphs on the left show that a higher concentration of acid gives a faster rate of reaction.

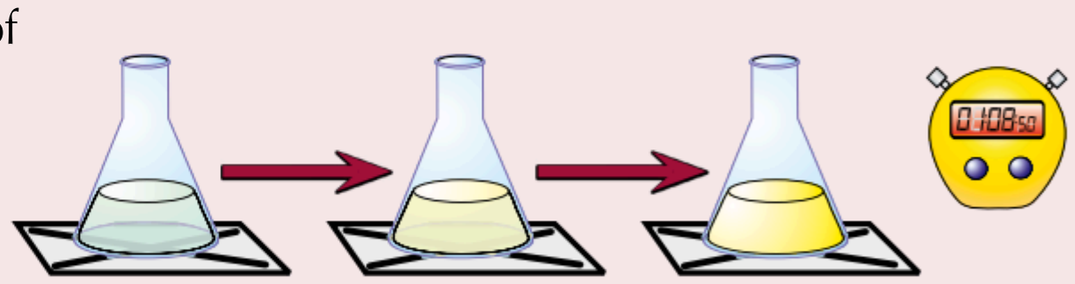
Sodium Thiosulfate and HCl Produce a Cloudy Precipitate

These two chemicals are both colourless solutions. They react together to form a yellow precipitate of sulfur.

1) Start by adding a set volume of dilute sodium thiosulfate to a conical flask.
2) Place the flask on a piece of paper with a black cross drawn on it. Add some dilute HCl to the flask and start the stopwatch.
3) Now time how long it takes the black cross to disappear through the cloudy sulfur.
4) The reaction can be repeated with solutions of either reactant at different concentrations.

This reaction releases sulfur dioxide, so the experiment should be carried out in a well-ventilated place.

Make sure you only change the concentration of one reactant at a time, and use the same depth of liquid each time, in order to make it a fair test — see page 142 for more.



- These results show the effect of increasing the concentration of HCl on the rate of reaction, when added to an excess of sodium thiosulfate.
- The higher the concentration, the quicker the reaction and therefore the less time it takes for the mark to disappear.
- One sad thing about this reaction is that it doesn't give you a nice set of rate of reaction graphs. All you get is a set of readings of how long it took till the mark disappeared for each concentration. Boring.

Concentration of HCl (mol/dm³)	0.5	1	1.5	2	2.5
Time taken for mark to disappear (s)	193	184	178	171	164

Although you could plot the data in the table as a graph, which'll show the time taken decreasing as the concentration increases.

Don't forget about any safety precautions you need to take

Learn both methods on this page for investigating rate, but don't forget that other methods can also be used (e.g. the change in mass method on the previous page). You need to be able to choose the right method for your reaction.

Quick Quiz

Finding Reaction Rates from Graphs

*You might remember how to **interpret** reaction rate graphs from page 80 — now it's time to put that to use...*

You can Calculate the **Mean Reaction Rate** from a **Graph**

- Remember, a rate of reaction graph shows the amount of product formed or amount of reactant used up on the *y*-axis and time on the *x*-axis.
- So to find the mean rate for the whole reaction, you just work out the overall change in the *y*-value and then divide this by the total time taken for the reaction.
- You can also use the graph to find the mean rate of reaction between any two points in time:

EXAMPLE

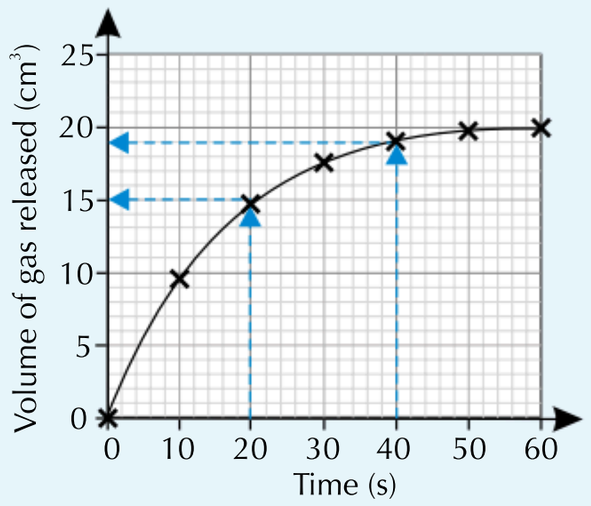
The graph shows the volume of gas released by a reaction, measured at regular intervals. Find the mean rate of reaction between 20 s and 40 s.

1) Use the graph to find the change in *y* and change in *x*.

$19 - 15 = 4$ cm^3
$40 - 20 = 20$ s

2) Find the mean rate of reaction by calculating change in *y* ÷ change in *x*.

4 cm^3 ÷ 20 s
$= 0.2$ cm^3/s (1 s.f.)



MATHS SKILLS

If you're asked to find the mean rate of reaction for the whole reaction, remember that the reaction finishes as soon as the line on the graph goes flat.

Draw a **Tangent** to Find the **Reaction Rate** at a **Particular Point**

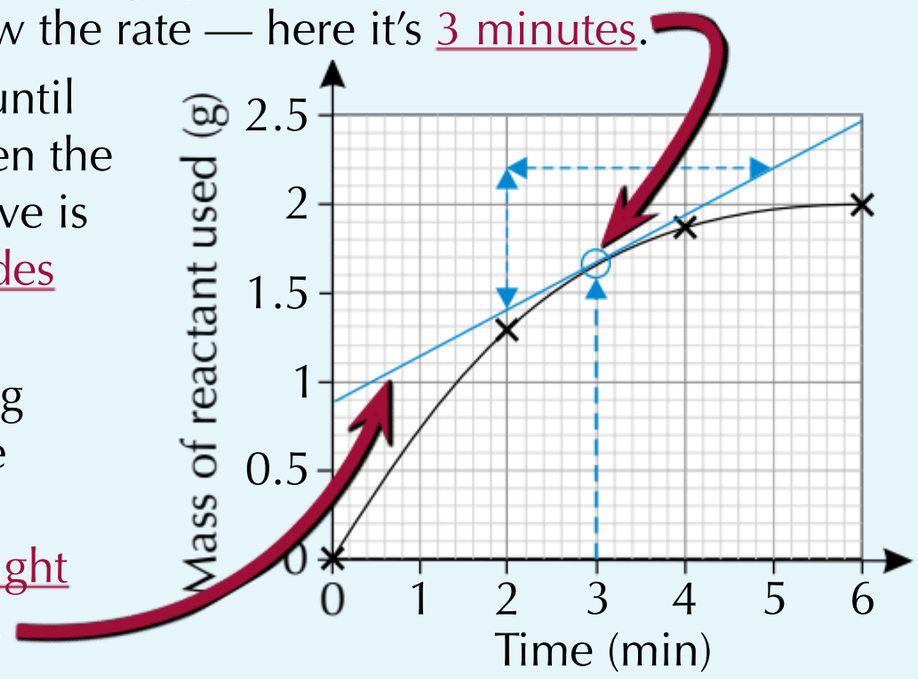
If you want to find the rate of the reaction at a particular point in time, you need to find the gradient (slope) of the curve at that point. The easiest way to do this is to draw a tangent to the curve — a straight line that touches the curve at one point and doesn't cross it. You then work out the gradient of the tangent. It's simpler than it sounds, honest...

EXAMPLE

The graph below shows the mass of reactant used up measured at regular intervals during a chemical reaction. What is the rate of reaction at 3 minutes?

MATHS SKILLS

1) Position a ruler on the graph at the point where you want to know the rate — here it's 3 minutes.
2) Adjust the ruler until the space between the ruler and the curve is equal on both sides of the point.
3) Draw a line along the ruler to make the tangent. Extend the line right across the graph.



4) Pick two points on the line that are easy to read. Use them to calculate the gradient of the tangent in order to find the rate:

gradient = change in *y* ÷ change in *x*
$= (2.2 - 1.4) \div (5.0 - 2.0)$
$= 0.8 \div 3.0$
$= 0.266...$
$= 0.27$ (2 s.f.)

So, the rate of reaction at 3 minutes was 0.27 g/min.

A tangent touches a curve but doesn't cross it

There's only one way to learn this stuff properly — practise. So you'd better get going with this question.

Q1 Magnesium powder was added to a conical flask containing dilute H_2SO_4. H_2 was produced and collected in a gas syringe. The volume of gas released was recorded at 10 second intervals in the following table:

Time (s)	10	20	30	40	50	60
Volume of H_2 (cm^3)	18	28	34	38	40	41

a) Plot these results on a graph and draw a line of best fit. [3 marks]
b) Find the rate of the reaction at time = 25 s. [4 marks]

Q1 Video Solution

PRACTICAL

Warm-Up & Exam Questions

It's easy to think that you've understood something when you've just read through it. These questions should test whether you really understand the previous chunk of pages, and get you set for the next bit.

Warm-Up Questions

1) Give an example of a reaction that happens very slowly, and one that is very fast.
2) Why does increasing the concentration of solutions increase the rate of a reaction?
3) What units of rate would be used for a reaction where the change in mass of a reaction vessel (measured in grams) has been recorded over time (measured in seconds)?
4) Describe how you could measure the rate of a precipitation reaction.

Exam Questions

1 **Figure 1** shows one method of measuring the rate of a reaction which produces a gas.

GRADE 4-6

1.1 What piece of apparatus necessary for measuring the rate of this reaction, is missing from **Figure 1**?

[1 mark]

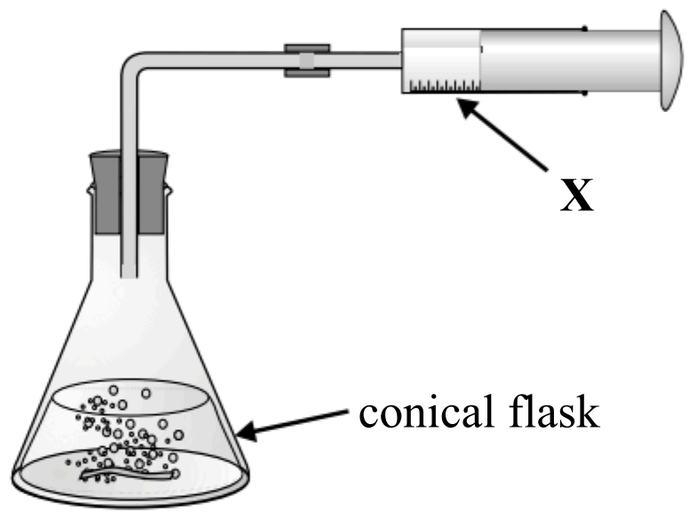
1.2 Name of the piece of apparatus in **Figure 1** labelled **X**.

[1 mark]

1.3 Describe **one** other method of measuring the rate of a reaction which produces a gas.

[2 marks]

Figure 1



2 Set volumes of sodium thiosulfate and hydrochloric acid were reacted at different temperatures. The time taken for a black cross to be obscured by the sulfur precipitate was measured at each temperature. The results are shown in **Table 1**.

GRADE 6-7

Table 1

Temperature (°C)	Time (s)
55	6
36	11
24	17
16	27
9	40
5	51

2.1 Give **two** variables that should be kept constant in this experiment.

[2 marks]

2.2 Plot the results on a graph (with time on the *x*-axis) and draw a line of best fit.

[4 marks]

2.3 Describe the relationship illustrated by your graph.

[1 mark]

2.4 Describe how the results would change if the experiment was repeated with a **lower** concentration of sodium thiosulfate.

[1 mark]

2.5 Explain how using a lower reactant concentration affects the rate of a reaction.

[3 marks]

2.6 Suggest how you could assess if the results of the experiment are repeatable.

[2 marks]

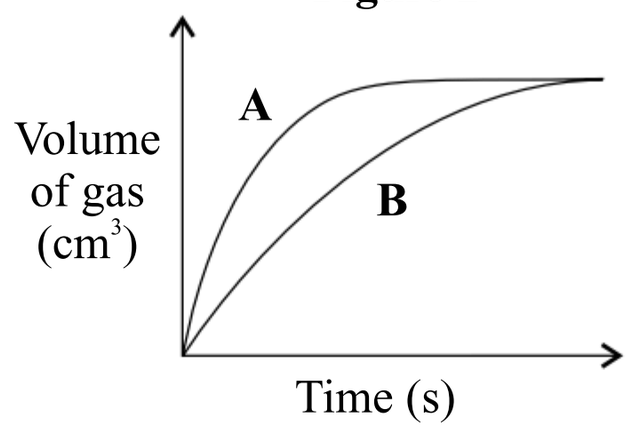
Exam Questions

PRACTICAL

3 A teacher demonstrated an experiment to investigate the effect of temperature on the rate of a reaction. The teacher added dilute hydrochloric acid at 20 °C to marble chips and measured the volume of gas produced at regular time intervals. The teacher then repeated the experiment at 30 °C using the same mass of marble chips of the same size. The results are shown in **Figure 2**.

GRADE 6-7

Figure 2



3.1 On **Figure 2**, which curve, **A** or **B**, shows the result of the experiment at 30 °C?

[1 mark]

3.2 Sketch the curve you would expect if you repeated the experiment at 25 °C onto **Figure 2**. Label it **C**.

[1 mark]

3.3 The teacher made sure that the same volume and concentration of acid was used in each repeat. Explain why these variables needed to be controlled.

[1 mark]

3.4 Does **Figure 2** suggest that the teacher successfully controlled these variables? Explain your answer.

[1 mark]

3.5 Which of the following methods could also be used to measure the rate of this reaction?
Tick **one** box.

Measuring how quickly the reaction loses mass. ☐

Timing how long the reaction takes to go cloudy. ☐

Timing how long the reaction takes to start. ☐

[1 mark]

4 Calcium carbonate powder was added to a conical flask containing dilute HCl. CO_2 was produced and collected in a gas syringe. The volume of gas released was recorded at 10 second intervals in **Table 2**:

GRADE 6-7

Table 2

Time (s)	0	10	20	30	40	50	60
Volume of CO_2 (cm^3)	0	24	32	36	38	39	40

4.1 Calculate the mean rate of reaction between 0 and 60 seconds.

[2 marks]

4.2 Plot these results on a graph and draw a line of best fit.

[4 marks]

4.3 Find the rate of the reaction 25 seconds after starting the experiment.

[4 marks]

5* Hydrogen gas and ethene gas react to form ethane. Nickel can be used as a catalyst for this reaction.

GRADE 7-9

Using your knowledge of collision theory, explain how the rate of this reaction can be increased.

[6 marks]

Reversible Reactions

Quick Quiz

In a reversible reaction, both the forward and the reverse reactions are happening ***at the same time****.*
Certain conditions can be ***more favourable*** *to one reaction than the other, making it happen faster.*

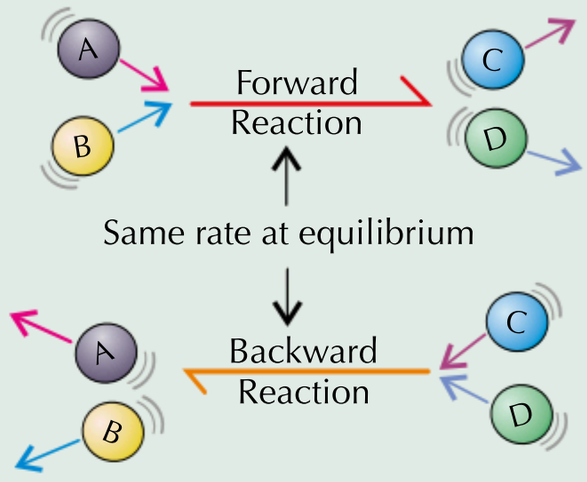
Reversible Reactions Will Reach Equilibrium

The ⇌ shows the reaction goes both ways.

This equation shows a reversible reaction — the products (C and D) can react to form the reactants (A and B) again:

$$A + B \rightleftharpoons C + D$$

1) As the reactants react, their concentrations fall — so the forward reaction will slow down (see p.81). But as more products are made and their concentrations rise, the backward reaction will speed up.
2) After a while the forward reaction will be going at exactly the same rate as the backward one — the system is at equilibrium.
3) At equilibrium, both reactions are still happening, but there's no overall effect (it's a dynamic equilibrium). This means the concentrations of reactants and products have reached a balance and won't change.



Equilibrium is only reached if the reversible reaction takes place in a 'closed system'.
A closed system just means that none of the reactants or products can escape and nothing else can get in.

The **Position of Equilibrium** Can be on the **Right** or the **Left**

When a reaction's at equilibrium it doesn't mean the amounts of reactants and products are equal.

- If the equilibrium lies to the right, the concentration of products is greater than that of the reactants.
- If the equilibrium lies to the left, the concentration of reactants is greater than that of the products.

The position of equilibrium depends on the following conditions (as well as the reaction itself):

1) the temperature,
2) the pressure (this only affects equilibria involving gases),
3) the concentration of the reactants and products.

E.g. ammonium chloride ⇌ ammonia + hydrogen chloride
Heating this reaction moves the equilibrium to the right (more ammonia and hydrogen chloride) and cooling it moves the equilibrium to the left (more ammonium chloride).

The next page tells you why these things affect equilibrium position.

Reversible Reactions Can Be **Endothermic** and **Exothermic**

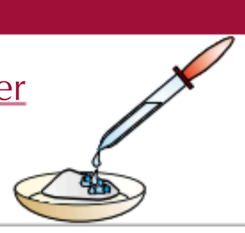
- In reversible reactions, if the reaction is endothermic in one direction, it will be exothermic in the other.
- The energy transferred from the surroundings by the endothermic reaction is equal to the energy transferred to the surroundings during the exothermic reaction.
- A good example is the thermal decomposition of hydrated copper sulfate:

See page 72 for more on endothermic and exothermic reactions.

If you heat blue hydrated copper(II) sulfate crystals, it drives the water off and leaves white anhydrous copper(II) sulfate powder. This is endothermic.

$$\text{hydrated copper sulfate} \rightleftharpoons \text{anhydrous copper sulfate} + \text{water}$$

If you then add a couple of drops of water to the white powder you get the blue crystals back again. This is exothermic.



'Anhydrous' just means 'without water', and 'hydrated' means 'with water'.

Dynamic equilibrium — lots of activity, but not to any great effect

The idea of dynamic equilibrium is something that you need to get to grips with, as things are about to get a bit trickier... Have another read of this page and let it all sink in for a minute before you flick over to the next page.

Quick Quiz

Le Chatelier's Principle

*Reversible reactions **don't like change**. So if you change something, the system will respond to undo the change.*

Reversible Reactions Try to **Counteract Changes**...

Le Chatelier's Principle is the idea that if you change the conditions of a reversible reaction at equilibrium, the system will try to counteract that change.

It can be used to predict the effect of any changes you make to a reaction system.

Video

...Such as Changes to the **Temperature**...

Change to conditions	Equilibrium moves in the direction of...	Result
Temperature is increased.	...the endothermic reaction.	More heat taken in, so temperature decreases.
Temperature is decreased.	...the exothermic reaction.	More heat is released, so temperature increases.

$N_2 + 3H_2 \rightleftharpoons 2NH_3$

Here the forward reaction is exothermic, so a decrease in temperature moves the equilibrium to the right (more NH_3).

So the original change to the conditions (the temperature decrease) has been counteracted.

...**Pressure**...

Changing the pressure only affects an equilibrium involving gases.

Change to conditions	Equilibrium moves in the direction of...	Result
Pressure is increased.	...the side with fewer molecules of gas.	Fewer gas molecules present, so pressure decreases.
Pressure is decreased.	...the side with more molecules of gas.	More gas molecules present, so pressure increases.

$N_2 + 3H_2 \rightleftharpoons 2NH_3$

There are 4 moles of gas on the left (1 of N_2 and 3 of H_2) but only 2 on the right. So, if you increase the pressure, the equilibrium shifts to the right (more NH_3).

You can use the balanced symbol equation for a reaction to see which side has more molecules of gas.

...or **Concentration**

If you change the concentration of either the reactants or the products, the system will no longer be at equilibrium. So the system responds to bring itself back to equilibrium again.

Change to conditions	Equilibrium moves in the direction of...	Result
Concentration of reactants is increased.	...the products.	Amount of reactants is reduced, system reaches equilibrium again.
Concentration of products is decreased.	...the products.	More product is made, system reaches equilibrium again.

$N_2 + 3H_2 \rightleftharpoons 2NH_3$

If more N_2 or H_2 is added, the forward reaction increases to produce more NH_3.

So, you do one thing, and the reaction does the other...

Q1 Video Solution

This sounds like it'd be a pain, but in reality it's what gives you control. In industry, that control means profit.

Q1 For each of the following reactions, state the effect of an increase in pressure on the amount of products at equilibrium.

a) $C_2H_{4\,(g)} + H_2O_{(g)} \rightleftharpoons C_2H_5OH_{(g)}$ [1 mark]

b) $N_2O_{4\,(g)} \rightleftharpoons 2NO_{2\,(g)}$ [1 mark]

c) $2CO_{(g)} + O_{2\,(g)} \rightleftharpoons 2CO_{2\,(g)}$ [1 mark]

Warm-Up & Exam Questions

Not long now till this section's over, but first there are some questions for you to tackle.

Warm-Up Questions

1) What can you say about forward and backward reaction rates at equilibrium?
2) For a reversible reaction, what is the effect on equilibrium of removing some of the reactants from the reaction mixture?
3) For the reaction, $N_{2(g)} + O_{2(g)} \rightleftharpoons 2NO_{(g)}$, what would be the effect on the equilibrium of changing the gas pressure?

Exam Questions

1 In the reaction below, substances A and B react to form substances C and D. GRADE 4-6

$$2A + B \rightleftharpoons 2C + D$$

1.1 What can you deduce about this reaction from the symbol $\rightleftharpoons$?

[1 mark]

1.2 What is meant by the term **dynamic equilibrium**?

[1 mark]

2 This question is about how pressure affects the position of equilibrium. GRADE 6-7

Reaction 1: $N_2O_{4(g)} \rightleftharpoons 2NO_{2(g)}$

Reaction 2: $ClNO_{2(g)} + NO_{(g)} \rightleftharpoons NO_{2(g)} + ClNO_{(g)}$

2.1 For Reaction 1, explain the effect of an **increase** in pressure on the amount of products at equilibrium.

[2 marks]

2.2 For Reaction 2, explain the effect of a **decrease** in pressure on the amount of products at equilibrium.

[2 marks]

3 When calcium carbonate is heated to a high temperature in a closed system, an equilibrium is reached: GRADE 6-7

$$CaCO_{3(s)} \rightleftharpoons CaO_{(s)} + CO_{2(g)}$$

The forward reaction is endothermic.

3.1 Does the reverse reaction take in or give out energy? Explain your answer.

[2 marks]

3.2 Explain why changing the temperature of a reversible reaction always affects the position of the equilibrium.

[2 marks]

3.3 For the reaction shown above, describe what would happen to the equilibrium position if the temperature was raised.

[1 mark]

Revision Summary Test for Topic 6

I'd rate this topic ten out of ten. To make sure you've understood it all, you can:

- Tackle the revision summary test below, or scan the QR code to do it online.
  The questions are hard, but they'll show you how well you really know your stuff.
- Track your progress online and see which areas need more work.
- Compare your answers with sample answers for the test here: cgpbooks.co.uk/Base

Summary Test

Rates of Reaction and Factors Affecting Them (p.80-81) ☐

1) What is meant by the 'rate of reaction'? ☐
2) On a rate of reaction graph, what is the relationship between the steepness of the line and the rate? ☐
3) On a graph showing the amount of product made in a reaction plotted against time, how can you tell the point when the reaction has finished? ☐
4) According to collision theory, what needs to happen in order for two particles to react? ☐
5) What are the four factors that affect the rate of a chemical reaction? ☐
6) Why does increasing the temperature of a reaction mixture increase the rate of a reaction? ☐
7) For a reaction between a solution and a solid, describe and explain the effect of each of these changes on the rate of reaction:
   a) Decreasing the concentration of the solution. ☐
   b) Grinding up the solid reactant into a fine powder. ☐
8) Explain what would happen to the rate of a reaction between two gases if you increased the pressure. ☐
9) What is a catalyst? How does it increase the rate of a reaction? ☐

Measuring and Calculating Rates of Reaction (p.82-84) ☐

10) What's the formula you could use to calculate the mean rate of a reaction? ☐
11) Give three possible units for the rate of a chemical reaction. ☐
12) Describe how you could measure the rate of a reaction between two colourless solutions, in which the product formed is a precipitate. Give a disadvantage of this method. ☐
13) How could you use a mass balance to measure the rate of a reaction that produces a gas, if the reaction is between a solid reactant and a solution? ☐
14) a) Give two ways to measure the volume of gas produced by a reaction over time.
    b) How could you use one of those methods to investigate the effect of increasing the concentration of HCl on the rate of reaction between HCl and Mg? ☐
15) Describe how you could use a graph to find the mean rate of a reaction between two points in time. How could you use it to find the mean rate for the whole reaction? ☐
16) How would you find the rate of reaction at a particular point on a curved rate of reaction graph? ☐

Reversible Reactions and Le Chatelier's Principle (p.87-88) ☐

17) What is a reversible reaction? ☐
18) Describe how a reversible reaction reaches equilibrium. ☐
19) What does it mean if the position of an equilibrium is said to lie to the right? ☐
20) What colour change and temperature change would you observe when water is added to anhydrous copper(II) sulfate? ☐
21) State Le Chatelier's Principle. ☐
22) If the forward reaction in a reversible reaction is exothermic, what effect will decreasing the temperature have on the position of equilibrium? ☐
23) How can you predict the effect of changing the pressure on a reversible reaction involving gases? ☐
24) In a reversible reaction, how will decreasing the concentration of the products of the forward reaction affect the position of equilibrium? ☐

Hydrocarbons

Quick Quiz

*Organic chemistry is about compounds that contain **carbon**. **Hydrocarbons** are the simplest of the lot.*

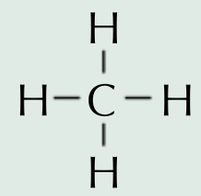
Hydrocarbons Only Contain **Hydrogen** and **Carbon** Atoms

A hydrocarbon is any compound that is formed from carbon and hydrogen atoms only.
So $C_{10}H_{22}$ (decane, an alkane) is a hydrocarbon, but $CH_3COOC_3H_7$ (an ester) is not — it contains oxygen.

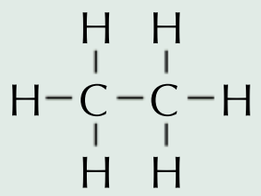
Alkanes Are the **Simplest** Type of Hydrocarbon

Alkanes have the general formula C_nH_{2n+2}. They are saturated compounds — each carbon atom forms four single covalent bonds. The first four alkanes are methane, ethane, propane and butane.

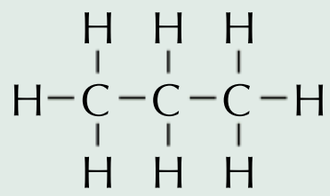
Methane: CH_4



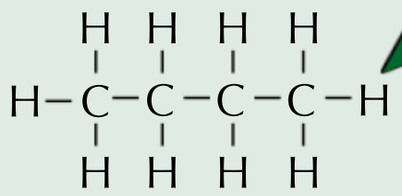
Ethane: C_2H_6



Propane: C_3H_8



Butane: C_4H_{10}



This is a displayed formula (see p.30).

Hydrocarbon **Properties Change** With the **Chain Length**

The shorter the carbon chain:

1) The more runny a hydrocarbon is — that is, the less viscous (gloopy) it is.
2) The more volatile it is, i.e. it has a lower boiling point than hydrocarbons with longer chains.
3) The more flammable (easier to ignite) the hydrocarbon is.

The properties of hydrocarbons affect how they're used for fuels. E.g. short chain hydrocarbons with lower boiling points are used as 'bottled gases' — stored under pressure as liquids in bottles.

Complete Combustion Occurs When There's Plenty of **Oxygen**

- The complete combustion of any hydrocarbon in oxygen releases lots of energy — this makes them useful as fuels. The waste products are carbon dioxide and water vapour.

hydrocarbon + oxygen → carbon dioxide + water (+ energy)

- During combustion, both carbon and hydrogen from the hydrocarbon are oxidised.
- Here's how you'd work out a balanced equation for the complete combustion of a simple hydrocarbon:

Oxidation can be defined as the gain of oxygen.

EXAMPLE

Write a balanced equation for the complete combustion of methane (CH_4).

1) First write down the unbalanced equation, showing the formulas for the reactants and products.

$?CH_4 + ?O_2 \rightarrow ?CO_2 + ?H_2O$

2) On the left hand side, there's one carbon atom, so only one molecule of CO_2 is needed to balance this.

$CH_4 + ?O_2 \rightarrow CO_2 + ?H_2O$

3) On the left hand side, there are four hydrogen atoms, so two water molecules are needed to balance them.

$CH_4 + ?O_2 \rightarrow CO_2 + 2H_2O$

4) There are four oxygen atoms on the right hand side of the equation. Two oxygen molecules are needed on the left to balance them.

$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$

Alkanes are useful fuels as they release energy when burnt

The chain lengths of hydrocarbons have a big effect on their properties and how useful they are.

Q1 Write a balanced symbol equation for the complete combustion of ethane, C_2H_6. [2 marks]

Q2 A student has two alkanes, C_5H_{12} and $C_{10}H_{22}$. Compare the following properties of the alkanes: a) viscosity b) boiling point c) flammability [3 marks]

Q1 Video Solution

Fractional Distillation

Quick Quiz

Crude oil *can be used to make loads of useful things, such as fuels. But it needs to be* ***separated out*** *first.*

Crude Oil is Made Over a **Long Period of Time**

- Crude oil is a fossil fuel. It's formed from the ancient remains of plants and animals, mainly plankton, that died and were buried in mud. Over millions of years, with high temperature and pressure, the remains turn to crude oil, which can be drilled up from the rocks where it's found.
- It's a finite resource (see p.125) — it's used up much faster than it's formed, so one day it'll run out.

Fractional Distillation can be Used to Separate **Hydrocarbon Fractions**

Crude oil is a mixture of lots of different hydrocarbons, most of which are alkanes.
The different compounds in crude oil are separated by fractional distillation.

Video

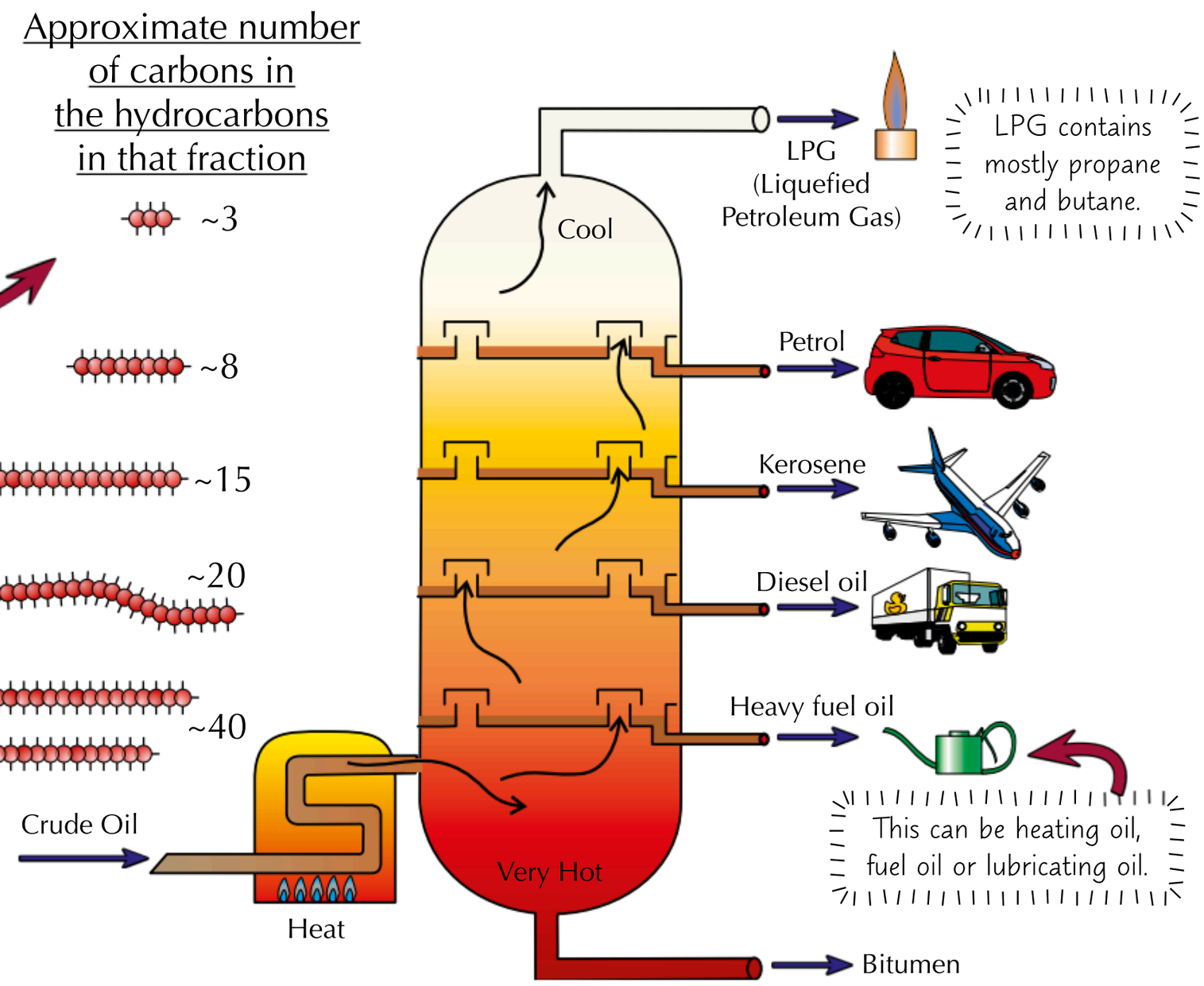
Here's how it works:

1) The oil is heated until most of it has turned into gas. The gases enter a fractionating column (and the liquid bit is drained off).

2) In the column there's a temperature gradient — it's hot at the bottom and gets cooler as you go up.

3) The shorter hydrocarbons have lower boiling points. They condense back into liquids and drain out near to the top of the column where it's cooler.

4) The longer hydrocarbons have higher boiling points. They condense and drain out of the column early on, when they're near the bottom.



5) You end up with the crude oil mixture separated out into different fractions. Each fraction contains a mixture of hydrocarbons that all contain a similar number of carbon atoms, so have similar boiling points.

Fractional distillation separates crude oil into useful compounds

Q1 Video Solution

Learn how fractional distillation works — it might just save your life... OK, maybe not, but it'll help in the exams.

Q1 Petrol drains further up a fractionating column than diesel. Use the diagram of the fractionating column to explain why the boiling point of petrol is lower than that of diesel. [1 mark]

Q2 Describe the temperature gradient in a fractionating column used for fractional distillation. [1 mark]

Uses and Cracking of Crude Oil

Quick Quiz

*Crude oil has fuelled **modern civilisation** — it would be a very different world if we hadn't discovered oil.*

Crude Oil has Various Uses Important in Modern Life

- Oil provides the fuel for most modern transport — cars, trains, planes, the lot.
- The petrochemical industry uses some of the hydrocarbons from crude oil as a feedstock to make new compounds for use in things like polymers, solvents, lubricants, and detergents.
- All the products you get from crude oil are examples of organic compounds (compounds containing carbon atoms).

The reason you get such a large variety of products is because carbon atoms can bond together to form different groups called homologous series. These groups contain similar compounds with many properties in common. Alkanes are a homologous series, as are alkenes and the other families you'll meet in this topic.

Cracking Means **Splitting Up** Long-Chain Hydrocarbons

- Short-chain hydrocarbons are in higher demand than long-chain hydrocarbons, because they are flammable and make good fuels.
- So a lot of the longer alkane molecules produced from fractional distillation are turned into smaller, more useful ones using a process called cracking.
- As well as alkanes, cracking also produces another type of hydrocarbon called alkenes. Alkenes are used as a starting material when making lots of other compounds and can be used to make polymers (see p.98).
- Some of the products of cracking are useful as fuels, e.g. petrol for cars and paraffin for jet fuel.

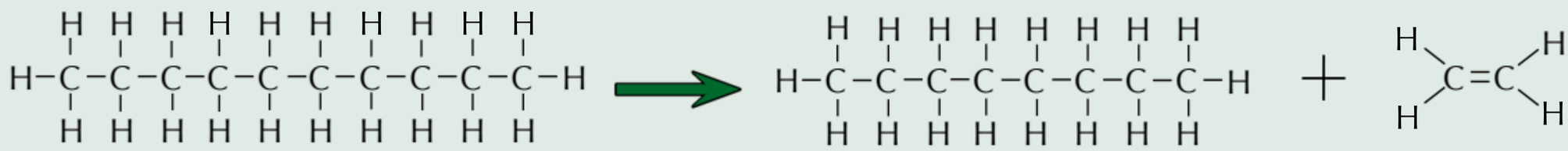
There are **Different Methods** of **Cracking**

Cracking is a thermal decomposition reaction — breaking molecules down by heating them. There are two types of cracking you need to know about:

1) Catalytic Cracking — The long-chain hydrocarbons are heated to vaporise them (turn them into a gas). Then the vapour is passed over a hot powdered aluminium oxide catalyst and the molecules split apart on its surface.
2) Steam Cracking — The long-chain hydrocarbons are vaporised and mixed with steam. They're then heated to a very high temperature so they split apart.

You need to be able to balance chemical equations for cracking. For example:

Long-chain hydrocarbon molecule → Shorter alkane molecule + alkene



E.g. decane (ten C atoms) → octane (eight C atoms) + ethene (two C atoms)
(too much of this in crude oil) (useful for petrol) (for making plastics)

Make sure that there are the same number of carbon and hydrogen atoms on both sides of the equation.

Cracking breaks down long hydrocarbons into shorter ones

Q1 Video Solution

Hydrocarbons are used in all areas of industry. Unfortunately only the smaller molecules are really useful. Luckily, we have cracking to get as many useful products from crude oil as possible.

Q1 Pentane, C_5H_{12}, can be cracked into ethene and one other hydrocarbon.
Give the balanced symbol equation for the cracking reaction. [1 mark]

Warm-Up & Exam Questions

Hydrocarbons contain only hydrogen and carbon atoms. This page contains only Warm-Up and Exam Questions. Time to get thinking.

Warm-Up Questions

1) Name the first four alkanes.
2) What kind of bonds are present in alkanes?
3) Why are alkanes often used as fuels?
4) Suggest a use of crude oil fractions that modern society depends on.
5) What sort of hydrocarbon molecules are cracked?

Exam Questions

1 Choose the statement which best describes the bonding in alkanes. GRADE 4-6 GRADE
Tick **one** box.

Carbon-carbon single bonds and carbon-hydrogen single bonds. ☐

Carbon-carbon single bonds and carbon-hydrogen double bonds. ☐

Carbon-carbon double bonds and carbon-hydrogen single bonds. ☐

Carbon-carbon double bonds and carbon-hydrogen double bonds. ☐

[1 mark]

2 Alkanes are a homologous series of hydrocarbons made up of chains of carbon atoms surrounded by hydrogen atoms. GRADE 4-6 GRADE

Pentane is an alkane with the formula C_5H_{12}.

2.1 State the name of the alkane that comes before pentane in the homologous series.

[1 mark]

2.2 State the formula of the alkane that comes before pentane in the homologous series.

[1 mark]

2.3 Suggest **two** materials produced from hydrocarbons by the petrochemical industry.

[2 marks]

3 Draw **one** line from each property of hydrocarbons to its trend with changing molecular size. GRADE 4-6 GRADE

Property	Trend
Flammability	Increases as the molecules get bigger
Viscosity	
Boiling point	Decreases as the molecules get bigger

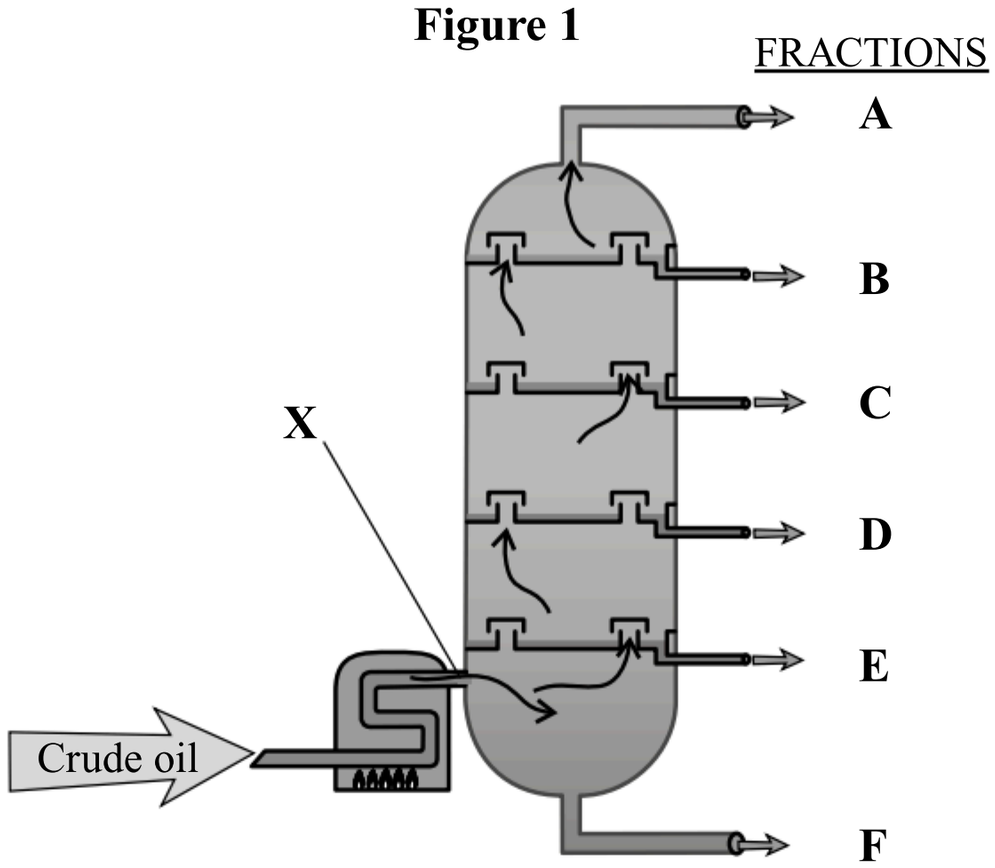
[3 marks]

Exam Questions

4 Crude oil can be separated into a number of different compounds in a fractional distillation column. **Figure 1** shows a fractional distillation column.

GRADE 6-7 GRADE

Figure 1



4.1 Which letter, **A-F**, represents the fraction with the longest hydrocarbon molecules?

[1 mark]

4.2 Which letter, **A-F**, represents the fraction with the lowest boiling point?

[1 mark]

4.3 Gaseous crude oil enters near the bottom of the fractional distillation column (point **X** in **Figure 1**). Explain why different fractions exit the column at different points and how this relates to their structure.

[3 marks]

5 Alkanes burn in excess oxygen by complete combustion.

GRADE 6-7 GRADE

5.1 What two products are made by the complete combustion of an alkane?

[2 marks]

5.2 Complete the equation below for the complete combustion of propane (C_3H_8):

$$C_3H_8 + ...O_2 \rightarrow ..... + ...$$

[2 marks]

5.3 Carbon and hydrogen atoms gain oxygen during the combustion of hydrocarbons. What type of reaction is this?

[1 mark]

6 Cracking is a process used to break longer chain molecules in crude oil down to shorter ones.

GRADE 7-9 GRADE

6.1 Why is cracking an important process in the petrochemical industry?

[1 mark]

6.2 Octane (C_8H_{18}) can be cracked to form two products, ethane and propene (C_3H_6). Complete the equation below for this reaction:

$$C_8H_{18} \rightarrow ..... + ...C_3H_6$$

[2 marks]

6.3 Describe the **two** methods used by the petrochemical industry for cracking hydrocarbons.

[4 marks]

Alkenes

Quick Quiz

__Alkenes__ look and sound a bit like the alkanes on page 91, but they have a __double__ carbon-carbon bond.

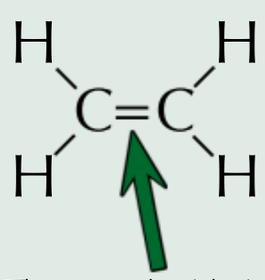
__Alkenes__ Have a __C=C Double Bond__

1) Alkenes are hydrocarbons which have a double bond between two of the carbon atoms in their chain.
2) The C=C double bond means that alkenes have two fewer hydrogens compared with alkanes containing the same number of carbon atoms. This makes them unsaturated.
3) Straight-chain alkenes have twice as many hydrogen atoms as carbon:

General formula for alkenes = C_nH_{2n}

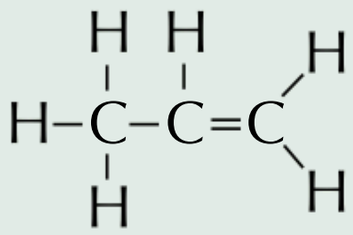
4) The first four alkenes are ethene (with two carbon atoms), propene (three Cs), butene (four Cs) and pentene (five Cs).

Ethene: C_2H_4

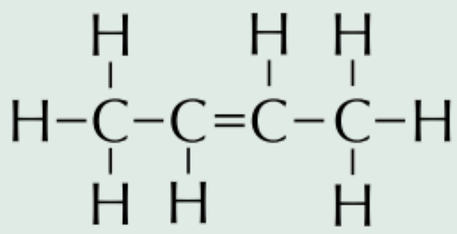


This is a double bond.

Propene: C_3H_6



Butene: C_4H_8



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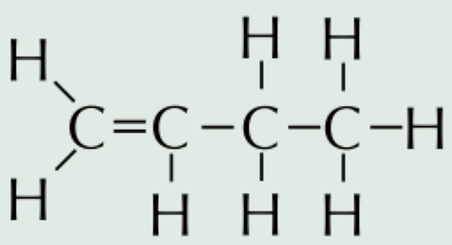
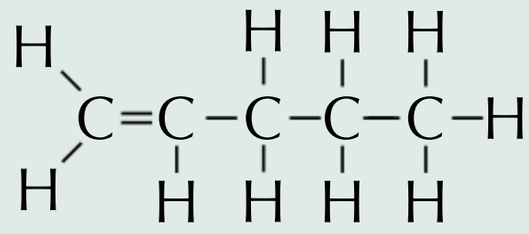


Pentene: C_5H_{10}



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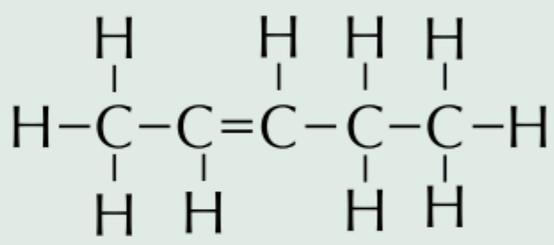


There are two different structures for butene (C_4H_8) and pentene (C_5H_{10}) as the double bond can be in two different places.

5) The C=C double bond can open up to make a single bond, allowing the two carbon atoms to bond with other atoms (see the next page). This makes alkenes reactive — far more reactive than alkanes.

__Alkenes__ Burn With a __Smoky Flame__

- In a large amount of oxygen, alkenes combust completely to produce only water and carbon dioxide (see p.91).
- However, there isn't enough oxygen in the air for this, so when you burn them they tend to undergo incomplete combustion. Carbon dioxide and water are still produced, but you can also get carbon and carbon monoxide (CO), which is a poisonous gas.

alkene + oxygen → carbon + carbon monoxide + carbon dioxide + water (+ energy)

- Incomplete combustion results in a smoky yellow flame, and less energy being released compared to complete combustion of the same compound.
- Here's an example of an equation for incomplete combustion:

$$C_4H_8 + 5O_2 \rightarrow 2CO + 2CO_2 + 4H_2O$$

This is just one possibility — the products depend on how much oxygen is present. E.g. you could also have:
$C_4H_8 + 3O_2 \rightarrow 2C + 2CO + 4H_2O$.
Whatever you have, the equation must be balanced.

That double bond makes all the difference...

Don't get alkenes confused with alkanes. Alkenes have a C=C bond, alkanes don't, which makes alkenes more reactive.

Reactions of Alkenes

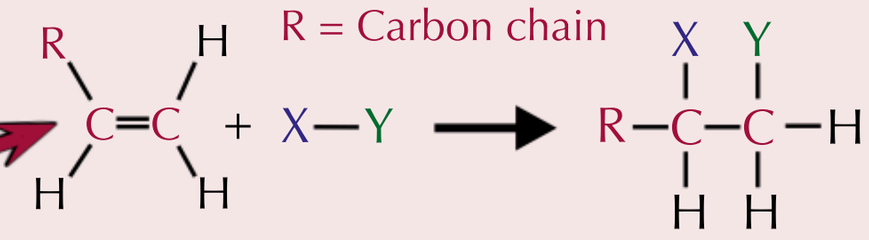
Quick Quiz

*Alkenes **react** with lots of compounds, which is great. Unless you have to learn the reactions that is...*

Alkenes Have a C=C Functional Group

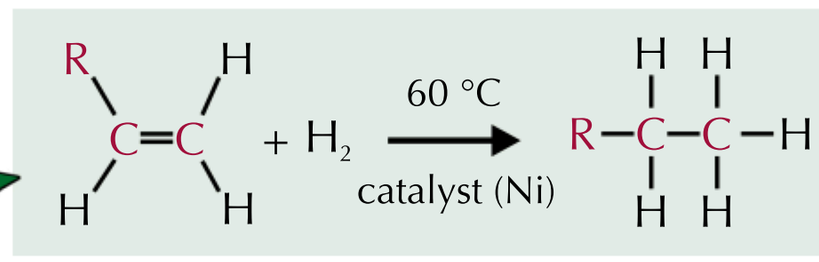
- A functional group is a group of atoms in a molecule that determine how that molecule typically reacts.
- All molecules in the same homologous series have the same functional group, so they all have similar properties and react in similar ways.
- All alkenes have the functional group 'C=C'. So you can suggest the products of a reaction involving an alkene based on your knowledge of how alkenes react in general.

Most of the time, alkenes react via addition reactions. The carbon-carbon double bond will open up to leave a single bond and a new atom is added to each carbon:



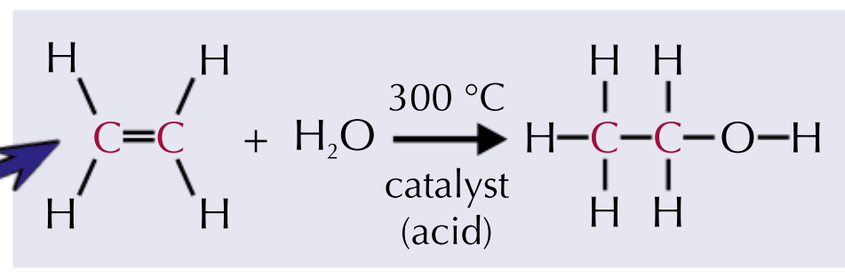
Hydrogen Reacts with Alkenes to Form Alkanes

Hydrogen can react with an alkene in the presence of a catalyst to form the equivalent, saturated, alkane.



Steam can React with Alkenes to Form Alcohols

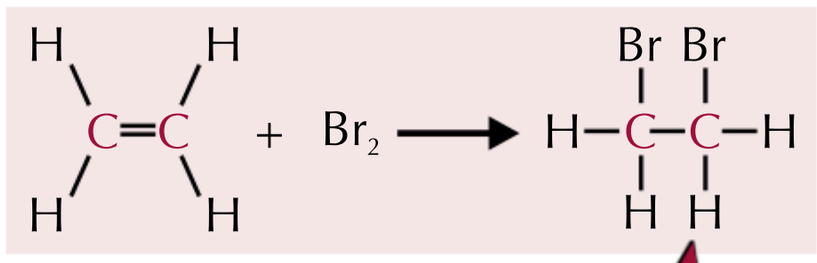
- When alkenes react with steam, water is added across the double bond and an alcohol is formed.
- For example, ethanol can be made by mixing ethene with steam and then passing it over a catalyst:



Have a look at p.100 for more on alcohols

Halogens can React with Alkenes

- Alkenes will also react in addition reactions with halogens such as bromine, chlorine and iodine. The molecules formed are saturated, with the C=C carbons each becoming bonded to a halogen atom.
- For example, bromine and ethene react together to form dibromoethane:

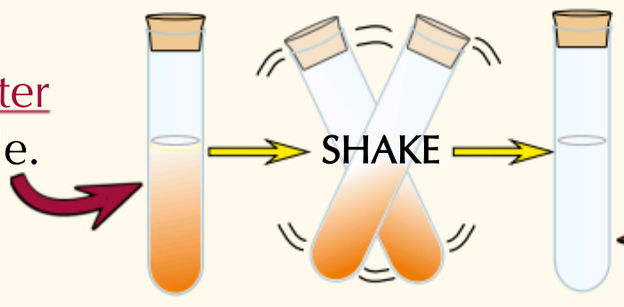


It's called dibromoethane because there are two bromine atoms.

This reaction is the same for Cl_2 or I_2 except the halogen added is different.

The addition of bromine to a double bond can be used to test for alkenes:

1) Add orange bromine water to the sample.



2) If the compound is saturated, like an alkane, no reaction will happen and it'll stay bright orange.

3) If the compound is unsaturated, like an alkene, a colourless dibromo compound is formed — so the bromine water is decolourised.

Addition reactions happen when the C=C bond opens up

These reactions aren't too bad to get your head around. Learn the reactants, products and conditions for each.

Q1 State what colour change occurs when butene, C_4H_8, reacts with bromine water. [1 mark]

Q2 Propene reacts with hydrogen in the presence of a nickel catalyst at 60 °C. Give the word equation for this reaction and the displayed formulas of propene and the product. [3 marks]

Q2 Video Solution

Addition Polymers

Quick Quiz

*There are two types of **polymers** you need to know about — this page is all about **addition** polymers. Oooh...*

Polymers are **Long-Chain** Molecules

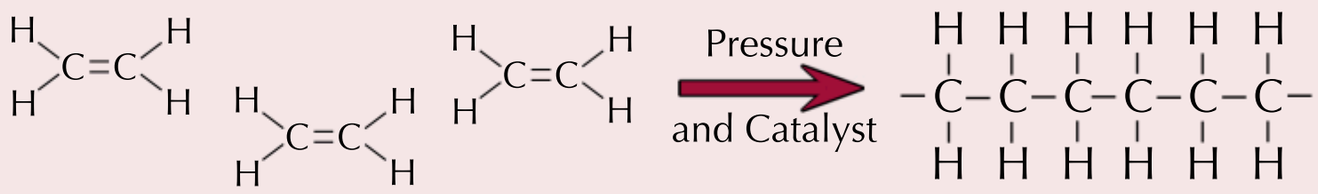
Polymers are long molecules formed when lots of small molecules called monomers join together. This reaction is called polymerisation — and it usually needs high pressure and a catalyst.

Plastics are made up of polymers. They're usually carbon based and their monomers are often alkenes.

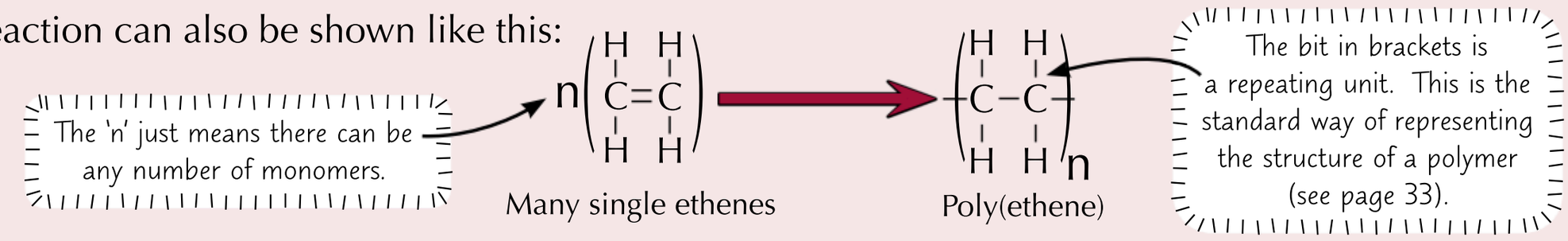
Addition Polymers are Made From **Unsaturated Monomers**

- The monomers that make up addition polymers have a double covalent bond.
- The monomers (e.g. alkenes) open up their double bonds and join together to form polymer chains. This is called addition polymerisation.

For example, lots of ethene molecules can react together to form poly(ethene) (or polythene):



The reaction can also be shown like this:

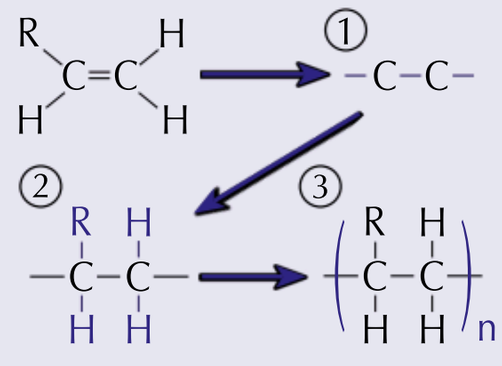


- When the monomers react in addition polymerisation reactions, the only product is the polymer, so an addition polymer contains exactly the same type and number of atoms as the monomers that formed it.
- The name of the polymer comes from the type of monomer it's made from — you just stick the word 'poly' in front of it and put the monomer name in brackets. So e.g. propene becomes poly(propene).

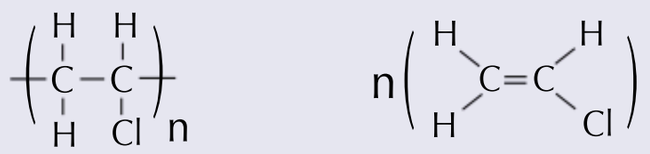
You May Need to Draw the **Repeating Unit** of a Polymer

Drawing the displayed formula of an addition polymer from the displayed formula of its monomer is easy:

1) Start by drawing the two alkene carbons, then replace the double bond with a single bond and add an extra single bond sticking out to the side of each carbon.
2) Then fill in the rest of the groups in the same way that they surrounded the double bond in the monomer.
3) Finally, stick a pair of brackets around the repeating bit, and put an 'n' after it (to show that there are lots of monomers).



To get from the displayed formula of the polymer to the displayed formula of the monomer, just do the reverse. Draw out the repeating bit of the polymer, get rid of the two bonds going out through the brackets and put a double bond between the carbons.



Addition polymer carbon chains have no C=C bonds

Make sure you can recognise and draw polymers and their monomers by practising lots of times.

Q1 Name the polymer that would be formed from the monomer bromoethene. [1 mark]

Q2 Two different polymers can be formed from butene. This is dependent on whether the double bond of the butene monomer is located in the middle or at the end of the carbon chain. Draw the two possible displayed formulas of the polymer made from butene molecules. [2 marks]

Q2 Video Solution

Warm-Up & Exam Questions

Alkenes and addition polymers are really useful compounds — you need to know how they are made, some of their reactions and how to draw them. Try these questions to test yourself out on.

Warm-Up Questions

1) What are alkenes?
2) Why is an alkene described as unsaturated?
3) What happens to alkenes when they react with halogens?
4) What will you observe if you add propene to bromine water?

Exam Questions

1 Complete the table to show the missing information for the two alkenes given. GRADE 4-6 GRADE

Name of alkene	Formula	Displayed formula
ethene		
	C_3H_6	

[4 marks]

2 Ethene can be used to make ethanol. GRADE 6-7 GRADE

2.1 Write a chemical equation for the conversion of ethene into ethanol. Include state symbols.

[2 marks]

2.2 State the name for this type of reaction.

[1 mark]

2.3 Ethene can also be used to make ethane.
Write a chemical equation for the conversion of ethene into ethane.

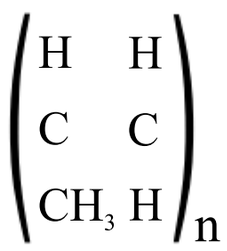
[1 mark]

2.4 State why alkenes tend to burn with a smoky yellow flame in air.

[1 mark]

3 Propene can undergo an addition polymerisation reaction to form a polymer. GRADE 6-7 GRADE

3.1 Draw the bonds to complete the displayed formula of the polymer formed in this addition polymerisation reaction.



[1 mark]

3.2 Name the polymer formed in this reaction.

[1 mark]

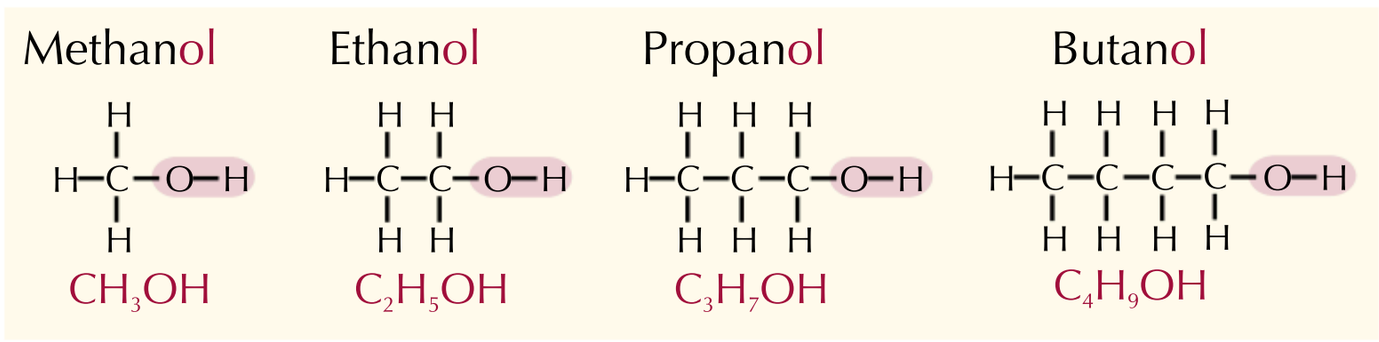
Quick Quiz

Alcohols

Alcohol *isn't just what your Nan has for special occasions. It's yet another* ***homologous series*** *of compounds.*

Alcohols Have an '**-OH**' Functional Group and End in '**-ol**'

- The general formula of an alcohol is $C_nH_{2n+1}OH$. So an alcohol with 2 carbons has the formula C_2H_5OH.
- All alcohols contain an -OH functional group. You need to know the first four in the homologous series:



- The basic naming system is the same as for alkanes — but replace the final '-e' with '-ol'.
- Don't write CH_4O instead of CH_3OH — it doesn't show the -OH group.

Make sure you can write balanced equations for the combustion of alcohols.

The **First Four Alcohols** Have **Similar Properties**

1) Alcohols are flammable. They undergo complete combustion in air to produce carbon dioxide and water. For example:

$$2CH_3OH_{(l)} + 3O_{2(g)} \rightarrow 2CO_{2(g)} + 4H_2O_{(g)}$$

2) Methanol, ethanol, propanol and butanol are all soluble in water. Their solutions have a neutral pH.
3) They also react with sodium. One of the products of this reaction is hydrogen.
4) Alcohols can be oxidised by an oxidising agent to produce a carboxylic acid (see the next page). Different alcohols form different carboxylic acids. For example, methanol is oxidised to methanoic acid, while ethanol is oxidised to ethanoic acid.

An oxidising agent is a substance that oxidises (see p.65) another substance in a reaction.

Alcohols are Used as **Solvents** and **Fuels**

- Alcohols such as methanol and ethanol are used as solvents in industry. This is because they can dissolve most things water can dissolve, but they can also dissolve substances that water can't dissolve — e.g. some hydrocarbons, oils and fats.
- The first four alcohols are used as fuels. For example, ethanol is used as a fuel in spirit burners — it burns fairly cleanly and it's non-smelly.

Ethanol can be Made by **Fermentation**

Ethanol is the alcohol found in alcoholic drinks such as wine or beer. It's usually made using fermentation.

- Fermentation uses an enzyme in yeast to convert sugars into ethanol. Carbon dioxide is also produced.

$$\text{sugar} \xrightarrow{\text{yeast}} \text{ethanol + carbon dioxide}$$

- The reaction occurs in solution so the ethanol produced is aqueous.
- Fermentation happens fastest at a temperature of around 37 °C, in a slightly acidic solution and under anaerobic conditions (no oxygen).
- The enzyme in yeast works best to convert the sugar to alcohol under these conditions. At a different temperature or pH, the enzyme could be denatured (destroyed) or could work at a much slower rate.

The first four alcohols behave in similar ways

Alcohols can do all sorts of things — they burn in air, are soluble in water, react with sodium and are oxidised to form carboxylic acids. Make sure you can describe the reactions and be able to write equations for the combustion of alcohols.

Carboxylic Acids

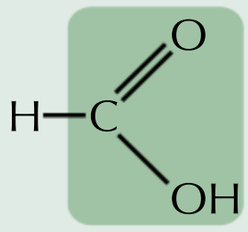
Quick Quiz

*It's time to dive deeper into the exciting and wondrous world of **carboxylic acids**. (And **esters**, too...)*

Carboxylic Acids Have the Functional Group **-COOH**

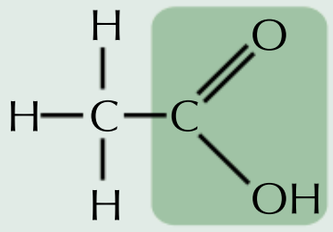
Carboxylic acids are a homologous series of compounds that all have '-COOH' as a functional group. You need to know the names and structures of these four:

Methanoic Acid



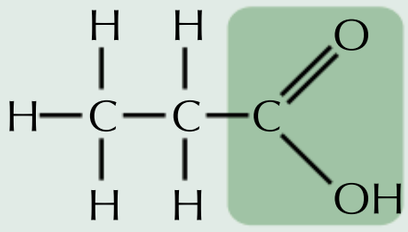
HCOOH

Ethanoic Acid



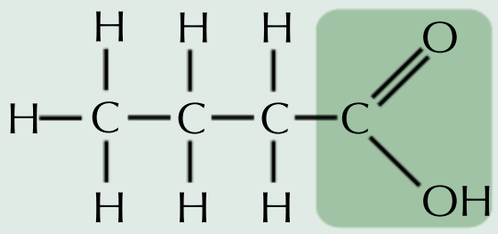
CH_3COOH

Propanoic Acid



C_2H_5COOH

Butanoic Acid



C_3H_7COOH

Carboxylic Acids React Like Other **Acids**

- They react (like any other acid) with carbonates to produce a salt, water and carbon dioxide.
- The salts formed in these reactions end in -anoate — e.g. methanoic acid will form a methanoate, ethanoic acid an ethanoate, etc. For example:

ethanoic acid + sodium carbonate → sodium ethanoate + water + carbon dioxide

- Carboxylic acids can dissolve in water. They are weak acids, so when they dissolve they partially ionise — only some of the molecules release H^+ ions.
- This means that aqueous solutions of carboxylic acids have a higher pH (are less acidic) than solutions of strong acids with the same concentration.

There's more about strong and weak acids on page 60.

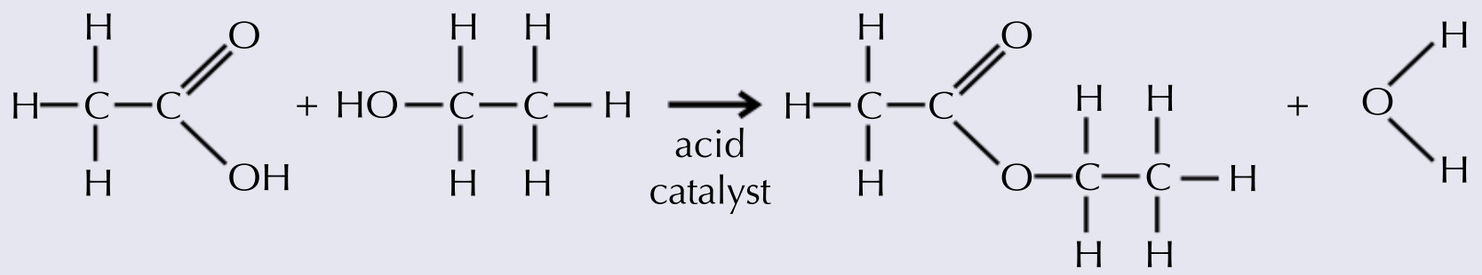
Esters can be Made from **Carboxylic Acids**

1) Esters have the functional group '-COO-'.
2) They are formed from an alcohol and a carboxylic acid.
3) An acid catalyst is usually used (e.g. concentrated sulfuric acid).

The names of esters can be a bit complicated and are often tongue twisters. The one you need to make sure you learn is ethyl ethanoate — the examiners could ask you about this one.

alcohol + carboxylic acid $\xrightarrow{\text{acid catalyst}}$ ester + water

For example, ethyl ethanoate can be made from ethanoic acid and ethanol with an acid catalyst:



CH_3COOH	C_2H_5OH	$CH_3COOC_2H_5$	H_2O
Ethanoic Acid	Ethanol	Ethyl ethanoate	Water

Carboxylic acids — not all that different to other acids...

...and they don't ionise completely, so are weak acids. They can react with alcohols to form esters.

Q1 Sodium carbonate is added to an aqueous solution of methanoic acid.
a) Describe and explain what you would see during this reaction. [2 marks]
b) Give the word equation for this reaction. [1 mark]

Q1 Video Solution

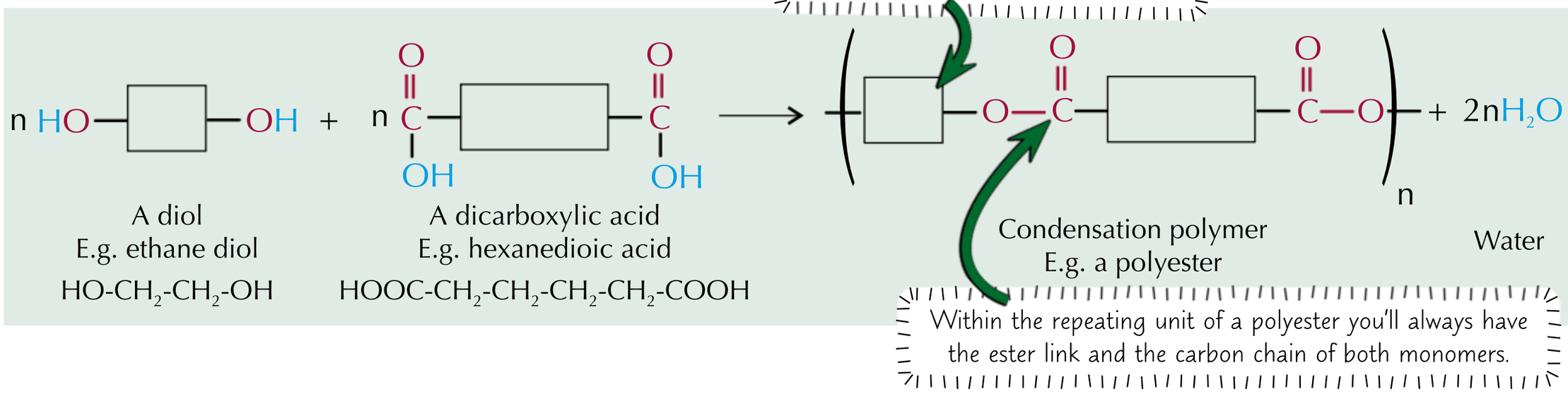
Condensation Polymers

Quick Quiz

*You might remember **addition polymerisation** from back on page 98, but there is **another** type of polymerisation that you could be asked about. This time the monomers need to have **TWO** functional groups.*

Polymers can be Made by Condensation Polymerisation

1) Condensation polymerisation involves monomers which contain different functional groups.
2) The monomers react together and bonds form between them, making polymer chains.
3) For each new bond that forms, a small molecule (for example, water) is lost.
   This is why it's called condensation polymerisation.
4) The simplest types of condensation polymers contain two different types of monomer, each with two of the same functional groups.
5) For example, here's how a polyester can be made by condensation polymerisation:



Addition and Condensation Polymerisation are Different

Both addition and condensation polymerisation produce polymers.
However, the products and reactants are very different:

	Addition Polymerisation	Condensation Polymerisation
Number of types of monomers	Only one monomer type containing a C=C bond.	Two monomer types each containing two of the same functional groups. or One monomer type with two different functional groups (see next page).
Number of products	Only one product formed.	Two types of product — the polymer and a small molecule (e.g. water).
Functional groups involved in polymerisation	Carbon-carbon double bond in monomer.	Two reactive groups on each monomer.

Condensation polymerisation results in the loss of a small molecule

You need to be able to recognise whether certain monomers will react by addition or condensation polymerisation. And if you're given the products of a polymerisation reaction, you should be able to tell whether the reaction was addition or condensation. That's no mean feat — it'd probably be a good idea to read over this page again.

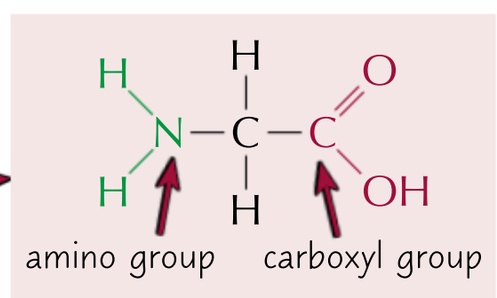
Naturally Occurring Polymers

Quick Quiz

Polymers *aren't just something created in the lab — they are also found in the* ***natural world***.

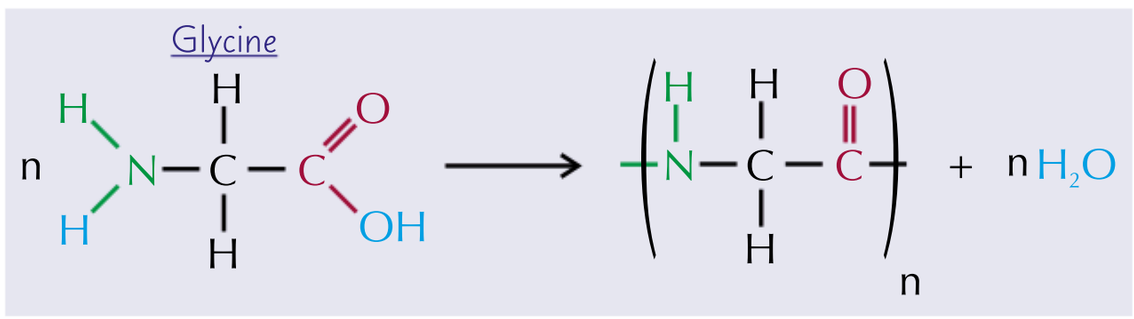
Amino Acids have an Amino Group and a Carboxyl Group

1) An amino acid contains two different functional groups — a basic amino group (NH_2) and an acidic carboxyl group (COOH).
2) An example of an amino acid is glycine — the smallest and simplest amino acid possible.



Proteins are Polymers of Amino Acids

1) Amino acids can form polymers known as polypeptides via condensation polymerisation.
2) The amino group of an amino acid can react with the carboxyl group of another, and so on, to form a polymer chain. For every new bond that is formed, a molecule of water is lost.



3) Proteins are made of one or more long chains of polypeptides. They have loads of important uses in the human body. For example, enzymes work as catalysts, haemoglobin transports oxygen, antibodies form part of the immune system, and the majority of body tissue is made from proteins.
4) Different polypeptides and proteins contain different amino acids in their polymer chains. The order of the amino acids is what gives proteins their different properties and shapes.

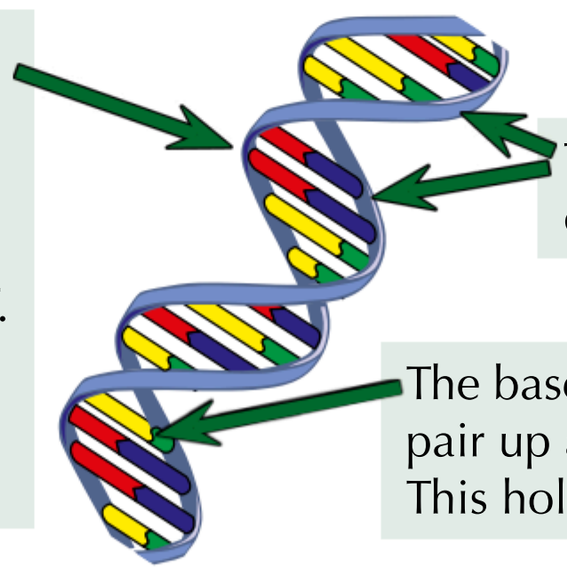
DNA Molecules are Made From Nucleotide Polymers

DNA (deoxyribonucleic acid) is found in every living thing and many viruses. It contains genetic instructions that allow the organism to develop and operate. It's a large molecule that takes a double helix structure.

Double Helix

The monomers are called 'nucleotides'.

- Each nucleotide contains a small molecule called a 'base'.
- There are four different bases, known by their initials — A, C, G, T.
- The order of bases acts as a code for the order of amino acids used to make proteins in the body.



There are two polymer chains (strands).

The bases on the different strands pair up and form cross links. This holds the two strands together.

Simple Sugars Can Form Polymers

1) Sugars are small molecules that contain carbon, oxygen and hydrogen.
2) Sugars can react together through polymerisation reactions to form larger carbohydrate polymers, e.g. starch, which living things use to store energy, and cellulose, which is found in plant cell walls.

Polymers are found everywhere in nature

Even the paper in this page is made of cellulose fibres, which are polymers. There really is no escape from them...

Warm-Up & Exam Questions

I'm sure you have everything on these pages learnt and understood. But just to check, try these questions.

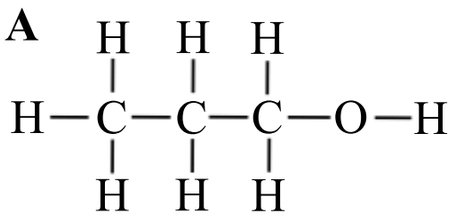
Warm-Up Questions

1) What is the general formula of an alcohol?
2) Name the first four carboxylic acids in the homologous series.
3) Give the functional group of an ester.
4) What type of polymerisation is involved in the production of a polyester?
5) Name the monomer that proteins are made of.

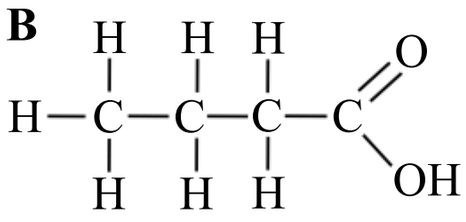
Exam Questions

1 Which of the following structures, **A**, **B**, **C**, or **D** shows butanol? GRADE 4-6

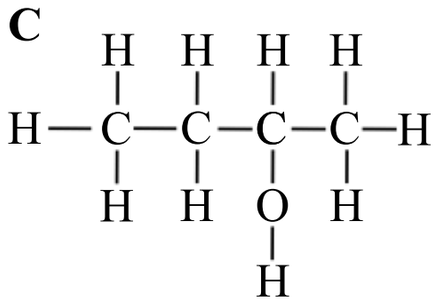
A



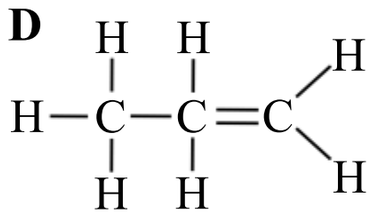
B



C



D



[1 mark]

2 Carboxylic acids are a widely used family of organic chemicals. GRADE 6-7

2.1 Draw bonds and missing atoms to complete the displayed formula of ethanoic acid.

C—C

[2 marks]

2.2 One use of carboxylic acids is in the production of esters.
Name the ester formed when ethanoic acid is reacted with ethanol.

[1 mark]

2.3 Carboxylic acids are weak acids. Explain what this means.

[1 mark]

3 Alcohols are an important group of organic chemicals. The most widely used alcohol is ethanol. GRADE 6-7

3.1 Name the alcohol in this homologous series that contains three carbon atoms.

[1 mark]

3.2 Ethanol burns in oxygen to give carbon dioxide and water.
Balance the symbol equation for this reaction.

$$.....C_2H_5OH + ...O_2 \rightarrow .....CO_2 + ...H_2O$$

[1 mark]

4* Polymers can be made by addition polymerisation and condensation polymerisation reactions. Discuss the differences between these two types of polymerisation. GRADE 7-9

[4 marks]

Revision Summary Test for Topic 7

Well that wraps up Topic 7. Lovely stuff. To make sure it's firmly in your brain, you can:

- Tackle the revision summary test below, or scan the QR code to do it online.
  The questions are hard, but they'll show you how well you really know your stuff.
- Track your progress online and see which areas need more work.
- Compare your answers with sample answers for the test here: cgpbooks.co.uk/Base

Summary Test

Hydrocarbons (p.91) ☑

1) What two elements do hydrocarbons contain?
2) What is the general formula for alkanes?
3) Draw the displayed formulas of: a) methane, b) ethane, c) propane, d) butane.
4) Describe how the viscosity, volatility and flammability of hydrocarbons change with their size.
5) What two waste products form from the complete combustion of hydrocarbons?

Crude Oil (p.92-93) ☑

6) How is crude oil formed?
7) Name the process used to separate the compounds in crude oil. How does it work?
8) Give two ways in which modern life depends on crude oil.
9) Why is cracking necessary? Describe two ways it can be done.

Alkenes and Addition Polymerisation (p.96-98) ☑

10) What is an alkene?
11) Give the general formula for an alkene.
12) Draw the displayed formulas for two different structures of butene.
13) What are the four possible products of the incomplete combustion of an alkene?
14) What is a functional group?
15) What type of molecule is formed when an alkene reacts with hydrogen? What about with steam?
16) Explain why bromine water can be used to test for alkenes.
17) Describe what happens in addition polymerisation.
18) Draw the displayed formula for the repeating unit of poly(ethene).

Alcohols and Carboxylic Acids (p.100-101) ☑

19) Give the names and molecular formulas of the first four alcohols.
20) Describe four properties that those four alcohols have in common.
21) Give two uses of alcohols.
22) Describe how ethanol can be made by fermentation, including the conditions for this process.
23) Name the carboxylic acid with the formula C_2H_5COOH.
24) Describe what happens when carboxylic acids: a) react with carbonates, b) dissolve in water.
25) Give the general word equation for the reaction of an alcohol with a carboxylic acid in the presence of an acid catalyst.

Condensation Polymerisation and Natural Polymers (p.102-103) ☑

26) What products are formed by the condensation polymerisation of a diol and a dicarboxylic acid?
27) What type of monomer makes up polypeptides? Name its two functional groups.
28) Describe the structure of DNA.
29) Name two natural polymers formed from sugars.

Purity and Formulations

Quick Quiz

*Substances are often not **100% pure** — they might have **other stuff** that you can't see mixed in with them. The purity of a substance might need to be **checked**, before, say, a drug is made from it.*

Purity is Defined Differently in Chemistry to Everyday

Usually when you say that a substance is 'pure' you mean that nothing has been added to it — it's in its natural state. For example: pure milk or pure beeswax.

In chemistry, a pure substance is something that only contains one compound or element throughout — not mixed with anything else.

The Boiling or Melting Point Tells You How Pure a Substance Is

1) A chemically pure substance will melt or boil at a specific temperature.
2) You can test the purity of a sample by measuring its melting or boiling point and comparing it with the melting or boiling point of the pure substance (which you can find in a data book).
3) The closer your measured value is to the actual melting or boiling point, the purer your sample is.

Impurities in a sample will lower the melting point and increase the melting range of your substance.
Impurities also increase the boiling point and may result in a sample boiling at a range of temperatures.

Formulations are Mixtures with Exact Amounts of Components

1) Formulations are useful mixtures with a precise purpose that are made by following a 'formula' (a recipe).
2) Each component in a formulation is present in a measured quantity, and contributes to the properties of the formulation so that it meets its required function.

Take a look at p.9 for more on mixtures.

For example, paints are formulations composed of:

- Pigment — gives the paint colour, e.g. titanium oxide in white paints.
- Solvent — dissolves the other components and changes the viscosity (thickness).
- Binder (resin) — forms a film that holds the pigment in place after it's been painted on.
- Additives — added to further change the properties of the paint.

The chemicals used and their amounts are changed, so each paint is right for its purpose.

3) Formulations are important in the pharmaceutical industry. For example, by altering the formulation of a pill, chemists can make sure it delivers a drug to the correct part of the body at the right concentration.
4) In everyday life, formulations can also be found in cleaning products, fuels, cosmetics, fertilisers (see p.136), metal alloys and even food and drink.
5) When you buy a product, you might find that it has information about its composition on the packaging. For example, the ratio or percentage of each component. This tells you the product's a formulation.

Make sure you can use data to identify pure and impure substances...

Knowing how pure a product is can be vital in industries such as pharmaceuticals and the food industry. If extra 'stuff' ends up in a product by mistake, it could change the properties of the product, or even make it dangerous.

Chromatography

Quick Quiz

You met chromatography on page 9, as a way of separating mixtures. Now it's time to see ***how it works****.*

Chromatography uses Two Phases

There are different types of chromatography, but they all have two 'phases':

Mobile phase	Where the molecules can move. It is always a liquid or a gas.
Stationary phase	Where the molecules can't move. It can be a solid or a really thick liquid.

1) During chromatography, the substances in the sample constantly swap between being dissolved in the mobile phase and being attracted (stuck) to the stationary phase. They are 'distributed' differently between the phases — some substances are more soluble in the solvent, so spend more time in the mobile phase, while other substances are more strongly attracted to the stationary phase, so spend more time stuck to it.
2) The mobile phase moves through the stationary phase — and anything that's dissolved in the mobile phase moves with it.
3) Substances that spend more time in the mobile phase move more quickly than substances that spend more time stuck to the stationary phase, so they travel further.
4) This cause the different substances to separate out.

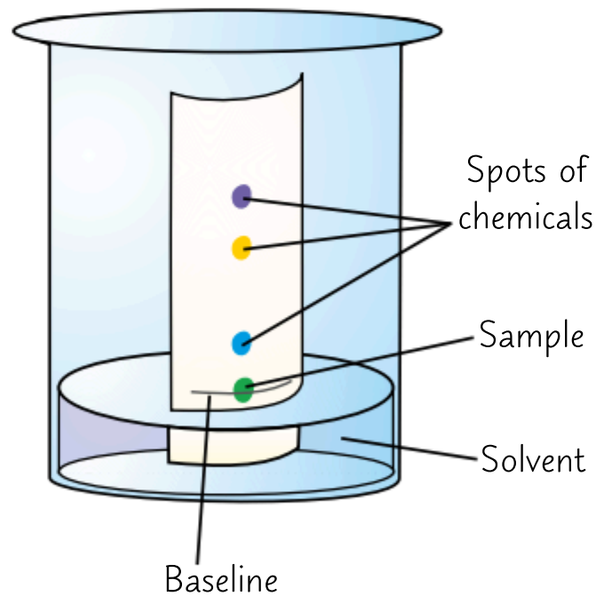
If two substances are distributed similarly between the phases, they may not separate out completely. In paper chromatography, this means that their spots would overlap and look like one spot. Changing the solvent changes their distribution so could cause them to separate out, producing a different number of spots. A pure substance will only ever form one spot in any solvent.

Paper Chromatography uses Paper and a Solvent

PRACTICAL

- In paper chromatography the stationary phase is the paper (often filter paper) and the mobile phase is the solvent (e.g ethanol or water).
- Page 9 already covered how you can use paper chromatography to separate the parts of a mixture, but here's a quick recap:

1) Place a spot of your mixture on a pencil baseline that has been drawn on a piece of filter paper.
2) Pop the paper into a beaker with some solvent in the bottom. (The baseline must be above the level of the solvent.)
3) Put a lid on the container and leave it for a while.
4) The solvent soaks up through the paper. As it moves, it carries the sample with it.
5) The parts of the mixture separate out into spots.
6) When the solvent has nearly reached the top of the paper, take the paper out of the beaker and leave it to dry.
7) The pattern of spots that you're left with on the paper is called a chromatogram.



The spot that moves the furthest is the substance that's most soluble in the solvent and least attracted to the paper.

Chromatography separates substances based on their solubilities

Chromatography works because each of the chemicals in a mixture spends different amounts of time dissolved in the mobile phase and stuck to the stationary phase. It's great — all you need is some paper and a bit of solvent. There's more about using chromatography to identify chemicals coming up on the next page.

Quick Quiz

R_f Values

PRACTICAL

*Now that you know a bit of the **theory** behind how paper chromatography works, here's how you can use a **chromatogram** to analyse a particular substance and find out what's in it. Exciting stuff, I know...*

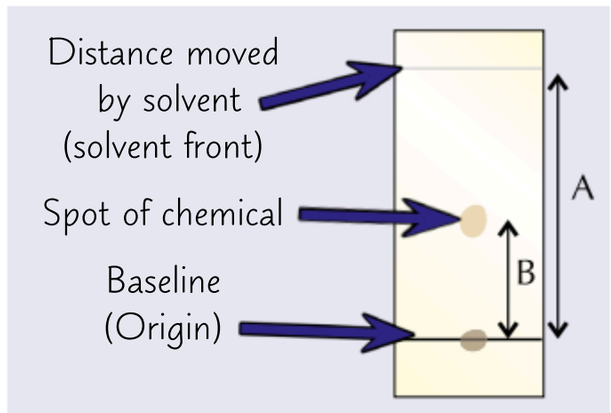
You can Calculate the R_f Value for Each Chemical

- An R_f value is the ratio between the distance travelled by the dissolved substance (the solute) and the distance travelled by the solvent.
- The further through the stationary phase a substance moves, the larger its R_f value will be.

You can calculate R_f values using the formula:

$$R_f = \frac{\text{distance travelled by substance (B)}}{\text{distance travelled by solvent (A)}}$$

This is the distance from the baseline to the centre of the spot.



EXAMPLE

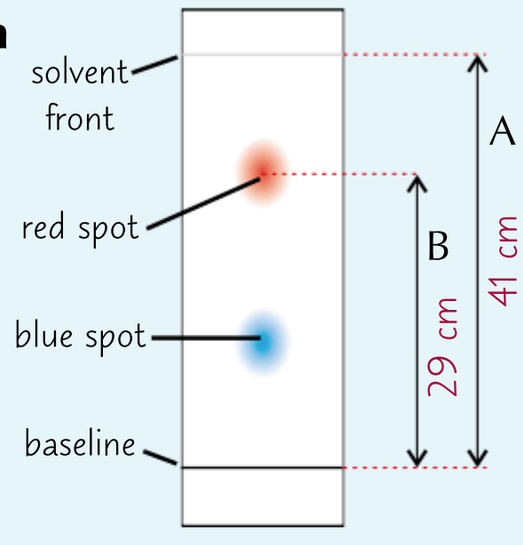
A chromatography experiment produced the chromatogram on the right. Calculate the R_f value of the red spot.

1) Find the distance moved by the red spot (B). 29 cm

2) Find the distance moved by the solvent (A). 41 cm

3) Calculate the R_f value:

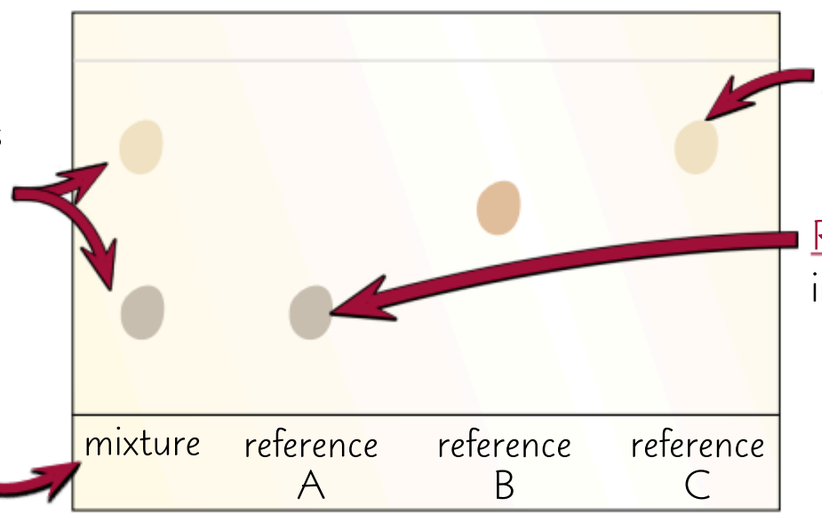
$$R_f = \frac{\text{distance moved by substance (B)}}{\text{distance moved by solvent (A)}} = \frac{29}{41} = 0.70731...$$
$$= 0.71 \text{ (2 s.f.)}$$



MATHS SKILLS

Add **References** to Your Experiment to Help **Identify** Substances

- You can use chromatography to see if a mixture contains a certain substance.
- To do this, you run a pure sample of a known substance (a reference) next to the mixture.
- If the R_f value of the reference matches one of the spots, that substance could be in the mixture. For example:



So, the mixture possibly contains substance A and substance C.

The R_f value of a substance depends on the solvent — if you change the solvent, the R_f value will change. If the R_f values match in one solvent, you can check to see if the chemicals are the same by repeating with a different solvent. If they match again, it's likely that they're the same.

You need to learn the formula for R_f

Sometimes, when you're doing paper chromatography, you'll end up with a spot left sitting on the baseline, even after your solvent has run all the way up the paper. Any substance that remains on the baseline is insoluble in that solvent. To identify this substance, you could try repeating the experiment using a different solvent.

Q1 Video Solution

Q1 A spot on a chromatogram moved 6.3 cm from the baseline.
The solvent front moved 8.4 cm. Calculate the R_f value. [1 mark]

Warm-Up & Exam Questions

Look, a chromatography question — those things are fun. Get your investigative hat on and get stuck in...

Warm-Up Questions

1) What is meant by a pure substance in chemistry?
2) What effect will impurities in a substance have on its boiling point?
3) Give an example of a formulation used in everyday life.
4) What effect does chromatography have on a mixture?

Exam Questions

1 The melting point of a sample of aspirin made by a student is measured as being between 128-132 °C. The melting point of pure aspirin is 136 °C. Give two reasons why the melting point measured shows that the sample is not pure.

GRADE 4-6

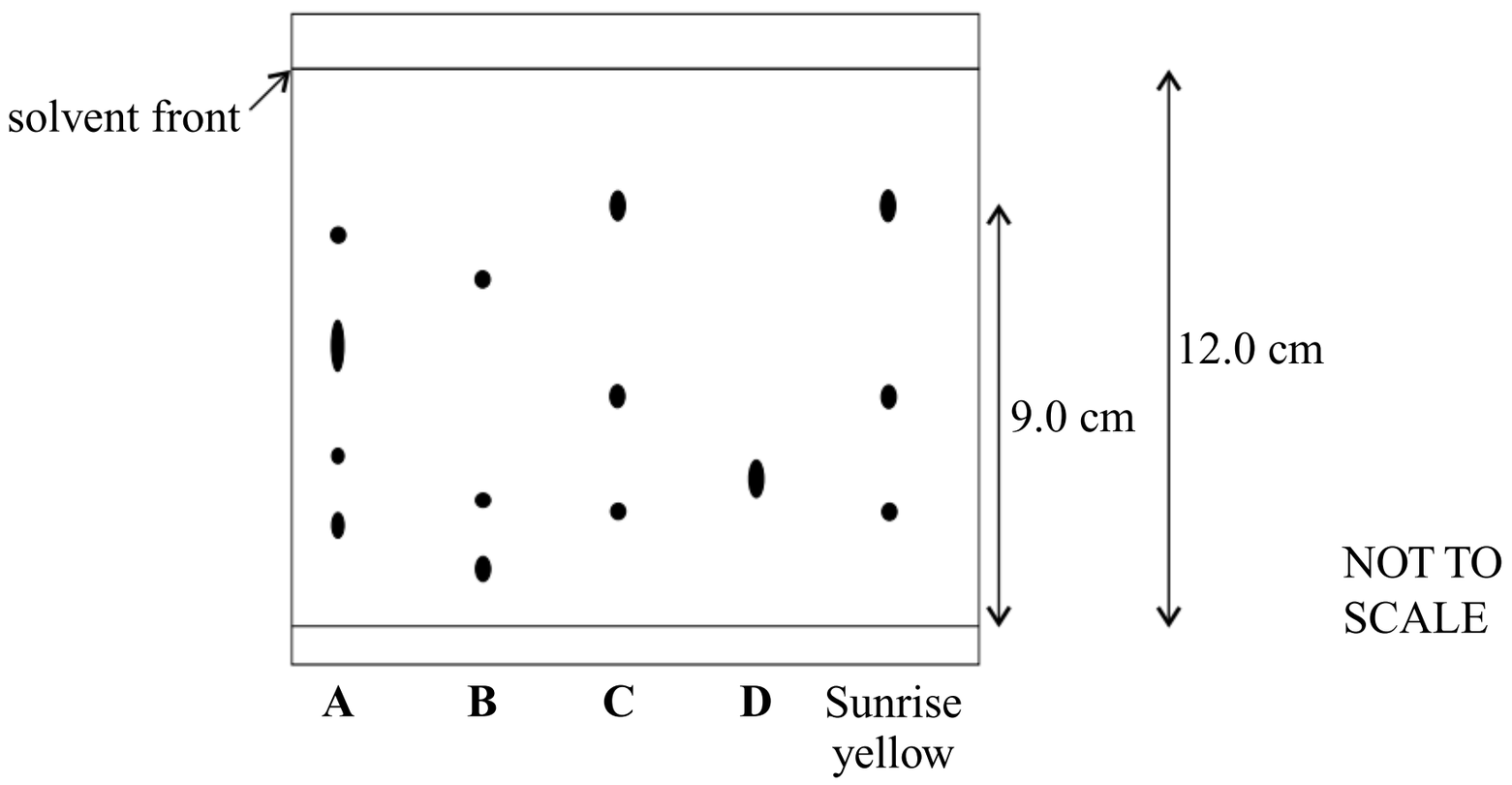
[2 marks]

PRACTICAL

2 A scientist used chromatography to analyse the composition of five food colourings. Four of the colourings were unknown (**A** – **D**). The other was sunrise yellow. The results are shown in **Figure 1**.

GRADE 6-7

Figure 1



2.1 Which food colouring is most likely to be a pure compound?

[1 mark]

2.2 Which food colouring definitely contains at least four different compounds?

[1 mark]

2.3 Which of the food colourings, **A**-**D**, could be the same as sunrise yellow?

[1 mark]

2.4 Calculate the R_f value for the spot of chemical in sunrise yellow which is furthest up the chromatogram.

[2 marks]

Tests for Gases and Anions

PRACTICAL

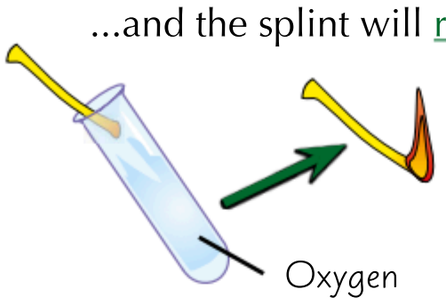
Quick Quiz

*Yep, that's right, you need to revise **tests** for your test (or exam, if you're being technical).*

There are **Tests** for **4 Common Gases**

OXYGEN

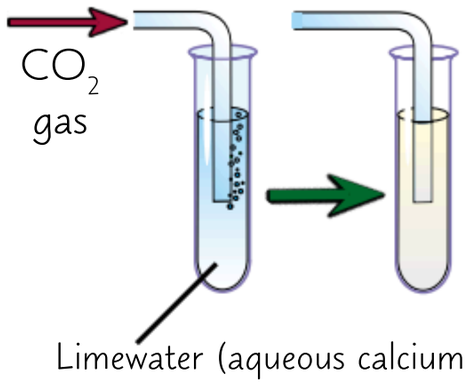
Put a glowing splint in a test tube of oxygen...



...and the splint will relight.

CARBON DIOXIDE

Bubble carbon dioxide through (or shake it with) limewater...

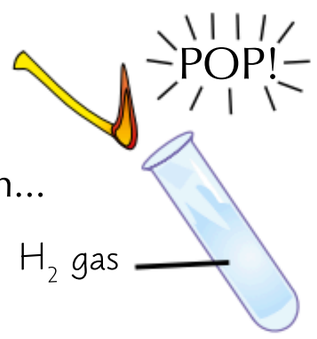


...and the solution will turn cloudy.

Video

HYDROGEN

Hold a lit splint at the end of a test tube of hydrogen...



...and you'll hear a "squeaky pop".

CHLORINE

Put a piece of damp litmus paper in a test tube containing chlorine...

...and the chlorine will bleach it (turn it white).

It may turn red for a moment first, because a solution of chlorine is acidic.

Test for **Carbonates** with **Dilute Acid** and **Limewater**

1) To test for carbonate ions (CO_3^{2-}), first use a dropping pipette to add a couple of drops of dilute acid to a sample of your mystery solution in a test tube.
2) Connect this test tube to another one containing limewater so that any gas produced will bubble through it. If carbonate ions are present, carbon dioxide will be released and you'll see bubbles — this will turn the limewater cloudy (as shown above).

For example: $Na_2CO_{3(aq)} + 2HCl_{(aq)} \rightarrow CO_{2(g)} + 2NaCl_{(aq)} + H_2O_{(l)}$

Test for **Sulfates** with HCl and **Barium Chloride**

1) To test for sulfate ions (SO_4^{2-}), first use a dropping pipette to add a couple of drops of dilute hydrochloric acid (HCl) to a sample of your mystery solution.
2) Then add a couple of drops of barium chloride solution ($BaCl_2$).
3) A white precipitate of barium sulfate will form if sulfate ions are present: $Ba^{2+}_{(aq)} + SO_4^{2-}{}_{(aq)} \rightarrow BaSO_{4(s)}$

In the tests for sulfates and halides, adding an acid gets rid of any carbonate ions before doing the test. The carbonate ions would also form a precipitate, so they'd confuse the results.

Test for **Halides (Cl^-, Br^-, I^-)** with **Nitric Acid** and **Silver Nitrate**

To identify a halide ion, add a couple of drops of dilute nitric acid (HNO_3), followed by a couple of drops of silver nitrate solution, $AgNO_3$, to your mystery solution.

Chloride ions	give a white precipitate of silver chloride	$Ag^+_{(aq)} + Cl^-_{(aq)} \rightarrow AgCl_{(s)}$
Bromide ions	give a cream precipitate of silver bromide	$Ag^+_{(aq)} + Br^-_{(aq)} \rightarrow AgBr_{(s)}$
Iodide ions	give a yellow precipitate of silver iodide	$Ag^+_{(aq)} + I^-_{(aq)} \rightarrow AgI_{(s)}$

These are all really useful tests to know...

There's quite a lot to learn here, so don't just stare at this page till your eyes swim and you don't want to see the word 'precipitate' ever again. My advice is to take each section one by one and learn the tests that way.

Q1 A student adds dilute nitric acid and silver nitrate solution to an aqueous solution of a calcium halide. A white precipitate forms. Give the balanced symbol equation for this reaction. [2 marks]

Q1 Video Solution

PRACTICAL

Tests for Cations

Quick Quiz

*I haven't met a person who doesn't like **flame tests** for positive ions — everybody loves 'em...*

Flame Tests Identify Metal Ions

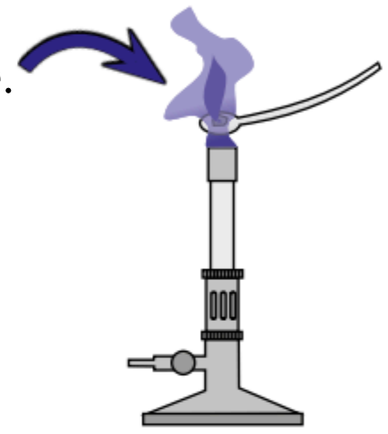
Compounds of some metals burn with a characteristic colour. You can test for these metal ions by heating your substance and seeing whether it burns with a coloured flame.

Here's how you can do a flame test:

1) Clean a platinum wire loop by dipping it in some dilute HCl and then holding it in a blue flame from a Bunsen burner until it burns without any colour.
2) Then, dip the loop into the sample you want to test and put it back in the flame.
3) Record the colour of the flame. Use it to detect and identify different ions.

Be careful though — this only works for samples that contain a single metal ion. If the sample tested contains a mixture of metal ions, the flame colours of some ions may be hidden by the colours of others.

Metal	Ion	Flame colour
lithium	Li^+	crimson
sodium	Na^+	yellow
potassium	K^+	lilac
calcium	Ca^{2+}	orange-red
copper	Cu^{2+}	green



Some Metals Form a Coloured Precipitate with NaOH

1) Many metal hydroxides are insoluble and precipitate out of solution when formed. Some of these hydroxides have a characteristic colour.
2) In this test, you add a few drops of sodium hydroxide solution to a solution of your mystery compound.
3) If a coloured insoluble hydroxide forms, you can often tell which metal was in the compound.

Metal ions	Colour of precipitate	Ionic equation for precipitate formation
Calcium, Ca^{2+}	White	$Ca^{2+}_{(aq)} + 2OH^-_{(aq)} \rightarrow Ca(OH)_{2\,(s)}$
Copper(II), Cu^{2+}	Blue	$Cu^{2+}_{(aq)} + 2OH^-_{(aq)} \rightarrow Cu(OH)_{2\,(s)}$
Iron(II), Fe^{2+}	Green	$Fe^{2+}_{(aq)} + 2OH^-_{(aq)} \rightarrow Fe(OH)_{2\,(s)}$
Iron(III), Fe^{3+}	Brown	$Fe^{3+}_{(aq)} + 3OH^-_{(aq)} \rightarrow Fe(OH)_{3\,(s)}$
Aluminium, Al^{3+}	White at first, but then redissolves in excess NaOH to form a colourless solution.	$Al^{3+}_{(aq)} + 3OH^-_{(aq)} \rightarrow Al(OH)_{3\,(s)}$
Magnesium, Mg^{2+}	White	$Mg^{2+}_{(aq)} + 2OH^-_{(aq)} \rightarrow Mg(OH)_{2\,(s)}$

You don't need to know the equation for how the aluminium hydroxide precipitate dissolves.

Don't mix up the flame colours and the precipitate colours...

When fireworks burn, it's compounds containing metal cations that give the explosions their pretty colours.

Q2 Video Solution

Q1 Why can you use a flame test to identify the metal ion in a compound? [1 mark]

Q2 Sodium hydroxide is added to a solution of an unknown metal chloride. A white precipitate is formed which redissolves following the addition of excess NaOH.

a) Suggest what the unknown metal ion is. [1 mark]

b) Give the balanced equation for the formation of the precipitate. [2 marks]

Quick Quiz

Flame Emission Spectroscopy

*As well as producing pretty colours, the science behind flame tests can be used to **identify** different metal ions in solution **accurately** and calculate their **concentrations**. Colours and science — this is my kind of page...*

Every **Metal Ion** Gives a Characteristic **Line Spectrum**

1) During flame emission spectroscopy, a sample of a solution containing metal ions is placed in a flame.
2) As the ions heat up, their electrons become excited (they gain energy and move to a higher energy level). When the electrons drop back to their original energy levels, they transfer energy as light.
3) The light that's given out by the ions is passed through a spectroscope.
This detects the wavelengths of the light being emitted and produces a line spectrum.

A line spectrum for an ion will look something like this:

4) The wavelengths of light emitted by an ion depend on its charge and its electron arrangement (see p.15).
Each ion will produce a different pattern of wavelengths — so each ion has a different line spectrum.
5) The intensity of the spectrum indicates the concentration of that ion in solution.
6) So, line spectrums can be used to identify ions in solution and calculate their concentrations.

Flame Emission Spectroscopy Works for **Mixtures**

Flame emission spectroscopy can be used to identify different ions in mixtures. This makes it more useful than flame tests, which only work for substances that contain a single metal ion.

For example, a solution containing different ions might give this spectrum:

Sample

You can compare this spectrum against reference spectra:

Spectrum for ion X

Spectrum for ion Y

The combination of the spectra for ion X and for ion Y gives the spectrum for the sample — this suggests the mixture contains ion X and ion Y.

Machines can **Analyse** Unknown Substances

Chemists often use instrumental analysis (i.e. tests that use machines), such as flame emission spectroscopy, instead of conducting tests by hand.

Advantages of Using Machines:

- Very sensitive — they can detect even the tiniest amounts of substances.
- Very fast and tests can be automated.
- Very accurate.

It's the same idea as a flame test, but you get more information

When you do a bog standard flame test for metal ions, you get a coloured flame. But determining what colour the flame is can be a bit subjective. Flame emission spectroscopy produces a line spectrum instead. The pattern on the line spectrum is unique for every ion — and it's easy to tell the different patterns of lines apart just by comparing them.

Warm-Up & Exam Questions

And now, some questions to hone your skills. There are quite a few picky little details to learn for all those ion tests, plus all the flame emission spectroscopy stuff, so make sure you've got it all before you move on.

Warm-Up Questions

1) Describe two tests for gases that use a splint.
2) What colour flame do potassium ions produce in a flame test?
3) a) What colour precipitate forms when you add sodium hydroxide to a solution containing copper(II) ions?
   b) Write an ionic equation for the reaction between copper ions (Cu^{2+}) and hydroxide ions (OH^-).
4) In flame emission spectroscopy, what does the intensity of the line spectrum tell you?

Exam Questions

PRACTICAL

1 Kelly carried out flame tests on compounds of four different metal ions. Complete **Table 1**, below, which shows her results.

GRADE 4-6 GRADE

Table 1

Flame colour	Metal ion
green	
	Li^+
yellow	
	Ca^{2+}

[4 marks]

2 The elements caesium and rubidium were both discovered by their line spectra.

GRADE 6-7 GRADE

2.1 Describe how flame emission spectroscopy can be used to produce a line spectrum for an element.

[3 marks]

2.2 Explain why caesium and rubidium have different line spectra.

[2 marks]

PRACTICAL

3 **Table 2** shows the results of a series of chemical tests conducted on two unknown compounds, **X** and **Y**.

GRADE 7-9 GRADE

Table 2

Test	Observation	
	Compound X	Compound Y
sodium hydroxide solution	white precipitate	green precipitate
hydrochloric acid and barium chloride solution	white precipitate	no precipitate
nitric acid and silver nitrate solution	no precipitate	yellow precipitate

3.1 What is the chemical name of compound **Y**?

[2 marks]

3.2 A student says "From these results, I can be certain that compound **X** is magnesium sulfate." State whether he is correct. Explain your answer.

[3 marks]

Revision Summary Test for Topic 8

Lots of tests in this section. Here's another one. To test that all the info has stuck, you can:

- Tackle the revision summary test below, or scan the QR code to do it online.
  The questions are hard, but they'll show you how well you really know your stuff.
- Track your progress online and see which areas need more work.
- Compare your answers with sample answers for the test here: cgpbooks.co.uk/Base

Summary Test

Purity and Formulations (p.106)

1) In chemistry, what does it mean when you say that a substance is 'pure'?
   How is this different to what 'pure' means as an everyday term?
2) Explain how you could test the purity of a chemical using its melting point.
3) What is a formulation? Give four examples of types of products that are formulations.

Chromatography and R_f Values (p.107-108)

4) What are the two different phases in chromatography?
5) What does it mean when you say that different substances are 'distributed' differently between the two phases in chromatography?
6) How could you recognise a pure substance in a chromatography experiment?
7) List the equipment you would need to do a paper chromatography experiment.
8) Give the formula for calculating the R_f value of a substance from a chromatogram.
9) How can you use reference compounds to identify substances in a mixture during chromatography?

Tests for Gases, Anions and Cations (p.110-111)

10) Describe the tests for: a) oxygen gas, b) hydrogen gas, c) carbon dioxide gas, d) chlorine gas.
11) Describe the test for carbonate ions.
12) A student has a sample of a mystery substance. They add a couple of drops of dilute hydrochloric acid to it, followed by a couple of drops of barium chloride solution.
    a) If the mystery sample is a sulfate, what would you expect to happen?
    b) Why did the student add dilute hydrochloric acid before adding the barium chloride solution?
13) How would you test a solution for halide ions?
14) Describe how you would carry out a flame test.
15) What metal ion is present if the sample in a flame test burns with a:
    a) yellow flame, b) lilac flame, c) orange-red flame, d) green flame, e) crimson flame?
16) Why are flame tests not useful for samples containing several metal ions?
17) A student adds a few drops of sodium hydroxide to a mystery solution containing metal ions.
    What metal ion is present if a precipitate forms that is: a) brown, b) blue, c) green?
18) If you add a few drops of sodium hydroxide to a solution containing aluminium ions, a white precipitate will form.
    a) Write an ionic equation for the reaction that happens.
    b) Solutions of calcium ions form a white precipitate when you add sodium hydroxide too.
    How could you show that the solution contained aluminium ions, not calcium ions?

Flame Emission Spectroscopy (p.112)

19) Briefly describe how flame emission spectroscopy is carried out and what the result looks like.
20) What two things can a flame emission spectrum tell you about a sample?
21) Give one way that flame emission spectroscopy is more useful than a flame test.
22) Give three advantages of using instrumental analysis, rather than testing substances by hand.

The Evolution of the Atmosphere

Quick Quiz

Theories for how the Earth's atmosphere ***evolved*** *have changed a lot — it's hard to get evidence from such a* ***long time period*** *and from* ***so long ago*** *(4.6 billion years). This page describes our best theory.*

Video

Phase 1 — **Volcanoes** Gave Out **Gases**

The first billion years of Earth's history were pretty explosive — the surface was covered in volcanoes that erupted and released lots of gases. We think this was how the early atmosphere was formed.

- The early atmosphere was probably mostly carbon dioxide, with virtually no oxygen. This is quite like the atmospheres of Mars and Venus today.
- Volcanic activity also released nitrogen, which built up in the atmosphere over time, as well as water vapour and small amounts of methane and ammonia.

Phase 2 — **Oceans, Algae** and **Green Plants Absorbed Carbon Dioxide**

1) When the water vapour in the atmosphere condensed, it formed the oceans.
2) Lots of carbon dioxide was removed from the early atmosphere as it dissolved in the oceans. It then went through a series of reactions to make carbonate precipitates that formed sediments on the seabed.
3) Algae evolved around 2.7 billion years ago. Then green plants evolved over the next billion years (ish). These organisms could carry out photosynthesis (see below) so absorbed more carbon dioxide.
4) As oxygen levels increased (also see below), more complex life could evolve — i.e. animals. When marine animals evolved, their shells and skeletons contained carbonates from the oceans.
5) All this means that much of the carbon in the early atmosphere got locked up in rocks and fossil fuels:
   - When plants, plankton and marine animals die, they fall to the seabed and get buried by layers of sediment. Over millions of years, they are compressed and form sedimentary rocks, oil and gas. This traps the carbon they contain, helping to reduce CO_2 levels in the atmosphere.
   - Limestone is a sedimentary rock mostly made of calcium carbonate from the shells and skeletons of marine organisms.
   - Coal is also a sedimentary rock made from thick plant deposits.
   - Crude oil and natural gas are formed from deposits of plankton that are subjected to very high temperature and pressure, forming reservoirs trapped in rocks under the seabed.
   - Coal, crude oil and natural gas are fossil fuels.

Phase 3 — Green Plants and Algae **Produced Oxygen**

1) As well as absorbing CO_2, green plants and algae produced oxygen by photosynthesis — this is when plants use light to convert carbon dioxide and water into sugars:

carbon dioxide + water $\xrightarrow{\text{light}}$ glucose + oxygen

$$6CO_2 + 6H_2O \xrightarrow{\text{light}} C_6H_{12}O_6 + 6O_2$$

2) So as well as carbon dioxide levels falling, oxygen gradually built up in the atmosphere.
3) Eventually, about 200 million years ago, the atmosphere reached a composition similar to what it is today: approximately 80% nitrogen, 20% oxygen and small amounts of other gases (together they make up less than 1% of the atmosphere), mainly carbon dioxide, noble gases and water vapour.

Before volcanic activity, the Earth didn't even have an atmosphere

One way scientists can get information about what Earth's atmosphere was like in the past is from Antarctic ice cores. Each year a layer of ice forms with tiny bubbles of air trapped in it. The deeper you go in the ice, the older the air is. So analysing the bubbles from different layers shows you how the atmosphere has changed over time.

Quick Quiz

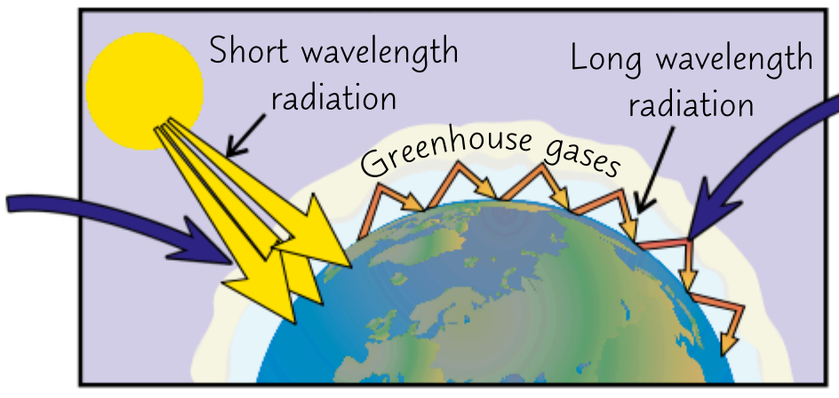
Greenhouse Gases and Climate Change

*Greenhouse gases are important but can also cause **problems** — it's all about keeping a delicate **balance**.*

Carbon Dioxide is a Greenhouse Gas

Greenhouse gases like carbon dioxide, methane and water vapour act like an insulating layer in the Earth's atmosphere — this, amongst other factors, allows the Earth to be warm enough to support life.
It's known as the greenhouse effect:

All particles absorb certain frequencies of radiation. Greenhouse gases don't absorb the incoming short wavelength radiation from the Sun, which is absorbed by the Earth and heats it.



The Earth re-emits some energy as long wavelength (thermal) radiation, which the greenhouse gases do absorb. They then re-radiate it in all directions — including back towards the Earth. This results in warming of the surface of the Earth.

Some human activities affect the amount of greenhouse gases in the atmosphere — e.g.:

- Deforestation: fewer trees means less CO_2 is removed from the atmosphere via photosynthesis.
- Burning fossil fuels: carbon that was 'locked up' in these fuels is released as CO_2.
- Agriculture: more farm animals produce more methane through their digestive processes. Methane is also given out in large quantities by the farming of rice in rice paddies.
- Creating waste: more landfill sites and more waste from agriculture means more CO_2 and methane released by decomposition of waste.

Increasing Carbon Dioxide is Linked to Climate Change

1) The Earth's temperature varies naturally, but recently the average temperature of the Earth's surface has been increasing. Scientists agree that extra carbon dioxide from human activity is causing this increase and that this is leading to climate change.
2) Evidence for this has been peer-reviewed (see p.139) — so you know the information out there is reliable.
3) Unfortunately, it's hard to fully understand the Earth's climate — this is because it's so complex, and there are so many variables, that it's very hard to make a model that isn't oversimplified.
4) This has led to speculation, particularly in the media — where stories may be biased or only some of the information given.

See page 140 for more on science in the media.

Climate Change Could Have Dangerous Consequences

The Earth's climate is complex, but it's still important to make predictions about the consequences of climate change so that policy-makers can make decisions now. For example:

- Higher global temperatures are causing the polar ice caps to melt — causing sea levels to rise. If sea levels keep rising, this will lead to increased flooding in coastal areas and coastal erosion.
- Changes in rainfall patterns (the amount, timing and distribution) are causing some regions to get too much or too little water. This, along with changes in temperature, may affect the ability of certain regions to produce food.
- The frequency and severity of storms may also increase.
- Changes in temperature and the amount of water available in different habitats affect the ability of wild species to survive. This is leading to changes in the distribution of some species, which can affect food chains.

Greenhouse gases aren't all bad — we need them to survive

Everyone's talking about climate change these days — it's pretty scary stuff, so make sure you get it.

Q1 Describe the greenhouse effect and how it affects global temperature. [4 marks]

Q1 Video Solution

Carbon Footprints

Quick Quiz

*Scientists believe that greenhouse gas emissions from **human activities** are causing **climate change**. Knowing what leads to a lot of emissions of carbon dioxide could be useful for **stopping** it happening.*

Carbon Footprints are Tricky to Measure

1) Carbon footprints are basically a measure of the amount of carbon dioxide and other greenhouse gases released over the full life cycle of something. That can be a service (e.g. the school bus), an event (e.g. the Olympics), a product (e.g. a toastie maker) — almost anything.
2) Measuring the total carbon footprint of something can be really hard, though — or even impossible.
3) That's because there are so many different factors to consider — for example, you would have to count the emissions released as a result of sourcing all the parts of your toastie maker and in making it, not to mention the emissions produced when you actually use it and finally dispose of it.
4) Still, a rough calculation can give a good idea of what the worst emitters are — so that people can avoid them in the future.

There are Ways of Reducing Carbon Footprints

It might be hard to measure carbon footprints exactly, but it's easier to think of ways to reduce them. Anything that reduces the amount of greenhouse gases (e.g. carbon dioxide or methane) given out by a process will reduce its carbon footprint. Here are some things that can be done:

Renewable energy sources or nuclear energy could be used instead of fossil fuels.

Using more efficient processes could conserve energy and cut waste. Lots of waste decomposes to release methane, so this will reduce methane emissions.

There's also technology that captures the CO_2 produced by burning fossil fuels before it's released into the atmosphere — it can then be stored deep underground, e.g. in old oil wells.

Governments could tax companies or individuals based on the amount of greenhouse gases they emit. E.g. taxing cars based on the amount of CO_2 they emit over a set distance could encourage people to buy ones that are more fuel-efficient and less polluting.

Governments can also put a cap on emissions of all greenhouse gases that companies make — then sell licences for emissions up to that cap.

But Making Reductions is Still Difficult

1) It's easy enough saying that we should cut emissions, but actually doing it — that's a different story.
2) For a start, there's still a lot of work to be done on alternative technologies that reduce CO_2 emissions.
3) A lot of governments are also worried that making these changes will impact on the economic growth of communities — which could be bad for people's well-being. This is particularly important for countries that are still developing.
4) Because not everyone is on board, it's hard to make international agreements to reduce emissions. Most countries don't want to sacrifice their economic development if they think others won't do the same.
5) It's not just governments, though — individuals in developed countries need to make changes to their lifestyles. But it might be hard to get people to make changes if they don't want to and if there isn't enough education provided about why the changes are necessary and how to make them.

Lifestyle changes can be tough, but they might be essential in future

So governments, as well as individuals, can take actions to reduce carbon emissions. But that doesn't mean it's easy — make sure you know some reasons why it can be hard to get countries and people to reduce their carbon footprints.

Air Pollution

Quick Quiz

Increasing carbon dioxide is causing climate change. But CO_2 isn't the only gas released when fossil fuels burn — you also get other nasties like ***oxides of nitrogen****,* ***sulfur dioxide*** *and* ***carbon monoxide****.*

Combustion of Fossil Fuels Releases Gases and Particles

- Fossil fuels, such as crude oil and coal, contain hydrocarbons. During combustion, the carbon and hydrogen in these compounds are oxidised, releasing carbon dioxide and water vapour into the atmosphere.
- When there's plenty of oxygen, all the fuel burns — this is called complete combustion (see p.91).
- If there's not enough oxygen, some of the fuel doesn't burn — this is incomplete combustion (see p.96).
- As well as carbon dioxide and water, incomplete combustion produces solid particles (called particulates) of soot (carbon) and carbon monoxide gas. Unburnt hydrocarbon fuel can also be released.
- Releasing these pollutants into the atmosphere can cause all sorts of problems:

Hydrocarbons are compounds that only contain hydrogen and carbon (see p.91).

Problems caused by particulates

- Particulates (and the clouds they help produce) reflect sunlight back into space so less light reaches the Earth — this is global dimming.
- If inhaled, they can get stuck in the lungs and cause damage, leading to respiratory problems.

Problems caused by carbon monoxide

- Carbon monoxide (CO) is really dangerous because it can stop your blood from carrying oxygen around your body.
- It does this by binding to the haemoglobin in your blood that normally carries oxygen.
- A lack of oxygen can lead to fainting, a coma or even death.
- Carbon monoxide doesn't have any colour or smell, so is very hard to detect, making it even more dangerous.

Sulfur Dioxide and Oxides of Nitrogen Cause Acid Rain

Video

- Sulfur dioxide (SO_2) is released during the combustion of fossil fuels, such as coal, that contain sulfur impurities — the sulfur in the fuel becomes oxidised.
- Nitrogen oxides are created from a reaction between the nitrogen and oxygen in the air, caused by the heat released during combustion. (This can happen in the internal combustion engines of cars.)

You can test for sulfur impurities in a fuel by bubbling the gases from combustion through a solution containing universal indicator — if the fuel contains sulfur, the gases will contain SO_2, which will dissolve to form an acidic solution and turn the universal indicator red.

1) When these gases mix with clouds they form dilute sulfuric acid or dilute nitric acid.
2) This then falls as acid rain.
3) Acid rain kills plants and aquatic life, damages buildings and statues, and makes metal corrode.

As if all that's not bad enough, sulfur dioxide and nitrogen oxides are bad for human health too — they cause respiratory problems if breathed in.

Fossil fuels are bad news — but we depend on them for many things...

...so a big reduction in their use isn't going to be easy to achieve. Make sure you know the different pollutants that are given out when fossil fuels burn, as well as the differences between complete and incomplete combustion.

Warm-Up & Exam Questions

There's lots of important information in this section, from the Earth's atmosphere to climate change and pollution. Answer these questions to see what you can remember and what you need to go over again.

Warm-Up Questions

1) Which gas do scientists think was most abundant in the Earth's early atmosphere?
2) Give an example of a sedimentary rock.
3) Give an example of a fossil fuel.
4) Suggest one way that an individual could reduce their personal carbon footprint.
5) Name two potential pollutants that could be released as a result of incomplete combustion of hydrocarbons, that wouldn't be released as a result of complete combustion.

Exam Questions

1 Use words from the box to complete the sentences below. GRADE 4-6

oxygen	carbon	carbon dioxide	photosynthesis	respiration

Once green plants had evolved, they thrived in Earth's early atmosphere which was rich in ..... . These plants also produced ..... by the process of ..... .

Some of the ..... from dead plants eventually became 'locked up' in fossil fuels.

[4 marks]

2 Burning fossil fuels can produce pollutants like carbon dioxide, sulfur dioxide and particulate matter. GRADE 6-7

2.1 Describe how sulfur dioxide can cause acid rain.

[2 marks]

2.2 Give **one** environmental problem caused by acid rain.

[1 mark]

2.3 Explain how burning fossil fuels can lead to global dimming.

[3 marks]

2.4 Carbon dioxide is a greenhouse gas. Describe how the greenhouse effect is responsible for warming the surface of the Earth.

[4 marks]

3 Many people believe that it is necessary for individuals to reduce their carbon footprints in order to prevent climate change from happening in the future. GRADE 7-9

3.1 Define the term 'carbon footprint'.

[1 mark]

3.2* Describe some of the effects that an increase in global temperature could have on the environment. Explain the impact these effects may have on humans.

[6 marks]

Revision Summary Test for Topic 9

Topic 9 was short, but not particularly sweet, sadly... Here's how you can test yourself on it:

- Tackle the revision summary test below, or scan the QR code to do it online.
  The questions are hard, but they'll show you how well you really know your stuff.
- Track your progress online and see which areas need more work.
- Compare your answers with sample answers for the test here: cgpbooks.co.uk/Base

Summary Test

The Evolution of the Atmosphere (p.115) ☑

1) Where do scientists think the gases in the early atmosphere came from?
2) Name five gases found in the early atmosphere, and one gas there was virtually none of.
3) What two things caused the level of carbon dioxide in the atmosphere to reduce over time?
4) Give two examples of sedimentary rocks that store carbon, and say what they are formed from.
5) Which fossil fuels are generally formed under the seabed from deposits of plankton?
6) What process caused the level of oxygen in the atmosphere to gradually build up?
7) Describe the composition of the Earth's atmosphere today.

Greenhouse Gases, Climate Change and Carbon Footprints (p.116-117) ☑

8) How does the greenhouse effect keep the Earth's surface warm?
9) Give three examples of human activities that increase the amount of greenhouse gases in the atmosphere, and explain how they cause this.
10) Describe the link between carbon dioxide and climate change.
11) Why is it hard to fully understand the Earth's climate?
12) Describe four possible consequences of climate change.
13) What is a carbon footprint? Give three kinds of things a carbon footprint can be calculated for.
14) Why is it difficult to accurately measure the total carbon footprint of something?
15) Describe four possible ways to reduce greenhouse gas emissions.
16) Explain why many countries are reluctant to make changes to reduce their greenhouse gas emissions.

Air Pollution (p.118) ☑

17) Give the two products of complete combustion of a hydrocarbon fuel.
    Now give three other things that can be released during incomplete combustion of a hydrocarbon fuel.
18) What is global dimming?
19) What causes global dimming? Describe another problem caused by these pollutants.
20) Explain why carbon monoxide is very dangerous to human health.
21) Describe how the following gases can be produced when burning fossil fuels:
    a) sulfur dioxide, b) oxides of nitrogen.
22) How do sulfur dioxide and nitrogen oxides cause acid rain?
23) Give three negative effects of acid rain.
24) Why is the release of sulfur dioxide and nitrogen oxides bad for human health?

Ceramics, Composites and Polymers

Quick Quiz

*Over time, humans have **developed** and **adopted** materials which we use to make our lives easier.*

Ceramics Come in Many **Different Forms**

Ceramics are non-metal solids with high melting points that aren't made from carbon-based compounds.

Clay ceramics:

- Clay is soft when it's dug up out of the ground, so can be moulded.
- When it's fired at high temperatures, it hardens to form a clay ceramic.
- Its ability to be moulded when wet and then hardened makes clay ideal for making pottery and bricks.

Glass:

- Most glass is soda-lime glass, made by heating a mixture of limestone, sand and sodium carbonate (soda) until it melts. When the mixture cools it forms glass.
- Borosilicate glass is made from sand and boron trioxide instead. It has a higher melting point than soda-lime glass.
- Glass is generally transparent, can be moulded when hot, but can be brittle when thin.

Composites are Generally Made of **Two Different Materials**

Composites are made of one material embedded in another. Fibres or fragments of a material (known as the reinforcement) are surrounded by a matrix acting as a binder. The properties of a composite depend on the properties of the materials it is made from, e.g.:

Fibreglass	Fibres of glass embedded in a polymer (plastic) matrix.	Has a low density (like plastic) but is very strong (like glass). Used for e.g. skis, boats, surfboards.
Carbon fibre	Reinforcement made from long chains of carbon atoms bonded together (carbon fibres) or from carbon nanotubes (p.34). Has a polymer matrix.	Very strong and light. Used in aeroplanes and sports cars.
Concrete	Aggregate (a mixture of sand and gravel) embedded in cement.	Very strong. Used as a building material, e.g. in skate parks.
Wood	A natural composite — made from cellulose fibres held together by an organic polymer matrix.	Light and very strong in one direction. Often used as a building material.

Polymers Can Have Very Different **Properties**

Two important things can influence the properties of a polymer — how it's made and what monomer is used.

The reaction conditions (i.e. temperature and pressure) and the catalyst used affect the properties of the polymer being produced. E.g. ethene can be used to produce two types of polymer:

- Low density (LD) poly(ethene) is produced at a moderate temperature under a high pressure. It's flexible and is used for bags and bottles.
- High density (HD) poly(ethene) is produced at a lower temperature and pressure with a catalyst. It's more rigid and is used for water tanks and drainpipes.

The monomers used to make a polymer affect the type and strength of bonds between polymer chains, e.g.:

- Thermosoftening polymers contain individual polymer chains with weak intermolecular forces between them. These polymers melt when they're heated and can be remoulded.
- Thermosetting polymers contain monomers that can form cross-links between the polymer chains. These are strong bonds that hold the chains together in a solid structure. Unlike thermosoftening polymers, these polymers don't melt or soften when heated. They're strong, hard and rigid.

The properties of a polymer depend on how it's made...

Low and high density poly(ethene) both come from ethene, but have different properties. You need to be able to explain how the conditions used to make these polymers affect their final properties — best have another read of this page...

Quick Quiz

Properties of Materials

It's all very well making something super cool, but the material it's made from needs to be ***fit for purpose****.*

Different **Materials** are Suited to Different **Jobs**

What materials are used for depends on their properties. You need to be able to interpret information about the properties of materials and decide how suitable these materials would be for different purposes.

Ceramics include glass and clay ceramics such as porcelain and bricks. They're insulators of heat and electricity, brittle (they aren't very flexible and break easily) and stiff.

Polymers are insulators, they can be flexible (i.e. they can be bent without breaking) and are easily moulded. They have many uses, including in clothing and insulators in electrical items.

The properties of composites depend on the matrix/binder and reinforcement used. They have many different uses.

Metals are malleable, good conductors, ductile (they can be drawn into wires), shiny and stiff (see p.19). They have many uses, e.g. in electrical wires, car bodywork and cutlery.

EXAMPLE

Suggest which of these materials would be best for making the hull of a large ship.

Material	Brittleness	Water resistance
Clay ceramic	brittle	porous (absorbs water)
Rigid PVC (a polymer)	brittle	non-porous, corrosion resistant
Low carbon steel	tough	non-porous, corrodes in water, but cheap to protect against corrosion

1) A ship's hull needs to be non-porous so it doesn't absorb water. Also, it shouldn't corrode.

Clay ceramic absorbs water so is unsuitable. Rigid PVC and low carbon steel are both non-porous. Low carbon steel corrodes, but is cheap to protect, so could still be suitable.

2) If the hull's brittle, it could suddenly break.

Rigid PVC is brittle so is unsuitable, but the steel is tough. If protected against corrosion, low carbon steel is the best choice.

Pure Metals Don't Always Have the **Properties Needed**

1) As you saw on p.35, pure metals are soft — often too soft for use in everyday life. Meanwhile, alloys (made of a metal plus other elements) are harder than pure metals.
2) For example, alloys of iron called steels are often used instead of pure iron. Steels are made by adding small amounts of carbon and sometimes other metals to iron.
3) Many other alloys are used in everyday life.

TYPE OF STEEL	PROPERTIES	USE
Low carbon steel (0.1–0.3% carbon)	easily shaped	car bodies
High carbon steel (0.22–2.5% carbon)	very strong, inflexible, brittle	bridges
Stainless steel (chromium added, and sometimes nickel)	corrosion-resistant, hard	cutlery

- Bronze = Copper + Tin: Bronze is harder than copper. It's used for e.g. medals, ornaments and statues.
- Brass = Copper + Zinc: Brass is more malleable than bronze and is used in situations where lower friction is required, such as in water taps and door fittings.
- Gold alloys: Pure gold is very soft, so metals such as zinc, copper, and silver are used to harden it to make it suitable for use in jewellery.

  The proportion of gold in the alloy is measured in carats. Pure gold = 24 carat, so e.g. 18 carat gold means 18 out of 24 parts of the alloy are gold. You can work out the percentage of gold like this: (18 ÷ 24) × 100% = 75%.

- Aluminium alloys: Aluminium has a low density which is very handy for aircraft manufacture. But pure aluminium is too soft, so it's alloyed with small amounts of other metals to make it stronger.

Alloys are really important in industry

Unlike pure metals, the properties of alloys can be varied, making them useful for a wide range of applications.

Q1 To the nearest percent, what percentage of pure gold is in 14 carat gold? [1 mark]

Q1 Video Solution

Corrosion

Quick Quiz

*Some metals **react** with substances in the environment (usually oxygen and water), and this can **weaken** them.*

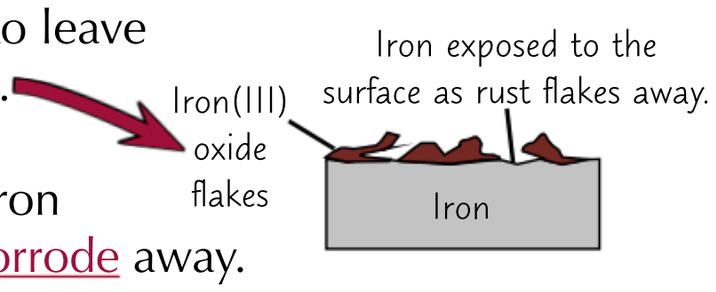
Rusting is the **Corrosion** of **Iron**

1) Corrosion is where materials such as metals react with substances in their environment and are gradually destroyed.
2) Iron corrodes easily. The corrosion of iron is also called rusting.
3) In order to rust, iron needs to be in contact with both oxygen and water:

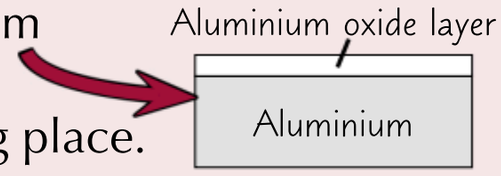
iron + oxygen + water → hydrated iron(III) oxide

Iron is oxidised in this reaction to form hydrated iron(III) oxide — also known as rust.

4) Oxygen and water are both present in air, so iron objects exposed to air will rust over time.
5) Only the surface of an object can corrode, as that's the bit in contact with the air. However, rust is a soft, crumbly solid that easily flakes off surfaces to leave more iron exposed. This means that eventually all the iron in an object will corrode away.

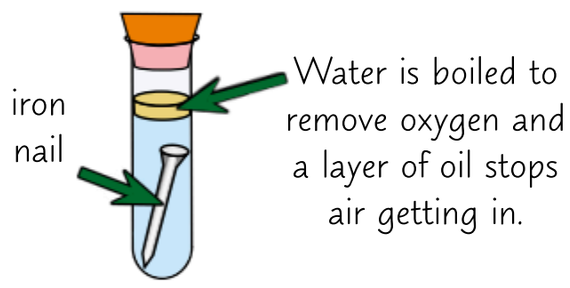


Aluminium also corrodes when exposed to air, but unlike iron objects, aluminium objects aren't completely destroyed by corrosion. The aluminium oxide formed by corrosion doesn't flake away like rust does. In fact, it forms a nice protective layer that sticks firmly to the aluminium below and stops any further reaction taking place.

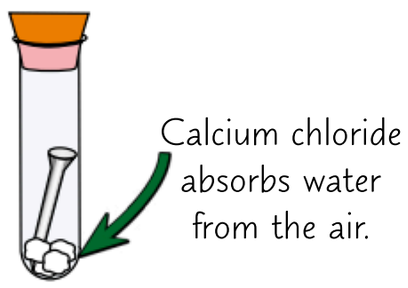


Experiments Show that Both **Air** and **Water** are Needed for **Iron** to **Rust**

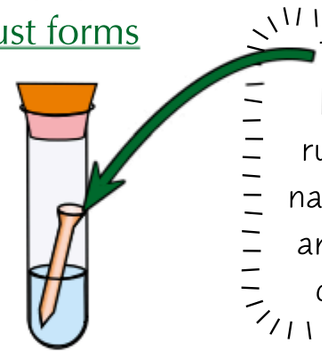
Just water — no rust forms



Just air — no rust forms



Water and air — rust forms



The mass of a rusty nail will be higher than it was before rusting. The iron atoms in the nail have now bonded to oxygen and water molecules, forming a compound heavier than iron.

There are **Two** Main Ways to **Prevent Rusting**

Barrier methods

Coating the iron with a barrier prevents rusting by keeping out the water and oxygen.

This can be done by:

- Painting/Coating with plastic — can be done on big or small structures, and can be decorative too.
- Electroplating — the iron item is used as an electrode in electrolysis (p.68-69) and gets coated with a layer of a metal that won't corrode.
- Oiling/Greasing — this is used to coat parts that need to be able to move, e.g. bike chains.

Sacrificial method

If a more reactive metal (e.g. zinc or magnesium) is placed in contact with the iron, it will react with water and oxygen before the iron does — the more reactive metal is 'sacrificed' to preserve the iron object.

Galvanisation uses both of these techniques. An object is galvanised by spraying it with a zinc coating. The zinc layer is a barrier, but if it gets scratched the zinc around the scratch will work as a sacrificial metal.

Coatings and the sacrificial method can be used to prevent rust

Make sure you can describe experiments that investigate rusting. Remember, for rusting to occur, both oxygen and water need to be in contact with iron or steel. There are a few ways to stop this happening, that you need to learn.

Warm-Up & Exam Questions

There's tonnes of important information on these pages. If you don't think you've got it covered, go back through and check any areas that you might be a bit unsure of, then have a go at the questions on this page.

Warm-Up Questions

1) What are ceramics?
2) Why don't thermosetting polymers melt?
3) What two metals is brass made from?
4) Write the word equation for the rusting of iron.
5) How does electroplating iron prevent it from rusting?

Exam Questions

1 An alloy is a mixture containing a metal and at least one other element. GRADE 4-6 GRADE

1.1 Why are alloys often used instead of pure metals?
Tick **one** box.

- ☐ they are more plentiful
- ☐ their properties make them more suitable for the application
- ☐ they have a more regular structure
- ☐ they are completely inert

[1 mark]

1.2 Low carbon steel and high carbon steel are two different alloys of iron.
State **one** difference in the properties of these two alloys.

[1 mark]

1.3 Give **one** use for high carbon steel.

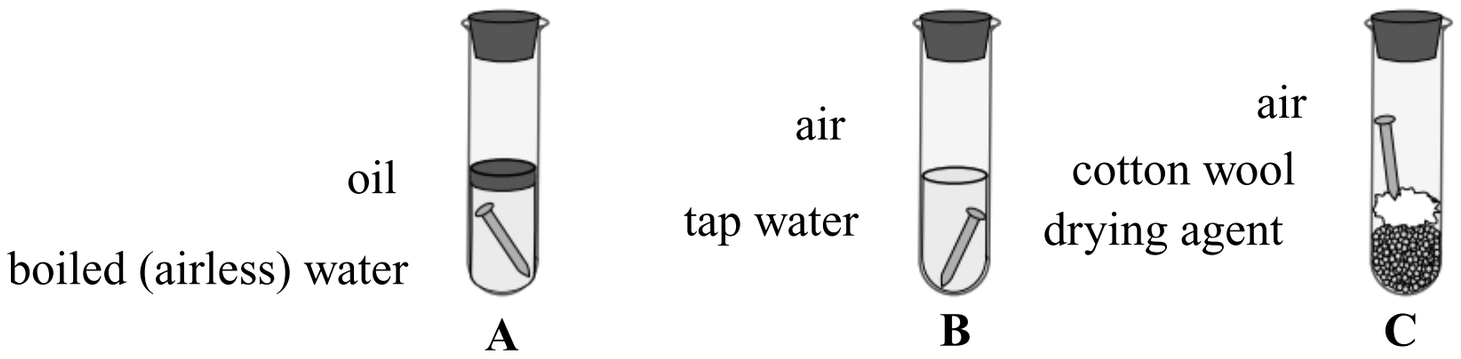
[1 mark]

1.4 Steel **X** is an alloy of iron that is resistant to corrosion. Name steel **X**.

[1 mark]

2 In an experiment to investigate rusting, three iron nails were placed into separate boiling tubes for ten days. The experimental set-up is shown in **Figure 1**. GRADE 6-7 GRADE

Figure 1



2.1 The nail in tube **C** doesn't rust. State which other tube would not contain a rusted nail.

[1 mark]

2.2 A fourth tube, **D**, was set up in the same way as tube **B** except the nail used was coated in paint.
Explain the difference in what would happen to the nail in tube **D** compared with the nail in tube **B**.

[2 marks]

2.3 Iron nails can be protected from rusting by coating them with zinc.
Explain how this method protects the nails from rust.

[3 marks]

Finite and Renewable Resources

Quick Quiz

*There are lots of different resources that humans use to provide **energy** for things like **heating** or **travelling**, as well as for **building materials** and **food**. Some of these resources get replaced, some don't.*

Natural Resources Form Without **Human** Input

1) Natural resources include anything that comes from the earth, sea or air.
2) Humans use them to provide things we need, such as warmth, shelter, clothing, food and fuel. E.g. cotton is used to make clothes and oil is used as a fuel for heating and transport.
3) Some of these natural products can be replaced by synthetic products or improved upon by man-made processes. For example, rubber is a natural product that can be extracted from the sap of a tree, however man-made polymers have now been made which can replace rubber in uses such as tyres.
4) Agriculture provides conditions where natural resources can be enhanced for our needs. E.g. the development of fertilisers has meant we can produce high yields of crops.

Some **Natural Resources** will **Run Out**

Renewable resources	Form faster than we use them (or at a similar rate).	E.g. water, food, timber (trees can be planted after a harvest and regrow in a few years).
Finite (non-renewable) resources	Not formed quickly enough to be replaceable — they will run out.	E.g. fossil fuels, nuclear fuels (e.g. uranium), metal ores, minerals.

After they've been extracted, many finite resources undergo man-made processes to provide fuels and materials necessary for modern life. For example, fractional distillation (see p.92) is used to produce usable products such as petrol from crude oil and metal ores are reduced to produce a pure metal (see p.64).

Tables, Charts and **Graphs** can Give You an **Insight** Into **Different Resources**

You may be asked to interpret information about resources in the exam.

EXAMPLE

This table shows data for two resources, coal and timber. Identify which resource is which.

	Energy Density (MJ/m^3)	Time it takes to form
Resource 1	7 600-11 400	10 years
Resource 2	23 000-26 000	10^6 years

10^6 is a shorthand way of showing 1 000 000.
$10^6 = 10 \times 10 \times 10 \times 10 \times 10 \times 10 = 1\,000\,000$

Resource 1 takes ten years to form while Resource 2 takes a million years to form, suggesting Resource 1 is renewable (timber) and Resource 2 is non-renewable (coal). Resource 1 also provides less energy than Resource 2 (for the same amount of fuel) so is more likely to be timber than coal.

Resource 1 is timber and Resource 2 is coal.

Extracting Finite Resources has **Risks**

1) Many modern materials are made from raw, finite resources, such as most plastics, metals, glass, clay ceramics and building materials.
2) People have to balance the social, economic and environmental effects of extracting finite resources.
3) For example, mining metal ores is good because useful products can be made. It also provides local people with jobs and brings money into the area. However, mining ores is bad for the environment as it uses loads of energy, scars the landscape, produces lots of waste and destroys habitats.

Natural resources can be either renewable or finite

Finite resources are ones that will eventually run out, whereas renewable resources are, well, renewable. Before you move on, make sure you know the difference between the two types of natural resources and learn a few examples of each.

Quick Quiz

Reuse and Recycling

*Many materials used in the modern world are **limited**. Once they're finished with, it's usually far better to **recycle** them than to use new finite resources which will eventually run out. Read on to find out why...*

Chemistry is **Improving Sustainability**

1) Sustainable development is an approach to development that takes account of the needs of present society while not damaging the lives of future generations.
2) Using, extracting and processing resources can be unsustainable. This could be because:
   - The resources are non-renewable (see previous page) so it's unsustainable to keep using them.
   - The amount of energy used and waste produced when extracting resources can make that part of the process unsustainable.
   - The amount of energy used can also make processing resources into useful materials (like glass or bricks) unsustainable, as the processes often use energy generated using finite resources.
3) One way of reducing the use of finite resources is for people to use less.
   This doesn't just reduce the use of that resource but also anything needed to produce it.
4) We can't stop using finite resources altogether, but chemists can develop and adapt processes that use lower amounts of finite resources and reduce damage to the environment. For example, chemists have developed catalysts that reduce the amount of energy required for certain industrial processes.

Recycling Metals is Important

Mining and extracting metals takes lots of energy, most of which is generated by burning fossil fuels.
It's usually better to recycle metals than to extract new metals:

- Recycling metals often uses much less energy than mining and extracting new metal.
- It helps conserve the finite amount of each metal in the earth.
- It also cuts down on the amount of waste getting sent to landfill.

Metals are usually recycled by separating out different metals, melting them down and then recasting or reforming them into a new product. Sometimes different metals don't need to be completely separated — the amount of separation needed depends on the metals and how the final product will be used.

For example, waste steel and iron can be kept together as they can both be added to iron in a blast furnace to reduce the amount of iron ore required.

A blast furnace is used to extract iron from its ore at a high temperature using carbon.

Glass can **Also** be **Recycled**

Glass recycling can help sustainability by reducing the amount of energy needed to make new glass products, and also the amount of waste created when used glass is thrown away.

1) Glass bottles can often be reused without reshaping.
2) Other forms of glass can't be reused so they're recycled instead. Usually the glass is separated by colour and chemical composition before being recycled.
3) The glass is crushed and then melted to be reshaped for use in glass products such as bottles or jars. It might also be used for a different purpose such as insulating glass wool for wall insulation in homes.

Recycling is key to sustainability — it's useful in lots of ways...

Remember that recycling doesn't just reduce the use of raw materials, it also reduces the amount of energy used, the amount of damage caused to the environment and the amount of waste produced. Sounds like a win to me...

Extracting Copper & Life Cycle Assessments

Quick Quiz

Getting more out *of resources and considering* ***environmental impacts*** *are both ways to improve sustainability.*

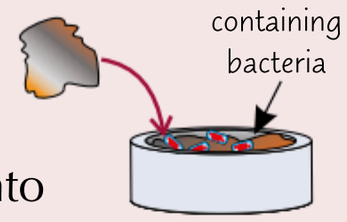
Copper-Rich Ores are in **Short Supply**

Copper is a finite resource. One way to improve its sustainability is to extract it from low-grade ores, which don't have much copper. There are some new ways of doing this:

Another way to improve sustainability is to reduce our need for copper-rich ores, e.g. by recycling more.

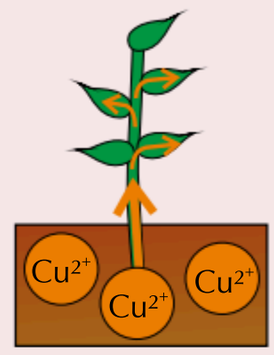
Bioleaching

- Copper ore is added to a solution with bacteria that can convert copper compounds in the ore into soluble copper compounds.
- A leachate solution is produced that contains copper ions.



Phytomining

- Plants are grown in soil that contains copper.
- The plants can't use or get rid of the copper so it gradually builds up in the leaves.
- The plants are harvested, dried and burned in a furnace. The ash contains soluble copper compounds.



Both processes produce a solution containing copper ions. Copper metal can be extracted from this by electrolysis (p.69) or displacement with a more reactive metal, e.g. scrap iron (p.66).

Traditional methods of copper mining are damaging to the environment. These new methods have a much smaller impact, but the disadvantage is that they're slow.

These methods can be used to extract other metals too.

Life Cycle Assessments Show Total Environmental Costs

A life cycle assessment (LCA) looks at all stages of a product's life to assess its impact on the environment.

1 Getting the Raw Materials:

- Extracting raw materials needed for a product can damage the local environment, e.g. mining metals. Extraction can also result in pollution due to the amount of energy needed.
- Raw materials often need to be processed to extract the desired materials and this can require large amounts of energy. E.g. extracting metals from ores or fractional distillation of crude oil.

2 Manufacture and Packaging:

- Manufacturing products and their packaging can use a lot of energy resources and can also cause a lot of pollution, e.g. harmful fumes such as carbon monoxide or hydrogen chloride.
- The reactions used to make the product can produce waste compounds that need to be disposed of. Some waste can be turned into other useful chemicals, reducing the amount that ends up polluting.

3 Using the Product:

- Using a product can damage the environment. E.g. burning fuels releases greenhouse gases and other harmful substances. Fertilisers can leach into streams and rivers, causing damage to ecosystems.
- How long a product is used for or how many uses it gets is also a factor — products that need lots of energy to produce but are used for ages mean less waste in the long run.

4 Product Disposal:

- Products are often disposed of in landfill sites. This takes up space and pollutes land and water.
- Energy is used to transport waste to landfill, causing pollutants to be released into the atmosphere.
- Products might be incinerated (burnt), which causes air pollution.

There are lots of ways to improve the sustainability of resources...

...and assessing a product's environmental impact can help us to identify areas where this impact could be reduced.

Quick Quiz

Using Life Cycle Assessments

*You can compare the life cycle assessments of **similar products** to see which has the **smallest effect** on the environment. You might want to do this to see which material will affect the environment the **least**.*

You Can **Compare** Life Cycle Assessments for **Plastic** and **Paper Bags**

When comparing life cycle assessments for different products, you need to compare the environmental impact at each stage of the products' lives and make an overall judgement on which is least harmful.

Life Cycle Assessment Stage	Plastic Bag	Paper Bag
Raw Materials	• Crude oil	• Timber
Manufacturing and Packaging	• The compounds needed to make the plastic are extracted from crude oil by fractional distillation, followed by cracking and then polymerisation. • Waste is reduced as the other fractions of crude oil have other uses.	• Pulped timber is processed using lots of energy. • Lots of waste is made.
Using the Product	• Can be reused. • Can be used for other things as well as shopping, for example bin liners.	• Usually only used once.
Product Disposal	• Recyclable • Not biodegradable — will take up space in landfill and pollute land.	• Can be recycled • Biodegradable • Non-toxic

- Life cycle assessments have shown that even though plastic bags aren't usually biodegradable, they may still be less harmful to the environment.
- This is because they take less energy to make and have a longer lifespan than paper bags.

There are **Problems** with **Life Cycle Assessments**

1) Some of the effects of a product over its lifetime can be easily quantified. For example:
   - The amount of energy used to make the product.
   - The natural resources used (and their amounts).
   - The amount of certain types of waste produced.
2) But it's harder to give a numerical value to other effects. E.g. it's difficult to apply a value to the negative visual effects of plastic bags in the environment compared to paper ones.
3) So, producing an LCA is not an objective method — it takes into account the values of the person carrying out the assessment. This means LCAs can be biased.
4) Selective LCAs only show some of the impacts of a product on the environment. These can be biased — e.g. they can be written to deliberately support the claims of a company, in order to give them positive advertising.

LCAs aren't all they cracked up to be...

In the exam, you may have to analyse LCAs comparing two similar products and come to a conclusion about which product is the most environmentally friendly. But remember, not all environmental impacts can be measured in an LCA. And the opinions or motives of the assessors can have an effect on the result of the assessment, making it biased.

Warm-Up & Exam Questions

That's some more revision done and dusted and now it's time to test yourself on how much you've taken in. Have a go at the questions on this page to see if you need to revisit any of the previous pages.

Warm-Up Questions

1) Give an example of a natural resource which has been replaced by a man-made alternative.
2) What is sustainable development?
3) Give three positive effects of recycling metals.
4) What are the four stages that need to be considered to conduct a life cycle assessment?

Exam Questions

1 Natural resources are formed without human input and are used for construction, fuel and food. (GRADE 4-6 GRADE)

1.1 What is a finite natural resource?
Tick **one** box.

- A natural resource that can never be remade. ☐
- A natural resource that will never run out. ☐
- A natural resource that doesn't renew itself quickly enough to be considered replaceable. ☐
- A natural resource that can only be used in certain conditions. ☐

[1 mark]

1.2 Aluminium is used to make soft drink cans. Extracting aluminium is a very energy intensive process.
Suggest **two** ways that the use of soft drink cans can be made more sustainable.

[2 marks]

2 Copper needs to be extracted from its ore before it can be used. (GRADE 6-7 GRADE)

2.1 Explain why scientists have developed ways of extracting copper from low-grade ores.

[1 mark]

2.2 Copper can be extracted from low-grade ores by a process called bioleaching.
Explain how the process works.

[2 marks]

2.3 Give **one** advantage of using processes such as bioleaching rather than traditional mining.

[1 mark]

2.4 Give **one** disadvantage of using processes such as bioleaching rather than traditional mining.

[1 mark]

3 A life cycle assessment looks at the environmental impact of a product over its lifetime. (GRADE 6-7 GRADE)

3.1 Suggest why it can be difficult to give a complete assessment of a product over its entire life cycle.

[1 mark]

3.2 A company carried out a LCA on one of their products but didn't take into account its disposal. However, an independent LCA found the disposal of the product to have the greatest environmental impact of any part of the product's life cycle.
Suggest why the LCA performed by the company didn't include details of the disposal of the product.

[2 marks]

Potable Water

Quick Quiz

*We all need safe drinking water. The **way** that water's made safe depends on **local conditions**.*

Potable Water is Water You Can **Drink**

- Potable water is water that's been treated or is naturally safe for humans to drink — it's essential for life.
- Chemists wouldn't call it pure, though. Pure water only contains H_2O molecules whereas potable water can contain lots of other dissolved substances.
- The important thing is that the levels of dissolved salts aren't too high, that it has a pH between 6.5 and 8.5 and also that there aren't any nasties (like bacteria or other microbes) swimming around in it.

The **Way** that Potable Water is **Produced** Depends on **Where You Are**

1) Rainwater is a type of fresh water. Fresh water is water that doesn't have much dissolved in it.
2) When it rains, water can either collect as surface water or as groundwater.

SURFACE WATER: collects in lakes, rivers and reservoirs.
GROUNDWATER: collects in rocks called aquifers that trap water underground.

3) In the UK, the source of fresh water used depends on location. Surface water tends to dry up first, so in warm areas, e.g. the south-east, most of the domestic water supply comes from groundwater.
4) In some very dry countries, e.g. Kuwait, there's not enough fresh water (surface or groundwater), so sea water must be treated to provide potable water:

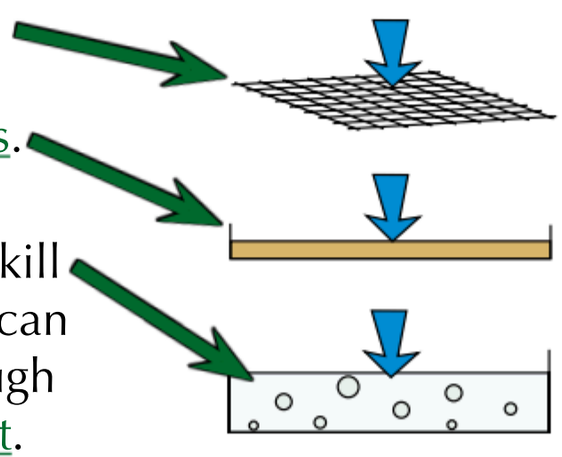
- Sea water has too much dissolved salt in it to be potable — it needs to be desalinated.
- Distillation is one way to do this (the next page shows how this can be achieved in the lab).
- Sea water can also be treated by processes that use membranes — like reverse osmosis. The salty water is passed through a membrane that only allows water molecules to pass through. Ions and larger molecules are trapped by the membrane and separated from the water.
- Both of these processes use loads of energy — they're expensive and impractical for producing large quantities of fresh water.

Most **Fresh Water** Needs to be **Treated** to Make it **Safe**

Even though it only has low levels of dissolved substances, water from fresh water sources still needs to be treated to make it safe before it can be used. This process includes:

Filtration — a wire mesh screens out large twigs etc, and then gravel and sand beds filter out any other solid bits.

Sterilisation — the water is sterilised to kill any harmful bacteria or microbes. This can be done by bubbling chlorine gas through it or by using ozone or ultraviolet light.



Chemicals can also be added to the water supply during water treatment, such as fluoride (which is good for teeth). This is controversial, because people aren't given any choice over whether they consume them or not.

The water that comes out of our taps has been treated

Location is a really important factor in determining how water is treated. For example, in the UK, there is a lot of fresh water available so this is filtered and then sterilised. However in very dry countries, where fresh water is less readily available, a more expensive process like distillation or desalination of sea water might have to be used.

PRACTICAL

Analysing and Purifying Water

Quick Quiz

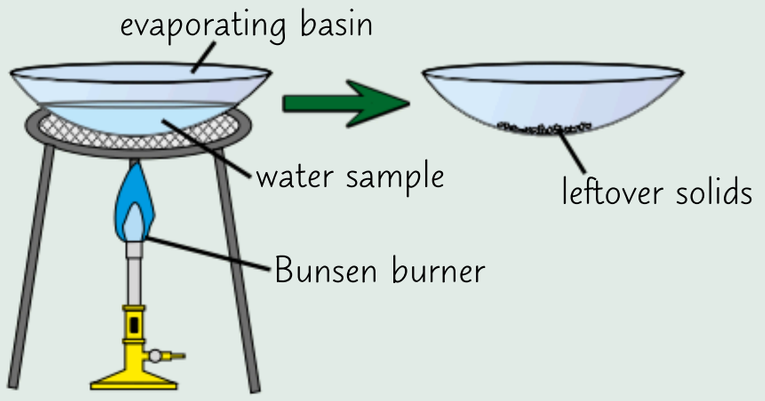
*You can do **tests** to see If water is pure. Grab your Bunsen burner, things are about to heat up...*

There are Several Ways to **Test** if **Water** is **Pure**

Pure water boils at 100 °C, has a pH of 7 and doesn't contain any dissolved solids — it's just water, H_2O. If you want to confirm that a sample of water is pure, you can check one or more of these properties:

One way you can check for dissolved solids is by boiling a sample of water to see if any solids are left behind:

1) Measure the mass of a clean, dry evaporating basin.
2) Add a known volume of your water sample to the basin.
3) Heat the basin (e.g. using a Bunsen burner) until all of the water has boiled off — it needs to be completely dry.
4) Once it's cooled down, measure the mass of the evaporating basin again and work out the change in mass. If there is an increase in mass, it means that the water sample did contain dissolved solids, which were left behind when it evaporated. (You can also look to see if there is any solid in the basin, but if the water sample only contained a small amount of dissolved solids, you might not be able to see them.)



You could use the mass of dissolved solids and the volume of water to work out the concentration of dissolved solids in your sample — see p.50.

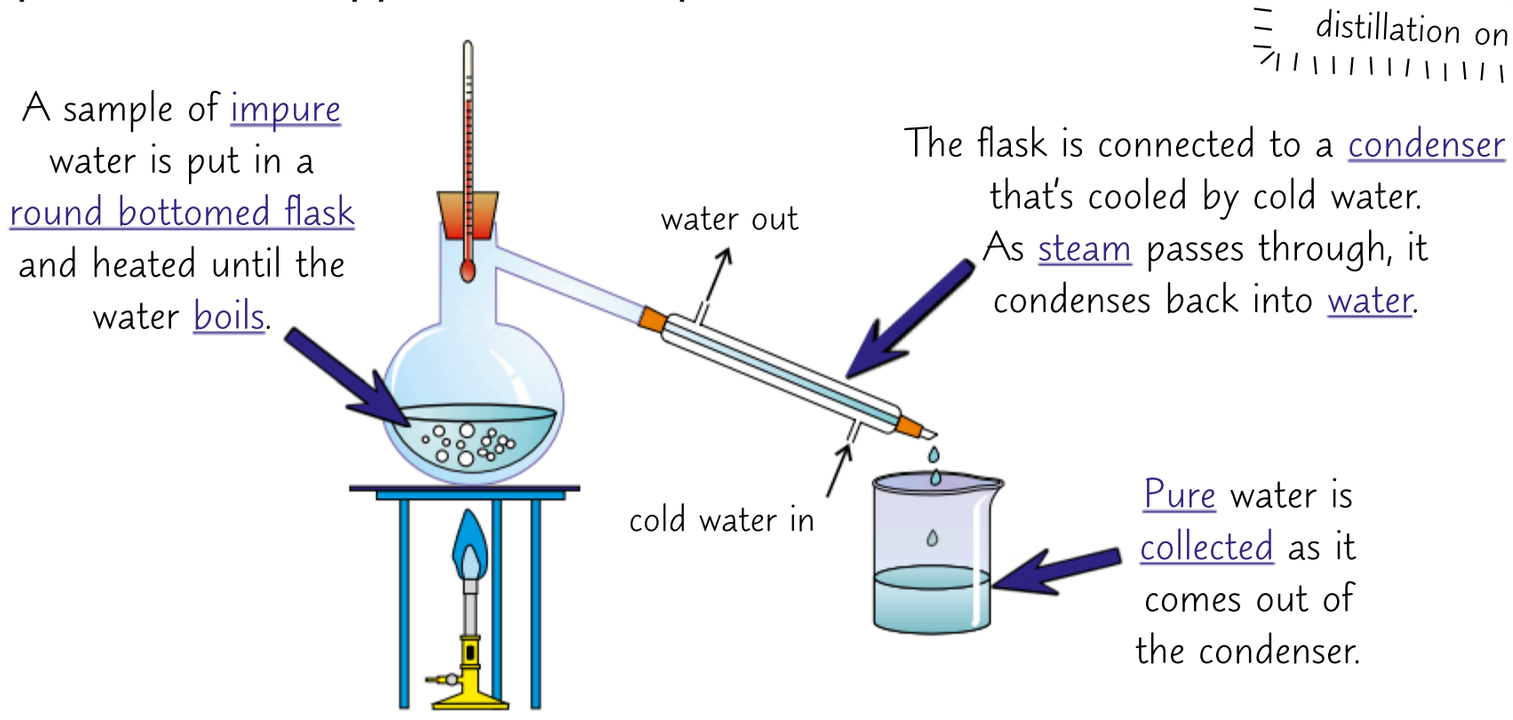
- Another way to test for dissolved solids is to perform tests for ions.
  E.g. you could test the water for the presence of sodium chloride (the main salt in seawater) — there are tests for sodium cations and chloride anions on pages 110-111.
- There are several ways you could check the pH of a water sample to see if it's around 7.
  E.g. you could measure it with a pH probe and meter, or use universal indicator paper or solution.
  There's more about measuring pH on page 152.
- As you saw on p.106, pure substances melt and boil at specific temperatures.
  So you could measure the temperature a sample of water boils at — if it's pure, it will boil at 100 °C.

Water Can Be **Purified** by **Distillation**

As you saw on the previous page, sea water can be desalinated by distillation — boiling the water and condensing it in a different container separates it from impurities, including dissolved solids.

You can carry out distillation in the lab to purify a sample of water.
Here's an example of how the apparatus is set up:

There's more on distillation on p.11.



Pure water does not contain any dissolved solids

Remember that you can test the purity of water by boiling it and looking for any solids left over after it has evaporated. You can also check whether the boiling point is higher than 100 °C, test for the presence of ions, or test the pH to assess purity.

Waste Water Treatment

Quick Quiz

It might not be pretty, but dealing with our waste is really important to ensure that we don't ***pollute*** *the natural environment — it also means that we can access* ***nice clean water****.*

How **Waste Water** is **Treated** Depends on its **Source**

1) Waste water can come from different sources, for example:

- Domestic — water from homes goes down the drain and into sewers, towards sewage treatment plants. This is water (and other waste) from e.g. washing-up, flushing the toilet or having a bath.
- Agricultural — farming waste water includes nutrient run-off from fields and slurry from animal farms.
- Industrial — factories produce a lot of waste water, e.g. from the industrial production of fertilisers (see p.136).

2) Sewage from domestic or agricultural sources has to be treated to remove any organic matter and harmful microbes before it can be put back into freshwater sources like rivers or lakes. Otherwise it would pollute the environment and pose health risks.

3) As well as organic matter, industrial waste water can also contain harmful chemicals, so it has to undergo additional stages of treatment before it is safe to release it into the environment.

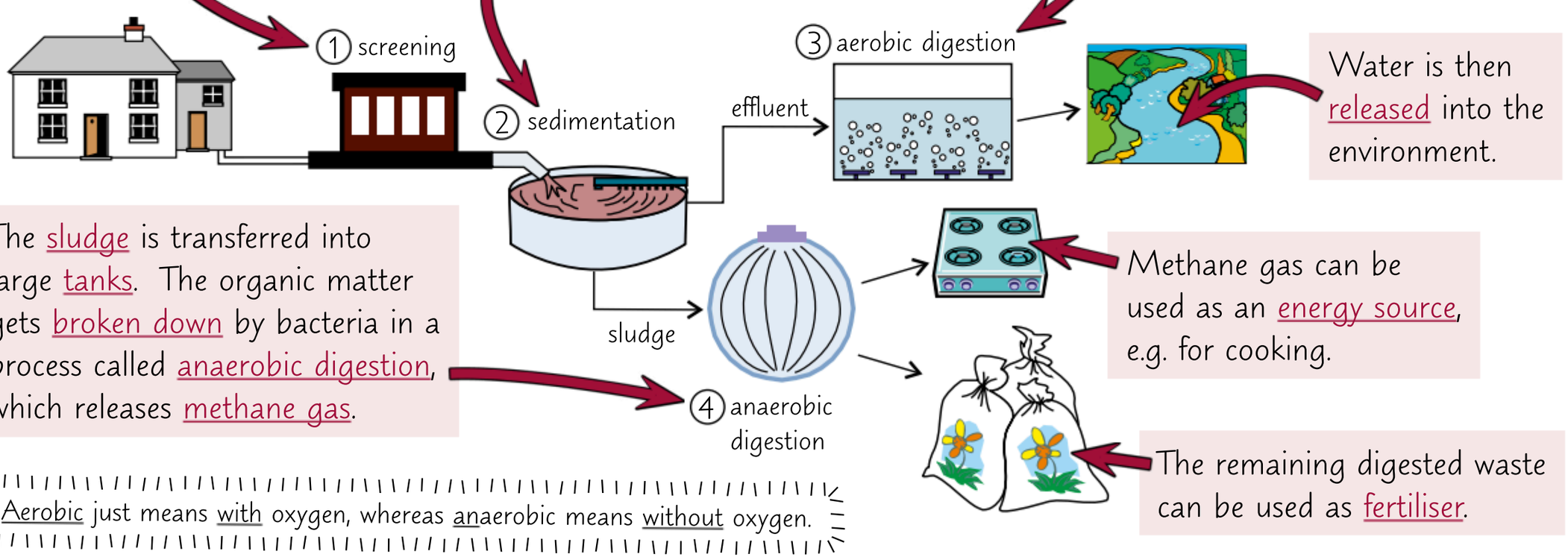
Sewage **Treatment** Happens in **Several Stages**

Before treatment, the sewage is screened — large bits of material (like twigs or plastic bags) and grit are removed.

Then it's allowed to stand in a settlement tank and undergoes sedimentation. Heavier suspended solids sink, producing sludge, while the lighter effluent floats on top.

The effluent is removed and treated by biological aerobic digestion. Air is pumped through it to encourage aerobic bacteria to break down the organic matter, including harmful microbes.

Video



Aerobic just means with oxygen, whereas anaerobic means without oxygen.

Sewage treatment requires more processes than treating fresh water but uses less energy than the desalination of salt water, so could be used as an alternative in areas where there's not much fresh water. For example, Singapore is treating waste water and recycling it back into drinking supplies. However, people don't like the idea of drinking water that used to be sewage.

For waste water containing toxic substances, additional stages of treatment may involve adding chemicals (e.g. to precipitate metals), or using UV radiation or membranes.

Waste water can be recycled to produce potable water

To learn the stages of water treatment you can try covering the diagram, writing out each step and then checking them.

Warm-Up & Exam Questions

Now you know all there is to possibly know about water, it's time to test yourself with some questions.

Warm-Up Questions

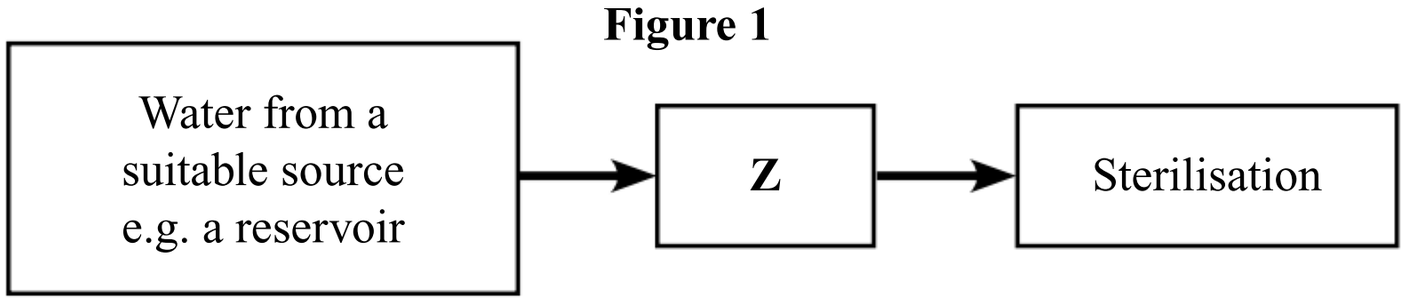
1) Describe two features of potable water that make it safe for drinking.
2) What is the first step in treating fresh water?
3) In addition to organic matter, what else may have to be removed from industrial waste water?
4) What is produced following the sedimentation stage of water treatment?

Exam Questions

1 **Figure 1** shows how water in some parts of the UK is treated during the production of potable water.

GRADE 6-7 GRADE

Figure 1



1.1 Describe the difference between potable water and pure water.

[1 mark]

1.2 Describe what happens during stage **Z** in **Figure 1**.

[2 marks]

1.3 Suggest **three** agents that could be used during the sterilisation stage of the process.

[3 marks]

2 Waste water from human activities must undergo treatment before being released into the environment. The stages of sewage treatment include screening, sedimentation, anaerobic digestion and aerobic digestion.

GRADE 6-7 GRADE

2.1 State the stage at which large pieces of material, such as twigs, are removed.

[1 mark]

2.2 During sedimentation, effluent is produced.
Describe what happens during the next step in the treatment of effluent following sedimentation.

[2 marks]

2.3 In areas where fresh water is limited, treated waste water and sea water can be used as drinking water.
Describe two disadvantages of using these processes compared to the treatment of fresh water.

[2 marks]

PRACTICAL

3 In some dry countries potable water is produced by the distillation of salt water.
A student carries out distillation of salt water in the lab.

GRADE 6-7 GRADE

3.1 Describe **two** tests which can be used to determine whether or not the distilled water contains any sodium or chloride ions.

[4 marks]

3.2 Suggest a further test that would need to be carried out before the water could be considered potable.

[1 mark]

Quick Quiz

The Haber Process

*This is an important industrial process. It produces **ammonia** (NH_3), which is used to make **fertilisers**.*

Nitrogen and **Hydrogen** are Needed to Make **Ammonia**

The Haber process is used to make ammonia from hydrogen and nitrogen using the following reaction:

$$N_{2\,(g)} + 3H_{2\,(g)} \rightleftharpoons 2NH_{3\,(g)}$$

nitrogen + hydrogen ⇌ ammonia

The forwards reaction is exothermic (it gives out heat). This is important for choosing the conditions for the industrial process — see the next page.

This reaction is well suited for an industrial scale as the reactants aren't too difficult or expensive to obtain:

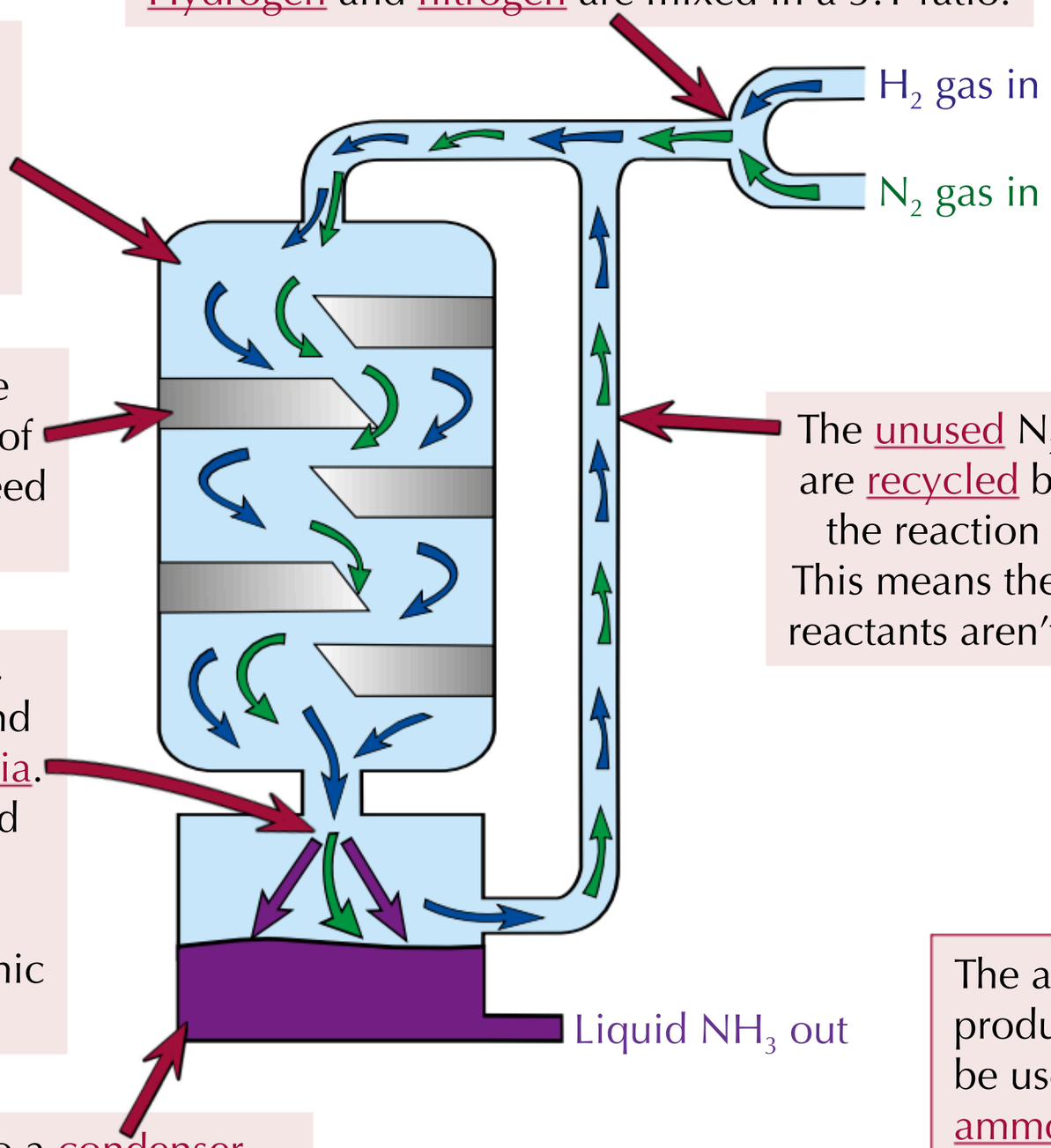
- The nitrogen is obtained easily from the air, which is 78% nitrogen.
- The hydrogen mainly comes from reacting methane (from natural gas) with steam to form hydrogen and carbon dioxide.

You Need to Know the **Conditions** Used in the Haber Process

Video

Hydrogen and nitrogen are mixed in a 3:1 ratio.

The reaction vessel uses a high temperature (450 °C) and a high pressure (200 atmospheres).



The reactants are passed over trays of iron catalyst to speed up the reaction.

The unused N_2 and H_2 are recycled back into the reaction vessel. This means the leftover reactants aren't wasted.

As the reaction is reversible, only some of the hydrogen and nitrogen react to form ammonia. Some of the ammonia formed will also break down again in the reverse reaction. (Eventually, it reaches a dynamic equilibrium — see p.87.)

The gases pass into a condenser. The gases are cooled enough that the ammonia condenses and is removed from the mixture, but the hydrogen and nitrogen stay as gases.

The ammonia produced can then be used to make ammonium nitrate — a very nitrogen-rich fertiliser (see p.136).

The Haber process happens at a high temperature and pressure

Unfortunately you just need to learn the conditions used in the Haber process, sorry. 450 °C, 200 atmospheres, iron catalyst. These conditions have been chosen for very specific reasons, as you're about to discover on the next page...

Conditions for the Haber Process

Quick Quiz

*As the Haber process is **reversible**, the position of equilibrium is vital in generating a large enough **yield**. Choosing the best reaction conditions to suit the aims of the reaction in industry can be tricky...*

The Reaction is **Reversible**, So There's a **Compromise** to be Made

- As you saw on page 88, the temperature and pressure that a reversible reaction takes place at affect the position of equilibrium. This means the conditions of the reversible reaction affect the yield (how much product you get).
- The thing is, these conditions are also factors that affect the rate of reaction — see page 81.
- This means that you sometimes end up with a trade-off between increasing the rate and maximising the yield of a reaction.
- The formation of ammonia in the Haber process is a reversible reaction — so choosing the conditions for this industrial process does involve compromises.

Make sure you're able to explain how different factors affect the position of equilibrium using Le Chatelier's Principle.

Temperature is a **Compromise** Between Yield and Speed

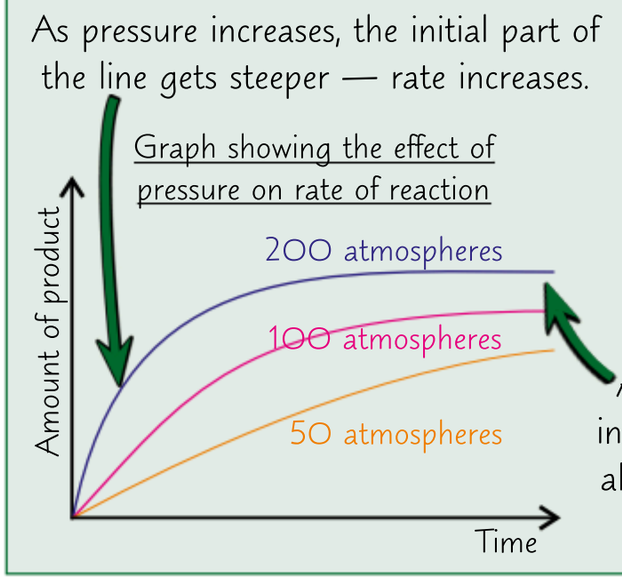
- The forward reaction in the Haber process is exothermic. That means that increasing the temperature will move the equilibrium the wrong way — away from ammonia and towards nitrogen and hydrogen. So the yield of ammonia would be greater at lower temperatures.
- The trouble is, lower temperatures mean a slower rate of reaction (so it takes longer to reach equilibrium and get that higher yield of ammonia).
- The actual temperature of 450 °C used in the Haber process is a compromise between maximum yield and speed of reaction. It's better to wait just 20 seconds for a 10% yield than to have to wait 60 seconds for a 20% yield, especially if you're able to recycle your unused reactants so they aren't wasted.

The iron catalyst also makes the reaction go faster, but doesn't affect the yield.

For **Pressure**, the **Compromise** is About Cost

- Increasing the pressure moves the position of equilibrium for the Haber process towards the products, since there are fewer molecules of gas on the right-hand side (2 on the right vs 4 on the left).
- Increasing the pressure also increases the rate of reaction. So for the Haber process, the higher the pressure, the higher the yield and the faster the rate — increasing the pressure is a win-win.
- However, there are drawbacks to increasing pressure — it's expensive to build and maintain reaction vessels that can operate safely at very high pressures.
- The actual pressure of 200 atmospheres is a compromise between getting the best yield and rate and keeping the process cheap and safe enough.

You need to be able to interpret graphs showing reaction conditions:



The conditions of the Haber process consider speed, yield and cost

Without the Haber process, we wouldn't have the fertiliser which has allowed the human population to grow as quickly as it has over the last 100 years. Make sure you learn the word equation, the sources of hydrogen and nitrogen for the process and the conditions for the reaction. Also understand how the conditions are a compromise between yield and rate.

Quick Quiz

NPK Fertilisers

Fertilisers *help us to produce* ***more food*** *by allowing us to keep growing crops on the* ***same land*** *every year.*

NPK Fertilisers Provide Plants with the Essential Elements for Growth

1) Farmers used to use manure to fertilise fields, but now artificial fertilisers are widely used.
2) The three main essential elements in fertilisers are nitrogen, phosphorus and potassium.
   If plants don't get enough of these elements, their growth and life processes are affected.
   These elements may be missing from the soil if they've been used up by a previous crop.
3) Fertilisers replace these missing elements or provide more of them. This helps to increase the crop yield, as the crops can grow faster and bigger. For example, fertilisers add more nitrogen to plant proteins, which makes the plants grow faster — increasing productivity.
4) NPK fertilisers are formulations containing salts of nitrogen (N), phosphorus (P) and potassium (K) (hence, "NPK") in appropriate amounts for the plants.

There's more on formulations on p.106.

Ammonia is Used to Produce Nitrogen-Containing Compounds

- Ammonia can be reacted with oxygen and water in a series of reactions to make nitric acid.
- You can also react ammonia with acids to get ammonium salts. Ammonia and nitric acid react to form ammonium nitrate — this is particularly good for fertilisers because it has nitrogen from two sources.

$$NH_{3\,(aq)} + HNO_{3\,(aq)} \rightarrow NH_4NO_{3\,(aq)}$$

Ammonia + Nitric acid → Ammonium nitrate

This reaction is carried out differently in industry to how it is done in the lab:

Industry

- Carried out in giant vats.
- It is very exothermic, and the high concentrations used cause lots of heat to be released. This heat evaporates water from the mixture to make a very concentrated ammonium nitrate product.

Laboratory

- Carried out on a much smaller scale — the reactants are at a much lower concentration than in industry, so less heat is produced and it's safer for a person to carry it out.
- The reaction is performed as a titration (p.59) and the mixture is then crystallised (p.10) to give pure ammonium nitrate crystals. Crystallisation isn't used in industry because it's very slow.

You could be asked to compare different methods in the exams. You'll be given all the details you need.

Phosphate and Potassium are Sourced from Mined Compounds

1) Potassium chloride and potassium sulfate can be mined and used as a source of potassium.
2) Phosphate rock is also mined. However, because the phosphate salts in the rock are insoluble, plants can't use them as nutrients.
3) Reacting phosphate rock with different acids produces soluble sources of phosphate:

Acid used	nitric acid	sulfuric acid	phosphoric acid
Products	calcium nitrate and phosphoric acid	calcium sulfate and calcium phosphate (mixture also called single superphosphate)	calcium phosphate (product also called triple superphosphate)

NPK = nitrogen + phosphorous + potassium

NPK fertilisers are made of compounds which contain all three elements. Getting the right percentages of each element in a formulation is important for delivering the correct nutrients to crops that farmers are growing. With the right mix of nutrients the growth of crops will be maximised, yields will be high and farmers will have lots to sell.

Warm-Up & Exam Questions

These pages have all the knowledge you need to know about the Haber process and fertilisers.
But don't assume you're the bee's knees — test yourself with this lovely selection of questions.

Warm-Up Questions

1) What two reactants are needed to manufacture ammonia?
2) True or false? The forward reaction in the Haber process is exothermic.
3) Name the catalyst used in the Haber process.
4) What are NPK fertilisers?
5) Name a compound commonly used as a source of potassium in NPK fertilisers.

Exam Questions

1 Farmers often use fertilisers. GRADE 4-6 GRADE

1.1 Fertilisers usually contain nitrogen. Name the **two** other elements they usually contain.

[2 marks]

1.2 Explain why farmers apply fertilisers to their crops.

[2 marks]

2 Fertilisers contain compounds derived from ammonia and phosphate rock. GRADE 6-7 GRADE

2.1 Ammonium nitrate is a fertiliser made by neutralising nitric acid and ammonia.
Suggest why ammonium nitrate is a particularly useful ingredient in a fertiliser.

[1 mark]

2.2 Phosphate rock can't be used as a fertiliser directly. Explain why.

[1 mark]

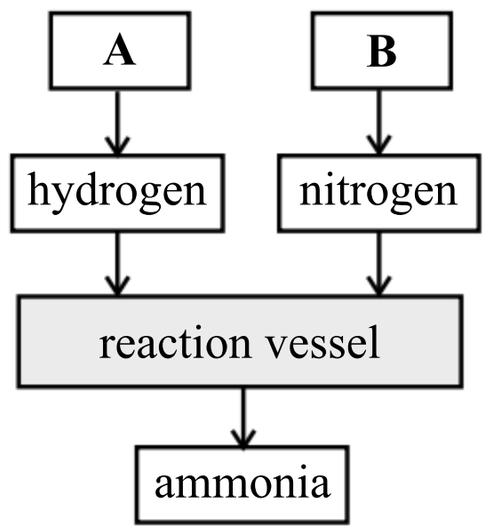
2.3 Name the compound commonly found in fertilisers made by reacting sulfuric acid with phosphate rock.

[1 mark]

3 Ammonia is made by combining nitrogen and hydrogen at a pressure of 200 atm, a temperature of 450 °C and in the presence of a catalyst. GRADE 6-7 GRADE

A flow diagram is shown for the reaction in **Figure 1**.

Figure 1



3.1 **A** and **B** represent the sources of nitrogen and hydrogen.
State appropriate sources for nitrogen and hydrogen used in the reaction.

[2 marks]

3.2 Balance the equation for the reaction shown below.

$$\ldots\ldots H_{2(g)} + \ldots\ldots N_{2(g)} \rightleftharpoons \ldots\ldots NH_{3(g)}$$

[1 mark]

3.3 The reaction is exothermic. What would happen to the yield of ammonia if the temperature was lowered? Explain your answer.

[2 marks]

3.4 Increasing the pressure increases the yield of the reaction.
Suggest why a pressure greater than 200 atm isn't used.

[1 mark]

Revision Summary Test for Topic 10

Well, that was an absolute whopper of a topic to end on. Here's what you could do next:

- Tackle the revision summary test below, or scan the QR code to do it online.
  The questions are hard, but they'll show you how well you really know your stuff.
- Track your progress online and see which areas need more work.
- Compare your answers with sample answers for the test here: cgpbooks.co.uk/Base

Summary Test

Materials and Corrosion (p.121-123) ☑

1) Name two types of glass and describe how they're made.
2) What is a composite material? Name three examples and give a use for each one.
3) Give the reaction conditions used to make both types of poly(ethene) and describe their properties.
4) How does using different monomers result in different properties for thermosetting and thermosoftening polymers?
5) Give three typical properties of: a) ceramics, b) polymers, c) metals.
6) Name three alloys and the elements used to make them. Give a use for each alloy.
7) Why are iron objects destroyed by corrosion but aluminium objects are not?
8) Explain how barrier and sacrificial methods of protection against corrosion work. How is galvanisation both a barrier and a sacrificial method?

Chemistry and Sustainability (p.125-128) ☑

9) Give an example of a natural product and a synthetic product that can be used to replace it.
10) What is meant by the terms 'renewable resource' and 'finite resource'?
11) What is sustainable development?
12) How can using, extracting and processing resources be unsustainable?
13) Describe how metals are usually recycled. How is glass usually recycled?
14) How do the processes of bioleaching and phytomining work to extract copper from low-grade ores?
15) What are the four stages that should be considered in a life cycle assessment?
16) Why could some plastic bags be considered less harmful to the environment than some paper bags?
17) Explain why LCAs can't be objective assessments of the environmental impact of a product.

Potable Water and Waste Water Treatment (p.130-132) ☑

18) What is potable water? Name three sources of water.
19) Describe the processes fresh water goes through to make it potable. Now name two ways water can be desalinated.
20) How could you measure the mass of dissolved solids in a sample of water? Give three other ways you could test the purity of a water sample.
21) Name three sources of waste water. Why do they need treatment?
22) Describe the following processes in waste water treatment:
    a) screening, b) sedimentation, c) aerobic digestion, d) anaerobic digestion.

The Haber Process and Fertilisers (p.134-136) ☑

23) What is the reaction equation for the Haber process? Where are the reactants obtained from?
24) What are the temperature, pressure and catalyst used in the Haber process? What happens to the gaseous reactants and product after they leave the reaction vessel?
25) Explain why compromises are required when choosing temperature and pressure for the Haber process.
26) What are the essential elements in NPK fertilisers? Why do fertilisers help increase crop yields?
27) How is ammonium nitrate produced in industry? How about in the lab?
28) Name three sources of elements for NPK fertilisers that are mined.

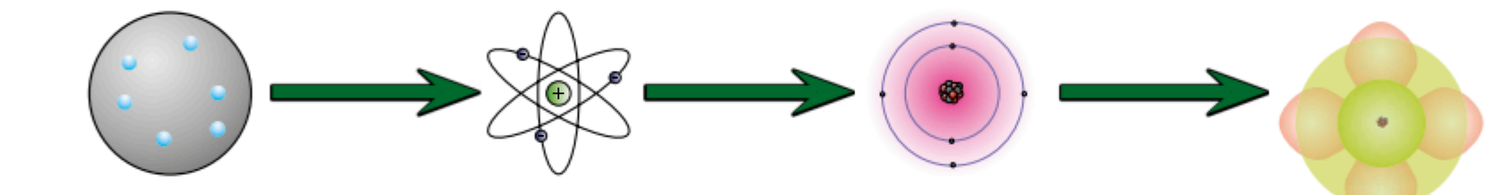
The Scientific Method

*This section **isn't** about how to 'do' science — but it does show you the way **most scientists** work.*

Scientists Come Up With **Hypotheses** — Then **Test** Them

1) Scientists try to explain things. They start by observing something they don't understand.
2) They then come up with a hypothesis — a possible explanation for what they've observed.
3) Next they make a prediction based on the hypothesis. They test the prediction by gathering evidence (i.e. data) from investigations. If the evidence backs up their prediction, then they're a step closer to figuring out if the hypothesis is true.
4) They can share their findings at conferences, or in peer-reviewed journals.
5) Once other scientists know about a hypothesis, they'll start basing their own predictions on it and carry out their own experiments. They'll also try to reproduce the original experiments to check the results — and if all the experiments in the world back up the hypothesis, then scientists start to think it is true.
6) However, if results from lots of other experiments don't fit with the hypothesis, then it may need to be modified or scrapped altogether.

Peer-review is where other scientists check that methods, results and scientific explanations seem sensible before they're published. It helps to detect errors and false claims, but it doesn't mean that findings are correct — just that they're not wrong in any obvious way.



About 100 years ago, scientists hypothesised that atoms looked like this.

New evidence from further experiments meant they changed their hypothesis to this.

Once even more evidence was gathered, scientists changed their hypothesis again.

Now we think it's more like this.

If **All** the **Evidence** Supports a Hypothesis, It's **Accepted** — For Now

Accepted hypotheses often become the basis of theories. Our currently accepted theories are the ones that have survived many rounds of testing over the years. However, theories never become totally indisputable fact. If new evidence comes along that can't be explained using the existing theory, then the hypothesising and testing is likely to start all over again.

Theories Can Involve **Different Types** of **Models**

Scientists use models to explain observations and make predictions, for example:

1) Representational models — simplified descriptions or pictures of what's going on in real life. E.g. the Bohr model of an atom is a simplified way of showing the arrangement of electrons in an atom (see p.14). It can be used to explain trends down groups in the periodic table.
2) Computational models — computers are used to make simulations of complex real-life processes (e.g. climate change). Computational models are used when there's lots of different variables to consider, and because you can easily change their design to take into account new data.

Scientists test models by carrying out experiments to check that the predictions made by the model happen as expected.

All models have limitations on what they can explain or predict. E.g. ball and stick models (a type of spatial model) can be used to show how ions are arranged in an ionic compound. One of their limitations is that they don't show the relative sizes of the ions (see p.28).

Scientific models are constantly being refined...

Aristotle (a Greek philosopher) was the first person to realise that scientific theories need to be based on observations. Muslim scholars then introduced the ideas of creating a hypothesis, testing it, and repeating work to check results. And Roger Bacon took things even further by writing down everything he did, so other scientists could reproduce his work.

Communication & Issues Created by Science

*Scientific developments can be great, but they can sometimes **raise more questions** than they answer...*

It's Important to **Communicate** Scientific Discoveries to the **Public**

Some scientific discoveries show that people should change their habits, or they might provide ideas that could be developed into new technology. So scientists need to tell the world about their discoveries.

Technologies are being developed that make use of fullerenes (see p.34). These include drug delivery systems for use in medicine. Information about these systems needs to be communicated to doctors so they can use them, and to patients, so they can make informed decisions about their treatment.

Scientific **Evidence** can be **Presented** in a **Biased Way**

1) Media reports of scientific discoveries aren't peer-reviewed. So, even though news stories are often based on data that has been peer-reviewed, the data might be presented in a way that is over-simplified or inaccurate, making it open to misinterpretation.
2) People who want to make a point can sometimes present data in a biased way (sometimes without meaning to). For example, a scientist might overemphasise a relationship in some of their data, or a newspaper article might report data supporting an idea without giving any evidence against it.

Scientific Developments are **Great**, but they can **Raise Issues**

Economic issues	Society can't always afford to do what scientists recommend (e.g. investing in alternative energy sources) without cutting back elsewhere.
Social issues	Decisions based on scientific evidence affect people — e.g. should fossil fuels be taxed more highly? These decisions can have negative effects on some people's lifestyles, and some groups of people might be affected more than others.
Personal issues	Some decisions will affect individuals. For example, someone might support renewable energy, but object if a wind farm is built next to their house.
Environmental issues	Human activity can affect the natural environment. For example, building a hydroelectric dam might destroy the local habitats of some species. But it also reduces our need for fossil fuels, helping to reduce climate change.

Science **Can't Answer Every Question** — Especially **Ethical** Ones

- To answer scientific questions, scientists need data to provide evidence for their hypotheses.
- Some questions can't be answered yet because there's not enough data to support a theory, or because the data can't currently be collected. As more evidence is collected, more questions will be answered (like what the impact of global warming on sea levels will be).

But there will always be questions that experiments can't help us to answer...

Question: Should new drugs that boost 'brain power' be available to the public?

PRO
- Could improve your concentration or memory.

CONS
- Could give people an unfair advantage in exams.
- People might be pressured into taking them so that they could work more effectively, and for longer hours.

There are often issues with new scientific developments...

The trouble is, there's often no clear right answer where these issues are concerned. Different people have different views, depending on their priorities. So science can advise, but ultimately it's down to you and society to decide.

Risk

Scientific discoveries *are often great, but they can prove* ***risky****. With dangers all around, you've got to be aware of hazards — this includes* ***how likely*** *they are to* ***cause harm*** *and* ***how serious*** *the effects may be.*

Every **Hazard** has a **Risk** Attached to it

HAZARD — something that could potentially cause harm.

RISK — the chance that a particular hazard will cause harm.

There are risks that seem pretty obvious, or we've known about for a while (e.g. the risk of having a car accident while in a moving car) and new risks that scientists are unsure of that arise from scientific advances, (e.g. whether nanoparticles used in cosmetics and sun cream might be harming the cells in our bodies).

You can estimate the size of a risk based on how many times something happens in a big sample over a given period. For example, you could assess the risk of a driver crashing by recording how many people in a group of 100 000 drivers crashed their cars over a year.

People Make Their **Own Decisions** About Risk

Something's generally considered high risk if there is a high chance of it happening and if its consequences are serious. But there are lots of factors that can affect how people feel about taking different risks:

Consequences

Most people are happier to accept a higher probability of an accident if the consequences are short-lived and fairly minor. E.g. you're likelier to cut your finger when using a knife than you are to die when scuba-diving — cutting your finger is fairly minor compared to death so people are more willing to take that risk.

Familiarity

How risky people think things are isn't always accurate. They tend to view familiar activities as low-risk and unfamiliar activities as high-risk — even if that's not the case (e.g. cycling on roads is often high-risk, but some people are happy to do it because it's a familiar activity).

Visibility

People may underestimate the risk of things with long-term or invisible effects, e.g. using tanning beds.

Choice

People tend to be more willing to accept a risk if they choose to do something, compared to having the risk imposed on them (e.g. having a nuclear power station built next door).

Investigations Can be **Hazardous**

Examples of Hazards in Experiments

Microorganisms — some bacteria can make you ill.

Chemicals — sulfuric acid can burn your skin and alcohols catch fire easily.

Fire — an unattended Bunsen burner is a fire hazard.

Electricity — faulty electrical equipment could give you a shock.

Part of planning an investigation is making sure that it's safe. You should always make sure that you identify all the hazards that you might encounter. Then you should think of ways of reducing the risks from these hazards. For example:

- If you're working with sulfuric acid, always wear gloves and safety goggles. This will reduce the risk of the acid coming into contact with your skin and eyes.
- If you're using a Bunsen burner, stand it on a heatproof mat. This will reduce the risk of starting a fire.

Risk is the chance that a hazard will cause harm

The world's a dangerous place and it's impossible to rule out the chance of an accident altogether. However, if you can recognise hazards, decide how to reduce their risks, and be happy to accept some risks, you're more likely to stay safe.

Designing Investigations

*The design of a scientific investigation is **important** — if it's not done well, the data collected is pretty useless...*

Investigations **Produce Evidence** to **Support** or **Disprove** a **Hypothesis**

1) Like all great scientists, you'll need to be able to come up with a hypothesis to test an observation (see p.139).
2) To determine whether a hypothesis is right, you need to do an investigation to gather evidence. To do this, you'll need to use your hypothesis to make a prediction — something you think will happen that you can test.
3) Investigations are used to see if there are patterns or relationships between two variables, e.g. to see if there's a relationship between the variables 'number of spots' and 'size of feet'.

1. Observation: People have big feet and spots.
↓
2. Hypothesis: Having big feet causes spots.
↓
3. Prediction: People who have bigger feet will have more spots.

Evidence Needs to be **Repeatable, Reproducible** and **Valid**

REPEATABLE — if the same person does the same experiment again, they'll get similar results.

REPRODUCIBLE — if someone else does the experiment, or a different method or piece of equipment is used, the results will still be similar.

VALID RESULTS — results that are both repeatable and reproducible AND they answer the original question. They come from experiments that were designed to be a fair test.

If data is repeatable and reproducible, it's reliable and scientists are more likely to have confidence in it.

To Make an Investigation a **Fair Test** You Have to **Control the Variables**

1) In a lab experiment you usually change one variable and measure how it affects another.
2) To make it a fair test, everything else that could affect the results should stay the same — otherwise you can't tell if the thing you're changing is causing the results or not.
3) Because you can't always control all the variables, you often need to use a control experiment — an experiment that's kept under the same conditions as the rest of the investigation, but doesn't have anything done to it. This is so that you can see what happens when you don't change anything at all.

The different kinds of variables are:

Independent	The variable that you change.
Dependent	The variable that you measure.
Control	The variables you keep the same.

E.g. you can find how temperature affects the rate of a reaction by measuring the volume of gas formed over time. Temperature is the independent variable. The volume of gas formed is the dependent variable. The control variables include the concentration and amounts of reactants, the time you measure for, etc.

The **Bigger** the **Sample Size** the **Better**

- A sample should represent the whole population — a small sample can't do that as well. It's also harder to spot anomalies if your sample size is too small.
- Scientists have to be realistic when choosing sample sizes, though. For example, if you were studying the effects of a chemical used to sterilise water on the people drinking it, it'd be great to study everyone who drinks the water (a huge sample), but it'd take ages and cost a tonne. It's more realistic to study a thousand people, with a range of ages, genders and races to represent the population as a whole.

Designing an investigation is an involved process...

So when designing an investigation, start by figuring out what your hypothesis is. Then you need to think about all the variables involved, so you can develop a method that'll make sure the test is fair. Oh, and sample size is important too.

Collecting Data

You've designed the perfect investigation — now it's time to get your hands mucky and ***collect some data****.*

Your Data Should be **Repeatable, Reproducible, Accurate** and **Precise**

- To check repeatability you need to repeat the readings and check that the results are similar. You need to repeat each reading at least three times.
- To make sure your results are reproducible you can cross check them by taking a second set of readings with another instrument (or a different observer).
- Your data also needs to be precise and accurate:

PRECISE RESULTS — where the data is all really close to the mean (average) of your repeated results (i.e. not spread out).

ACCURATE RESULTS — results that are really close to the true answer.

Repeat	Data set 1	Data set 2
1	12	11
2	14	17
3	13	14
Mean	13	14

Data set 1 is more precise than data set 2.

- The accuracy of your results usually depends on your method. E.g. estimating the amount of gas released from a reaction by counting bubbles isn't very accurate because you might miss bubbles and they will have different volumes. It'd be more accurate to measure the volume using a gas syringe (see p.151).

Your **Equipment** has to be **Right for the Job**

Sensitivity

The measuring equipment you use has to be sensitive enough to measure the changes you're looking for. E.g. if you need to measure changes of 1 cm^3 you need to use a measuring cylinder or burette that can measure in steps of 1 cm^3.

Resolution

Resolution is the smallest change a measuring instrument can detect. E.g. some mass balances have a resolution of 1 g, some have a resolution of 0.1 g, and some are even more sensitive.

Calibration

Equipment also needs to be calibrated by measuring a known value. If there's a difference between the measured and known value, you can use this to correct the inaccuracy of the equipment.

Look out for **Errors** and **Anomalous Results**

Type of error	Definition	Example	How to correct
Random	Unpredictable differences in results caused by things like human errors in measuring.	When reading from a burette, you have to estimate or round the level if it's between two marks.	By taking repeat readings and finding the mean.
Systematic	When measurements are wrong by a consistent amount every time.	Measuring from the end of a ruler each time, rather than from 0 cm.	By accounting for them, e.g. adding the extra length to each measurement.

A systematic error caused by using equipment that isn't zeroed properly is called a ZERO ERROR. E.g. if a mass balance always reads 1 gram with nothing on it, all your measurements will be 1 gram too heavy. You can correct this error by subtracting 1 gram from all your results for mass.

If there's no systematic error, then doing repeats and calculating a mean can make your results more accurate.

An ANOMALOUS RESULT is one that doesn't fit in with the rest of your data at all. You should investigate it and try to work out what happened. If you can work out what happened (e.g. you measured something totally wrong) you can ignore it when processing your results.

Errors in your data can come from all kinds of sources...

It might sound weird, but sometimes data can be really precise but not very accurate. For example, a fancy piece of lab equipment might give results that are really precise, but if it's not been calibrated properly those results won't be accurate.

Processing Data

*Processing your data means doing some **calculations** with it to make it more **useful**...*

Data Needs to be **Organised**

- Tables are dead useful for organising data.
- When you draw a table, use a ruler and make sure each column has a heading (including the units).
- Here's a table about the kinds of biscuits eaten by Cookies and Goods Production Ltd. employees:

The green row is the header row

Categories of biscuit go down the left side

Biscuit	Consumption (bpd)				
	Day 1	Day 2	Day 3	Day 4	Day 5
Bourbon	234	221	182	270	261
Custard cream	351	342	295	309	342
Pink wafer	121	137	97	82	101

Units — here it's biscuits per day (obviously)

Repeats go in the header row under a main header

You input your results in these rows underneath the header

You Might Have to **Process Your Data**

- When you've done repeats of an experiment you should always calculate the mean. To do this add together all the data values and divide by the total number of values in the sample.
- You might also need to calculate the range (how spread out the data is). To do this find the largest value and subtract the smallest value from it.
- You should leave out anomalous results when calculating the mean and the range.

When people say to find the 'average', they usually mean the mean.

EXAMPLE

The results of an experiment to find the mass of gas lost from two reactions are shown below. Calculate the mean and the range for the mass of gas lost in each reaction.

MATHS SKILLS

Test tube	Mass of gas lost (g)			Mean (g)	Range (g)
	Repeat 1	Repeat 2	Repeat 3		
A	28	37	32	(28 + 37 + 32) ÷ 3 = 32.33... = 32 (2 s.f.)	37 – 28 = 9
B	47	51	60	(47 + 51 + 60) ÷ 3 = 52.66... = 53 (2 s.f.)	60 – 47 = 13

Round to the **Lowest Number** of **Significant Figures**

The first significant figure of a number is the first digit that's not zero. The second and third significant figures come straight after (even if they're zeros). You should be aware of significant figures in calculations.

1) In any calculation, you should round to the lowest number of significant figures (s.f.) given.
2) Remember to write down how many significant figures you've rounded to after your answer.
3) If your calculation has multiple steps, only round the final answer, or it won't be as accurate.

EXAMPLE

The volume of one mole of gas is 24.0 dm^3 at room temperature and pressure. How many moles are there in 4.6 dm^3 of gas under the same conditions?

MATHS SKILLS

No. of moles of gas = 4.6 dm^3 ÷ 24.0 dm^3 = 0.19166... = 0.19 mol (2 s.f.)

4.6: 2 s.f. 24.0: 3 s.f.

Final answer should be rounded to 2 s.f.

Processing your data can make it easier to interpret

In the exam you could be given some data and be expected to process it in some way. Make sure you write down how many significant figures you've rounded your final answer to, just like writing down the rest of your working out.

Presenting Data

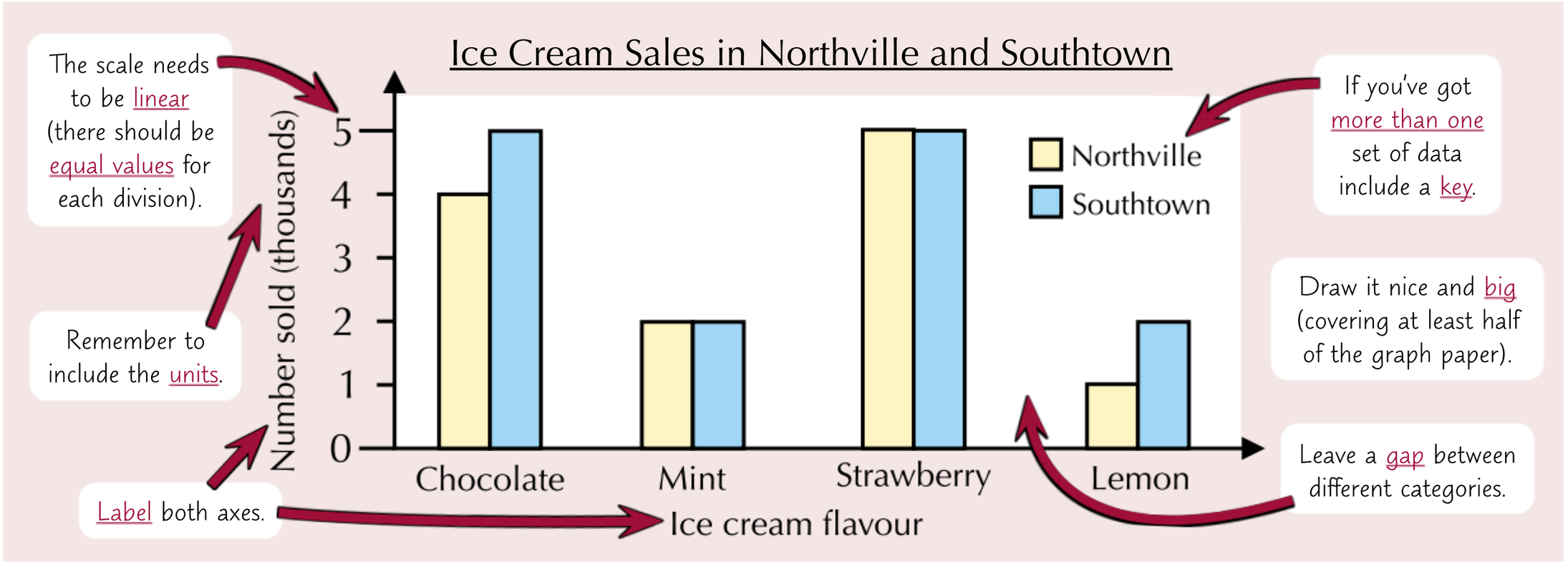
Once you've processed data, you can present your results in a nice ***chart*** *or* ***graph*** *to help you* ***spot any patterns*** *in your data. The type of data you record will affect how you should* ***present*** *it...*

If Your Data Comes in **Categories**, Present It in a **Bar Chart**

You should represent your data with a bar chart if your independent variable is:

- CATEGORIC — data that comes in distinct categories (e.g. solid, liquid, gas).
- DISCRETE — data that can be counted in chunks, where there's no in-between value (e.g. number of protons — you can't have half a proton).

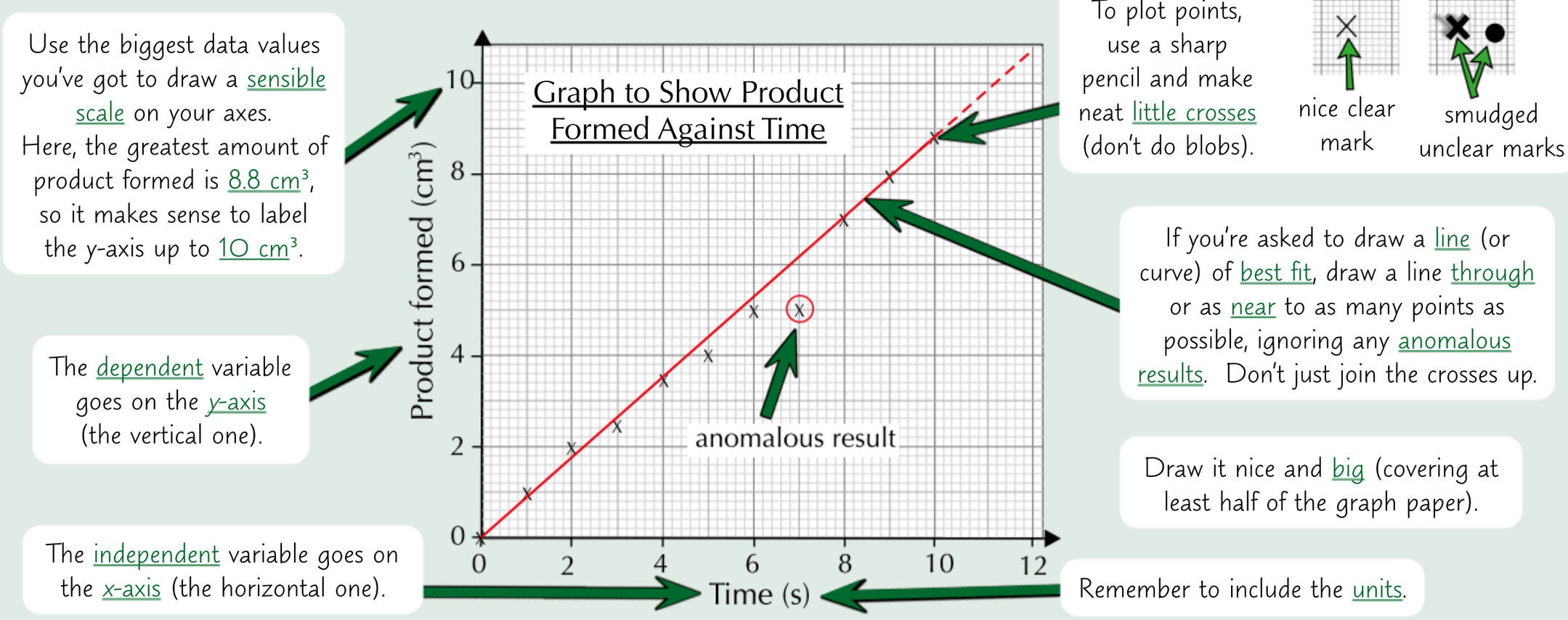
There are some golden rules you need to follow for drawing bar charts:



If Your Data is **Continuous**, Plot a **Graph**

If both variables are continuous (numerical data that can have any value within a range, e.g. length, volume, temperature) you should use a graph to display the data.

Here are the rules for plotting points on a graph:



Make sure you choose the right presentation method for your data

Remember that if your data comes in distinct categories, you should present it in a bar chart. If your data is continuous, present it in a graph. Don't get the two presentation methods muddled up, or you could lose marks in the exam.

Interpreting Graphs

*Graphs aren't just fun to plot, they're also really useful for showing **trends** in your data.*

Graphs Can Give You a Lot of **Information** About Your **Data**

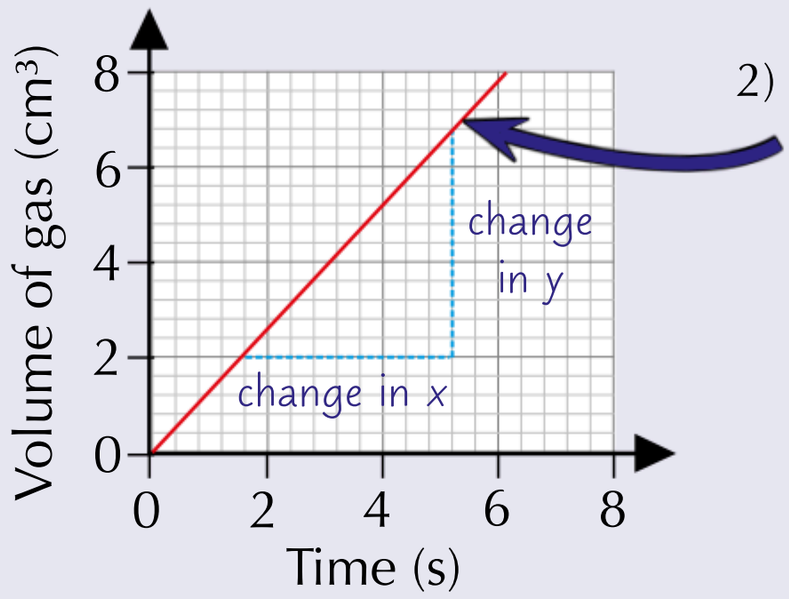
The gradient (slope) of a graph tells you how quickly the dependent variable changes if you change the independent variable.

$$\text{gradient} = \frac{\text{change in } y}{\text{change in } x}$$

This graph shows the volume of gas produced in a reaction against time. The graph is linear (it's a straight line graph), so you can calculate the gradient of the line to find out the rate of reaction.

1) To calculate the gradient, pick two points on the line that are easy to read and a good distance apart.

You can use this method to calculate other rates from a graph, so long as you always make x time.



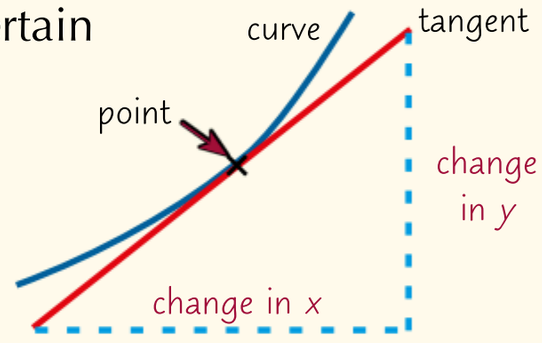
2) Draw a line down from the top point and a line across from the bottom point to make a triangle. The line drawn down the side of the triangle is the change in y and the line across the bottom is the change in x.

The units of the gradient are (units of y) ÷ (units of x).

$$\text{Rate} = \text{gradient} = \frac{\text{change in } y}{\text{change in } x} = \frac{6.8 - 2}{5.2 - 1.6} = \frac{4.8\text{ m}}{3.6\text{ s}} = 1.33\ldots = 1.3\text{ cm}^3/\text{s}\ (2\text{ s.f.})$$

The intercept of a graph is where the line of best fit crosses one of the axes. The x-intercept is where the line of best fit crosses the x-axis and the y-intercept is where it crosses the y-axis.

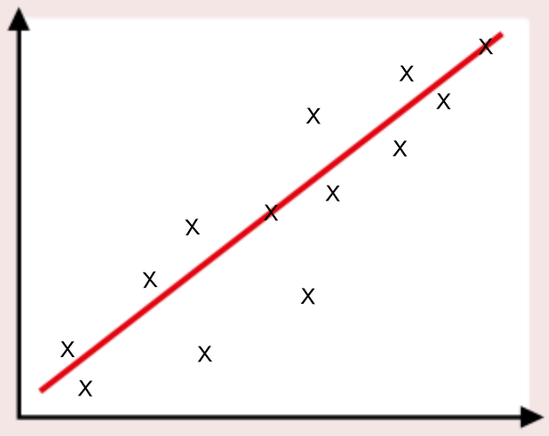
To find the gradient of a curve at a certain point, draw a tangent to the curve at that point. This is a straight line that touches the curve at that point only. Then just find the gradient of the tangent in the same way as above.



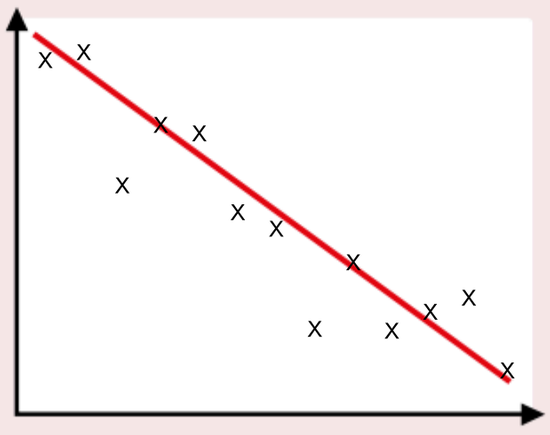
Graphs Show the **Relationship** Between Two Variables

You can get three types of correlation (relationship) between variables:

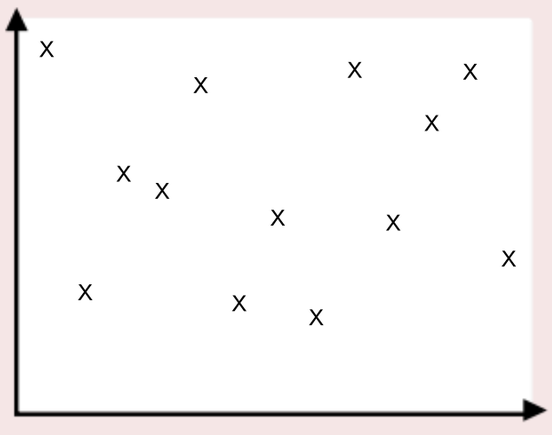
POSITIVE CORRELATION:
as one variable increases the other increases.



INVERSE (NEGATIVE) CORRELATION:
as one variable increases the other decreases.



NO CORRELATION:
no relationship between the two variables.



Just because there's correlation, it doesn't mean the change in one variable is causing the change in the other — there might be other factors involved (see page 149).

Draw a line of best fit to spot correlation in your data

If your graph's line of best fit has a positive gradient (the line goes up from left to right), then there's a positive correlation between your variables. If the gradient's negative (the line goes down), then there's a negative correlation between them.

Units

*You can **convert units** using **scaling prefixes**. This can save you from having to write loads and loads of 0s...*

S.I. Units Are Used All Round the World

It wouldn't be all that useful if I defined volume in terms of bath tubs, you defined it in terms of egg-cups and my pal Sarwat defined it in terms of balloons — it'd make it really hard to compare our data.

To stop this happening, scientists came up with a set of standard units, called S.I. units, that all scientists use to measure their data. Here are some S.I. units you'll see in chemistry:

Quantity	S.I. Base Unit
mass	kilogram, kg
length	metre, m
time	second, s
amount of a substance	mole, mol

Scaling Prefixes Can Be Used for Large and Small Quantities

1) Quantities come in a huge range of sizes. For example, the volume of a swimming pool might be around 2 000 000 000 cm^3, while the volume of a cup is around 250 cm^3.
2) To make the size of numbers more manageable, larger or smaller units are used. These are the S.I. base unit (e.g. metres) with a prefix in front:

prefix	tera (T)	giga (G)	mega (M)	kilo (k)	deci (d)	centi (c)	milli (m)	micro (μ)	nano (n)
multiple of unit	10^{12}	10^{9}	1 000 000 (10^{6})	1000	0.1	0.01	0.001	0.000001 (10^{-6})	10^{-9}

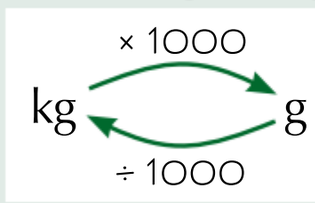
3) These prefixes tell you how much bigger or smaller a unit is than the base unit. So one kilometre is one thousand metres.
4) To convert between units, you need to know the conversion factor:

The conversion factor is the number of times the smaller unit goes into the larger unit.

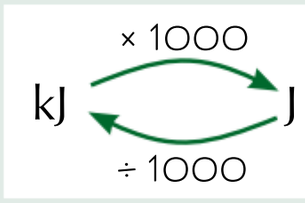
- To go from a bigger unit (like m) to a smaller unit (like cm), you multiply by the conversion factor.
- To go from a smaller unit (like g) to a bigger unit (like kg), you divide by the conversion factor.

5) Here are some conversions that'll be useful for GCSE chemistry:

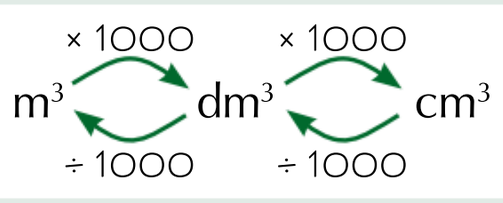
Mass can have units of kg and g.



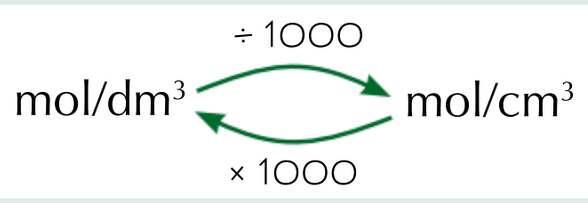
Energy can have units of J and kJ.



Volume can have units of m^3, dm^3 and cm^3.



Concentration can have units of mol/dm^3 and mol/cm^3.



Always Check The Values Used in Equations Have the Right Units

Always make sure the values you put into an equation or formula have the right units. In some cases you'll have to convert a value into different units before you can put it into the equation.

To make sure your units are correct, it can help to write them down on each line of your calculation.

EXAMPLE

10 g of sodium chloride is dissolved in 250 cm^3 of water. Calculate the concentration of the sodium chloride solution in g/dm^3.

MATHS SKILLS

1) Convert the volume measurement from cm^3 into dm^3 by dividing by the correct conversion factor.

250 cm^3 —÷ 1000→ 0.25 dm^3

2) Find the concentration using:
concentration = mass of solute ÷ volume of solvent.

concentration = 10 g ÷ 0.25 dm^3 = 40 g/dm^3

To convert between SI units, multiply or divide by 1000

It's easy to get in a muddle when converting between units, but there's a handy way to check you've done it right. If you're moving from a smaller unit to a larger unit (e.g. g to kg) the number should get smaller, and vice versa.

Maths Skills

No, you've not opened up the wrong book — this is indeed a page on ***maths****. Maths comes up* ***a lot*** *in chemistry, so you'd better be* ***comfortable*** *using it. Read on for some of the* ***key*** *maths skills you need...*

Standard Form Is Often Used For **Really Big** or **Really Small** Numbers

You need to know how to deal with numbers in standard form. They always look like this:

A is a number between 1 and 10, but not including 10.

$$A \times 10^{n}$$

n is the number of places the decimal point would move if you wrote the number out fully. It's negative for numbers less than 1, and positive for numbers greater than 1.

So 200 000 cm^3 would be written as 2×10^5 cm^3 in standard form. And 0.0000038 m is 3.8×10^{-6} m.

Do the **Same Thing** to **Both Sides** When **Rearranging Formulas**

Often you need to rearrange a formula to be able to solve for the variable you want.
You just need to do the same thing to each side until you get that variable on its own.

EXAMPLE

Rearrange the formula on the right to make the number of moles the subject.

$$\text{concentration} = \frac{\text{number of moles}}{\text{volume}}$$

1) You want to get the number of moles term on its own.

$$\text{concentration} = \frac{\text{number of moles}}{\text{volume}}$$

2) Multiply both sides by the volume.

$$\text{concentration} \times \text{volume} = \frac{\text{number of moles}}{\cancel{\text{volume}}} \times \cancel{\text{volume}}$$

3) Simplify to tidy things up.

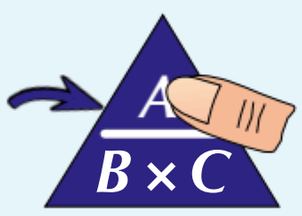
$$\text{concentration} \times \text{volume} = \text{number of moles}$$

4) Flip things around so the number of moles is on the left — ta da!

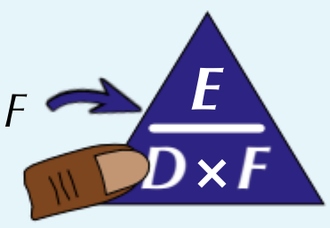
$$\text{number of moles} = \text{concentration} \times \text{volume}$$

You can put simple formulas into formula triangles if you find them helpful:

$A = B \times C$



$D = E \div F$



To use them, cover up what you want to find and write down what's left showing.

Make Sure You Know How to Use Your **Calculator** Correctly

Here are a few things to remember to make sure you don't lose marks for silly calculation mistakes:

- You need to make sure that what you put in your calculator is the same calculation as is on the page, even if it looks a bit different.
- For example, to put $\frac{11.5 + 6.8}{3}$ into any standard calculator, you type (11.5 + 6.8) ÷ 3. Don't forget the brackets.
  You could also use the fraction button on a scientific calculator.

Your calculator will follow the order of operations for what you've typed. If you forget the brackets, it won't do the calculation you want and you'll get 13.766... instead of 6.1.

Use exact values from previous calculations in your calculator rather than rounding (p.144). Make sure you know how to find these values on your calculator — e.g. by using the 'Ans' button or scrolling up.

Rearranging formulas is a really important skill...

Some people love formula triangles, some people not so much. You can use them as a tool if you want to, but you need to know how to rearrange formulas properly, not least because complicated formulas don't work in triangles.

Drawing Conclusions

Once you've carried out an experiment and processed your data, it's time to work out ***what your data means****...*

You Can **Only Conclude** What the Data Shows and **NO MORE**

1) Drawing conclusions might seem pretty straightforward — you just look at your data and say what pattern or relationship you see between the dependent and independent variables...

The table on the right shows the rate of a reaction in the presence of two different catalysts:

Catalyst	Rate of reaction (cm^3/s)
A	13.5
B	19.5
No catalyst	5.5

CONCLUSION: Catalyst B makes this reaction go faster than catalyst A.

2) ...but you've got to be really careful that your conclusion matches the data you've got and doesn't go any further:

You can't conclude that catalyst B increases the rate of any reaction more than catalyst A — the results might be completely different for other reactions.

3) You also need to be able to use your results to justify your conclusion (i.e. back up your conclusion with some specific data):

E.g. the rate of this reaction was 6 cm^3/s faster using catalyst B compared with catalyst A.

4) When writing a conclusion you need to refer back to the original hypothesis and say whether the data supports it or not:

If the hypothesis for this experiment was that catalyst B would make the reaction go quicker than catalyst A, then the data supports this.

Correlation DOES NOT Mean **Cause**

If two things are correlated (i.e. there's a relationship between them) it doesn't necessarily mean a change in one variable is causing the change in the other — this is really important, so don't forget it. There are three possible reasons for a correlation:

1 CHANCE: It might seem strange, but two things can show a correlation purely due to chance:

E.g. a study might find a correlation between people's hair colour and their frisbee skills. But, other scientists who investigate it don't get a correlation — the results of the first study are just a fluke.

2 LINKED BY A 3RD VARIABLE: A lot of the time it may look as if a change in one variable is causing a change in the other, but it isn't — a third variable links the two things:

E.g. there's a correlation between a higher water temperature and the amount of shark attacks. This isn't because warmer water makes sharks go crazy. The attacks are linked by a third variable — the number of people swimming (i.e. more people are swimming when the water's hotter, meaning more chance of a shark attack).

3 CAUSE: Sometimes a change in one variable does cause a change in the other. You can only make this conclusion when you've controlled all variables that could be affecting the result:

E.g. there is a correlation between smoking and lung cancer. This is because chemicals in tobacco smoke cause lung cancer. This conclusion was only made once other variables (such as age and exposure to other things that cause cancer) had been controlled. Once they were controlled, the effect of smoking was isolated and shown to increase the risk of someone getting lung cancer.

Make sure your conclusions match your data...

It can be easy to overextend your conclusions and claim things that aren't actually shown by your data. Make sure to double check that your conclusion is within the range of your results, and ensure that you can justify it with your data.

Uncertainties and Evaluations

Uncertainties *tell you how sure you can be about your data.* ***Evaluating*** *your investigation helps you improve it.*

Uncertainty is the Amount of Error Your Measurements Might Have

1) The measurements you make will have some uncertainty in them — for example, due to limits in the resolution of the equipment you use (see page 143).
2) When you repeat a measurement, you often get a slightly different figure each time you do it due to random error. So random errors are another source of uncertainty — they can cause uncertainty in the mean of a set of results.
3) You can calculate the uncertainty of a mean result using the equation:
4) The larger the range, the less precise your results are and the more uncertainty there will be in your results. Uncertainties are shown using the '±' symbol.

The range is the largest value minus the smallest value (p.144).

$$\text{uncertainty} = \frac{\text{range}}{2}$$

EXAMPLE

The table below shows the results of a titration experiment to determine the volume of 0.5 mol/dm³ sodium hydroxide solution needed to neutralise 25 cm³ of a solution of hydrochloric acid with unknown concentration. Calculate the uncertainty of the mean.

MATHS SKILLS

Repeat	1	2	3	mean
Volume of sodium hydroxide (cm³)	20.10	19.80	20.10	20.00

1) First work out the range:
Range = 20.10 – 19.80
= 0.30 cm³

2) Use the range to find the uncertainty:
Uncertainty = range ÷ 2 = 0.30 ÷ 2 = 0.15 cm³

So the uncertainty of the mean = 20.00 ± 0.15 cm³

5) Measuring a greater amount of something helps to reduce uncertainty. For example, in a rate of reaction experiment, measuring the amount of product formed over a longer period compared to a shorter period will reduce the percentage uncertainty in your results.

Evaluations — Describe How it Could be Improved

1) An evaluation is a critical analysis of the whole investigation:

- You should comment on the method — was it valid? Did you control all the other variables to make it a fair test?
- Comment on the quality of the results — was there enough evidence to reach a valid conclusion? Were the results repeatable, reproducible, accurate and precise?
- Were there anomalous results? If there were, try to explain them — were they caused by errors in measurement? Were there other variables that could have affected the results? You should comment on the level of uncertainty in your results too.

2) Answering these questions will allow you to say how confident you are that your conclusion is right.
3) Then you can suggest any changes to the method that would improve the quality of the results and give you more confidence in your conclusion. E.g. you might increase the number of measurements you took. Taking more measurements at narrower intervals could give you a more accurate result:

Enzymes have an optimum temperature (a temperature at which they work best). You could do an experiment to find an enzyme's optimum temperature and take measurements at 10 °C, 20 °C, 30 °C, 40 °C and 50 °C. Say the results tell you the optimum is 40 °C — you could then repeat the experiment, taking more measurements around 40 °C to get a more accurate value.

4) You could also make more predictions based on your conclusion, then further experiments could be carried out to test them.

When suggesting improvements to the investigation, always make sure that you say why you think this would make the results better.

Always look for ways to improve your investigations

So there you have it — Working Scientifically. Make sure you know this stuff like the back of your hand. It's not just in the lab that you'll need to know how to work scientifically. You can be asked about it in the exams as well. So swot up...

Practical Techniques

Safety specs out and lab coats on, it's time to find out about the skills you'll need in ***experiments****...*

Solids Should Be Measured Using a **Balance**

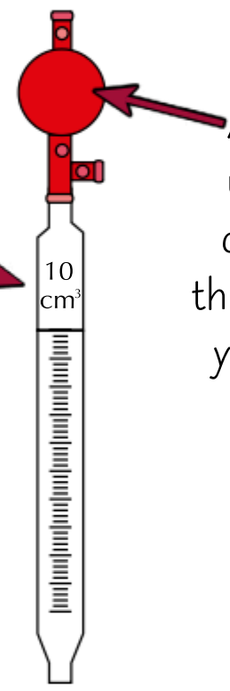
- To weigh a solid, start by putting the container you're weighing your substance into on the balance. Set the balance to exactly zero, then weigh out your substance. Easy-peasy.
- It's no good carefully weighing out your solid if you then just chuck it in your reaction vessel — some of it could get left behind. Here are a couple of methods you can use to make sure you know how much has made it into your experiment...

1) If you're dissolving a solid in a solvent to make a solution, you could wash any remaining solid into the new container using the solvent.
2) You could set the balance to zero before you put your weighing container on the balance. Then reweigh the weighing container after you've transferred the solid. Use the difference in mass to work out exactly how much solid you added to your experiment.

Three Ways to Measure **Liquids**:

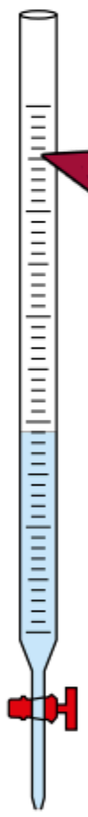
Pipettes

- Used to suck up an accurate volume of liquid and transfer it to another container.
- Often calibrated to allow for the fact that the last drop of liquid stays in the pipette when the liquid is ejected. This reduces transfer errors.



A pipette filler is used so that you can safely control the amount of liquid you're drawing up.

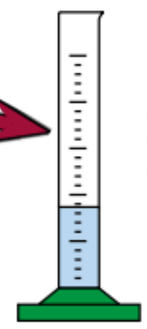
Burettes



- Measure from top to bottom (so the top of the scale reads zero).
- Have a tap at the bottom which you can use to release the liquid into another container (you can even release it drop by drop).
- To use it, take an initial reading, release as much liquid as you want, then take a final reading. The difference between the readings tells you how much liquid you used.

Measuring cylinders

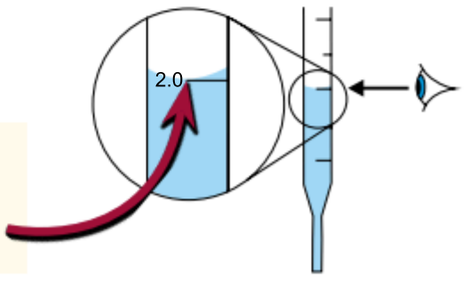
- The most common way to measure out a liquid.
- Come in all different sizes.



Make sure you choose the right size of measuring cylinder for the measurement you want to make. E.g. if you tried to use a 1000 cm^3 cylinder to measure out 2 cm^3 of a liquid, the graduations would be too big, so you'd end up with massive errors. It'd be much better to use a 10 cm^3 cylinder.

If you only want a couple of drops of liquid, and don't need it to be accurately measured, you can use a dropping pipette to transfer it. E.g. this is how you'd add a couple of drops of indicator into a mixture.

Whichever method you use, always read the volume from the bottom of the meniscus (the curved upper surface of the liquid) when it's at eye level.



Gas Syringes Measure Gas Volumes

- Gases can be measured with a gas syringe. The volume of a gas changes with temperature and pressure, so you should measure it at room temperature and pressure.
- You should also use a gas syringe that's the right size for the measurement you're making.
- Before you use the syringe, you should make sure it's completely sealed and that the plunger moves smoothly.

Practical Techniques

Measure **Temperature** Accurately

You can use a thermometer to measure the temperature of a substance:

1) Make sure the bulb of your thermometer is completely submerged in any mixture you're measuring.
2) If you're taking an initial reading, you should wait for the temperature to stabilise first.
3) Read your measurement off the scale on a thermometer at eye level to make sure it's correct.

You May Have to Measure the **Time Taken** for a Change

- You should use a stopwatch to time experiments. These measure to the nearest 0.1 s so are sensitive.
- Make sure you start and stop the stopwatch at exactly the right time. For example, if you're investigating the rate of a reaction, you should start timing at the exact moment you mix the reagents.

Measure **pH** to Find Out How **Acidic** or **Alkaline** a Solution Is

You need to be able to decide the best method for measuring pH, depending on what your experiment is.

Indicators	Dyes that change colour depending on whether they're in an acid or an alkali. You use them by adding a couple of drops of the indicator to the solution you're interested in. They're useful for titration reactions, when you want to find the point at which a solution is neutralised.
Universal indicator	A mixture of indicators that changes colour gradually as pH changes. It doesn't show a sudden colour change. It's useful for estimating the pH of a solution based on its colour.
Indicator paper	Useful when you don't want to change the colour of the substance, or if the substance is already coloured so might obscure the indicator colour. You use it by dropping a small amount of a solution onto the paper — it will change colour depending on the pH of the solution.
pH probes	Have a digital display that gives a numerical value for the pH of a solution. They're used to give an accurate value of pH.

There's loads more about pH on page 58.

Litmus paper turns red in acidic conditions and blue in basic conditions. Universal indicator paper can be used to estimate the pH based on its colour.

Be Careful When You **Handle** or **Mix** Substances

- There are lots of hazards in chemistry experiments, so before you start any experiment, you should read any safety precautions to do with your method or the chemicals you're using.
- The substances used in chemical reactions are often hazardous. For example, they might catch fire easily (they're flammable), or they might irritate or burn your skin if you come into contact with them.
- Whenever you're doing an experiment, you should wear a lab coat, safety goggles and gloves.
- Always be careful that the chemicals you're using aren't flammable before you go lighting any Bunsen burners, and make sure you're working in an area that's well ventilated.
- If you're doing an experiment that might produce nasty gases (such as chlorine), you should carry out the experiment in a fume hood so that the gas can't escape out into the room you're working in.
- Never directly touch any chemicals (even if you're wearing gloves). Use a spatula to transfer solids between containers. Carefully pour liquids between different containers, using a funnel to avoid spillages.
- Be careful when you're mixing chemicals, as a reaction might occur. If you're diluting a liquid, add the concentrated substance to the water (not the other way around) or the mixture could get very hot.

Have your stopwatch ready before you try to time something

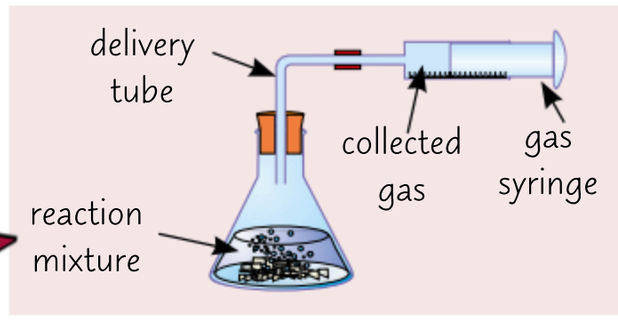
To make sure your results are reproducible and can be trusted by other scientists, all your measurements need to be accurate. Learn the tips on this page and the previous one and you'll be achieving lovely accurate results in no time.

Setting Up Equipment

*Setting up the equipment for an experiment correctly is **just as important** as making accurate measurements.*

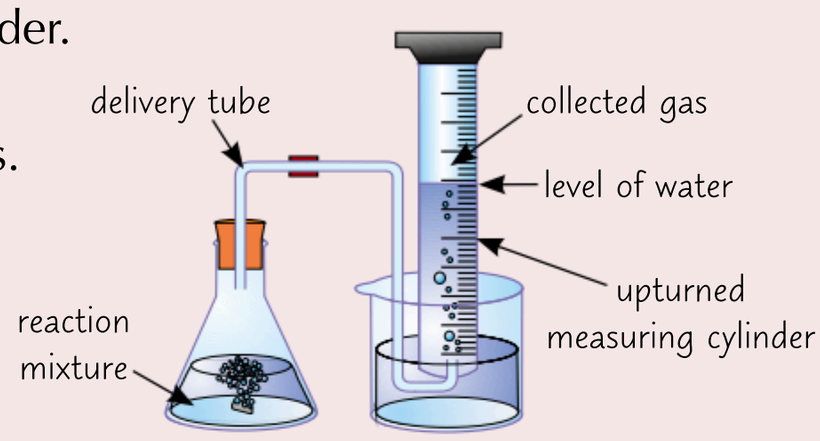
To Collect **Gases**, the System Needs to be **Sealed**

- There are times when you might want to collect the gas produced by a reaction. For example, to investigate the rate of a reaction.
- The most accurate way to measure the volume of a gas that's being produced is to collect it in a gas syringe (see page 151).



- You could also collect it by displacing water from a measuring cylinder. Here's how you do it...

1) Fill a measuring cylinder with water and carefully place it upside down in a container of water. Record the initial level of the water in the measuring cylinder.
2) Position a delivery tube coming from the reaction vessel so that it's inside the measuring cylinder, pointing upwards. Any gas that's produced will pass through the tube and into the measuring cylinder. This pushes the water out.
3) Record the level of water in the measuring cylinder again. The volume of gas produced is the difference between this value and the initial value.



If the delivery tube isn't completely inside the measuring cylinder, some of the gas might escape out into the air.

- This method is less accurate than using a gas syringe. This is because some gases can dissolve in water, so less gas ends up in the measuring cylinder than is actually produced.
- If you just want to collect a sample to test (and don't need to know the volume), you can collect it over water, as above, using a test tube. Once the test tube is full of gas, stopper it to store the gas.

You May Have to **Identify** the **Products** of Electrolysis

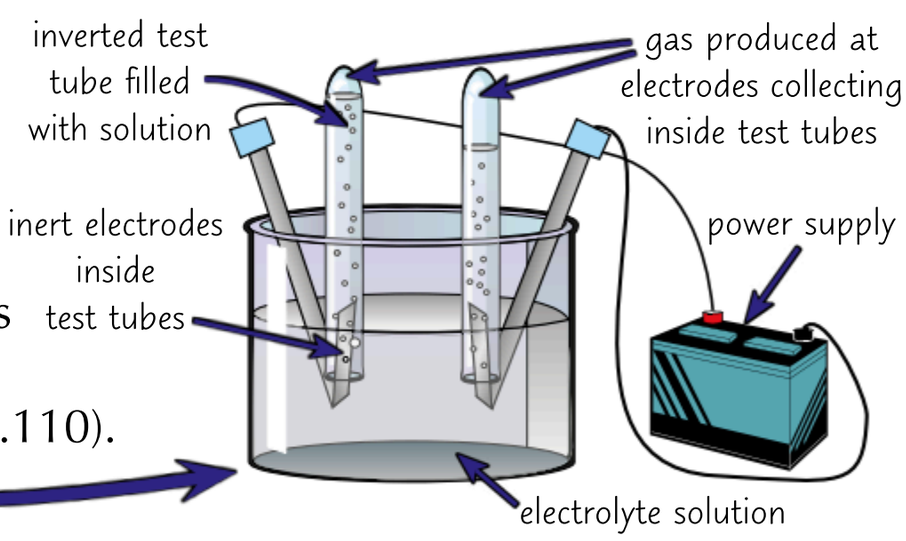
There's more about electrolysis on p.68-69.

1) When you electrolyse an aqueous solution, the products of electrolysis depend on how reactive the ions in the solution are compared to the H^+ and OH^- ions that come from water.

At the cathode, you'll either get a pure metal coating the electrode or bubbles of hydrogen gas.

At the anode, you'll get bubbles of oxygen gas unless a halide ion is present, when you'll get the halogen.

2) You may have to identify what's been made in an electrolysis experiment. To do this, you can collect any gases that are produced using test tubes and test them (using the tests on p.110).
3) Here's how you could set up the equipment to do that.



Make Sure You Can **Draw Diagrams** of Your Equipment

When you're writing out a method for your experiment, it's always a good idea to draw a labelled diagram showing how your apparatus will be set up. The easiest way to do this is to use a scientific drawing, where each piece of apparatus is drawn as if you're looking at its cross-section.

For example:

The pieces of glassware are drawn without tops, so they aren't sealed. If you want to draw a closed system, remember to draw a bung in the top.

You can test the gas you've collected to find out what it is

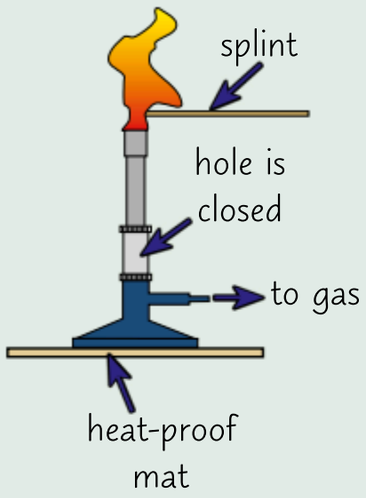
Having a good knowledge of practical techniques won't just make your investigations more reliable, it might come in handy for your exams. You could be asked to comment on how equipment has been set up (e.g. suggest improvements).

Heating Substances

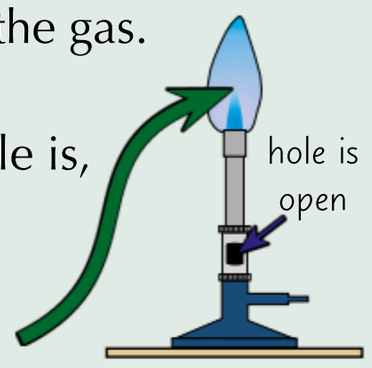
Heating a reaction isn't as simple as wrapping it up in a blanket and sitting it by the fireplace. There's more than one way to do it, and you need to be able to decide on the ***best****, and the* ***safest****, method.*

Bunsen Burners Have a Naked **Flame**

Bunsen burners are good for heating things quickly and you can easily adjust how strongly they're heating. But you need to use them safely:



1) Connect the Bunsen burner to a gas tap, and check that the hole is closed. Place it on a heat-proof mat.
2) Light a splint and hold it over the Bunsen burner. Now, turn on the gas. The Bunsen burner should light with a yellow flame.
3) Open the hole to the amount you want — the more open the hole is, the more strongly the Bunsen burner will heat your substance.
4) As you open the hole, the flame should turn blue. The hottest part is just above the blue cone, so you should heat things there.



- If your Bunsen burner is alight but not heating anything, make sure you close the hole so that the flame becomes yellow and clearly visible.
- If you're heating something in a container (e.g. a test tube) in the flame, you should hold the vessel at the top, furthest away from the substance (and so the flame) using a pair of tongs.
- If you're heating something over the flame (e.g. an evaporating dish), you should put a tripod and gauze over the Bunsen burner before you light it, and place the vessel on this.
- Don't use a Bunsen burner to heat flammable compounds — the flame means the substance would be at risk of catching fire.

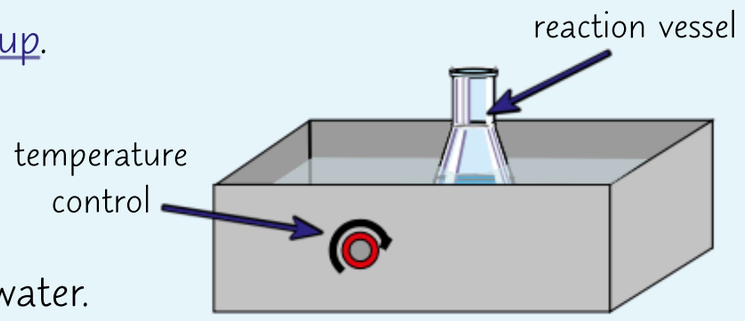
You'd use a Bunsen burner to carry out flame tests to identify metal ions in a compound (see p. 111). Place a sample of the compound on a metal wire and hold it just above the cone of the blue flame. The flame should change colour depending on what metal ion is in the sample.

The **Temperature** of **Water Baths** & **Electric Heaters** Can Be **Set**

1) A water bath is a container filled with water that can be heated to a specific temperature.

Method

1) Set the temperature on the water bath, and allow the water to heat up.
2) Place the vessel containing your substance in the water bath using a pair of tongs. The level of the water outside the vessel should be just above the level of the substance inside the vessel.
3) The substance will then be warmed to the same temperature as the water.



As the substance in the vessel is surrounded by water, the heating is very even. Water boils at 100 °C though, so you can't use a water bath to heat something to a higher temperature than this — the water won't get hot enough.

Handle any glassware you've heated with tongs until you're sure it's cooled down.

2) Electric heaters are often made up of a metal plate that can be heated to a certain temperature. The vessel containing the substance you want to heat is placed on top of the hot plate. You can heat substances to higher temperatures than you can in a water bath but, as the vessel is only heated from below, you'll usually have to stir the substance inside to make sure it's heated evenly.

You might not expect the blue flame to be hotter, but it is

Always be careful when heating things and make sure you choose the most appropriate method to heat your reaction mixture — e.g. use a water bath if you need to heat a flammable substance that might be ignited by a Bunsen burner.

Practice Exams

Once you've been through all the questions in this book, you should feel pretty confident about the exams. As final preparation, here is a set of **practice exams** to really get you set for the real thing. The time allowed for each paper is 1 hour 45 minutes. These papers are designed to give you the best possible preparation for your exams.

CGP Practice Exam Paper GCSE Chemistry

GCSE Chemistry

Paper 1

Higher Tier

In addition to this paper you should have:
- A ruler.
- A calculator.
- A periodic table.

Centre name					
Centre number					
Candidate number					

Time allowed:
- 1 hour 45 minutes

Surname
Other names
Candidate signature

Instructions to candidates
- Write your name and other details in the spaces provided above.
- Answer **all** questions in the spaces provided.
- Do all rough work on the paper.
- Cross out any work you do not want to be marked.

Information for candidates
- The marks available are given in brackets at the end of each question.
- There are 100 marks available for this paper.
- You are allowed to use a calculator.
- You should use good English and present your answers in a clear and organised way.

Advice to candidates
- In calculations show clearly how you worked out your answers.

For examiner's use							
Q	Attempt Nº			Q	Attempt Nº		
	1	2	3		1	2	3
1				6			
2				7			
3				8			
4				9			
5							
Total							

1 The elements in Group 1 of the periodic table are known as the alkali metals.

1.1 Group 1 metals can react with non-metals to form ionic compounds.
What is the charge on a Group 1 ion in an ionic compound?
Tick **one** box.

- ☐ +2
- ☐ +3
- ☐ –1
- ☐ +1

[1 mark]

1.2 A white solid is formed when sodium metal reacts with fluorine gas.
What is the chemical formula of this compound?
Tick **one** box.

- ☐ Na_2F
- ☐ NaF
- ☐ NaF_2
- ☐ NaF_3

[1 mark]

A student watched a teacher carefully place small pieces of lithium, sodium and potassium into cold water. The student's observations are recorded in **Table 1**.

Table 1

Metal	Observations
lithium	Fizzes, moves across surface.
sodium	Fizzes strongly, moves quickly across surface, melts.
potassium	Fizzes violently, moves very quickly across surface, melts and a flame is seen.

The student decides that the order of reactivity of the three metals is:

- potassium (most reactive)
- sodium
- lithium (least reactive)

1.3 Give **two** pieces of evidence from **Table 1** that support the student's conclusion.

.....

.....

.....

.....

[2 marks]

1.4 Explain the pattern of reactivity that the student has noticed in terms of the outer electrons of the atoms.

.....

.....

.....

.....

[3 marks]

1.5 Complete and balance the chemical equation for the reaction between lithium and water.

$2Li + 2H_2O \rightarrow$ ..... + ...

[2 marks]

1.6 Choose the statement that explains why the solution produced when lithium reacts with water is alkaline.
Tick **one** box.

- ☐ It contains lithium ions.
- ☐ It contains water.
- ☐ It contains an ionic compound.
- ☐ It contains hydroxide ions.

[1 mark]

Turn over for the next question

2 Atoms contain protons, neutrons and electrons.

2.1 Whose experimental work led to the discovery of neutrons within the nucleus?
Tick **one** box

- ☐ Niels Bohr
- ☐ Ernest Rutherford
- ☐ James Chadwick
- ☐ Ernest Marsden

[1 mark]

2.2 Complete **Table 2** to show the relative charges and relative masses of protons, neutrons and electrons.

Table 2

Particle	Relative mass	Relative charge
Proton	1	
Neutron		
Electron	Very small	

[2 marks]

2.3 **Table 3** shows the numbers of protons, neutrons and electrons present in six different atoms.

Table 3

Atom	Number of protons	Number of neutrons	Number of electrons
A	5	6	5
B	7	7	7
C	6	8	6
D	6	6	6
E	10	10	10
F	4	5	4

Which **two** atoms are isotopes of the same element? Explain your answer.

.....

.....

.....

[2 marks]

2.4 The nuclear symbol for an atom of zinc is shown below.

$$^{64}_{30}Zn$$

How many protons, neutrons and electrons are there in this atom of zinc?

Protons = ...

Neutrons = ...

Electrons = ...

[3 marks]

Atoms can bond with each other to form compounds.

2.5 What is a compound?

.....

.....

[2 marks]

2.6 Iron(II) sulfate is a compound with the formula $FeSO_4$.

Calculate the relative formula mass of iron(II) sulfate.

.....

.....

relative formula mass = ...

[2 marks]

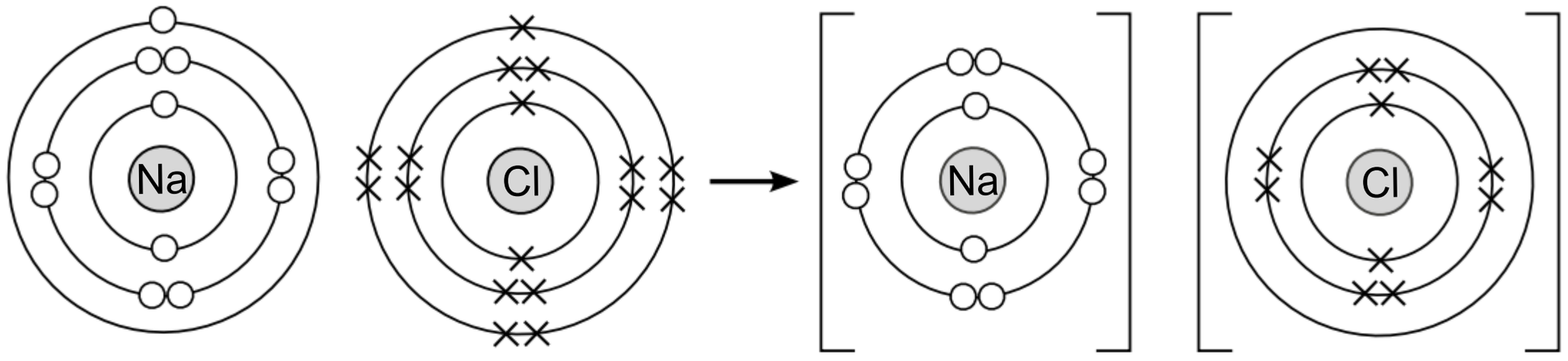
Turn over for the next question

3 The structure and bonding of elements and compounds affects their properties.

3.1 Sodium and chlorine can react together to form an ionic compound (sodium chloride).
Figure 1, below, is a dot and cross diagram illustrating this reaction.

Complete the right-hand side of **Figure 1** by adding the charges of both ions and adding the electrons to the outer shell of the chloride ion.

Figure 1



[2 marks]

3.2 Describe the structure of sodium chloride.
Include a description of the forces that hold the structure together.

.....

.....

.....

[2 marks]

Table 4 shows some properties of five substances, **A-E**.

Table 4

Substance	Melting point in °C	Boiling point in °C	Does it conduct electricity when solid?	Does it conduct electricity when dissolved or molten?
A	–210	–196	No	No
B	–219	–183	No	No
C	801	1413	No	Yes
D	115	445	No	No
E	1083	2567	Yes	Yes

3.3 Substance **A** consists of small, covalently bonded molecules.
Explain why it has a relatively low melting point.

.....

.....

[2 marks]

3.4 Look at **Table 4**. One of the substances, **A**-**E**, is an ionic compound.
Use the information in the table to suggest which of the substances is ionic.
Explain your answer.

.....

.....

.....

[2 marks]

3.5 **Table 5** contains information about some of the properties of diamond and graphite.

Table 5

	Hardness	**Melting point**	**Conducts electricity?**
Diamond	Hard	High	No
Graphite	Soft	High	Yes

Explain these properties of diamond and graphite
in terms of their structure and bonding.

.....

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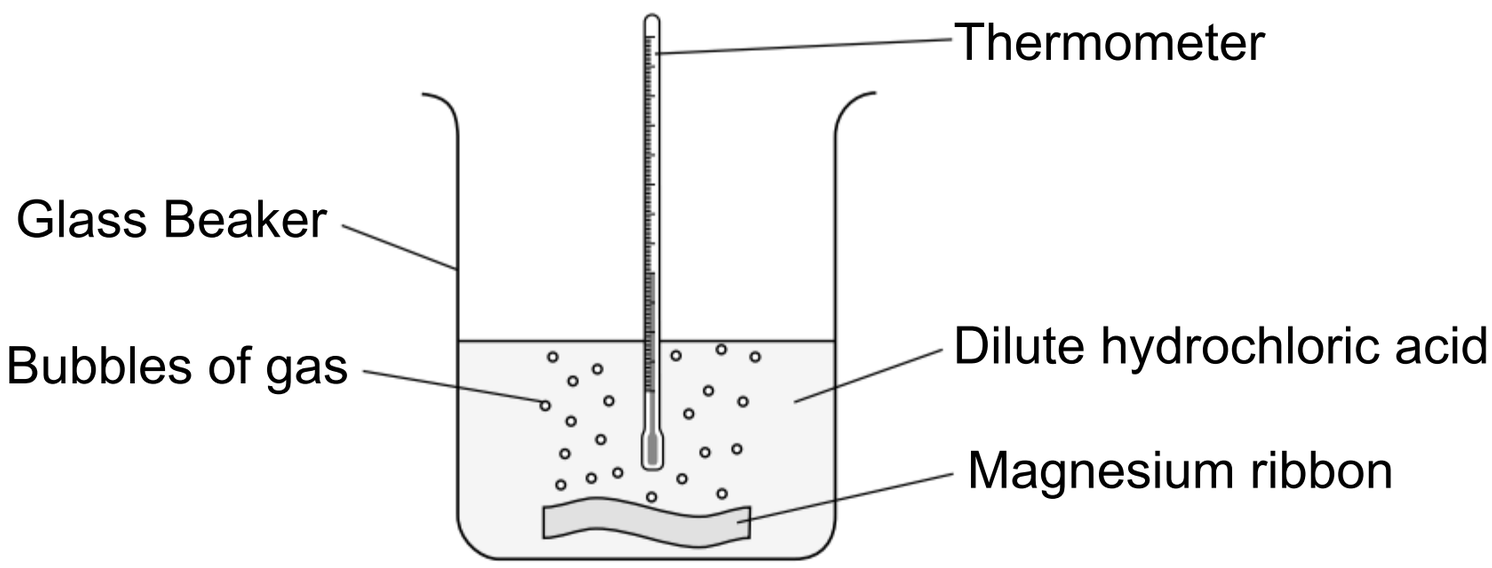
[6 marks]

Turn over for the next question

4 A student added a piece of magnesium ribbon to dilute hydrochloric acid and measured the temperature change that occurred. **Figure 2** shows the apparatus the student used.

Figure 2



4.1 Suggest **two** changes that the student could make to the apparatus in order to reduce heat loss from the experiment.

.....

.....

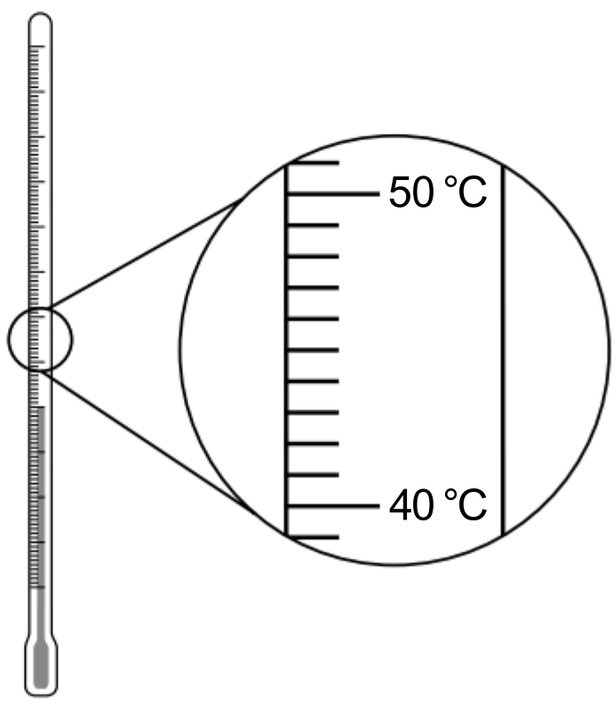
.....

.....

[2 marks]

4.2 A close up of the thermometer used during the experiment is shown in **Figure 3.**

Figure 3



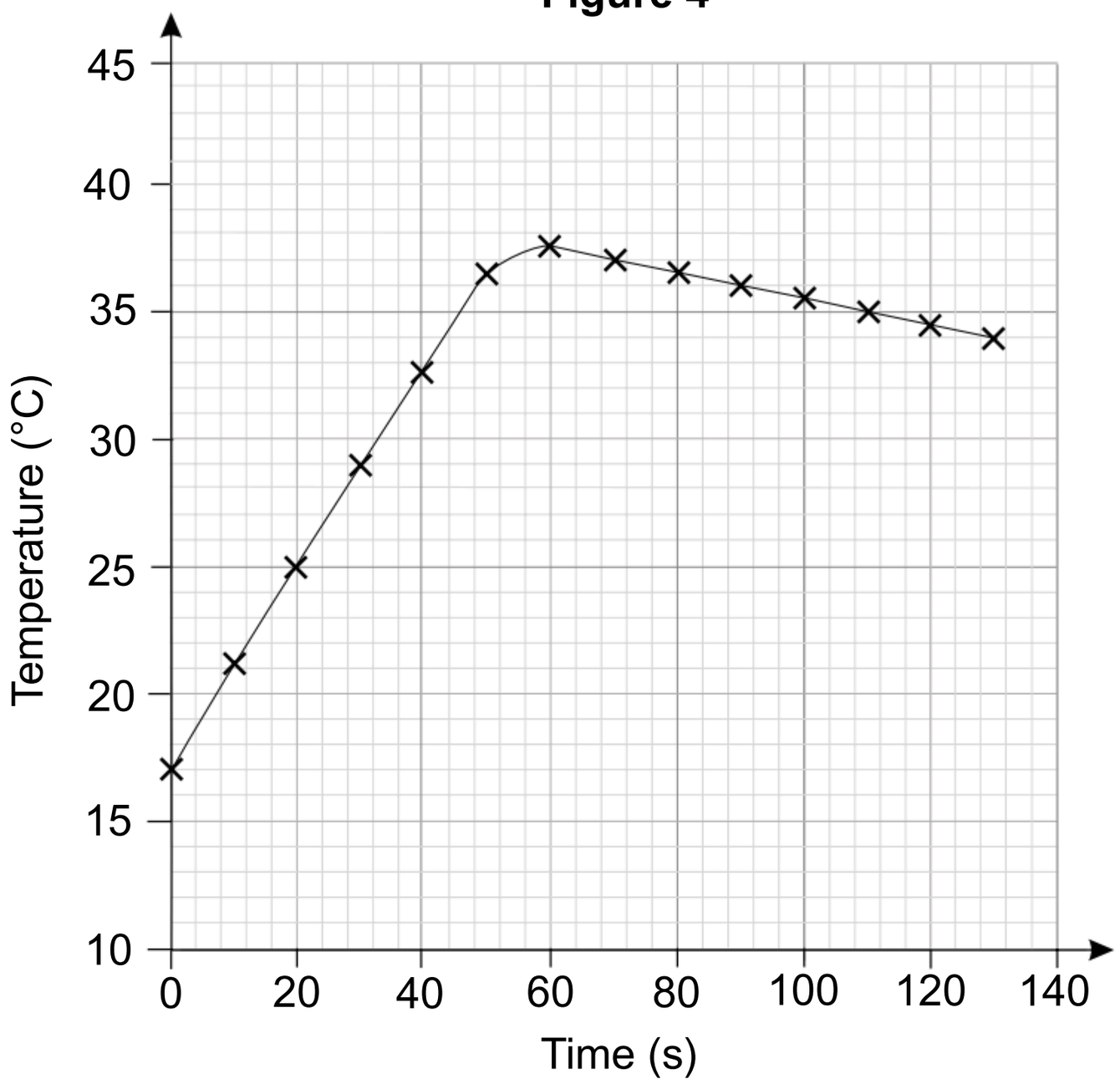
What value does each **small division** on the scale of the thermometer represent?
Tick **one** box.

- ☐ 0.1 °C
- ☐ 10 °C
- ☐ 1 °C
- ☐ 2 °C

[1 mark]

The student recorded the initial temperature of the dilute hydrochloric acid.
The student added the magnesium ribbon to the acid and then measured the temperature of the reaction mixture every 10 seconds.
The results are shown on the graph in **Figure 4**.

Figure 4



4.3 Using the graph in **Figure 4**, give the highest temperature of the mixture that the student recorded.

Highest temperature = ... °C

[1 mark]

4.4 The initial temperature of the acid was 17.0 °C.
Use this information and your answer from **4.3** to estimate the total change in temperature of the reaction mixture.

.....

Temperature change = ... °C

[2 marks]

4.5 State whether this reaction was exothermic or endothermic. Explain your answer.

.....

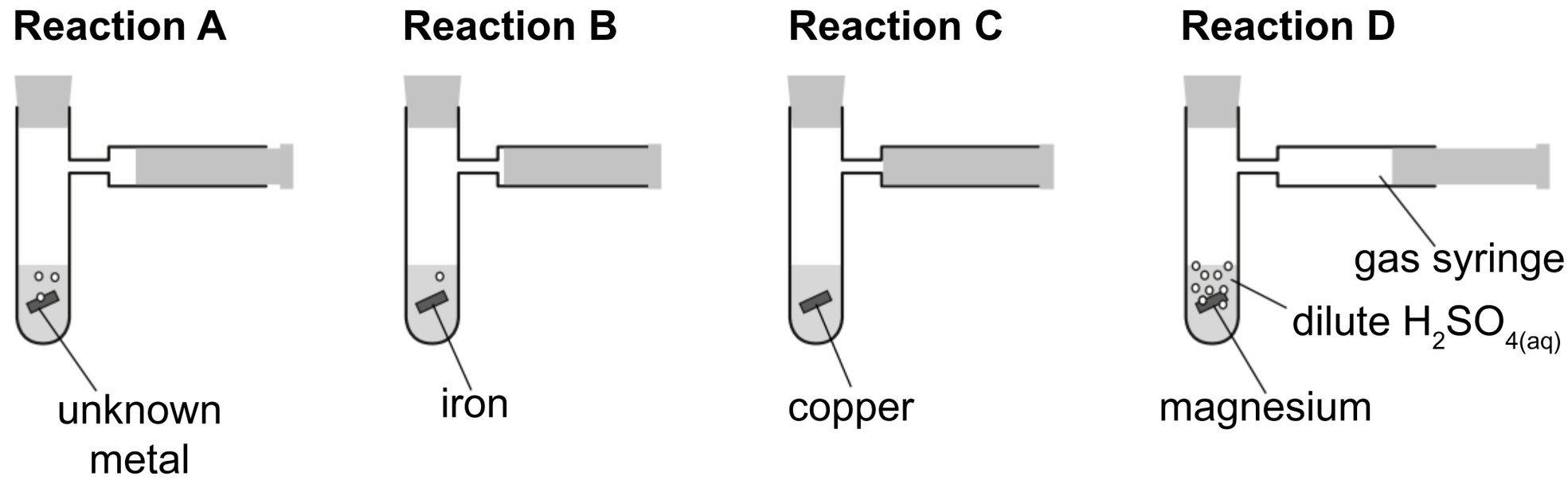
.....

[1 mark]

Turn over for the next question

5 A student reacts four different metals with dilute sulfuric acid.
The student controls all of the relevant variables to make sure that the test is fair.
The gas given off by each reaction is collected in a gas syringe.
Figure 5 shows all four reactions after 30 seconds.

Figure 5



5.1 Name the gas that is being collected in the gas syringes.

.....

[1 mark]

5.2 Which reaction, **A**, **B**, **C**, or **D**, contains the **most reactive** metal?
Explain how you can tell.

.....

.....

[3 marks]

5.3 Use your knowledge of the reactivity series to suggest a possible identity for the unknown metal used in reaction **A**.

.....

[1 mark]

5.4 The student wanted to include sodium in the experiment, but a teacher said this would not be safe.
Suggest why it would not be safe to include sodium in the experiment.

.....

[1 mark]

5.5 In another experiment, the student placed pieces of different metals in metal salt solutions and left them for 10 minutes. The student then recorded whether any reaction had occurred. The results of this experiment are shown in **Table 6**.

Table 6

	Did any reaction occur with:		
	iron sulfate	**magnesium sulfate**	**copper sulfate**
iron	No	No	.....
magnesium	.....	No	Yes
copper	No	No	No

Complete **Table 6** by filling in the gaps.

[2 marks]

5.6 The equation for the reaction between magnesium and copper sulfate solution is:

$$Mg_{(s)} + CuSO_{4(aq)} \rightarrow MgSO_{4(aq)} + Cu_{(s)}$$

Write the ionic equation for this reaction.

.....

[2 marks]

5.7 In the reaction described in **5.6**, which species was oxidised?

.....

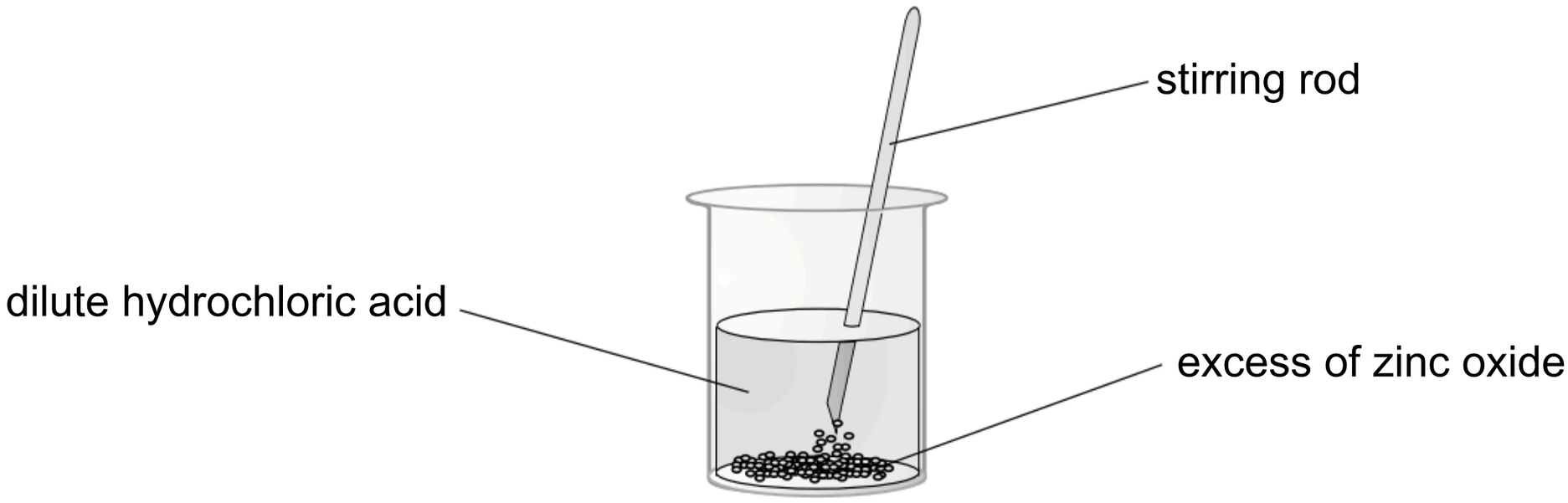
[1 mark]

Turn over for the next question

6 A student added an excess of zinc oxide (ZnO) to dilute hydrochloric acid (HCl). They reacted to produced zinc chloride, a soluble salt.

6.1 **Figure 6** shows the student's experiment just after the zinc oxide was added.

Figure 6



How could the student tell when all of the acid had been neutralised?

.....

.....

[1 mark]

6.2 In addition to the zinc chloride, one other product is formed by this reaction.
Name the other product of the reaction.

.....

[1 mark]

6.3 Give the chemical formula of zinc chloride.

.....

[1 mark]

6.4 Describe how you could produce pure, dry crystals of zinc chloride from the reaction mixture after the reaction had finished.

.....

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.....

.....

[4 marks]

6.5 Another student added solid sodium carbonate (Na_2CO_3) to dilute hydrochloric acid.
Name the salt formed by this reaction.

.....

[1 mark]

6.6 Which gas was given off by the reaction described in **6.5**?
Tick **one** box.

- ☐ Hydrogen
- ☐ Oxygen
- ☐ Chlorine
- ☐ Carbon Dioxide

[1 mark]

6.7 The solution formed at the end of the reaction described in **6.5** is neutral.
Suggest a method you could use to test the pH of the solution.
State what you would expect to observe.

.....

.....

.....

[2 marks]

Turn over for the next question

7 Iron can be extracted from iron(III) oxide, Fe_2O_3, by reduction, using carbon.
Here is the equation for this reaction:

$$2Fe_2O_3 + 3C \rightarrow 4Fe + 3CO_2$$

7.1 Give **one** reason why this reaction is described as a reduction reaction.

.....

[1 mark]

7.2 Calculate the mass of iron that could be extracted from 40 g of iron(III) oxide.
Relative atomic masses, A_r: Fe = 56, O = 16.

.....

.....

.....

.....

.....

Mass of iron = ... g

[4 marks]

7.3 Explain why reduction with carbon isn't used to extract aluminium from its ore.

.....

.....

[2 marks]

7.4 Small amounts of carbon can be added to iron to make an alloy called steel.
Explain why steel is harder than pure iron.

.....

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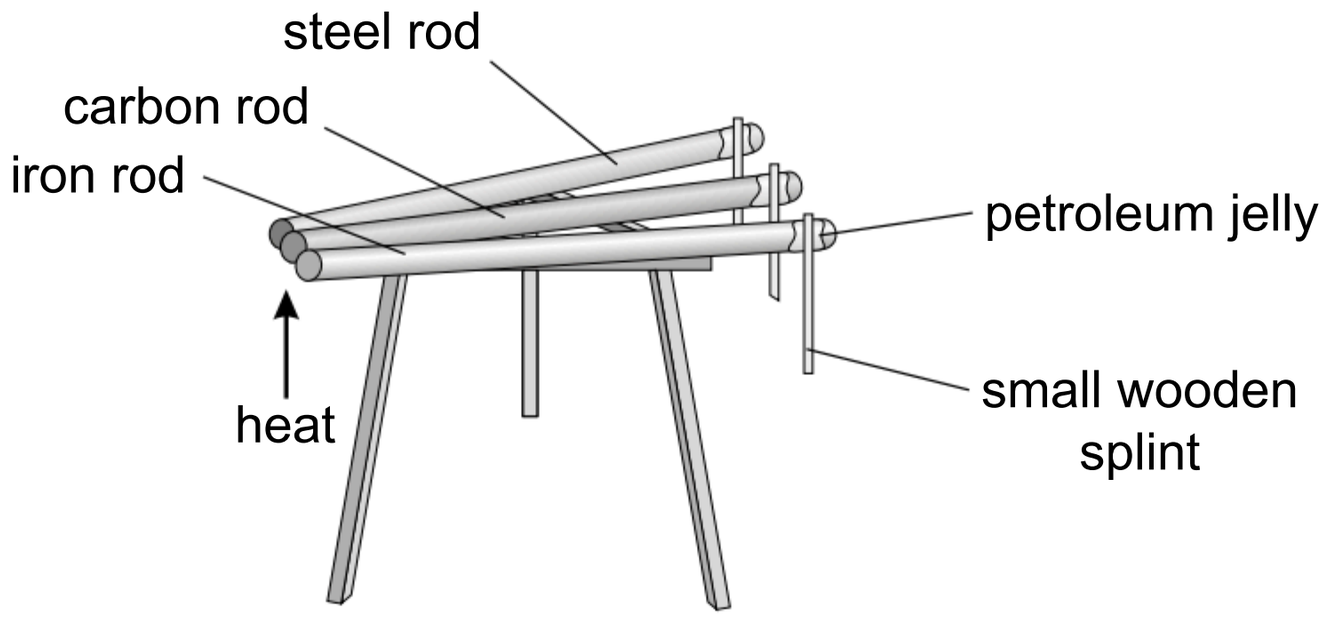
.....

.....

[4 marks]

7.5 A student used the apparatus shown in **Figure 7** to compare how well steel, carbon and iron conduct thermal energy.

Figure 7



The student placed one rod of each material on a tripod, with a small wooden splint stuck to one end with petroleum jelly. The student then heated the rods at the other end and timed how long it took for each wooden splint to fall off as the petroleum jelly melted.

Describe **three** things the student should keep the same for each material to make the experiment a fair test.

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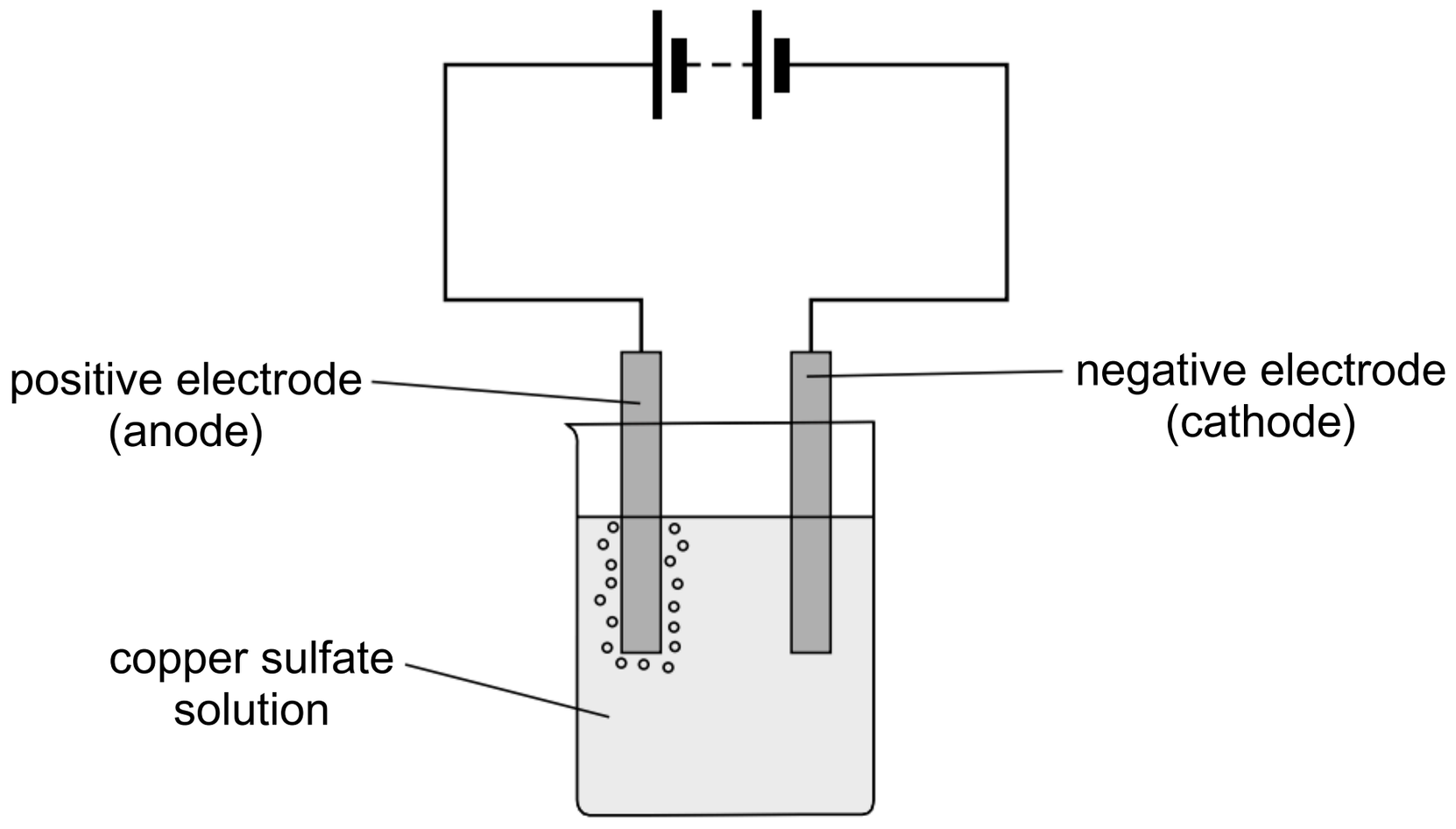
[3 marks]

Turn over for the next question

8 A student carried out an experiment to investigate the electrolysis of copper sulfate solution, $CuSO_{4\,(aq)}$, using inert carbon electrodes. The diagram in **Figure 8** shows how the experiment was set up.

Figure 8



8.1 Explain why a solution of copper sulfate can conduct electricity, but solid copper sulfate cannot.

.....

.....

[2 marks]

8.2 List all **four** of the ions that are present in aqueous copper sulfate solution.

.....

.....

[1 mark]

8.3 Complete the half equation shown below for the reaction taking place at the positive electrode (anode).

..... OH^- $\rightarrow$ O_2 + ... H_2O + 4 .....

[1 mark]

The student repeated the experiment three times, running the electrolysis for exactly 30 minutes each time. The dry electrodes were weighed before starting each run. At the end of each run, both electrodes were dried and weighed again.

Table 7 shows the change in mass of both electrodes for each of the three runs.

Table 7

	Change in mass of electrode in g			
	Run 1	**Run 2**	**Run 3**	**Mean**
Positive electrode (anode)	0.00	0.00	0.00	0.00
Negative electrode (cathode)	2.36	2.71	2.55	.....

8.4 Complete **Table 7** by finding the mean change in mass for the negative electrode.

[1 mark]

8.5 Explain why the negative electrode increased in mass in this experiment.
Your answer should include the half equation for the reaction taking place at the negative electrode.

.....

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.....

.....

[3 marks]

8.6 The student decided to run the experiment for 60 minutes instead of 30 minutes. Predict the effect that this would have on the change in mass of the negative electrode.

.....

[1 mark]

Turn over for the next question

9 Atom economy and percentage yield are both important factors that are considered when chemical firms decide how to produce a chemical industrially.

9.1 Discuss why it is better to use reactions with high atom economies than reactions with low atom economies in industry.
In your answer you should talk about:

- the environmental benefits of using high atom economy reactions,
- the economic benefits of using high atom economy reactions.

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[6 marks]

9.2 Lithium chloride can be made by reacting lithium carbonate with hydrochloric acid.
The equation for this reaction is:.

$$Li_2CO_3 + 2HCl \rightarrow 2LiCl + H_2O + CO_2$$

Calculate the atom economy for producing lithium chloride using this reaction.
Give your answer to 2 significant figures.
Relative formula masses, M_r: Li_2CO_3 = 74, HCl = 36.5, LiCl = 42.5, H_2O = 18, CO_2 = 44.

.....

.....

.....

.....

..... %

[4 marks]

9.3 Lithium chloride can also be formed following the reaction of lithium metal with chlorine gas. The equation for this reaction is shown below.

$$2Li + Cl_2 \rightarrow 2LiCl$$

What is the atom economy for the above reaction? Tick **one** box.

- ☐ 100%
- ☐ 50%
- ☐ 200%
- ☐ 10%

[1 mark]

9.4 A student carried out the reaction shown in **9.2** using 185 g of lithium carbonate and hydrochloric acid in excess. The theoretical yield of this reaction was 212.5 g.
The student actually produced 170 g of lithium chloride.
Calculate the percentage yield of the reaction.

.....

.....

..... %

[1 mark]

END OF QUESTIONS

CGP Practice Exam Paper GCSE Chemistry

GCSE Chemistry

Paper 2

Higher Tier

In addition to this paper you should have:
- A ruler.
- A calculator.
- A periodic table.

Centre name					
Centre number					
Candidate number					

Time allowed:
- 1 hour 45 minutes

Surname
Other names
Candidate signature

Instructions to candidates
- Write your name and other details in the spaces provided above.
- Answer **all** questions in the spaces provided.
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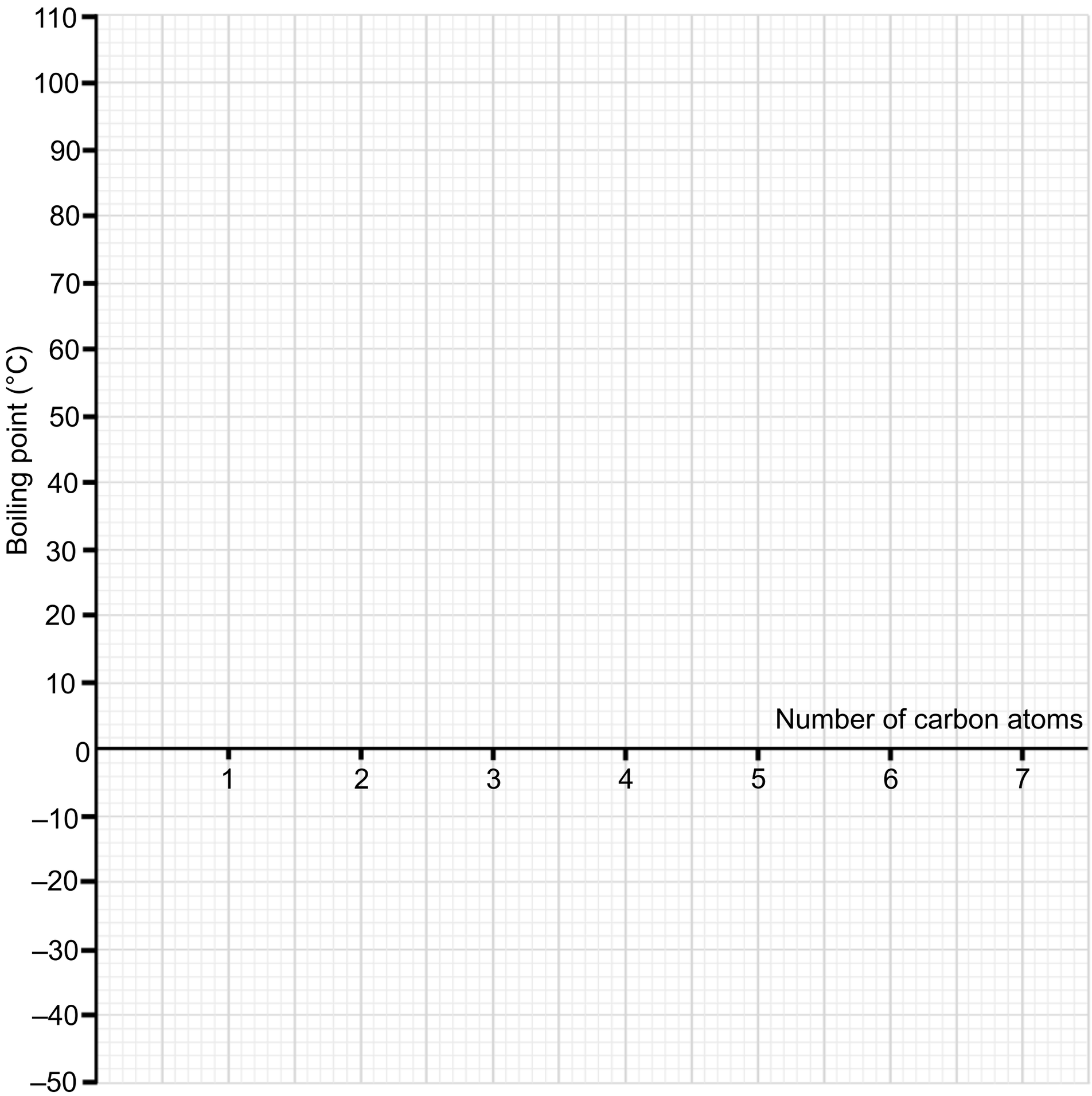
	1	2	3		1	2	3
1				6			
2				7			
3				8			
4				9			
5							
Total							

1 Alkanes are hydrocarbon compounds found in crude oil. **Table 1** shows how the boiling points of some alkanes change as the molecules get bigger.

Table 1

Alkane	Molecular formula	Boiling point (°C)
Propane	C_3H_8	–42
Butane	C_4H_{10}	–0.5
Pentane	C_5H_{12}	
Hexane	C_6H_{14}	69
Heptane	C_7H_{16}	98

1.1 Using the data in **Table 1**, plot a graph of the number of carbon atoms in an alkane molecule against boiling point on the axes below.
Draw a smooth curve through the points you have plotted.



[2 marks]

1.2 Use your graph to estimate the boiling point of pentane.

..... °C

[1 mark]

Question 1 continues on the next page

1.3 Describe what the graph shows about the size of alkane molecules and their boiling points.

.....

.....

[1 mark]

1.4 What is the general formula of the alkanes? Tick **one** box.

- ☐ C_nH_{2n}
- ☐ C_nH_{2n+1}
- ☐ C_nH_{2n+2}
- ☐ C_nH_{2n-1}

[1 mark]

1.5 Propane is an alkane with three carbon atoms.
Draw the displayed formula of propane.

[1 mark]

1.6 Propene is an alkene with three carbon atoms.
Describe a test you could use to distinguish between propene and propane.
Say what you would observe in each case.

.....

.....

.....

[3 marks]

1.7 Propane burns in the presence of oxygen.
Complete this equation for the complete combustion of propane.

$$C_3H_8 + 5O_2 \rightarrow 3CO_2 + ...$$

[2 marks]

1.8 Decane, $C_{10}H_{22}$, is an alkane with ten carbon atoms.
Which of the following statements about decane is correct? Tick **one** box.

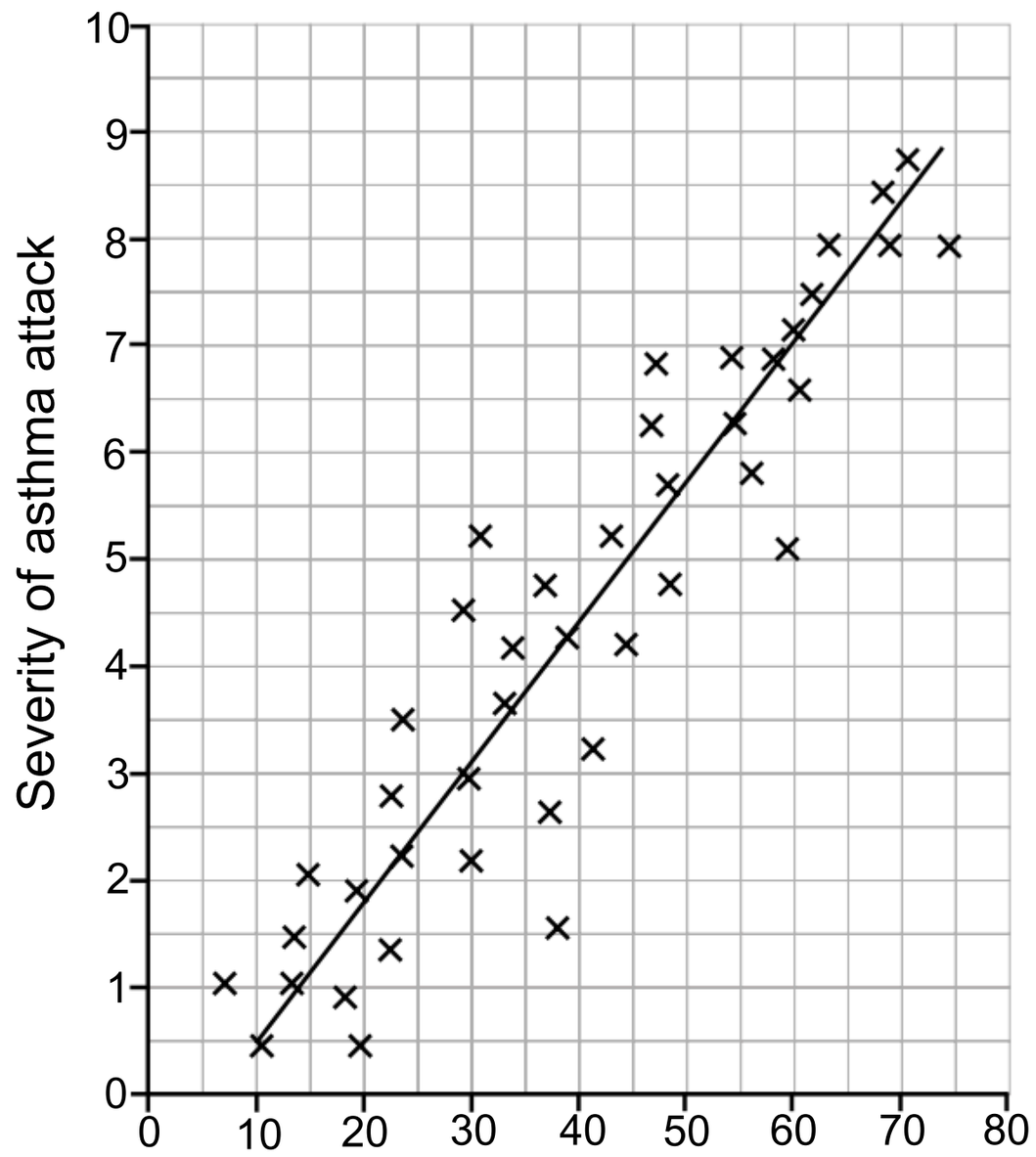
- ☐ Decane is more flammable than methane.
- ☐ The complete combustion of decane results in the formation of carbon monoxide.
- ☐ Decane is less viscous than propane.
- ☐ Decane has a higher boiling point than pentane.

[1 mark]

2 Nitrogen dioxide is an atmospheric pollutant that irritates the respiratory system. It is thought that there is a link between exposure to nitrogen dioxide and the severity of asthma attacks in people with asthma.

Figure 1 shows the results of a study carried out by a group of scientists that compared atmospheric nitrogen dioxide levels with the severity of asthma attacks suffered by men under the age of 40 working in a city centre.

Figure 1



2.1 Describe the relationship between the level of nitrogen dioxide in the air and the severity of asthma attacks shown in **Figure 1**.

.....

.....

[1 mark]

2.2 The scientists decided it would not be possible to draw a general conclusion about the link between nitrogen dioxide levels and the severity of asthma attacks that would apply to everyone.

Which of these reasons **does not** describe why this might be the case? Tick **one** box.

- ☐ The scientists would need to collect data from women too.
- ☐ The severity of asthma attacks might be affected by another pollutant which happens to be abundant in the same areas as nitrogen dioxide.
- ☐ The sample of people that the scientists collected data from was too big.
- ☐ The scientists would need to collect data from other age groups too.

[1 mark]

Question 2 continues on the next page

2.3 Where do the oxides of nitrogen in the air come from?
Suggest why levels of nitrogen oxides can be particularly high in cities.

.....

.....

.....

.....

[3 marks]

2.4 Nitrogen dioxide dissolves in water to form acid rain.
What is the substance formed when nitrogen dioxide dissolves in water?

.....

[1 mark]

2.5 Name **one** other pollutant gas that can cause acid rain.

.....

[1 mark]

2.6 The scientists who collected the data shown on the graph in **Figure 1** also found higher than normal levels of carbon monoxide in the samples taken in a city centre.
Explain why carbon monoxide is a toxic gas.

.....

.....

[2 marks]

3 In the UK, the majority of our drinking water is produced from treating groundwater or surface water. Drinking water can also be made by treating sea water or waste water.

3.1 Producing water that is safe to drink from sea water is expensive.
Suggest why some countries produce drinking water by this method.

.....

[1 mark]

3.2 What is the name of the process used to remove salt from sea water?
Tick **one** box

- ☐ Filtration
- ☐ Desalination
- ☐ Sterilisation
- ☐ Cracking

[1 mark]

3.3 A teacher gives a student a sample of sea water, and asks the student to produce a sample of pure water from it.
Outline a method that the student could use to remove the salt from the sea water.

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[6 marks]

Question 3 continues on the next page

Turn over ▶

3.4 Sewage treatment plants process sewage and release clean, treated water back into the environment. The first step in the treatment of sewage is screening.
What happens in the screening step?
Tick **one** box.

- ☐ Chlorine is added to the sewage to kill microorganisms.
- ☐ The sewage is placed into large storage tanks and allowed to settle.
- ☐ Anaerobic digestion is used to break down the sewage.
- ☐ Large bits of material (such as grit) are removed from the sewage.

[1 mark]

3.5 Following screening, sedimentation is used to separate the effluent from the sludge.
Describe what happens to the effluent before it can be returned to the environment.

.....

.....

.....

[2 marks]

4 A student is investigating the properties of vinegar. Ethanoic acid is found in vinegar.

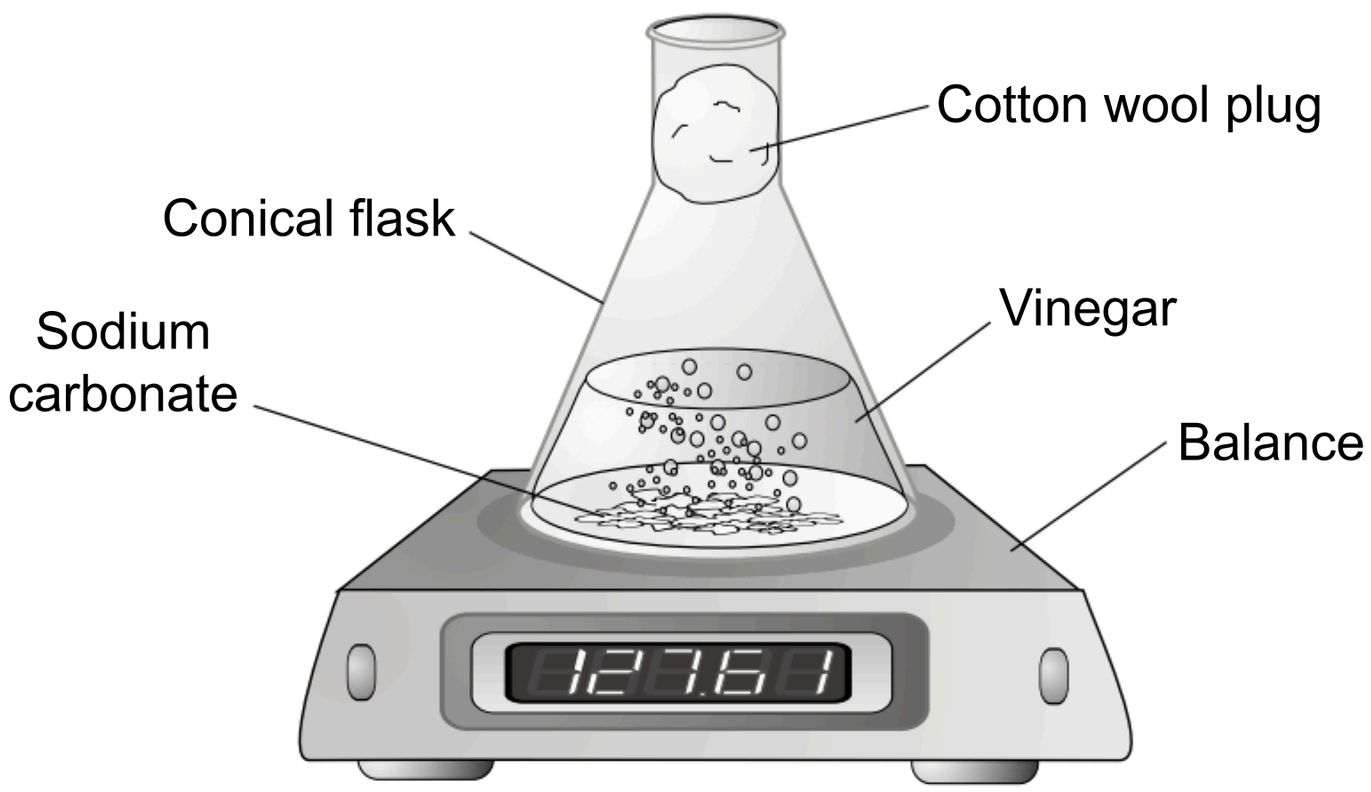
4.1 Give the functional group of the homologous series that ethanoic acid belongs to.

.....

[1 mark]

The student reacted some of the vinegar with solid sodium carbonate.
The apparatus used is shown in **Figure 2**.

Figure 2



The student recorded the initial mass of the flask and then recorded it again after 60 s.
The experiment was repeated three times. The results are shown in **Table 2**.

Table 2

	Initial Mass in g	Final Mass in g	Loss of Mass in g
Run 1	128.00	127.45	0.55
Run 2	128.00	127.61	0.39
Run 3	128.35	127.90	0.45

4.2 Calculate the mean loss of mass after 60 s.

.....

mean mass lost = ... g

[2 marks]

Question 4 continues on the next page

4.3 Why would it be inappropriate to use a mass balance that has a resolution of 1 g for this experiment?

.....

.....

[1 mark]

4.4 The student repeats the experiment a fourth time, but this time the gas released by the reaction is collected in a gas syringe.
Name the gas collected in the gas syringe.
Describe a test that could be used to confirm the identity of the gas.

.....

.....

.....

[3 marks]

4.5 The teacher asks the student to test the solid sodium carbonate used in the experiment to confirm that it contains sodium ions.
Suggest what kind of test the student could perform and state what observation you would expect the student to make.

.....

.....

[2 marks]

5 When a company is choosing what materials to use to make a particular product, they will consider the properties, availability and cost of the materials that they could use.

5.1 **Table 3** shows some properties of four different materials.

Table 3

Material	Density (g/cm^3)	Electrical conductivity	Flexibility	Corrosion resistance
Steel	7.8	High	Low	Low
Aluminium alloy	3	Very High	Low	High
PVC	1.4	Very Low	High	High
Glass	2.5	Very Low	Low	High

Use data from the table to explain why PVC is particularly suitable for covering electrical wiring.

.....

.....

.....

[2 marks]

5.2 What kind of material is glass?

Tick **one** box.

- ☐ Polymer
- ☐ Ceramic
- ☐ Composite
- ☐ Alloy

[1 mark]

5.3 Steel and aluminium alloy are both very strong. Many aeroplane parts need to withstand high forces, so they must be made from strong materials.
Using the information in **Table 3**, explain why aluminium alloy is often used to make aeroplane parts instead of steel.

.....

.....

.....

[2 marks]

Question 5 continues on the next page

5.4 Copper is used to make electrical wiring due to its high electrical conductivity.
Phytomining can be used to extract copper from low-grade ores.
Describe how copper is extracted from low-grade ores using phytomining.

.....

.....

.....

.....

.....

[4 marks]

5.5 A sports company is deciding on the best material to use for making the shaft of a professional golf club.
Look at **Table 4**, below. Discuss which of the four materials shown in the table would be the most suitable choice for making the golf club shaft.
Use the information in the table to justify your choice.

Table 4

Material	Density in g/cm^3	Strength in MPa	Corrosion resistance	Cost
Carbon fibre	1.5	4100	Good	High
Iron	7.8	200	Poor	Low
Stainless Steel	7.8	780	Good	Low
Lead	11.3	12	Good	Low

.....

.....

.....

.....

.....

.....

.....

.....

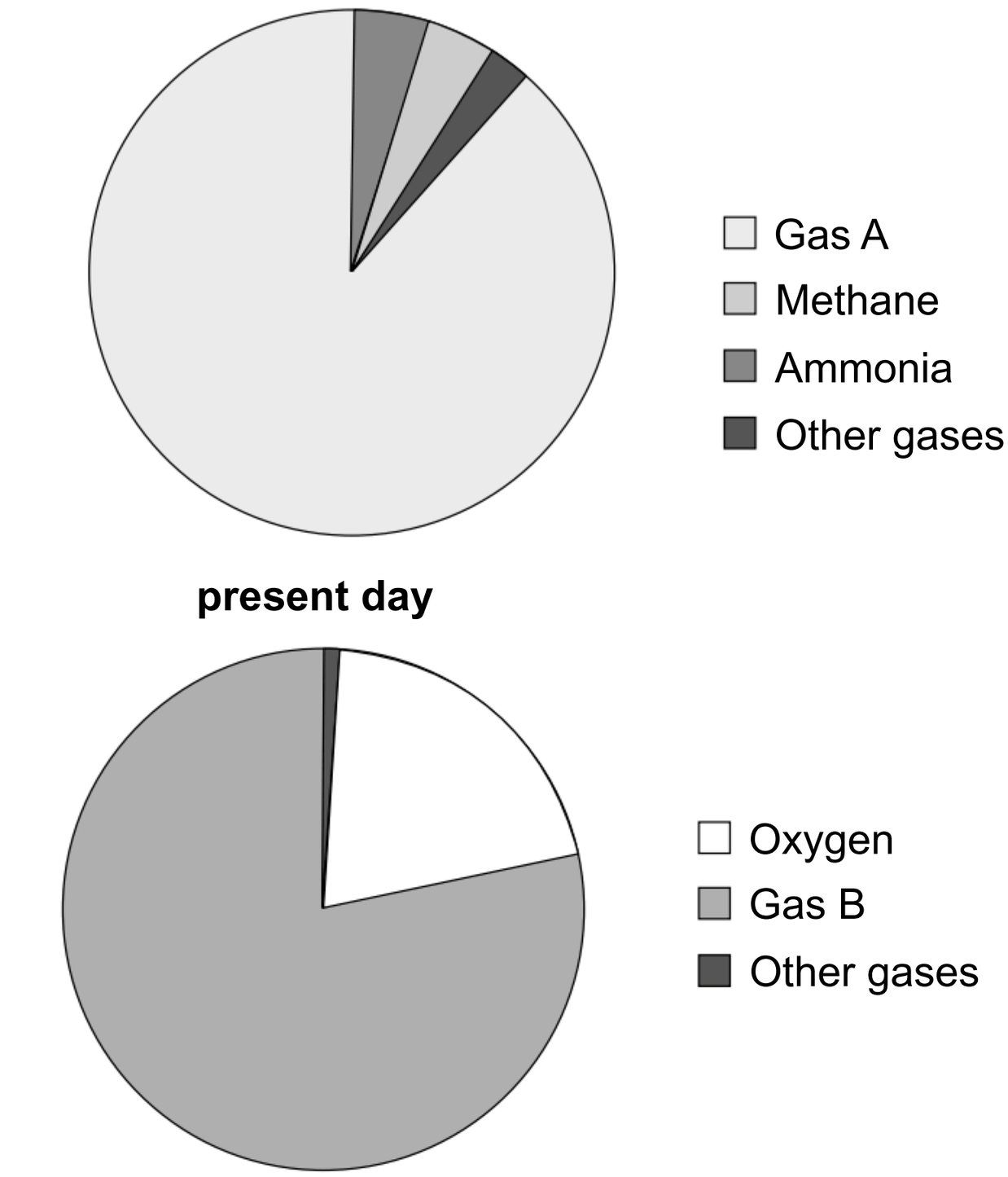
.....

[6 marks]

6 This question is about the evolution of the atmosphere.

The pie charts in **Figure 3** show the composition of Earth's atmosphere as it may have been 4 billion years ago and as it is at the present time.

Figure 3



6.1 Name Gas A.

.....

[1 mark]

6.2 Name Gas B.

.....

[1 mark]

6.3 The evolution of algae and plants affected the composition of gases in the atmosphere. Explain how algae and plants changed the atmosphere.

.....

.....

.....

.....

[3 marks]

Question 6 continues on the next page

Turn over ▶

6.4 Human activity is affecting the composition of the atmosphere. Suggest **one** reason why each of the following factors affects the level of carbon dioxide in the atmosphere.

Deforestation .....

.....

Increased energy consumption .....

.....

Increasing population .....

.....

[3 marks]

6.5 A student says:

"I think that if the amount of carbon dioxide in the atmosphere continues to rise, this may lead to a drop in global food production in the future."

Explain why the student may be correct.

.....

.....

.....

.....

.....

.....

[3 marks]

7 A student is investigating how the rate of the reaction between calcium carbonate and hydrochloric acid is affected by the concentration of the acid.

The student uses the following method:

- Weigh out 0.7 g of calcium carbonate.
- Add the calcium carbonate to an excess of 0.2 mol/dm^3 hydrochloric acid in a conical flask.
- Use a gas syringe to collect the gas given off by the reaction.
- Measure and record the volume of gas produced every 10 s.
- Repeat the experiment using hydrochloric acid with a different concentration.

7.1 Suggest **two** variables other than the mass of calcium carbonate that the student would have to keep the same for each run to make it a fair test.

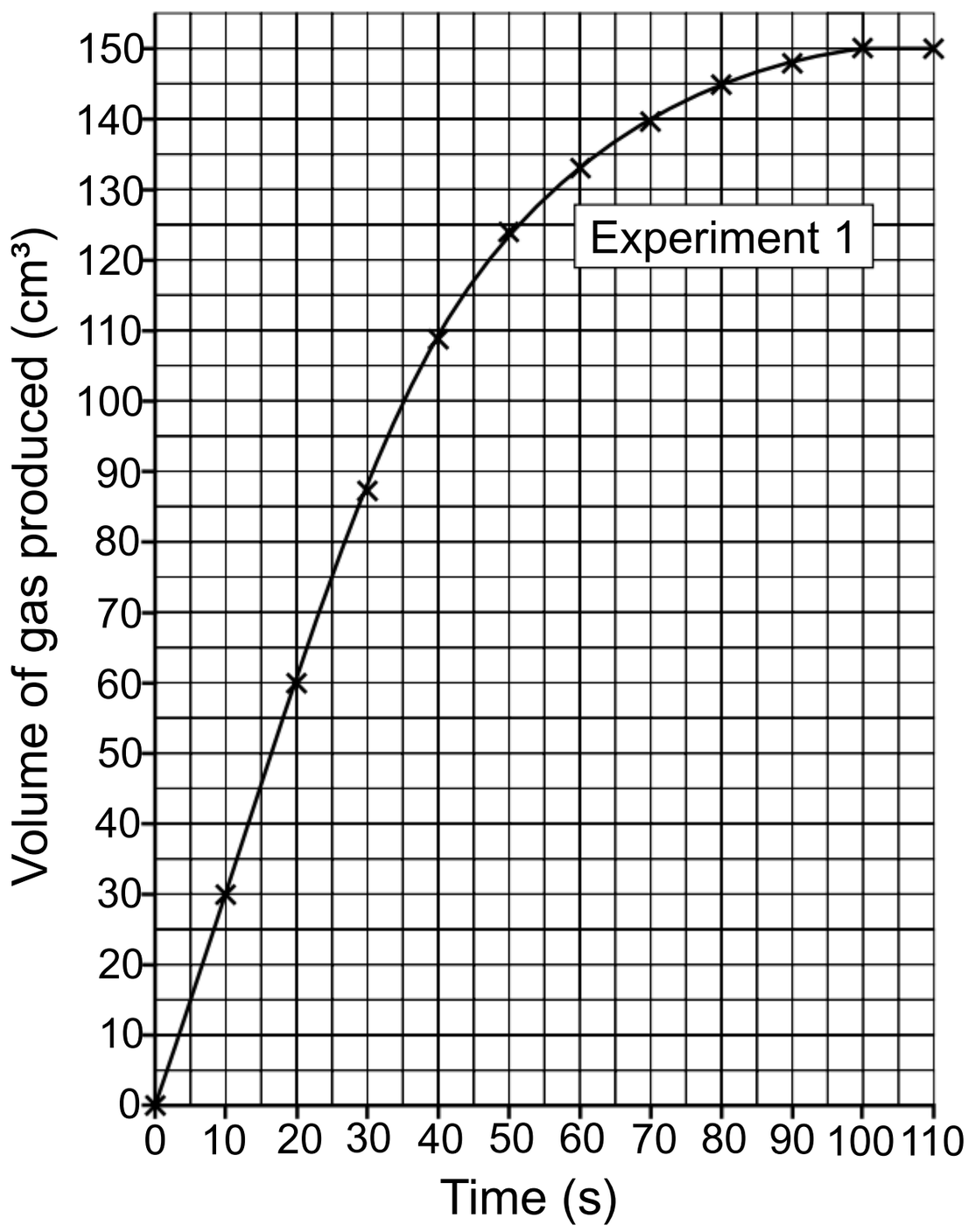
.....

.....

[2 marks]

Figure 4 shows the student's results for the first experiment, using 0.2 mol/dm^3 hydrochloric acid.

Figure 4



Question 7 continues on the next page

7.2 What does the graph in **Figure 4** show about how the rate of reaction changes as the reaction proceeds?

.....

.....

[1 mark]

7.3 Draw a tangent to the graph in **Figure 4** at 50 seconds.
Using the gradient of the tangent, find the rate of the reaction at this time.
Include the units of the rate in your answer.

.....

.....

.....

.....

.....

Rate of reaction = ... Unit = ...

[4 marks]

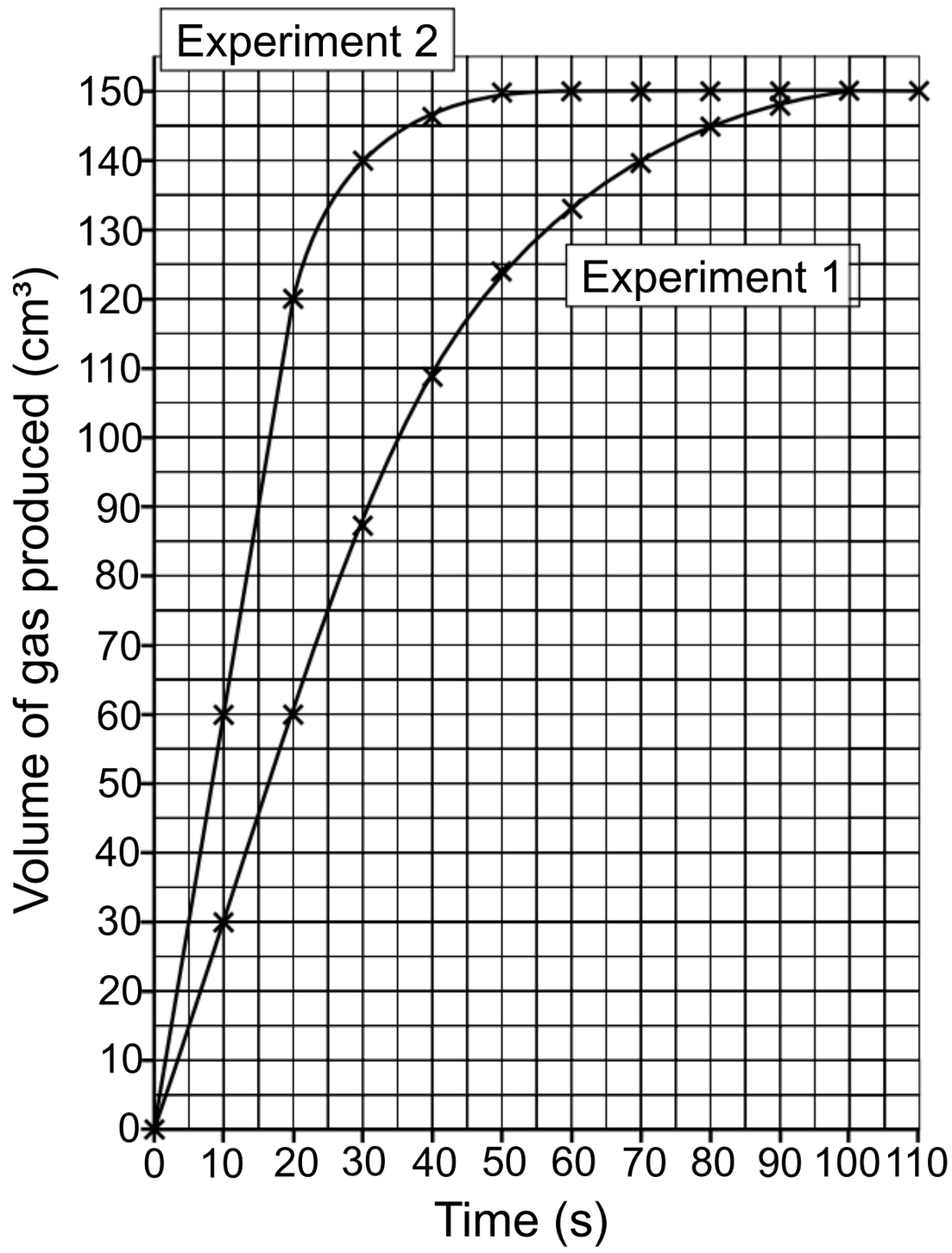
7.4 The student's teacher repeated the experiment using 0.35 g of calcium carbonate and 0.2 mol/dm^3 hydrochloric acid.
Predict the total amount of gas produced in this experiment.
You can assume that the acid was in excess in both experiments.

.....

[1 mark]

The student repeated the experiment with the same mass of calcium carbonate, but using 0.4 mol/dm^3 hydrochloric acid**.**
The student added the results of this second experiment to the graph from the first experiment. This graph is shown in **Figure 5**, below.

Figure 5



7.5 Explain how the student could tell from **Figure 5** that increasing the concentration of the acid had increased the reaction rate.

.....

.....

[1 mark]

7.6 In terms of collision theory, explain why concentration affects the rate of a reaction.

.....

.....

.....

.....

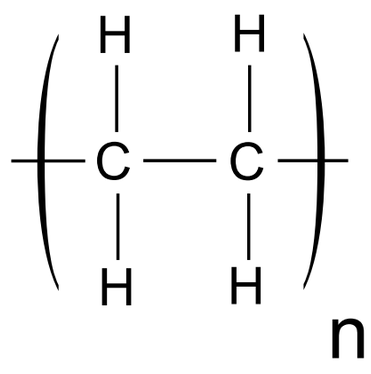
[2 marks]

Turn over for the next question

8 This question is about synthetic and natural polymers.

8.1 **Figure 6** shows the displayed formula of the polymer poly(ethene).

Figure 6



Draw the displayed formula of the monomer used to make this polymer.

[1 mark]

8.2 Which homologous series does the monomer molecule belong to?

.....

[1 mark]

8.3 Some synthetic polymers are thermosetting, while others are thermosoftening.
Explain the difference in structure between thermosetting and thermosoftening polymers.
Describe how this affects the way they respond to being heated.

.....

.....

.....

.....

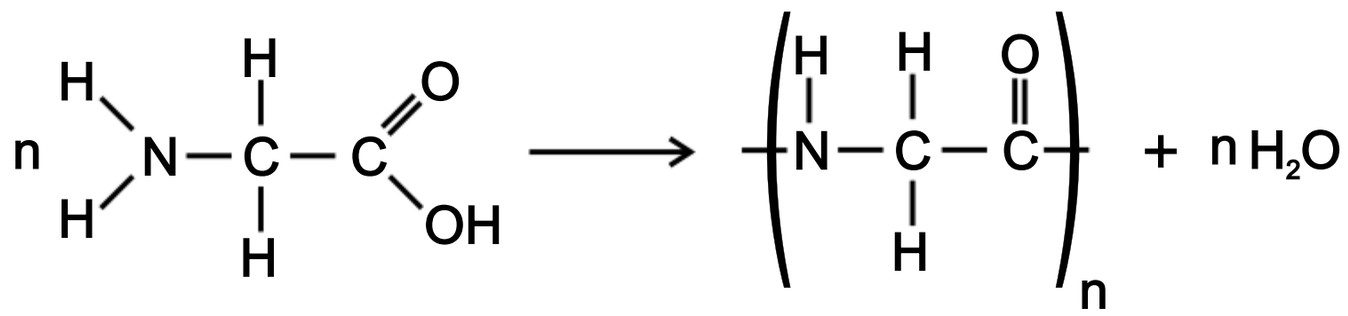
.....

.....

[4 marks]

Some polymers are naturally occurring. **Figure 7** shows the polymerisation of the molecule glycine to form a polypeptide.

Figure 7



8.4 Name the type of polymerisation shown in **Figure 7.**

Explain how you can tell that the type of polymerisation you have named is occurring.

.....

.....

[2 marks]

8.5 What type of molecule is glycine?

.....

[1 mark]

8.6 Name **one** other naturally occurring polymer.

.....

[1 mark]

Turn over for the next question

9 A company uses a process called the contact process to make sulfuric acid.
The first step of the process is the exothermic reaction of sulfur dioxide and oxygen to form sulfur trioxide. The equation for the reaction is shown below.

$$2SO_{2(g)} + O_{2(g)} \rightleftharpoons 2SO_{3(g)}$$

9.1 This reaction can reach equilibrium.
When a reaction is at equilibrium, what does this tell you about the forward and reverse reactions?

.....

.....

[1 mark]

9.2 You can predict how changing the conditions will affect this reaction using Le Chatelier's principle.
What is Le Chatelier's principle?

.....

.....

[1 mark]

9.3 State what will happen to the yield of sulfur trioxide in this reaction if the pressure inside the reaction vessel is increased.
Explain your answer.

.....

.....

.....

.....

[3 marks]

9.4 In order to increase the rate of the reaction, the company decides to increase the temperature of the reaction vessel.
State what effect this will have on the yield of sulfur trioxide. Explain your answer.

.....

.....

.....

.....

[3 marks]

9.5 A measured mass of vanadium oxide was added to the reaction vessel at the start of the process. The same mass of vanadium oxide is present at the end of the reaction.
Suggest what role vanadium oxide is playing in this reaction.

.....

[1 mark]

9.6 The company used to produce 7500 kg of sulfur trioxide each day.
After modifying the plant to allow the sulfur trioxide to be removed from the reaction vessel as soon as it is made, the yield of the plant increased by 15%.
How much sulfur trioxide can the plant make in five days now?

.....

.....

.....

mass made in five days = ... kg

[3 marks]

END OF QUESTIONS

Answers

Topic 1a — Atoms, Elements, Compounds and Mixtures

Page 3
Atoms

Q1 protons = atomic number = **31** ***[1 mark]***
electrons = protons = **31** ***[1 mark]***
neutrons = mass number – atomic number
= 70 – 31 = **39** ***[1 mark]***

Page 4
Elements

Q1 E.g. it's the number of protons in an atom that determines what type of atom it is, so if all the atoms have the same number of protons then the substance is an element ***[1 mark]***.

Q2 Relative atomic mass =
$$\frac{(92.2 \times 28) + (4.7 \times 29) + (3.1 \times 30)}{92.2 + 4.7 + 3.1}$$ ***[1 mark]***
$$= \frac{2581.6 + 136.3 + 93}{100} = \frac{2810.9}{100}$$
= 28.109 = **28.1 (1 d.p.)** ***[1 mark]***

Page 5
Compounds

Q1 (2 × Na) + (1 × C) + (3 × O) = 6 ***[1 mark]***

Q2 Aluminium — 2 atoms, sulfur — 3 atoms and oxygen — 12 atoms ***[1 mark]***.

Page 6
Chemical Equations

Q1 $2Fe + 3Cl_2 \rightarrow 2FeCl_3$ ***[1 mark]***

Q2 a) water → hydrogen + oxygen ***[1 mark]***

b) $2H_2O \rightarrow 2H_2 + O_2$
[1 mark for correct products and reactants, 1 mark for correctly balancing]

Pages 7-8
Warm-Up Questions

1) The total number of protons and neutrons.

2) E.g. an element is a substance that consists of atoms with the same number of protons in their nucleus.

3) A compound is a substance made from two or more elements in fixed proportions throughout the compound and they're held together by chemical bonds.

4) $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$

Exam Questions

1.1

Name of particle	Relative charge
Proton	**+1**
Neutron	**0**

[2 marks — 1 mark for each correct charge]

1.2 nucleus ***[1 mark]***

1.3 8 ***[1 mark]***

1.4 1 ***[1 mark]***

2.1 14 – 7 = **7** ***[1 mark]***

2.2 The number of protons is the same as the atomic number ***[1 mark]***.

2.3 N ***[1 mark]***

3.1 methane and oxygen ***[1 mark]***

3.2 carbon dioxide and water ***[1 mark]***

3.3 oxygen/O_2 ***[1 mark]***

3.4 $CH_4 + \mathbf{2O_2} \rightarrow CO_2 + \mathbf{2}H_2O$
[1 mark for correct missing reactant, 1 mark for correctly balancing the equation]

4.1 $\mathbf{H_2SO_4} + \mathbf{2NH_3} \rightarrow (NH_4)_2SO_4$
[1 mark for each correct reactant, 1 mark for correctly balancing the equation]

4.2 15 ***[1 mark]***
There are eight atoms of hydrogen, one atom of sulfur, four atoms of oxygen, and two atoms of nitrogen.

4.3 8 ***[1 mark]***
There are four hydrogen atoms in NH_4 and there are two of these groups.

5.1 An isotope is a different atomic form of the same element ***[1 mark]***, which has the same number of protons ***[1 mark]*** but a different number of neutrons ***[1 mark]***.

5.2 Protons = 6 ***[1 mark]***, neutrons = 13 – 6 = **7** ***[1 mark]***, electrons = 6 ***[1 mark]***

5.3 It has a different atomic number from carbon ***[1 mark]***.

5.4 sum of (isotope abundance × isotope mass number) =
(79 × 13) + (21 × 14) = 1321
Relative atomic mass = $\frac{1321}{100}$ = 13.21 = **13.2 (1 d.p.)**
[4 marks for correct answer to 1 d.p. Otherwise 1 mark for finding sum of (isotope abundance × isotope mass number), 1 mark for correctly substituting numbers into the formula and 1 mark for correct unrounded answer.]

Page 11
Distillation

Q1 Ethanol ***[1 mark]***. Ethanol has the second lowest boiling point and will be collected once all the methanol has been distilled off and the temperature increased ***[1 mark]***.

Page 12
Warm-Up Questions

1) Two or more elements or compounds mixed together.

2) fractional distillation

Exam Questions

1.1 E.g. put the paper in a beaker of solvent, e.g. water ***[1 mark]***.

1.2 Pencil marks are insoluble (so won't dissolve into the solvent) ***[1 mark]***.

2.1 E.g. mix the lawn sand with water to dissolve the ammonium sulfate ***[1 mark]***. Filter the mixture using filter paper to remove the sharp sand ***[1 mark]***. Pour the remaining solution into an evaporating dish and slowly heat it to evaporate the water until you have dry crystals of ammonium sulfate ***[1 mark]***.

2.2 E.g. the products were not completely dry ***[1 mark]***.

3.1 The boiling points of water and methanoic acid are too close together to allow them to be separated by simple distillation ***[1 mark]***.

3.2

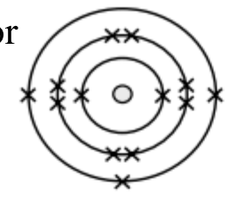
Temperature on thermometer	Contents of the flask	Contents of the beaker
30 °C	both liquids	no liquid
65 °C	water	propanone
110 °C	no liquid	both liquids

[3 marks for whole table correct, otherwise 1 mark for each correct row]

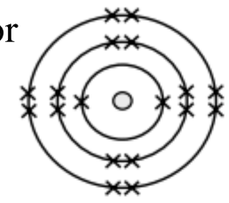
Topic 1b — The Periodic Table

Page 15

Electronic Structure

Q1 2,8,3 or 

[1 mark]

Q2 2,8,8 or 

[1 mark]

Page 18

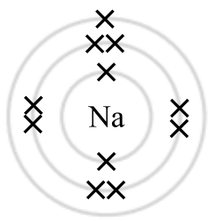
Warm-Up Questions

1) The 'plum pudding model' — a ball of positive charge with electrons stuck in it.

2) a) 2
b) 8

3) The discovery of isotopes.

Exam Questions

1.1 

[1 mark]

1.2 Any one of: lithium / potassium / rubidium / caesium / francium ***[1 mark]***.

1.3 1 ***[1 mark]***

2.1 By atomic weight ***[1 mark]***.

2.2 By atomic number ***[1 mark]***.

2.3 They have the same number of electrons in their outer shell ***[1 mark]***.

3 How to grade your answer:

Level 0: No description is given. ***[No marks]***
Level 1: Brief description of how the theory of atomic structure has changed. ***[1 to 2 marks]***
Level 2: Some detail given of how the theory of atomic structure has changed. ***[3 to 4 marks]***
Level 3: A clear and detailed description of how the theory of atomic structure has changed. ***[5 to 6 marks]***

Here are some points your answer may include:
At the start of the 19th century John Dalton described atoms as solid spheres that make up the different elements.

In 1897, JJ Thomson showed that atoms contained smaller, negatively charged particles (electrons). He suggested that an atom was a ball of positive charge, with electrons dotted through it. He called this the 'plum pudding model'.

In 1909, Rutherford and Marsden conducted their gold foil experiments, firing positively charged particles at a thin sheet of gold. If the plum pudding model was right, all of the particles should have passed straight through the sheet, or been only slightly deflected. But some were deflected straight back. So they concluded that the atom must have a positively charged nucleus at the centre, surrounded by a 'cloud' of negative electrons.

Niels Bohr proposed a new nuclear model of the atom. He suggested that electrons can only exist in fixed orbits, or shells.

Further experimentation by Rutherford and others showed that the nucleus could be divided into smaller particles, each with the same charge as a hydrogen atom. These became known as protons.

James Chadwick carried out an experiment which provided evidence of the existence of neutral particles within the nucleus, which became known as neutrons.

Page 20

Group 1 Elements

Q1 lithium + water $\rightarrow$ lithium hydroxide + hydrogen ***[1 mark]***

Page 21

Group 7 Elements

Q1 $Br_2 + 2NaI \rightarrow 2NaBr + I_2$ ***[1 mark]***

Pages 23-24

Warm-Up Questions

1) positive ions / cations

2) In the centre block / between Group 2 and Group 3.

3) a non-metal

4) potassium chloride / KCl

5) negative

6) Group 0

Exam Questions

1.1 Highest (X): fluorine ***[1 mark]***.
Lowest (Y): iodine ***[1 mark]***.

1.2 iodine ***[1 mark]***

2.1 transition metals ***[1 mark]***

2.2 To the right of the line ***[1 mark]***. Since it does not conduct electricity, it must be a non-metal ***[1 mark]***.

3.1 ionic bonds ***[1 mark]***

3.2 E.g. displacement reaction ***[1 mark]***

3.3 iodine/I_2 ***[1 mark]*** as it is displaced from potassium iodide ***[1 mark]***.

3.4 The products of the reaction will be potassium chloride/KCl and bromine/Br_2 ***[1 mark]***. Chlorine is more reactive than bromine ***[1 mark]***, so chlorine displaces bromine from potassium bromide ***[1 mark]***.

3.5 Group 0 elements are generally inert ***[1 mark]***. This is because they are energetically stable and don't need to lose or gain electrons to have a full outer shell ***[1 mark]***.

4.1 They have a single outer electron which is easily lost so they are very reactive ***[1 mark]***.

4.2 As you go down Group 1, the outer electron is further from the nucleus ***[1 mark]***. So the attraction between the nucleus and the electron decreases ***[1 mark]***. This means the outer electron is more easily lost and the metal is more reactive ***[1 mark]***.

4.3 hydrogen ***[1 mark]*** and potassium hydroxide ***[1 mark]***

4.4 alkaline ***[1 mark]***

Topic 2a — Types of Bonding

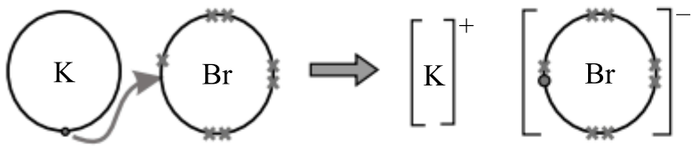
Page 26
Formation of Ions

Q1 Noble gas electronic structures have a full shell of outer electrons ***[1 mark]***, which is a very stable structure ***[1 mark]***.

Page 27
Ionic Bonding

Q1 The transfer of one electron from one atom to another ***[1 mark]***.

Q2



[1 mark for arrow showing electron transferred from potassium to bromine, 1 mark for correct outer shell electron configurations (with or without inner shells), 1 mark for correct charges]

Page 28
Ionic Compounds

Q1 a) It will have a high melting point ***[1 mark]*** because a lot of energy is needed to break the strong electrostatic attraction between the ions/the strong ionic bonds ***[1 mark]***.

b) When melted, the ions are free to move, so they can carry electric charge ***[1 mark]***.

c) MgS ***[1 mark]***.

Page 29
Warm-Up Questions

1) A charged atom or group of atoms.
2) 2+
3) 1–
4) high
5) When ionic compounds dissolve in water, the ions separate and are all free to move in the solution, so they'll carry an electric charge.
6) $Al(OH)_3$

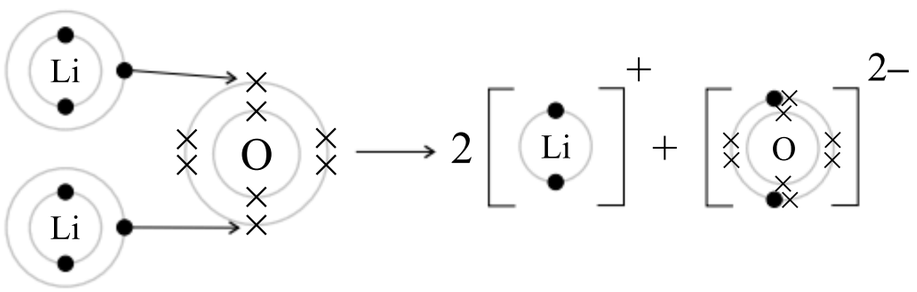
Exam Questions

1.1 Sodium will lose the electron from its outer shell to form a positive ion ***[1 mark]***. Fluorine will gain an electron to form a negative ion ***[1 mark]***.

1.2 Ionic compounds have a giant ionic lattice structure ***[1 mark]***.
The ions form a closely packed regular lattice arrangement ***[1 mark]***, held together by strong electrostatic forces of attraction between oppositely charged ions ***[1 mark]***.

2.1 lithium oxide ***[1 mark]***

2.2



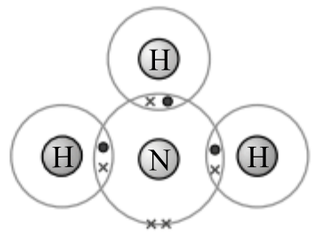
[1 mark for arrows shown correctly, 1 mark for correct electron arrangement and charge on lithium ion, 1 mark for correct electron arrangement and charge on oxygen ion]

2.3 When molten, the Li^+ and O^{2-} are able to move ***[1 mark]***. These ions are able to carry electric charge ***[1 mark]***.

2.4 The formula of the compound is **LiCl** ***[1 mark]***. Lithium is in Group 1, so forms 1+ ions. Chlorine is in Group 7, so forms 1– ions. Therefore there needs to be one lithium ion for every chlorine ion for the compound to be neutral ***[1 mark]***.

Page 30
Covalent Bonding

Q1



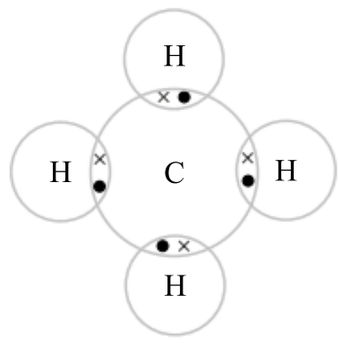
[1 mark for 3 shared pairs of electrons, 1 mark for correct number of electrons in outer shell of each atom (with or without inner shells on nitrogen)]

Page 32
Warm-Up Questions

1) A chemical bond made by the sharing of a pair of electrons between two atoms.
2) 3
3) gas or liquid
4) covalent bonds
5) intermolecular forces

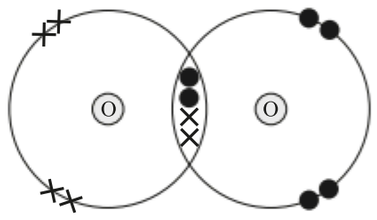
Exam Questions

1



[1 mark for showing a pairs of shared electrons between C and H. 1 mark for no other electrons shown]

2.1



[1 mark for showing 2 pairs of shared electrons between O atoms. 1 mark for showing two non-bonding pairs of electrons on each oxygen]

2.2 3 ***[1 mark]***

3.1 Hydrogen chloride doesn't contain any ions or delocalised electrons to carry a charge ***[1 mark]***.

3.2 E.g. a pair of electrons (one from the hydrogen atom and one from the chlorine atom) is shared between the two atoms ***[1 mark]***. The atoms are held together by the strong attraction between this shared pair of negatively charged electrons and the positively charged nuclei of the atoms ***[1 mark]***.

3.3 Chlorine is a larger molecule than hydrogen chloride ***[1 mark]*** and therefore there are greater intermolecular forces between molecules ***[1 mark]***. As the intermolecular forces are greater between chlorine molecules, its boiling point is higher than hydrogen chloride ***[1 mark]***.

Chlorine has three shells of electrons, whereas hydrogen only has one. So Cl_2 is a bigger molecule than HCl.

Page 33
Polymers and Giant Covalent Structures

Q1 $(C_2H_3Cl)_n$ ***[1 mark]***

Page 36

Warm-Up Questions

1) By covalent bonding

2) solid

3) E.g. diamond is much harder than graphite. Diamond doesn't conduct electricity whereas graphite does.

4) Buckminsterfullerene

5) The layers of atoms in metals are able to slide over each other.

Exam Questions

1.1 In a giant covalent structure, all of the atoms are bonded to each other with strong covalent bonds ***[1 mark]***. It takes lots of energy to break these bonds and melt the solid ***[1 mark]***.

1.2 E.g. diamond / graphite / silicon dioxide ***[1 mark]***

2.1 Graphite is formed of layers of carbon atoms that are held together weakly ***[1 mark]***. The layers can easily slide over each other, making graphite soft and slippery ***[1 mark]***.

2.2 In diamond, each carbon atom forms four covalent bonds in a very rigid structure ***[1 mark]***. This makes diamond very hard, so it would be good at cutting other substances ***[1 mark]***.

2.3 Any one from: e.g. delivering drugs / catalysts / lubricants / strengthening materials ***[1 mark]***

2.4 E.g. each carbon atom has a delocalised electron that is able to carry the charge ***[1 mark]***.

3.1 E.g.

[1 mark for showing two different sizes of atoms, 1 mark for showing irregular arrangement]

3.2 The regular arrangement of atoms in iron means that they can slide over each other, resulting in iron being soft ***[1 mark]***. Steel contains different sized atoms which distort the layers of iron atoms ***[1 mark]*** making it more difficult for them to slide over each other ***[1 mark]***.

Topic 2b — States of Matter and Nanoparticles

Page 39

Changing State

Q1 a) solid ***[1 mark]***

b) liquid ***[1 mark]***

c) liquid ***[1 mark]***

d) gas ***[1 mark]***

Page 40

Nanoparticles

Q1 a) Surface area of one face
= 50 nm × 50 nm = 2500 nm^2

Total surface area = 6 × 2500 nm = 15 000 nm^2 ***[1 mark]***

Volume = 50 nm × 50 nm × 50 nm
= 125 000 nm^3 ***[1 mark]***

Surface area to volume ratio
= surface area ÷ volume
= 15 000 nm^2 ÷ 125 000 nm^3
= **0.12 nm^{-1}** ***[1 mark]***

b) It would decrease ***[1 mark]*** by a factor of ten ***[1 mark]***.

Page 42

Warm-Up Questions

1) solid

2) The particles move faster.

3) $Na^+_{(aq)}$

4) E.g. coarse particles (between 2500 nm and 10 000 nm), fine particles (between 100 nm and 2500 nm) and nanoparticles (between 1 nm and 100 nm).

5) The surface area to volume ratio increases.

Exam Questions

1.1 In solids, there are **strong** ***[1 mark]*** forces of attraction between particles, which hold them in fixed positions in a **regular** ***[1 mark]*** arrangement. The particles don't **move** ***[1 mark]*** from their positions, so solids keep their shape. The **hotter** ***[1 mark]*** the solid becomes, the more the particles in the solid vibrate.

1.2 The particles gain energy, so they move faster ***[1 mark]***. The intermolecular bonds are weakened and, when they have enough energy, they break ***[1 mark]***. This means the particles are free to move far apart from each other and the liquid becomes a gas ***[1 mark]***.

2 E.g. some people are concerned about the use of nanoparticles as not enough is known about their effects on the body following long-term use ***[1 mark]***. Testing over a long time period could be carried out before products containing nanoparticles are used by humans to help minimise the risk ***[1 mark]***.

3.1 The particles are free to move about / have virtually no forces of attraction between them ***[1 mark]***, so they move randomly, spreading out to fill the container ***[1 mark]***.

3.2 gas ***[1 mark]***

Topic 3 — Quantitative Chemistry

Page 44

Relative Formula Mass

Q1 a) A_r of H = 1 and A_r of O = 16
M_r of H_2O = (2 × 1) + 16 = **18** ***[1 mark]***

b) A_r of Li = 7, A_r of O = 16 and A_r of H = 1
So M_r of LiOH = 7 + 16 + 1 = **24** ***[1 mark]***

c) A_r of H = 1, A_r of S = 32 and A_r of O = 16
M_r of H_2SO_4 = (2 × 1) + 32 + (4 × 16) = **98** ***[1 mark]***

Q2 A_r of K = 39, A_r of O = 16 and A_r of H = 1
M_r of KOH = 39 + 16 + 1 = 56 ***[1 mark]***

$\frac{39}{56}$ × 100 = 69.6428... = **70%** (to the nearest %) ***[1 mark]***

Page 45

The Mole

Q1 M_r of H_2O = 16 + (2 × 1) = 18 ***[1 mark]***

number of moles = mass ÷ M_r
number of moles = 90 g ÷ 18 = **5 moles** (1 s.f.) ***[1 mark]***

Q2 M_r of KBr = 39 + 80 = 119 ***[1 mark]***
mass = number of moles × M_r
mass = 0.20 × 119 = **24 g** (2 s.f.) ***[1 mark]***

Page 46

Conservation of Mass

Q1 Total mass on the left hand side
$= M_r(H_2SO_4) + 2 \times M_r(NaOH)$

M_r of $H_2SO_4 = (2 \times 1) + 32 + (4 \times 16) = 98$
$2 \times M_r$ of $NaOH = 2 \times (23 + 16 + 1) = 80$

So total mass on the left hand side = 98 + 80
= 178 ***[2 marks for 178, 1 mark for either 98 or 80]***

Total mass on right hand side
$= M_r(Na_2SO_4) + 2 \times M_r(H_2O)$
M_r of $Na_2SO_4 = (2 \times 23) + 32 + (4 \times 16) = 142$
$2 \times M_r$ of $H_2O = 2 \times [(2 \times 1) + 16] = 36$

142 + 36 = 178
[2 marks for 178, 1 mark for either 142 or 36]
The total M_r on the left-hand side is equal to the total M_r on the right-hand side, so mass is conserved ***[1 mark]***.

Page 47

Warm-Up Questions

1) relative formula mass

2) Avogadro constant

3) A mole is an amount of a substance that contains as many particles as the Avogadro constant (6.02×10^{23}).

4) 56 g

5) One mole of O_2 weighs $16 \times 2 = 32$ g

Exam Questions

1.1 M_r of $BF_3 = 11 + (19 \times 3) =$ **68** ***[1 mark]***

1.2 M_r of $B(OH)_3 = 11 + (17 \times 3) =$ **62** ***[1 mark]***

2 number of moles $= \dfrac{\text{mass in grams}}{M_r}$ ***[1 mark]***

3 M_r of KOH $= (39 + 16 + 1) = 56$ ***[1 mark]***
Mass of 4 moles of KOH $= 4 \times 56 = 224$ g ***[1 mark]***
Extra mass needed = 224 – 140 = **84 g** ***[1 mark]***

4.1 Oxygen in the air, which previously couldn't be weighed, has now reacted with metal **X** to form a solid oxide ***[1 mark]***.

4.2 100 – 40 = **60%** ***[1 mark]***

4.3 Mass of metal **X** in 8.0 g of mixture
$= 8.0 \times (24 \div 100) = 1.92$ g ***[1 mark]***
The metal oxide contains 60% metal **X** by mass, so the mass of metal oxide used to make the mixture
$= 1.92 \div (60 \div 100) =$ **3.2 g** (2 s.f.) ***[1 mark]***.

Page 48

The Mole and Equations

Q1 a) N_2: $\dfrac{84}{28} =$ **3 mol** ***[1 mark]***

H_2: $\dfrac{18}{2} =$ **9 mol** ***[1 mark]***

NH_3: $\dfrac{102}{17} =$ **6 mol** ***[1 mark]***

b) Divide by the smallest number of moles (3):
N_2: $\dfrac{3}{3} = 1$ H_2: $\dfrac{9}{3} = 3$ NH_3: $\dfrac{6}{3} = 2$ ***[1 mark]***
The balanced symbol equation is:
$N_2 + 3H_2 \rightarrow 2NH_3$ ***[1 mark]***

Page 49

Limiting Reactants

Q1 $M_r(KBr) = 39 + 80 = 119$ and
$M_r(KCl) = 39 + 35.5 = 74.5$ ***[1 mark]***
No. of moles of KBr = 23.8 ÷ 119 = 0.200 mol ***[1 mark]***
From the reaction equation, 2 moles of KBr react to form 2 moles of KCl. So 0.200 moles of KBr reacts to form 0.200 moles of KCl ***[1 mark]***.
Mass KCl = 74.5 × 0.200 = **14.9 g** (3 s.f.) ***[1 mark]***

Page 51

Concentration Calculations

Q1 Moles of KOH: 0.150 × (22.5 ÷ 1000)
= 0.003375 ***[1 mark]***
From the reaction equation, 1 mole of KOH reacts with 1 mole of HNO_3, so 0.003375 mol of KOH reacts with 0.003375 mol of HNO_3 ***[1 mark]***.
Concentration of HNO_3
= 0.003375 ÷ (25.0 ÷ 1000) ***[1 mark]***
= **0.135 mol/dm^3** (3 s.f.) ***[1 mark]***

Pages 52-53

Warm-Up Questions

1) The number of moles of each substance taking part in or made by the reaction.

2) To make sure that all of the other reactant reacts.

3) It halves.

4) Number of moles of gas = volume of gas (in dm^3) ÷ 24
= 2.4 ÷ 24 = **0.10 moles** (2 s.f.)

5) 160 ÷ 1000 = 0.160 dm^3
concentration = mass ÷ volume = 20.0 ÷ 0.160
= **125 g/dm^3** (3 s.f.)

6) number of moles = concentration × volume
= 0.10 × (25 ÷ 1000) = **0.0025 mol** (3 s.f.)

7) $M_r(Na_2CO_3) = (2 \times 23) + 12 + (3 \times 16) = 106$
Concentration in grams per dm^3 = 0.0250 × 106
= **2.65 g/dm^3** (3 s.f.)

Exam Questions

1.1 Li: 3.5 ÷ 7 = **0.5 mol**
O_2: 4.0 ÷ 32 = **0.125 mol**
Li_2O: 7.5 ÷ 30 = **0.25 mol**
[2 marks for all three correct answers, otherwise 1 mark for two correct answers]

1.2 Divide by the smallest number of moles (0.125):
Li: $\dfrac{0.5}{0.125} = 4$ O_2: $\dfrac{0.125}{0.125} = 1$ Li_2O: $\dfrac{0.25}{0.125} = 2$ ***[1 mark]***
The balanced symbol equation is:
$4Li + O_2 \rightarrow 2Li_2O$ ***[1 mark]***

2.1 M_r of $NaHCO_3 = 23 + 1 + 12 + (3 \times 16) = 84$ ***[1 mark]***
no. of moles of $NaHCO_3 = 6.0 \div 84 = 0.0714...$ mol ***[1 mark]***
1 mole of Na_2SO_4 is made for every 2 moles of $NaHCO_3$ that reacts, so no. of moles of $Na_2SO_4 = 0.0714... \div 2$
$= 0.0357...$ mol ***[1 mark]***
M_r of $Na_2SO_4 = (2 \times 23) + 32 + (4 \times 16) = 142$ ***[1 mark]***
Mass of $Na_2SO_4 = 142 \times 0.0357... = 5.071... =$ **5.1 g** (2 s.f.) ***[1 mark]***

2.2 The limiting reactant is the reactant that gets used up and therefore limits the amount of product formed ***[1 mark]***.

3 $M_r(CaCO_3) = 40 + 12 + (3 \times 16) = 100$ ***[1 mark]***
no. of moles of $CaCO_3 = 25 \div 100 = 0.25$ mol ***[1 mark]***
1 mole of CO_2 is formed for every mole of $CaCO_3$ that reacts.
So 0.25 moles of $CaCO_3$ reacts to form 0.25 moles of CO_2.
Volume of $CO_2 = 0.25 \times 24$ dm^3 ***[1 mark]***
= **6.0 dm^3** (2 s.f.) ***[1 mark]***

4.1 number of moles = concentration × volume

number of moles = 1.00 × (30.4 ÷ 1000)
= **0.0304 mol** (3 s.f.) ***[1 mark]***

4.2 H_2SO_4 and NaOH react in a 1:2 ratio, so number of moles of H_2SO_4

= 0.0304 ÷ 2 = **0.0152 mol** (3 s.f.) ***[1 mark]***

4.3 concentration = number of moles ÷ volume ***[1 mark]***

volume = 25 ÷ 1000 = 0.025

concentration = 0.0152 ÷ 0.025
= **0.608 mol/dm³** (3 s.f.) ***[1 mark]***

4.4 M_r of H_2SO_4 = (2 × 1) + 32 + (4 × 16) = 98 ***[1 mark]***
Concentration in g/dm³ = moles × M_r

Concentration in g/dm³ = 0.608 × 98
= 59.584
= **59.6 g/dm³** (3 s.f.) ***[1 mark]***

5 Moles of NaOH = concentration × volume
= 0.10 × (9.0 ÷ 1000) = 0.00090 mol ***[1 mark]***

HA and NaOH react in a 1:1 ratio, so moles of HA = 0.00090 moles ***[1 mark]***

Concentration of HA = number of moles ÷ volume ***[1 mark]***

volume = 25 ÷ 1000 = 0.025

Concentration of HA = 0.00090 ÷ 0.025
= **0.036 mol/dm³** (2 s.f.) ***[1 mark]***

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Atom Economy

Q1 a) Reaction B, because it only has one product / the atom economy is 100% ***[1 mark]***.

b) Total mass of all reactants:
$M_r(NH_4Cl)$ = 14 + (4 × 1) + 35.5 = 53.5 ***[1 mark]***
$M_r(CaO)$ = 40 + 16 = 56 ***[1 mark]***

(2 × 53.5) + (2 × 56) = 219 ***[1 mark]***
Total mass of desired product
= 2 × $M_r(NH_3)$ = 34 ***[1 mark]***

Atom economy = $\frac{34}{219}$ × 100 = 15.525...
= **16%** (to the nearest %) ***[1 mark]***

Page 55

Percentage Yield

Q1 $\frac{2.31}{2.72}$ × 100 = 84.926... = **84.9%** (3 s.f.) ***[1 mark]***

Page 56

Warm-Up Questions

1) 100%

All reactions with one product will have 100% atom economy.

2) Because a low atom economy means resources are used up quickly, so any non-renewable resources are likely to run out. It also means lots of waste is likely to be produced, and this has to be stored somewhere.

3) percentage yield = (4 ÷ 5) × 100 = **80%**

Exam Questions

1 To calculate the atom economy of a reaction, you need to know the mass of products you expected to form, and the mass of products that actually formed ***[1 mark]***.

2 M_r of ethanol = (12 × 2) + 6 + 16 = 46,
M_r of ethene = (12 × 2) + 4 = 28 ***[1 mark]***
Atom economy = (28 ÷ 46) × 100 ***[1 mark]***
= 60.8695... = **61 %** (2 s.f.) ***[1 mark]***

3 How to grade your answer:

Level 0: No reasons why yields are always less than 100% are given. ***[No marks]***

Level 1: Brief description of one reason why yields are less than 100% is given. ***[1 to 2 marks]***

Level 2: Two reasons why yields are always less than 100% are described clearly. ***[3 to 4 marks]***

Level 3: Three reasons why yields are always less than 100% are described clearly. ***[5 to 6 marks]***

Here are some points your answer may include:

Some reactions are reversible. Some of the reactants will never be completely converted to products because the reaction goes both ways.

If a liquid is filtered to remove solid particles, some of the liquid or solid could be lost when it's separated from the reaction mixture.

Transferring solutions from one container to another often leaves behind traces on the containers.

There may be unexpected reactions happening resulting in extra products forming other than the ones you wanted.

4.1 M_r(CuO) = 63.5 + 16 = 79.5 ***[1 mark]***
So moles of CuO = 4.0 ÷ 79.5 = 0.0503... mol

From the equation, 1 mole of Cu is produced for every 1 mol of CuO that reacts. So 0.0503... mol CuO reacts to form 0.0503... mol Cu ***[1 mark]***.

So theoretical yield of Cu = number of moles × M_r
= 0.0503... × 63.5 = 3.1949... = **3.2 g** (2 s.f.) ***[1 mark]***

4.2 Percentage yield = (2.8 ÷ 3.2) × 100 ***[1 mark]***
= 87.5 = **88%** (2 s.f.) ***[1 mark]***

Topic 4 — Chemical Changes

Page 60

Strong Acids and Weak Acids

Q1 Any one of, e.g. sulfuric acid / nitric acid / hydrochloric acid ***[1 mark]***.

Q2 Change in pH = 3 – 6 = –3
Change in concentration of $H^+ = 10^{-(-3)} = 10^3 = 1000$
So, the concentration of H^+ is 1000 times greater at pH 3 than at pH 6 ***[1 mark]***.

Page 61

Reactions of Acids

Q1 calcium carbonate + hydrochloric acid → calcium chloride + carbon dioxide + water ***[1 mark for calcium chloride, 1 mark for carbon dioxide and water]***.

$CaCO_3 + 2HCl \rightarrow CaCl_2 + CO_2 + H_2O$
[1 mark for the correctly balanced equation]

Page 62

Warm-Up Questions

1) 0-14

2) neutral

3) Strong acids ionise completely in water, so all of the acid particles dissociate to release H^+ ions. Weak acids do not fully ionise, so only some of the acid particles dissociate to release H^+ ions.

4) Difference in pH = 5 – 2 = 3
Factor H^+ ion concentration changes by = 10^{-3} = **0.001**

You could give your answer here as '10^{-3}', '0.001' or 'it decreased by a factor of 1000' — they all mean the same thing.

5) copper nitrate and water

Exam Questions

1.1 H^+ ions/hydrogen ions ***[1 mark]***

1.2 neutralisation ***[1 mark]***

2.1 alkalis ***[1 mark]***

2.2 $2HNO_3 + Mg(OH)_2 \rightarrow \mathbf{Mg(NO_3)_2} + \mathbf{2H_2O}$
[1 mark for formulas of both products correct, 1 mark for putting a 2 in front of H_2O to balance the equation]

3.1 E.g. the excess silver carbonate will sink to the bottom of the flask and stay there / the mixture will stop producing bubbles (of carbon dioxide) ***[1 mark]***.

3.2 filtration ***[1 mark]***

3.3 Heat the silver nitrate solution gently using a water bath/electric heater to evaporate some of the water ***[1 mark]***. Leave the solution to cool until crystals form ***[1 mark]***. Filter out the crystals and dry them ***[1 mark]***.

4 How to grade your answer:

Level 0: There is no relevant information. ***[No marks]***

Level 1: There is a brief explanation of how to carry out a titration, but some major details are missing. ***[1 to 2 marks]***

Level 2: There is a good explanation of how to carry out a titration. Some minor details may be missing. ***[3 to 4 marks]***

Level 3: There is a clear and detailed explanation of how to carry out a titration. ***[5 to 6 marks]***

Here are some points your answer may include:
Use a pipette (and pipette filler) to measure out 25.0 cm^3 of the alkali and put it into a conical flask.
Add two or three drops of an indicator / phenolphthalein / litmus / methyl orange to the alkali.
Use a funnel to fill a burette with the acid.
Record the initial volume of the acid in the burette.
Use the burette to slowly add the acid to the alkali.
Swirl the flask regularly.
Go especially slowly as you approach the end-point from the rough titration.
Add the acid drop-by-drop, so you can stop immediately after the colour change happens.
The indicator will change colour when all the alkali has been neutralised/at the end-point.
Record the final volume of acid in the burette.

Page 63

The Reactivity Series

Q1 $2Na_{(s)} + 2H_2O_{(l)} \rightarrow 2NaOH_{(aq)} + H_{2(g)}$
[1 mark for correct reactants and products, 1 mark for balancing, 1 mark for state symbols]

Page 64

Separating Metals From Metal Oxides

Q1 $2PbO + C \rightarrow 2Pb + CO_2$
[1 mark for the correct products, 1 mark for the correctly balanced equation]

Q2 Carbon is less reactive than calcium and therefore will not reduce calcium oxide. / Calcium is more reactive than carbon, so calcium oxide won't be reduced by carbon ***[1 mark]***.

Page 66

Displacement Reactions & Ionic Equations

Q1 a) $Zn_{(s)} + Fe^{2+}_{(aq)} \rightarrow Zn^{2+}_{(aq)} + Fe_{(s)}$ ***[1 mark]***

b) $Zn_{(s)}$ is being oxidised ***[1 mark]***.
$Fe^{2+}_{(aq)}$ is being reduced ***[1 mark]***.

Page 67

Warm-Up Questions

1) How easily it loses electrons to form positive ions.

2) magnesium chloride

3) A metal hydroxide and hydrogen.

4) Hydrogen ions/H^+ are being reduced.
Lithium metal/Li is being oxidised.

5) a) $Zn_{(s)} + Cu^{2+}_{(aq)} \rightarrow Zn^{2+}_{(aq)} + Cu_{(s)}$

b) zinc / Zn

Exam Questions

1.1 Any one from: potassium / sodium / calcium ***[1 mark]***.

1.2 copper ***[1 mark]***

Metals below hydrogen in the reactivity series don't react with acids.

1.3 E.g. iron and dilute acid would produce bubbles of hydrogen gas ***[1 mark]*** more slowly than zinc and dilute acid / very slowly ***[1 mark]***.

You don't need to use these exact words — the two points you need to cover are that it would still produce hydrogen gas, but at a slower rate than with zinc.

2.1 $2Fe_2O_3 + 3C \rightarrow 4Fe + 3CO_2$
[1 mark for formulas of all reactants and products correct, 1 mark for correctly balancing the equation]

2.2 Oxygen is being lost from iron oxide / the iron(III) ions in the iron oxide are gaining electrons ***[1 mark]***.

3 In A there will be no change ***[1 mark]***. In B the solution will have changed from blue to green ***[1 mark]*** and the grey metal will be coated in orange metal ***[1 mark]***. Iron is more reactive than copper ***[1 mark]***. So in A, the copper cannot displace the iron from iron sulfate ***[1 mark]***, but in B, the iron does displace the copper from copper sulfate ***[1 mark]***.

Page 68

Electrolysis

Q1 a) chlorine gas/Cl_2 ***[1 mark]***

b) calcium atoms/Ca ***[1 mark]***

Page 69

Electrolysis of Aqueous Solutions

Q1 a) bromine gas/Br_2 ***[1 mark]***

b) copper atoms/Cu ***[1 mark]***

Page 70

Warm-Up Questions

1) the cathode / negative electrode

2) zinc metal and chlorine gas

3) reduced

4) hydrogen gas

Exam Questions

1.1 To lower the melting point of the aluminium oxide ***[1 mark]***.

1.2 $\mathbf{Al^{3+}} + 3e^- \rightarrow \mathbf{Al}$ ***[1 mark]***

1.3 $\mathbf{2O^{2-}} \rightarrow \mathbf{O_2} + 4e^-$ ***[1 mark]***

1.4 The oxygen made at the positive electrode reacts with the carbon in the electrode to make carbon dioxide ***[1 mark]***, wearing the electrode away ***[1 mark].***

2.1 hydrogen ***[1 mark]***

2.2 $2H^+ + 2e^- \rightarrow H_2$ ***[1 mark]***

2.3 chlorine ***[1 mark]***

2.4 $2Cl^- \rightarrow Cl_2 + 2e^-$ / $2Cl^- - 2e^- \rightarrow Cl_2$ ***[1 mark]***

2.5 Sodium is more reactive than hydrogen, so the sodium ions from the sodium chloride stay in solution ***[1 mark]***. Hydroxide ions are also left in solution when hydrogen is produced from water ***[1 mark]***.

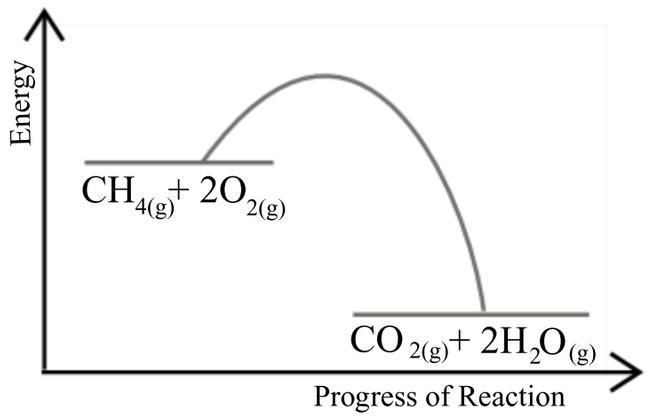
2.6 Copper is less reactive than hydrogen ***[1 mark]***.

Topic 5 — Energy Changes

Page 73

More Exothermic & Endothermic Reactions

Q1



[1 mark for correct axes, 1 mark for correct energy levels of reactants and products, 1 mark for correct shape of curve linking the reactants to the products]

Page 74

Bond Energies

Q1 Energy required to break original bonds:
$(1 \times N{\equiv}N) + (3 \times H{-}H)$
$= 941 \text{ kJ/mol} + 1308 \text{ kJ/mol} = 2249 \text{ kJ/mol}$ ***[1 mark]***
Energy released by forming new bonds:
$(6 \times N{-}H) = 2346 \text{ kJ/mol}$ ***[1 mark]***
Overall energy change:
$= 2249 \text{ kJ/mol} - 2346 \text{ kJ/mol} =$ **–97 kJ/mol** ***[1 mark]***

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Warm-Up Questions

1) An exothermic reaction is one which transfers energy to the surroundings.

2) An endothermic reaction is one which takes in energy from the surroundings.

3) When bonds are formed.

4) The activation energy is the minimum amount of energy the reactants need to react when they collide with each other.

Exam Questions

1 B ***[1 mark]***

2.1 Exothermic, because the temperature of the mixture has increased ***[1 mark]***, therefore the particles have transferred energy to the reaction mixture ***[1 mark]***.

2.2 A lid was placed on the cup to reduce energy lost to the surroundings (by evaporation) ***[1 mark]***.

3.1 C – H and O = O ***[1 mark]***

3.2 $(4 \times 414) + (2 \times 494) = 2644 \text{ kJ/mol}$

This is the amount of energy required to break the bonds in CH_4 and $2O_2$.

$(2 \times 800) + (4 \times 459) = 3436 \text{ kJ/mol}$

This is the amount of energy released by forming the bonds in CO_2 and $2H_2O$.

2644 – 3436 = **–792 kJ/mol** ***[3 marks for correct answer, otherwise 1 mark for finding the energy needed to break the bonds in the reactants, 1 mark for finding the energy released by forming bonds in the products.]***

Page 76

Cells and Batteries

Q1 Metal 2, Metal 3, Metal 1 ***[1 mark]***.
Iron is less reactive than all three metals, so the most reactive metal will make the cell with the greatest voltage, since there will be the biggest difference in reactivity ***[1 mark]***. So the order of reactivity from lowest to highest is the order of the size of voltage from lowest to highest ***[1 mark]***.

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Warm-Up Questions

1) A liquid that contains ions which react with the electrodes.

2) You connect them together in series.

3) water (H_2O)

4) e.g. potassium hydroxide/phosphoric acid

Exam Questions

1 voltmeter ***[1 mark]***

2.1 In the fuel cell, there is reduction at the cathode ***[1 mark]*** and oxidation at the anode ***[1 mark]***.

2.2 Reduction, because the reactant/oxygen is gaining electrons ***[1 mark]***.

2.3 $2H_2 + O_2 \rightarrow 2H_2O$ ***[1 mark]***

2.4 How to grade your answer:

Level 0: There is no relevant information. ***[No marks]***

Level 1: There is a brief explanation of some relevant advantages of using fuel cells rather than rechargeable batteries to power cars. ***[1 to 2 marks]***

Level 2: There is clear and detailed explanation of several advantages of using fuel cells rather than rechargeable batteries to power cars. ***[3 to 4 marks]***

Here are some points your answer may include:

Hydrogen-oxygen fuel cells are less polluting to dispose of than rechargeable batteries because rechargeable batteries are made from toxic metal compounds.
Fuel cells can be reused many times but there is a limit to how often batteries in electric cars can be recharged.
Fuel cells are less expensive than batteries to make.
Fuel cells store more energy than batteries and so batteries need to be recharged more often, which can take a long time.

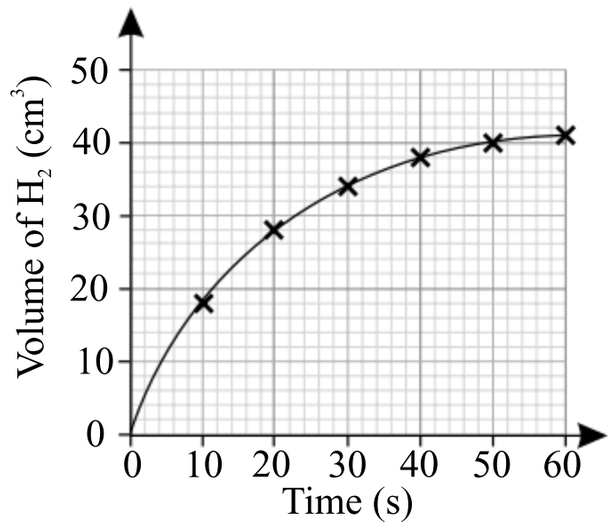
3.1 Metal 2, Metal 3, Metal 1 ***[1 mark]***. Zinc is less reactive than all 3 metals, so the most reactive metal will make the cell with the greatest voltage, since there will be the biggest difference in reactivity ***[1 mark]***. So the order of reactivity is simply the order of the size of voltage ***[1 mark]***.

3.2 E.g. join together three cells in series ***[1 mark]***, using Metal 3 and zinc for the electrodes in each cell ***[1 mark]***. / Join together two cells in series, using Metal 2 and zinc as the electrodes ***[1 mark]*** and one cell using Metal 1 and zinc as the electrodes ***[1 mark]***.

Topic 6 — The Rate and Extent of Chemical Change

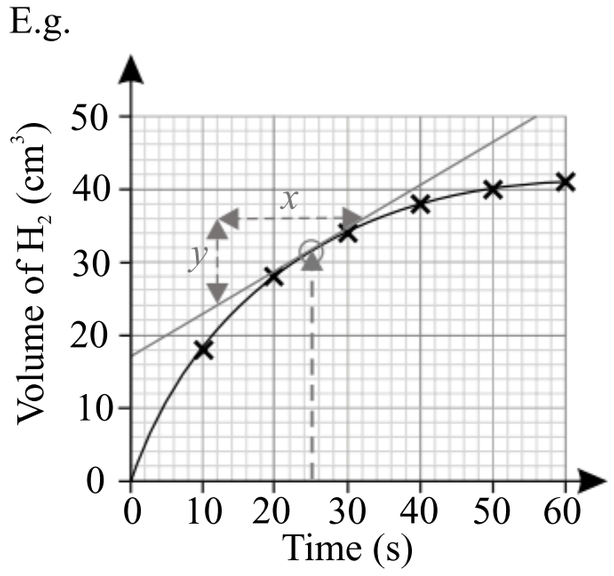
Page 84
Finding Reaction Rates from Graphs

Q1 a) E.g.



[1 mark for correctly marking on all 6 points, 1 mark for choosing a sensible scale for the axes, 1 mark for drawing a line of best fit.]

b) E.g.



change in $y = 36 - 24 = 12$,
change in $x = 32 - 12 = 20$,
rate = change in $y \div$ change in x
$= 12 \div 20 =$ **0.6 cm^3/s**

[1 mark for drawing a tangent at 25 s, 1 mark for correctly calculating a change in y from the tangent, 1 mark for correctly calculating a change in x from the tangent and 1 mark for a rate between 0.5 cm^3/s and 0.7 cm^3/s]

Pages 85-86
Warm-Up Questions

1) E.g. the rusting of iron is a reaction that happens very slowly. Explosions are very fast reactions.

2) There will be more particles in the same volume, which will lead to more frequent collisions, so the rate of reaction will increase.

3) g/s

4) E.g. observe a mark through the solution in which the precipitate forms and measure how long it takes for it to disappear.

Exam Questions

1.1 stopwatch/stopclock/timer ***[1 mark]***

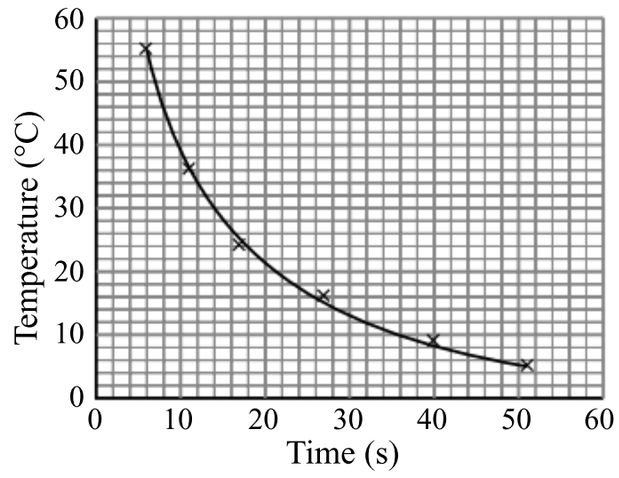
1.2 gas syringe ***[1 mark]***

1.3 E.g. place a conical flask on a mass balance ***[1 mark]*** and record the change in mass at regular intervals as the gas leaves the flask ***[1 mark]***.

2.1 Any two from: e.g. the concentration of sodium thiosulfate/ hydrochloric acid / the depth of liquid / the person judging when the black cross is obscured / the black cross used (size, darkness etc.).
[2 marks — 1 mark for each correct answer.]

Judging when a cross is completely obscured is quite subjective — two people might not agree on exactly when it happens. You can try to limit this problem by using the same person each time, but you can't remove the problem completely. The person might have changed their mind slightly by the time they do the next experiment — or be looking at it from a different angle, be a bit more bored, etc.

2.2



[1 mark for correctly drawn axes with a sensible scale, 2 marks for all points plotted correctly, otherwise 1 mark if 5 of 6 points plotted correctly, 1 mark for a suitable line of best fit]

2.3 As the temperature decreases the time increases, meaning that the reaction is slower as the temperature decreases ***[1 mark]***.

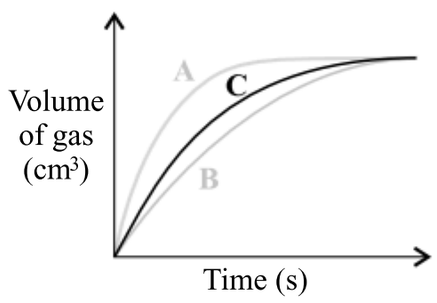
2.4 At each temperature it would take longer for the reaction to complete ***[1 mark]***.

2.5 A lower concentration of reactant means there are fewer particles in the same volume of solution ***[1 mark]***. This makes collisions between the reactant particles less frequent ***[1 mark]***, and therefore decreases the rate of reaction ***[1 mark]***.

2.6 E.g. by repeating the experiment ***[1 mark]***. If the results gained are similar then the experiment is repeatable ***[1 mark]***.

3.1 A ***[1 mark]***

3.2



Curve C, as left. The curve should be between curves A and B ***[1 mark]***.

3.3 The variables are controlled so that you can tell if the variable you're changing/the independent variable is causing the results seen ***[1 mark]***.

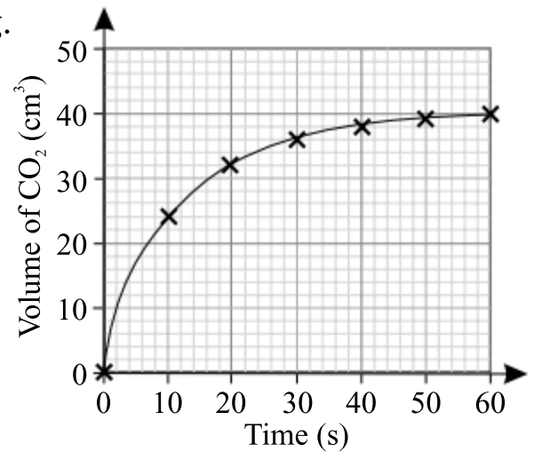
3.4 E.g. yes, because the same volume of gas was produced in each experiment ***[1 mark]***.

This suggests that the same concentration and volume of acid (or an excess of acid) was used in both experiments.

3.5 Measuring how quickly the reaction loses mass ***[1 mark]***.

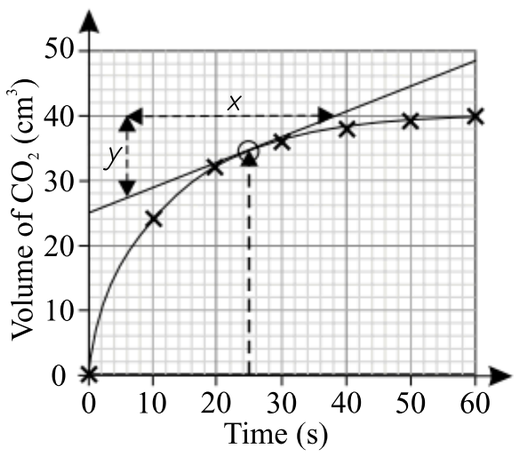
4.1 Mean rate = amount of product formed ÷ time
$= 40 \div 60 = 0.666... =$ **0.67 cm^3/s** (2 s.f.)
[2 marks for correct answer to 2 s.f., otherwise 1 mark for correct working]

4.2 E.g.



[1 mark for correctly drawn axes with a sensible scale, 2 marks for correctly marking on all 6 points, otherwise 1 mark if 5 of 6 points plotted correctly, 1 mark for a suitable line of best fit.]

4.3 E.g.



rate at 25 s = gradient of tangent at 25 s
change in y = 40 – 27 = 13, change in x = 38 – 6 = 32,
rate = change in y ÷ change in x = 13 ÷ 32 = **0.4 cm^3/s**

[1 mark for drawing a tangent at 25 s, 1 mark for correctly calculating a change in y from the tangent, 1 mark for correctly calculating a change in x from the tangent and 1 mark for a rate between 0.3 cm/s and 0.5 cm/s]

5 How to grade your answer:

Level 0: No relevant information is given. ***[No marks]***

Level 1: One or two ways of increasing the rate are described and there's reference to collision theory. ***[1 to 2 marks]***

Level 2: At least two ways of increasing the rate are given with appropriate reference to collision theory. ***[3 to 4 marks]***

Level 3: There is a clear and detailed discussion of three ways by which the rate can be increased, which includes relevant references to collision theory. ***[5 to 6 marks]***

Here are some points your answer may include:

Collision theory says that the rate of reaction depends on how often and how hard the reacting particles collide with each other.
If the particles collide hard enough (with enough energy) they will react.
Increasing the temperature makes particles move faster, so they collide more often and with greater energy.
This will increase the rate of reaction.
If the surface area of the nickel catalyst is increased then the particles around it will have more area to work on.
This increases the frequency of successful collisions and will increase the rate of reaction.
Increasing the pressure of the hydrogen will mean the particles are more squashed up together. This will increase the frequency of the collisions and increase the rate of reaction.

Page 88

Le Chatelier's Principle

Q1 a) More C_2H_5OH would be produced ***[1 mark]***.

b) Less NO_2 would be produced ***[1 mark]***.

c) More CO_2 would be produced ***[1 mark]***.

Page 89

Warm-Up Questions

1) They are the same.

2) The equilibrium will move to the left/towards reactants and the amount of reactants will increase.

3) It would have no effect (because there are equal numbers of gas molecules on both sides).

Exam Questions

1.1 The reaction is reversible ***[1 mark]***.

1.2 Both (the forward and reverse) reactions are taking place at exactly the same rate ***[1 mark]***.

2.1 Less $NO_{2(g)}$ will be produced ***[1 mark]***, because there are more molecules of gas on the right hand side of the equation ***[1 mark]***.

2.2 It will have no effect ***[1 mark]***, because there are the same number of molecules of gas on both sides of the equation ***[1 mark]***.

3.1 It gives out energy ***[1 mark]***, because it's exothermic ***[1 mark]***.

As the forward reaction is endothermic you know that the reverse reaction is exothermic and therefore gives out energy.

3.2 A reversible reaction is always exothermic in one direction and endothermic in the other direction ***[1 mark]***, so a change in temperature will always favour one reaction more than the other ***[1 mark]***.

3.3 It would move to the right ***[1 mark]***.

Topic 7 — Organic Chemistry

Page 91

Hydrocarbons

Q1 $C_2H_6 + 3½O_2 \rightarrow 2CO_2 + 3H_2O$ or
$2C_2H_6 + 7O_2 \rightarrow 4CO_2 + 6H_2O$
[1 mark for correct reactants and products, 1 mark for correctly balancing]

Q2 a) $C_{10}H_{22}$ is more viscous than C_5H_{12} ***[1 mark]***.

b) $C_{10}H_{22}$ has a higher boiling point than C_5H_{12} ***[1 mark]***.

c) $C_{10}H_{22}$ is less flammable than C_5H_{12} ***[1 mark]***.

Page 92

Fractional Distillation

Q1 The diagram shows that hydrocarbons in petrol have a shorter chain length than the hydrocarbons in diesel, so petrol has a lower boiling point than diesel ***[1 mark]***.

Q2 It's hot at the bottom and cooler at the top ***[1 mark]***.

Page 93

Uses and Cracking of Crude Oil

Q1 $C_5H_{12} \rightarrow C_2H_4 + C_3H_8$ ***[1 mark]***

Pages 94-95

Warm-Up Questions

1) methane, ethane, propane, butane

2) single covalent bonds

3) When burnt they release energy.

4) e.g. fuel for transport / feedstock for making new compounds

5) long-chain hydrocarbons

Exam Questions

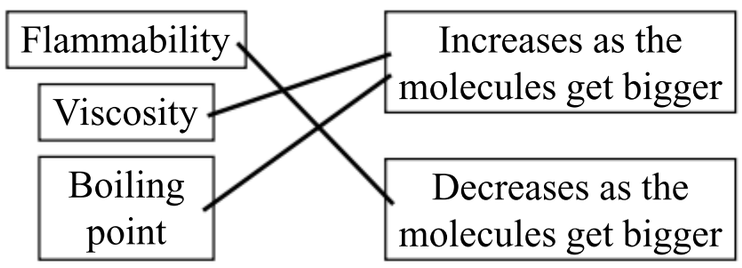
1 Carbon-carbon single bonds and carbon-hydrogen single bonds ***[1 mark]***.

2.1 butane ***[1 mark]***

2.2 C_4H_{10} ***[1 mark]***

2.3 Any two from: e.g. polymers / solvents / lubricants / detergents ***[2 marks — 1 mark for each correct answer]***

3



[1 mark for each correctly drawn line]

4.1 **F** ***[1 mark]***

4.2 **A** ***[1 mark]***

4.3 Inside the fractionating column there is a temperature gradient with the hottest part at the bottom and coolest at the top ***[1 mark]***. Crude oil that is gaseous moves up the column and when hydrocarbons in the gas reach a part of the column where the temperature is lower than their boiling points they condense and drain out of the column ***[1 mark]***. Hydrocarbons with longer chain lengths have higher boiling points so condense lower down the column whereas hydrocarbons with shorter chain lengths have lower boiling points so condense higher up the column ***[1 mark]***.

5.1 carbon dioxide/CO_2 ***[1 mark]*** and water/H_2O ***[1 mark]***

5.2 $C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$ ***[1 mark for correct reactants and products, 1 mark for correct balancing]***

5.3 oxidation ***[1 mark]***

6.1 Because shorter-chain hydrocarbons generated by cracking tend to be more useful than longer ones ***[1 mark]***.

6.2 $C_8H_{18} \rightarrow \mathbf{C_2H_6} + \mathbf{2}C_3H_6$ ***[1 mark for correct products, 1 mark for correct balancing]***

6.3 E.g. catalytic cracking is where long-chain hydrocarbons are vaporised ***[1 mark]*** and then passed over a hot powdered aluminium catalyst ***[1 mark]***. Or they can be cracked by steam cracking which is where long-chain hydrocarbons are vaporised and mixed with steam ***[1 mark]***. They are then heated to a very high temperature ***[1 mark]***.

You don't have to mention the names of the different cracking methods (catalytic and steam) but it might help structure your answer.

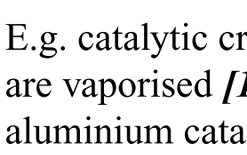
Page 97

Reactions of Alkenes

Q1 The solution turns from orange to colourless ***[1 mark]***.

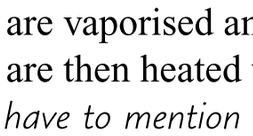
Q2 propene + hydrogen → propane ***[1 mark]***

Displayed formula of propene:



[1 mark]

Displayed formula of product (propane):



[1 mark]

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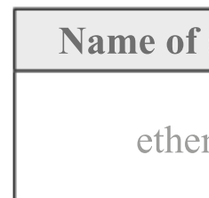
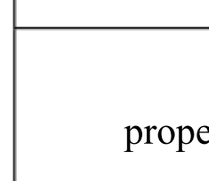
Addition Polymers

Q1 poly(bromoethene) ***[1 mark]***

Q2



[1 mark]



[1 mark]

Page 99

Warm-Up Questions

1) Alkenes are hydrocarbons (compounds that only contain hydrogen and carbon atoms) and have the functional group C=C.

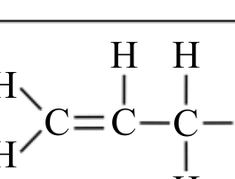
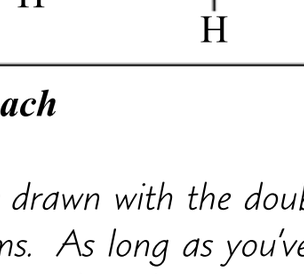
2) It contains two fewer hydrogen atoms than the alkane with the same number of carbon atoms.

3) The C=C bond opens up to make a single bond and the two carbon atoms can each bond to a halogen atom.

4) The bromine water will go colourless.

Exam Questions

1

Name of alkene	Formula	Displayed formula
ethene	C_2H_4	
propene	C_3H_6	

[4 marks available — 1 mark for each correct name or formula]

The displayed formula for propene could also be drawn with the double bond between the second and third carbon atoms. As long as you've got the right number of hydrogen atoms attached to each carbon atom, it doesn't matter which two carbons the double bond is between.

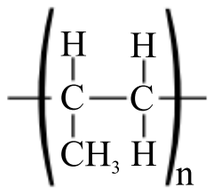
2.1 $C_2H_{4(g)} + H_2O_{(g)} \rightarrow C_2H_5OH_{(g)}$ ***[1 mark for correct reactants and products, 1 mark for correct state symbols]***

2.2 addition ***[1 mark]***

2.3 $C_2H_4 + H_2 \rightarrow C_2H_6$ ***[1 mark for correct reactants and products]***

2.4 Alkenes undergo incomplete combustion ***[1 mark]***.

3.1



[1 mark for all bonds drawn correctly]

3.2 poly(propene) ***[1 mark]***

Page 101

Carboxylic Acids

Q1 a) You would see bubbles of gas form ***[1 mark]*** as carbon dioxide is produced ***[1 mark]***.

b) sodium carbonate + methanoic acid → sodium methanoate + water + carbon dioxide ***[1 mark]***

Page 104

Warm-Up Questions

1) $C_nH_{2n+1}OH$

2) methanoic acid, ethanoic acid, propanoic acid, butanoic acid

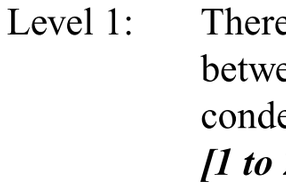
3) -COO-

4) condensation polymerisation

5) amino acids

Exam Questions

1 **C** ***[1 mark]***

2.1 

[1 mark for showing the –COOH functional group correctly, 1 mark for showing three hydrogens surrounding the carbon atom that doesn't have the functional group attached]

2.2 ethyl ethanoate ***[1 mark]***

2.3 Carboxylic acids don't ionise completely in water/ not all acid molecules release their H^+ ions ***[1 mark]***. The have a higher pH than solutions of strong acids in the same concentration ***[1 mark]***.

3.1 propanol ***[1 mark]***

3.2 $C_2H_5OH + 3O_2 \rightarrow 2CO_2 + 3H_2O$

[1 mark for correctly balanced equation]

4 How to grade your answer:

Level 0: There is no relevant information. ***[0 marks]***

Level 1: There is a brief description of some differences between addition polymerisation and condensation polymerisation. ***[1 to 2 marks]***

Level 2: There is a detailed description of all the key differences between addition polymerisation and condensation polymerisation. ***[3 to 4 marks]***

Here are some points your answer may include:

Addition polymerisation only has one monomer type containing a carbon-carbon double bond whereas condensation polymerisation includes two monomer types or one type of monomer type with two different functional groups.

Condensation polymerisation produces two products, the polymer and a small molecule but addition polymerisation only produces one product.

Addition polymerisation involves only one functional group in the monomer whereas condensation polymerisation occurs between two functional groups on each monomer.

Topic 8 — Chemical Analysis

Page 108

R_f Values

Q1 $R_f = \frac{6.3}{8.4} =$ **0.75** ***[1 mark]***

Page 109

Warm-Up Questions

1) A pure substance is a substance that only contains one compound or one element.

2) Impurities in a substance will increase the boiling point of the substance and may result in it boiling over a wider temperature range.

3) E.g. paint / drugs / cleaning products / fuels / cosmetics / fertilisers / metal alloys / food / drink.

4) Chromatography separates out the substances in a mixture.

Exam Questions

1 The sample melts over a range of temperatures ***[1 mark]***. The melting point is lower than that of pure aspirin ***[1 mark]***.

2.1 D ***[1 mark]***

2.2 A ***[1 mark]***

2.3 C ***[1 mark]***

2.4 R_f = distance travelled by substance in sunrise yellow ÷ distance travelled by solvent

$R_f = 9.0 \div 12.0 =$ **0.75**

[2 marks for correct answer, otherwise 1 mark for using the correct formula to calculate R_f]

Page 110

Tests for Gases and Anions

Q1 $2AgNO_3 + CaCl_2 \rightarrow 2AgCl + Ca(NO_3)_2$

[1 mark for correct reactants and products, 1 mark for correctly balancing]

Page 111

Tests for Cations

Q1 The metal ions burn with a distinctive colour ***[1 mark]***.

Q2 a) Al^{3+} ***[1 mark]***

b) $AlCl_3 + 3NaOH \rightarrow Al(OH)_3 + 3NaCl$

[1 mark for correct reactants and products, 1 mark for correctly balancing]

Page 113

Warm-Up Questions

1) Hydrogen:
Light a splint and hold it at the open end of a test tube containing the gas. If the gas is hydrogen, a squeaky pop is produced.
Oxygen:
Hold a glowing splint in a test tube containing your gas of interest. If oxygen is present, the splint will relight.

2) lilac

3) a) blue

b) $Cu^{2+}_{(aq)} + 2OH^-_{(aq)} \rightarrow Cu(OH)_{2(s)}$

4) The concentration of the ion in the solution.

Exam Questions

1

Flame colour	Metal ion
green	**Ba^{2+}**
crimson	Li^+
yellow	**Na^+**
orange-red	Ca^{2+}

[1 mark for each correct answer.]

2.1 A sample of the element is put into a flame ***[1 mark]***. The sample gives out light when heated / Heating the sample excites the electrons in the atoms. When the electrons drop back to their original energy levels, they release energy as light ***[1 mark]***. This light is passed through a spectroscope, which produces the line spectrum ***[1 mark]***.

2.2 Caesium and rubidium have different arrangements of electrons ***[1 mark]***, so emit different wavelengths of light when heated ***[1 mark]***.

3.1 iron(II) iodide ***[1 mark for iron(II), 1 mark for iodide]***

3.2 No, the student is not correct. The sample produced a white precipitate with hydrochloric acid and barium chloride solution, so he can be sure that it contains sulfate ions ***[1 mark]***. Magnesium does produce a white precipitate with sodium hydroxide ***[1 mark]***, but so do other metal ions / aluminium and calcium, so he cannot be sure that it contains magnesium ions ***[1 mark]***.

Topic 9 — Chemistry of the Atmosphere

Page 116
Greenhouse Gases and Climate Change

Q1 The Sun gives out short wavelength radiation ***[1 mark]*** which is absorbed by the Earth and then re-emitted as long wavelength radiation ***[1 mark]***. This radiation is absorbed by greenhouse gases in the atmosphere ***[1 mark]***. Greenhouse gases give out the long wavelength radiation in all directions including back towards the Earth, causing the temperature to rise ***[1 mark]***.

Page 119
Warm-Up Questions

1) carbon dioxide

2) E.g. limestone / coal

3) E.g. coal / crude oil / natural gas

4) E.g. walk, cycle, or take public transport instead of using their car for a journey / reduce their use of air travel / turn down their heating at home.

5) Any two from: e.g. carbon monoxide / soot / unburned hydrocarbons / particulates.

Exam Questions

1 Once green plants had evolved, they thrived in Earth's early atmosphere which was rich in **carbon dioxide** ***[1 mark]***. These plants also produced **oxygen** ***[1 mark]*** by the process of **photosynthesis** ***[1 mark]***. Some of the **carbon** ***[1 mark]*** from dead plants eventually became 'locked up' in fossil fuels.

2.1 When sulfur dioxide gas reacts with water in clouds ***[1 mark]*** it forms dilute sulfuric acid, which falls as rain ***[1 mark]***.

2.2 E.g. it kills plants / damages buildings and statues / makes metals corrode ***[1 mark]***.

2.3 Incomplete combustion of fossil fuels produces particulates ***[1 mark]***. Particulates (and the clouds they help to produce) reflect sunlight back into space ***[1 mark]*** so less light reaches the Earth ***[1 mark]***.

2.4 Short wavelength radiation from the Sun passes through the Earth's atmosphere ***[1 mark]***. It is absorbed by the Earth's surface and is re-emitted as long wavelength radiation ***[1 mark]***. This long wavelength radiation is absorbed by greenhouse gases in the atmosphere ***[1 mark]***. They re-radiate it in all directions, including back towards the Earth, warming its surface ***[1 mark]***.

3.1 A measure of the amount of carbon dioxide and other greenhouse gases released over the full life-cycle of a product, service or event. ***[1 mark]***.

3.2 How to grade your answer:

Level 0: There is no relevant information. ***[No marks]***

Level 1: There is a brief description of at least one potential effect of an increase in global temperature. Some attempt is made to explain what impact this effect could have on humans. ***[1 to 2 marks]***

Level 2: There is some description of at least two potential effects of an increase in global temperature and an explanation of the potential impact of these effects on humans. ***[3 to 4 marks]***

Level 3: There is a clear and detailed explanation of at least three effects of an increase in global temperature and a clear explanation of the potential impact of these effects on humans. ***[5 to 6 marks]***

Here are some points your answer may include:

Effect: An increase in global temperature is leading to polar ice caps melting. This is causing a rise in sea levels.
Impacts: If sea levels continue to rise, this will lead to increased flooding in coastal areas and cause coastal erosion.
Flooding and coastal erosion could make people who live near the coast homeless / cause costly damage to buildings and roads / pose a threat to human life.
Effect: Climate change is leading to changes in rainfall patterns.
Impacts: Some regions may have too much or too little water.
Having too little water may affect the ability of certain regions to produce enough food for the human population. This could lead to famine or even starvation.
Too much water could lead to flooding.
Flooding could make people homeless / cause costly damage to buildings and roads / pose a threat to human life.
Effect: Climate change may lead to an increase in both the frequency and severity of storms.
Impacts: Storms can cause damage to buildings and roads and can pose a direct threat to human life.
Effect: Rising temperatures can directly affect the human and wildlife populations in an area.
Impacts: An increase in temperature may affect what food plants can be grown in a certain region.
This could lead to famine or even starvation.

Topic 10 — Using Resources

Page 122
Properties of Materials

Q1 $(14 \div 24) \times 100 = 58.33...$
$= \mathbf{58\%}$ (to the nearest %) ***[1 mark]***

Page 124
Warm-Up Questions

1) Non-metal solids with high melting points that aren't made from carbon-based compounds.

2) They contain cross-links between polymer chains that hold the chains together in a solid structure.

3) copper and zinc

4) iron + oxygen + water → hydrated iron(III) oxide

5) It coats the iron with a layer of different metal that won't be corroded.

Exam Questions

1.1 Their properties make them more suitable for the application ***[1 mark]***.

1.2 E.g. high carbon steel is strong/brittle whereas low carbon steel is softer/more easily shaped ***[1 mark]***.

1.3 e.g. bridges ***[1 mark]***

1.4 stainless steel ***[1 mark]***

2.1 A ***[1 mark]***

2.2 The nail in tube D would not rust whereas the nail in tube B would ***[1 mark]***. The paint covering the nail in tube D creates a barrier that keeps out water and oxygen and so prevents rust from forming ***[1 mark]***.

2.3 Zinc is a more reactive metal than iron ***[1 mark]***, so water and oxygen will react with this before iron ***[1 mark]***. Also, the coating of zinc acts as a barrier that keeps out water and oxygen ***[1 mark]***.

Page 129

Warm-Up Questions

1) E.g. natural rubber

2) Development that takes account of the needs of present society while not damaging the lives of future generations.

3) E.g. saves energy needed to extract metals from the earth / conserves limited supplies of metals from the earth / cuts down on the amount of waste going to landfill.

4) E.g. getting the raw materials / manufacturing and packaging / using the product / product disposal.

Exam Questions

1.1 A natural resource that doesn't renew itself quickly enough to be considered replaceable ***[1 mark]***.

1.2 Any two from: e.g. reuse the cans / recycle the aluminium / use fewer cans ***[2 marks]***.

2.1 E.g. because the supply of copper-rich ores is limited ***[1 mark]***.

2.2 Bacteria convert copper compounds in the ore into soluble copper compounds which separate out from the ore ***[1 mark]***. This produces a (leachate) solution containing copper ions, from which copper can be extracted by electrolysis/displacement with a more reactive metal e.g. scrap iron ***[1 mark]***.

2.3 E.g. these processes have a smaller environmental impact ***[1 mark]***.

2.4 E.g. these processes are slower ***[1 mark]***.

3.1 The effect of some pollutants produced in the life cycle of a product can be difficult to quantify ***[1 mark]***.

3.2 E.g. the person carrying out the company's LCA may have purposefully not included details of the disposal of the product ***[1 mark]***, because the company wanted to use the LCA to support that their product was environmentally friendly for advertising reasons ***[1 mark]***.

Page 133

Warm-Up Questions

1) Any two from: e.g. the levels of dissolved salts in the water aren't too high / the water has a pH of between 6.5 and 8.5 / there aren't any bacteria or other microbes in the water.

2) filtration

3) harmful chemicals

4) effluent and sludge

Exam Questions

1.1 Potable water contains dissolved substances whereas pure water only contains water molecules ***[1 mark]***.

1.2 First, large bits of material are removed by a wire mesh ***[1 mark]***, then sand and gravel beds filter out any other solid bits ***[1 mark]***.

1.3 E.g. chlorine / ozone / ultraviolet light ***[1 mark for each correct answer up to a maximum of three marks.]***

2.1 screening ***[1 mark]***

2.2 The effluent undergoes aerobic digestion ***[1 mark]***. During aerobic digestion air is pumped through the effluent which encourages aerobic bacteria to break down any organic matter in the water ***[1 mark]***.

2.3 E.g. processes used to treat waste and sea water use more energy than treating fresh water and are therefore more expensive ***[1 mark]***. People may not like the idea of drinking treated waste water ***[1 mark]***.

3.1 E.g. sodium ions can be tested for by carrying out a flame test using a sample of water ***[1 mark]***. If sodium ions are present the flame will turn yellow ***[1 mark]***. To test for chloride ions a few drops of dilute nitric acid should be added to a sample of water, followed by a few drops of silver nitrate solution ***[1 mark]***. If chloride ions are present, a white precipitate will form ***[1 mark]***.

3.2 E.g. the pH of the water would need to be tested to see if it is between 6.5 and 8.5 ***[1 mark]***.

Page 137

Warm-Up Questions

1) hydrogen and nitrogen

2) true

3) iron

4) Fertilisers that are formulations containing salts of nitrogen, phosphorus and potassium.

5) potassium chloride / potassium sulfate

Exam Questions

1.1 phosphorus ***[1 mark]*** / potassium ***[1 mark]***

1.2 E.g. fertilisers provide/replace missing elements in the soil that are essential for plant growth ***[1 mark]***. This helps to increase crop yield ***[1 mark]***.

2.1 It contains nitrogen from two sources — the ammonia and the nitric acid ***[1 mark]***.

2.2 Phosphate salts in the rock are insoluble so plants can't directly absorb them and use them as nutrients ***[1 mark]***.

2.3 calcium sulfate and calcium phosphate ***[1 mark]***

3.1 **A**: natural gas ***[1 mark]*** **B**: air ***[1 mark]***

3.2 $\mathbf{3}H_{2(g)} + N_{2(g)} \rightleftharpoons \mathbf{2}NH_{3(g)}$ ***[1 mark]***

3.3 It would increase ***[1 mark]***. Lowering the temperature moves the position of equilibrium towards the products ***[1 mark]***.

3.4 If the pressure was higher the process may become too dangerous/expensive to build and maintain ***[1 mark]***.

Practice Paper 1

Pages 155-173

1.1 +1 ***[1 mark]***

1.2 NaF ***[1 mark]***

1.3 Any two from: e.g. potassium fizzes more than sodium, which fizzes more than lithium / potassium moves more quickly than sodium, which moves more quickly than lithium / potassium melts and a flame is seen, sodium melts but no flame is seen and lithium does not melt ***[1 mark for each correct observation]***.

1.4 As you go down Group 1 the outer electron gets further from the nucleus ***[1 mark]***. This means that the outer electron is more easily lost because it feels less attraction from the nucleus ***[1 mark]***. So reactivity increases as you go down Group 1/ from lithium to sodium to potassium ***[1 mark]***.

1.5 $2Li + 2H_2O \rightarrow \mathbf{2LiOH} + \mathbf{H_2}$
[1 mark for formulas of both products correct, 1 mark for correct balancing]

1.6 It contains hydroxide ions ***[1 mark]***.

2.1 James Chadwick ***[1 mark]***

2.2

Particle	Relative mass	Relative charge
Proton	1	**+1**
Neutron	**1**	**0**
Electron	Very small	**–1**

[2 marks for all correct, 1 mark for at least two correct]

2.3 C and D ***[1 mark]***. They are the only pair to have the same number of protons but a different number of neutrons ***[1 mark]***.

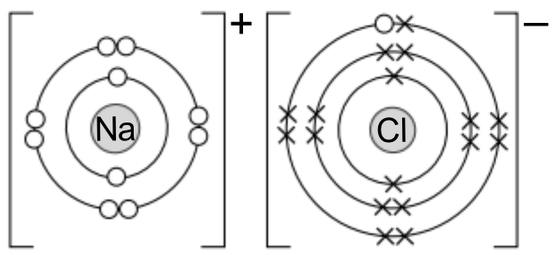
2.4 Protons = 30 ***[1 mark]***
Neutrons = 64 – 30 = **34** ***[1 mark]***
Electrons = 30 ***[1 mark]***

Remember: number of protons = atomic number, number of neutrons = mass number – atomic number, number of electrons (in a neutral atom) = number of protons.

2.5 A compound is a substance made of two or more different elements ***[1 mark]*** that are present in fixed proportions and chemically bonded together ***[1 mark]***.

2.6 $M_r = 56 + 32 + (4 \times 16) =$ **152**
[2 marks for correct answer, otherwise 1 mark for writing an expression that could be used to calculate the relative formula mass of iron(II) sulfate.]

3.1



[1 mark for adding seven crosses and one dot to outer shell of Cl^- ion, 1 mark for correct charge on both ions.]

3.2 Sodium chloride has a giant lattice structure ***[1 mark]*** held together by strong electrostatic forces of attraction/ bonds between positive and negative ions ***[1 mark]***.

3.3 The molecules in substance A are only attracted to each other by weak intermolecular forces ***[1 mark]*** which don't need much energy to break/overcome ***[1 mark]***.

3.4 Substance C ***[1 mark]***. It has a high melting and boiling point and can conduct electricity when molten or dissolved but not when solid ***[1 mark]***.

3.5 How to grade your answer:

Level 0: There is no relevant information. ***[No marks]***

Level 1: A brief attempt is made to explain one or two of these properties in terms of structure and bonding. ***[1 to 2 marks]***

Level 2: Some explanation of three or four of the properties, in terms of their structure and bonding, is given. ***[3 to 4 marks]***

Level 3: Clear and detailed explanation of five or all of the properties, in terms of their structure and bonding, is given. ***[5 to 6 marks]***

Here are some points your answer may include:

Diamond

Each carbon atom in diamond forms four covalent bonds in a rigid giant covalent structure/ tetrahedral lattice, making it very hard.

Because it is made up of lots of covalent bonds, which take a lot of energy to break, diamond has a very high melting point.

There are no free/delocalised electrons in the structure of diamond, so it can't conduct electricity.

Graphite

Each carbon atom in graphite forms three covalent bonds, creating sheets of carbon atoms that can slide over each other. The carbon layers are only held together weakly, which is what makes graphite soft and slippery.

The covalent bonds between the carbon atoms take a lot of energy to break, giving graphite a very high melting point.

Only three out of each carbon's four outer electrons are used in bonds, so graphite has lots of free/delocalised electrons and can conduct electricity.

4.1 Any two from: e.g. have a lid on the beaker / use insulation (e.g. cotton wool) around the beaker / use a polystyrene beaker ***[1 mark for each correct answer]***.

4.2 1 °C ***[1 mark]***

4.3 37.5 °C ***[1 mark]***

4.4 Temperature change = final temperature – initial temperature
= 37.5 – 17.0 = **20.5 °C** (3 s.f.)
[2 marks for correct answer, otherwise 1 mark for writing a correct expression for calculating the temperature change.]

If your answer to 4.3 was wrong, you can still have both marks for correctly subtracting 17.0 from it to find the temperature change.

4.5 The reaction was exothermic as the temperature of the surroundings increased during the reaction ***[1 mark]***.

5.1 hydrogen/H_2 ***[1 mark]***

5.2 Reaction D ***[1 mark]***. The most reactive metal will react fastest with the acid ***[1 mark]***. In reaction D the largest volume of gas has been collected in the syringe / the most bubbles are being given off ***[1 mark]***.

5.3 e.g. zinc ***[1 mark]***

You would get this mark if you named any metal between magnesium and iron in the reactivity series (for example, aluminium would be fine here too).

5.4 Sodium is very reactive and would react explosively with acid ***[1 mark]***.

5.5 Gap for reaction of copper sulfate with iron: yes ***[1 mark]***

Gap for reaction of iron sulfate with magnesium: yes ***[1 mark]***

5.6 $Mg_{(s)} + Cu^{2+}_{(aq)} \rightarrow Mg^{2+}_{(aq)} + Cu_{(s)}$
[1 mark for the correct reactants, 1 mark for the correct products]

5.7 Mg / magnesium ***[1 mark]***

Remember, when you're talking about oxidation and reduction in terms of electrons, oxidation is the loss of electrons. In this reaction, the magnesium went from being neutral atoms to positively charged ions — so they must have lost electrons.

6.1 E.g. the solid will stop dissolving and some solid will still remain at the bottom of the beaker ***[1 mark]***.

6.2 water ***[1 mark]***

6.3 $ZnCl_2$ ***[1 mark]***

6.4 Filter the solution to remove the excess solid zinc oxide ***[1 mark]***, then heat the solution and allow some of the water to evaporate ***[1 mark]***. Leave the solution to cool until crystals form ***[1 mark]***. Filter the crystals from the remaining solution and allow them to dry ***[1 mark]***.

6.5 sodium chloride ***[1 mark]***

6.6 carbon dioxide ***[1 mark]***

6.7 E.g. test a sample of the solution using universal indicator paper / test the solution using a pH probe. If the paper turns green / the probe shows a pH of seven, then the solution is neutral. ***[1 mark for any sensible method you could use to test the pH of the solution, 1 mark for a correct, matching observation of what that method would show if the solution was neutral.]***

7.1 Oxygen is being removed from the metal oxide / Fe^{3+} ions (iron(III) ions) are gaining electrons ***[1 mark]***.

7.2 M_r of $Fe_2O_3 = (2 \times 56) + (3 \times 16) = 160$
Moles of Fe_2O_3 = mass ÷ M_r = 40 ÷ 160 = 0.25
From the balanced equation you know that 2 moles of Fe_2O_3 produce 4 moles of Fe. So 0.25 moles of Fe_2O_3 will produce $0.25 \times 2 = 0.50$ moles of Fe.
Mass of Fe = moles × A_r = 0.50 × 56 = **28 g**

[4 marks for correct answer, otherwise 1 mark for M_r of Fe_2O_3, 1 mark for number of moles of Fe_2O_3, 1 mark for number of moles of Fe.]

7.3 Aluminium is more reactive than carbon/above carbon in the reactivity series ***[1 mark]***. Reduction with carbon can only be used to extract metals that are less reactive than carbon/below carbon in the reactivity series ***[1 mark]***.

7.4 In pure iron, the atoms are arranged in layers ***[1 mark]***. These layers can slide over each other freely ***[1 mark]***. Steel also contains carbon atoms, which are a different size to the iron atoms ***[1 mark]***.
The carbon atoms distort the layers, so they can't slide over one another so easily ***[1 mark]***.

7.5 Any three from: e.g. the rods should be the same size / the rods should be the same distance away from the heat / the same amount of petroleum jelly should be used on each rod / the wooden splints should be the same distance from the end of each rod / the wooden splints should be the same size ***[1 mark for each correct answer]***.

8.1 In a solution of copper sulfate, the ions are free to move and carry an electric charge ***[1 mark]***. When copper sulfate is solid, the ions are held in fixed positions and can't move ***[1 mark]***.

8.2 Cu^{2+}, SO_4^{2-}, H^+, OH^- ***[1 mark]***

The Cu^{2+} and SO_4^{2-} ions come from the ionic compound.
The H^+ and OH^- ions come from the water.

8.3 $\mathbf{4}OH^- \rightarrow O_2 + \mathbf{2}H_2O + 4\mathbf{e}^-$ ***[1 mark]***

8.4 Mean change of mass for negative electrode = **2.54** ***[1 mark]***

8.5 At the negative electrode, copper ions/Cu^{2+} ions are being reduced to copper metal/Cu ***[1 mark]***. The half equation for this reaction is: $Cu^{2+} + 2e^- \rightarrow Cu$ ***[1 mark]***. So a layer of copper metal forms on the negative electrode during the electrolysis ***[1 mark]***.

8.6 The negative electrode would change in mass by twice as much ***[1 mark]***.

9.1 How to grade your answer:

Level 0: There is no relevant information. ***[No marks]***

Level 1: A brief attempt is made to explain one or two advantages of using high atom economy reactions in industry. ***[1 to 2 marks]***

Level 2: A good explanation of at least one environmental benefit and one economic benefit of using high atom economy reactions in industry is given. ***[3 to 4 marks]***

Level 3: A clear and detailed explanation of at least two environmental benefits and two economic benefits of using high atom economy reactions in industry is given. ***[5 to 6 marks]***

Here are some points your answer may include:

<u>Environmental benefits</u>
Reactions with high atom economies use less of their raw materials/use up raw materials more slowly than reactions with low atom economies.
Less raw materials will need to be extracted for a high atom economy reaction, reducing the amount of energy used and the pollution caused by extracting the raw material.
Reactions with high atom economies produce less waste than reactions with low atom economies.
This means that using a reaction with a high atom economy sends less waste to landfill and creates less pollution.
Using a high atom economy reaction also reduces the amount of energy used to treat waste made by the reaction.
Reactions with high atom economies are more sustainable than reactions with low atom economies.

<u>Economic benefits</u>
Since high atom economy reactions use less of their raw materials/use up raw materials more slowly, it is usually cheaper to buy the raw materials needed for a high atom economy reaction.
As high atom economy reactions produce less waste, a company using a high atom economy reaction will spend less money on treating waste.
A company using a high atom economy reaction will spend less money on energy bills, since they will use less energy extracting raw materials/treating waste.

9.2 M_r of desired product = 2 × M_r of LiCl = 2 × 42.5 = 85
M_r of all reactants = M_r of Li_2CO_3 + (2 × M_r of HCl)
= 74 + (2 × 36.5) = 147

$$\text{Atom economy} = \frac{M_r \text{ of desired products}}{M_r \text{ of all reactants}} \times 100$$
= (85 ÷ 147) × 100 = 57.8... = **58%** (2 s.f.)

[4 marks for the correct answer, but deduct 1 mark if answer is not given to 2 s.f. Otherwise 1 mark for finding the M_r of the desired product, 1 mark for finding the M_r of all reactants and 1 mark for substituting these values correctly into the atom economy formula.]

9.3 100% ***[1 mark]***

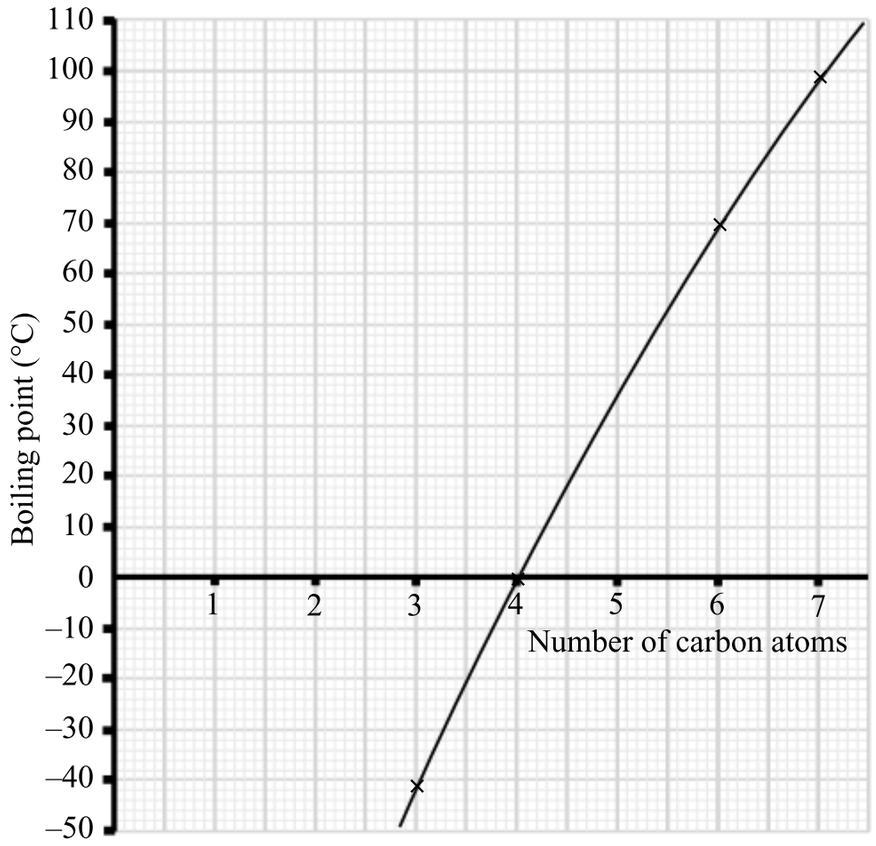
If a reaction only has one product, its atom economy must be 100%.

9.4 percentage yield = (actual yield ÷ theoretical yield) × 100
= (170 ÷ 212.5) × 100 = **80%** ***[1 mark]***

Practice Paper 2

Pages 174-193

1.1



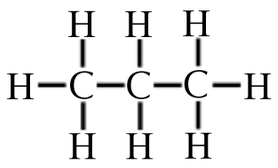
[1 mark for all four points correctly plotted, 1 mark for a smooth curve that passes through all the points]

1.2 36 °C ***[1 mark for any answer in the range 34-38 °C]***

1.3 The larger the alkane molecule, the higher its boiling point / the smaller the alkane, the lower its boiling point ***[1 mark]***.

1.4 C_nH_{2n+2} ***[1 mark]***

1.5



[1 mark]

1.6 Add a few drops of bromine water to both compounds and shake ***[1 mark]***. With propane, nothing will happen ***[1 mark]***. Propene will decolourise the bromine water / turn the solution from orange to colourless ***[1 mark]***.

1.7 $C_3H_8 + 5O_2 \rightarrow 3CO_2 + \mathbf{4H_2O}$

[1 mark for correctly giving H_2O as the missing product, 1 mark for correct number of moles of H_2O.]

1.8 Decane has a higher boiling point than pentane ***[1 mark]***.

2.1 As the level of nitrogen dioxide increases, the severity of asthma attacks increases / there is a positive correlation between nitrogen dioxide levels and severity of asthma attacks ***[1 mark]***.

2.2 The sample of people that the scientists collected data from was too big ***[1 mark]***.

2.3 Oxides of nitrogen are formed when nitrogen and oxygen from the air react at high temperatures ***[1 mark]***. This happens in car engines (which operate at high enough temperatures for nitrogen oxides to form) ***[1 mark]***. In cities, there are lots of cars, so the levels of nitrogen oxides tend to be higher ***[1 mark]***.

2.4 nitric acid ***[1 mark]***

2.5 e.g. sulfur dioxide ***[1 mark]***

2.6 Carbon monoxide binds with haemoglobin in the blood ***[1 mark]***. This lowers the amount of oxygen that can be transported around the body ***[1 mark]***.

3.1 E.g. they don't have enough fresh water / groundwater / surface water to treat and use as drinking water ***[1 mark]***.

3.2 Desalination ***[1 mark]***

3.3 How to grade your answer:

Level 0: There is no relevant information ***[No marks]***

Level 1: There is a brief description of the method to distil seawater, but many details are missing or incorrect. ***[1 to 2 marks]***

Level 2: There is some explanation of the method used to distil sea water. Most of the information is correct, but a few small details may be incorrect or missing. ***[3 to 4 marks]***

Level 3: There is a clear, detailed and fully correct explanation of the method used to distil sea water in the lab. ***[5 to 6 marks]***

Here are some points your answer may include:

Pour the sea water into a (round bottom) flask.
Attach the flask to a condenser and secure both with clamps.
Connect a supply of cold water to the condenser.
Place a beaker under the condenser to collect the fresh water.
Place a Bunsen burner under the round bottom flask and heat the sea water slowly.
The water will boil and form steam.
The steam will condense back to pure liquid water in the condenser and be collected in the beaker as it runs out.
Continue to heat the (round bottom) flask until you have collected a reasonable amount of pure water.

3.4 Large bits of material are removed from the sewage ***[1 mark]***.

3.5 The effluent undergoes aerobic biological treatment ***[1 mark]***. Air is added to encourage aerobic bacteria to break down any organic matter that is present ***[1 mark]***.

4.1 –COOH ***[1 mark]***

4.2 $(0.55 + 0.39 + 0.45) \div 3 = 0.463... = \mathbf{0.46\ g}$ (2 s.f.)

[2 marks for correct answer, otherwise 1 mark for writing a correct expression that could be used to find the mean]

4.3 The loss of mass for each run is less than 1 g ***[1 mark]***.

4.4 The gas is carbon dioxide ***[1 mark]***. The student should bubble the gas through/shake the gas with an aqueous solution of calcium hydroxide/lime water ***[1 mark]***.
If carbon dioxide is present the solution will turn cloudy ***[1 mark]***.

4.5 The student could perform a flame test ***[1 mark]***.
If sodium ions are present the substance would burn with a yellow flame ***[1 mark]***.

5.1 Any two from: e.g. PVC is an electrical insulator/does not conduct electricity, so it makes the wires safe to touch / PVC is very flexible, so wires that are coated in it can bend / PVC is light, so it won't weigh down the wires / PVC has high corrosion resistance, so it protects the wires inside from corrosion/exposure ***[1 mark for each correct reason]***.

5.2 ceramic ***[1 mark]***

5.3 Aluminium alloys are far less dense than steel (3 g/cm^3 compared to 7.8 g/cm^3), so the parts are much lighter ***[1 mark]***. They are also more resistant to corrosion ***[1 mark]***.

5.4 Plants are grown in soil containing copper compounds ***[1 mark]***. The plants don't use much of the copper so it accumulates in the leaves ***[1 mark]***. The plants are then dried and burnt, leaving ash containing soluble copper compounds ***[1 mark]***. The copper can then be extracted by electrolysis / displacement using scrap iron ***[1 mark]***.

5.5 How to grade your answer:

Level 0: There is no relevant information ***[No marks]***

Level 1: A sensible material has been chosen. An attempt has been made to justify the choice by referring to one of the properties in the table. ***[1 to 2 marks]***

Level 2: A sensible material has been chosen, and there is a good justification of the choice that refers to two or three of the properties shown in the table. ***[3 to 4 marks]***

Level 3: A sensible material has been chosen, and there is a clear and detailed justification of the choice that refers to all four of the properties given in the table. ***[5 to 6 marks]***

Here are some points your answer may include:

The golf club shaft needs to be strong to withstand forces applied when the ball is hit.
Carbon fibre and steel have the greatest strength (4100 MPa and 780 MPa, respectively). Neither iron (200 MPa) or lead (12 MPa) would be strong enough.
The shaft needs to be as lightweight as possible, since the club will be both carried around and lifted into the air.
Carbon fibre has the lowest density of all the materials (1.5 g/cm^3), so would make the lightest club.
Stainless steel and iron are both denser than carbon fibre, but with a density of 7.8 g/cm^3 they would make lighter clubs than lead.
The club will be used outdoors in all weather, so it should be corrosion resistant.
Carbon fibre, stainless steel and lead are all corrosion resistant, but iron is not, so would not be suitable.
The cost of carbon fibre is very high. Any of the other materials shown would be cheaper.
E.g. though carbon fibre is expensive, because the golf club is a professional golf club, the cost of the club can be high, so carbon fibre would be the best choice.

Make sure that you remember to say what you think is the best choice of material in questions like this. It would have been OK to pick stainless steel here too (as long as you justified your choice clearly), because it would still be a suitable material. It wouldn't be OK to pick lead or iron because they are unsuitable (neither of them is strong enough, iron is not corrosion resistant and lead is very dense).

6.1 carbon dioxide ***[1 mark]***

6.2 nitrogen ***[1 mark]***

6.3 Algae and plants evolved to carry out photosynthesis ***[1 mark]***. During photosynthesis carbon dioxide is absorbed, so the amount of carbon dioxide in the atmosphere decreased ***[1 mark]***. Photosynthesis also produces oxygen, so the amount of oxygen in the atmosphere increased ***[1 mark]***.

6.4 Deforestation — e.g. plants take carbon dioxide out of the atmosphere, so removing trees causes carbon dioxide levels to rise ***[1 mark]***.

Increased energy consumption — e.g. burning fossil fuels releases carbon dioxide into the atmosphere ***[1 mark]***.

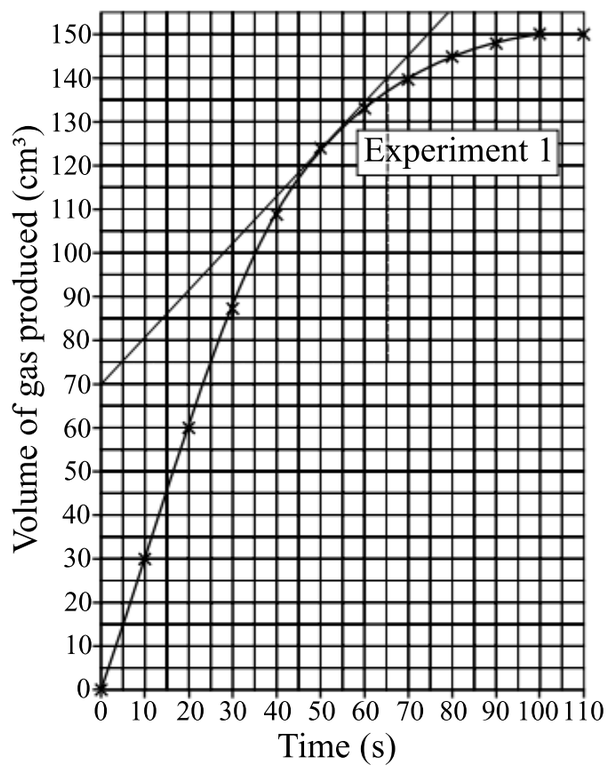
Increasing population — any one from: e.g. more people will be respiring and releasing carbon dioxide / more energy will be used (and carbon dioxide is released when fuels are burnt) / more land will be needed, so trees will have to be cut down, and less carbon dioxide will be removed from the air ***[1 mark]***.

6.5 E.g. carbon dioxide is a greenhouse gas ***[1 mark]***, so an increase in carbon dioxide in the atmosphere could lead to an increase in global temperatures/global warming ***[1 mark]***. As temperatures rise, global food production could decrease due to more extreme weather events / loss of farming land available / water shortages ***[1 mark]***.

7.1 Any two from: e.g. volume of acid / size of solid particles / temperature ***[1 mark for each correct answer]***

7.2 The rate of the reaction decreases as the reaction proceeds ***[1 mark]***.

7.3 Draw a tangent to the graph at 50 s, e.g.:



Then find the gradient of your tangent, e.g.:

$$\text{Gradient of tangent} = \frac{\text{change in } y}{\text{change in } x} = \frac{(140 - 75)}{(65 - 5)}$$

$= 65 \div 60 = 1.08...$

units = cm^3 ÷ s = cm^3/s

rate of reaction at 50 s = **1.1 cm^3/s** (2 s.f.)

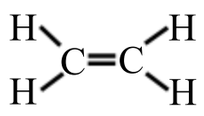
[4 marks in total — 1 mark for a correctly drawn tangent, 1 mark for gradient calculation, 1 mark for correct answer in the range 1.0-1.2, 1 mark for correct units]

7.4 75 cm^3 ***[1 mark]***

You can see from the graph that the first experiment produced 150 cm^3 of gas in total. Calcium carbonate must be the limiting reactant (as the acid is in excess), so if you use half the mass, you'll produce half the amount of gas.

7.5 The gradient of the graph at the start of the reaction is steeper for Experiment 2 ***[1 mark]***.

7.6 Increasing the concentration of a reactant increases the number of particles of that substance in the reaction mixture ***[1 mark]***. This increases the frequency of collisions between particles of the reactants, increasing the rate of the reaction ***[1 mark]***.

8.1  ***[1 mark]***

8.2 alkene ***[1 mark]***

8.3 The polymer chains in thermosetting polymers have cross-links between them ***[1 mark]***. The polymer chains in thermosoftening polymers are separate, and only held together by weak (intermolecular) forces ***[1 mark]***. When thermosetting polymers are heated the cross-links stop the polymer from melting ***[1 mark]***. When thermosoftening polymers are heated, the weak intermolecular forces holding the chain together are broken and the polymer melts ***[1 mark]***.

8.4 Condensation polymerisation ***[1 mark]***. You can tell this is condensation polymerisation because water/a small molecule is lost as the polymer is formed ***[1 mark]***.

8.5 amino acid ***[1 mark]***

8.6 Any one from: e.g. DNA / starch / cellulose / protein ***[1 mark]***

9.1 The forward and reverse reactions are happening at exactly the same rate ***[1 mark]***.

9.2 Le Chatelier's principle states that if you change the conditions of a reversible reaction at equilibrium the system will try to counteract that change ***[1 mark]***.

9.3 The yield of sulfur trioxide will increase ***[1 mark]***. When pressure is increased the position of equilibrium moves towards the side with fewer gas molecules ***[1 mark]***. There are two molecules of gas on the right, compared to three on the left, so the position of equilibrium will move to the right ***[1 mark]***.

9.4 The yield of sulfur dioxide will decrease ***[1 mark]***. Increasing the temperature will move the position of equilibrium towards the endothermic reaction ***[1 mark]***. As the reaction to form sulfur trioxide / the forward reaction is exothermic, the position of equilibrium will move to the left to counteract the increase in temperature ***[1 mark]***.

9.5 catalyst ***[1 mark]***

9.6 E.g. 15% of 7500 kg = 7500 × 0.15 = 1125 kg
new mass made in a day = 7500 + 1125 = 8625 kg
new mass made in 5 days = 8625 × 5 = 43 125 kg
= **43 000 kg** (2 s.f.)

[3 marks for correct answer, either rounded to 2 significant figures or unrounded. Otherwise 1 mark for finding the new mass made in a day and 1 mark for multiplying it by five to find the new mass made in five days.]

Glossary

A

Accurate result A result that is close to the true answer.

Acid A substance with a pH of less than 7 that forms H^+ ions in water.

Activation energy The minimum amount of energy that reactant particles must have when they collide in order to react.

Addition polymer A large molecule made from alkene monomers which bond together by opening up their double bonds and forming long chains.

Addition reaction Reactions where the double bond in an alkene opens up to leave a single bond and a new atom is added to each carbon.

Alcohol A compound that contains an –OH functional group.

Alkali A substance with a pH of more than 7 that forms OH^- ions in solution.

Alkali metal An element in Group 1 of the periodic table. E.g. sodium, potassium etc.

Alkane A saturated hydrocarbon with the general formula C_nH_{2n+2}. E.g. methane, ethane etc.

Alkene An unsaturated hydrocarbon that contains a carbon-carbon double bond and has the general formula C_nH_{2n}. E.g. ethene, propene etc.

Alloy A metal that is a mixture of two or more metals, or a mixture involving metals and non-metals.

Amino acid An organic molecule that contains an amino group and a carboxylic acid group.

Anion A particle with a negative charge, formed when one or more electrons are gained.

Anomalous result A result that doesn't fit with the rest of the data.

Atmosphere The layer of air that surrounds a planet.

Atom A neutral particle made up of protons and neutrons in the nucleus, with electrons surrounding the nucleus.

Atom economy The percentage of the mass of the reactants that ends up as the intended product when manufacturing a chemical.

Atomic number The number of protons in the nucleus of an atom. It's also known as proton number.

Avogadro constant The number of particles in one mole of a substance, which is 6.02×10^{23}.

B

Base A substance that reacts with acids in neutralisation reactions.

Battery Two or more cells connected together in series.

Bias Unfairness in the way data is presented, possibly because the presenter is trying to make a particular point (sometimes without knowing they're doing it).

Bioleaching The process by which a metal is separated from its ore using bacteria.

Bond energy The amount of energy required to break a bond (or the amount of energy released when a bond is made).

By-product A substance produced by a chemical reaction that isn't the desired product.

C

Calibrate Measure something with a known quantity and set the instrument being used to that quantity.

Carbon footprint A measure of the amounts of greenhouse gases released by a product, a service or an event.

Carboxylic acid A compound that contains a –COOH functional group.

Catalyst A substance that can speed up a reaction without being changed or used up in the reaction.

Categoric data Data that comes in distinct categories, e.g. types of metals etc.

Cation A particle with a positive charge, formed when one or more electrons are lost.

Cell A system that contains chemicals which react and produce electricity.

Ceramic A non-metallic solid with a high melting point that isn't made from carbon-based compounds.

Chemical bond The attraction of two atoms for each other, caused by the sharing or transfer of electrons.

Chromatogram The pattern of spots formed as a result of separating a mixture using chromatography.

Chromatography An analytical method used to separate the substances in a mixture based on how the components interact with a mobile phase and a stationary phase.

Climate change A change in the Earth's climate. E.g. global warming, changing rainfall patterns etc.

Collision theory The theory that in order for a reaction to occur, particles must collide with sufficient energy.

Combustion An exothermic reaction between a fuel and oxygen.

Composite A material made of one or more materials embedded in another with properties different to the individual components.

Compound A substance made up of atoms of at least two different elements, chemically joined together.

Concentration The amount of a substance in a certain volume of solution, given in units of 'units of amount of substance'/'units of volume'.

Condensation polymer A large molecule made from monomers with at least two different functional groups that react together to form chains. A small molecule is lost for each bond formed.

Continuous data Numerical data that can have any value within a range (e.g. length, volume or temperature).

Control variable A variable in an experiment that is kept the same.

Conversion factor A number which you must multiply or divide a unit by to convert it to a different unit.

Glossary

Correlation A relationship between two variables.

Corrosion The process of metals being gradually destroyed by reacting with substances in the environment.

Covalent bond A chemical bond formed when atoms share a pair of electrons.

Covalent substance A substance where the atoms are held together by covalent bonds.

Cracking The process that is used to break long-chain hydrocarbons down into shorter, more useful hydrocarbons. Two types of cracking are catalytic cracking and steam cracking.

Crystallisation The formation of solid crystals as water evaporates from a solution. For example, salt solutions undergo crystallisation to form solid salt crystals.

D

DNA The molecule in cells that stores genetic information.

Delocalised electron An electron that isn't associated with a particular atom or bond and is free to move within a structure.

Dependent variable The variable in an experiment that is measured.

Discrete data Numerical data that can only take a certain value, with no in-between value, e.g. number of people.

Displacement reaction A reaction where a more reactive element replaces a less reactive element in a compound.

Displayed formula A chemical formula that shows the atoms in a covalent compound and all the bonds between them.

Distillation A way of separating out a liquid from a mixture. You heat the mixture until the bit you want evaporates, then cool the vapour to turn it back into a liquid.

E

Electrode An electrical conductor which is submerged in the electrolyte during electrolysis.

Electrolysis The process of breaking down a substance using electricity.

Electrolyte A liquid or solution used in electrolysis to conduct electricity between the two electrodes.

Electron shell A region of an atom that contains electrons. It's also known as an energy level.

Electronic structure The number of electrons in an atom (or ion) of an element and how they are arranged.

Electrostatic force A force of attraction between opposite charges.

Element A substance that is made up only of atoms with the same number of protons.

Empirical formula A chemical formula showing the simplest possible whole number ratio of atoms in a compound.

End-point The point at which an acid or alkali is completely neutralised during a titration.

Endothermic reaction A reaction which takes in energy from the surroundings.

Energy level A region of an atom that contains electrons. It's also known as an electron shell.

Enzyme A biological catalyst.

Equilibrium The point at which the rates of the forward and backward reactions in a reversible reaction are the same, and so the amounts of reactants and products in the reaction container don't change.

Ester A compound that contains a –COO– functional group.

Exothermic reaction A reaction which transfers energy to the surroundings.

F

Fair test A controlled experiment where the only thing that changes is the independent variable.

Feedstock A raw material used to produce other substances through industrial processes.

Fermentation The process by which an enzyme in yeast converts sugars to ethanol by anaerobic respiration.

Filtration A physical method used to separate an insoluble solid from a liquid.

Finite resource A resource that isn't replaced at a quick enough rate to be considered replaceable.

Flame emission spectroscopy An analytical technique which can be used to identify and find the concentration of metal ions in solution based on line spectra.

Flammability How easy it is to ignite a substance.

Formulation A useful mixture with a precise purpose made by following a formula.

Fraction A group of hydrocarbons that condense together when crude oil is separated using fractional distillation. E.g. petrol, naphtha, kerosene etc.

Fractional distillation A process that can be used to separate substances in a mixture according to their boiling points.

Fuel cell An electrical cell that's supplied with fuel and oxygen and uses energy from the reaction between them to generate electricity.

Functional group A group of atoms that are responsible for the chemical properties of a compound. E.g. the –OH group in an alcohol.

G

General formula A formula that can be used to find the molecular formula of any member of a homologous series.

Giant covalent structure A large molecule made up of a very large number of atoms held together by covalent bonds (also known as a macromolecule).

Global dimming The decrease in the amount of sunlight reaching the Earth's surface due to an increase in the amount of particulates in the atmosphere.

Global warming The increase in the average temperature of the Earth.

Glossary

Greenhouse effect When greenhouse gases in the atmosphere absorb long wavelength radiation and re-radiate it in all directions, including back towards Earth, helping to keep the Earth warm.

Greenhouse gas A gas that can absorb long wavelength radiation.

Group A column in the periodic table.

H

Haber process A process used to make ammonia by reacting nitrogen with hydrogen.

Half equation An equation which shows how electrons are transferred when a substance is reduced or oxidised. E.g. at an electrode during electrolysis.

Hazard Something that has the potential to cause harm.

Homologous series A group of chemicals that react in a similar way because they have the same functional group. E.g. the alcohols or the carboxylic acids.

Hydrocarbon A compound that is made from only hydrogen and carbon.

Hypothesis A possible explanation for a scientific observation.

I

In excess A reactant that is not used up during a reaction.

Incomplete combustion When a fuel burns but there isn't enough oxygen for it to burn completely. Products can include carbon monoxide and carbon particulates. Also known as partial combustion.

Independent variable The variable in an experiment that is changed.

Indicator A substance that changes colour above or below a certain pH.

Insoluble A substance is insoluble if it does not dissolve in a particular solvent.

Intermolecular force A force of attraction that exists between molecules.

Ion A charged particle formed when one or more electrons are lost or gained from an atom or molecule.

Ionic bond A strong attraction between oppositely charged ions.

Ionic compound A compound that contains positive and negative ions held together in a regular arrangement (a lattice) by electrostatic forces of attraction.

Ionic equation An equation that shows only the particles that react and the products they form.

Ionic lattice A closely-packed regular arrangement of particles held together by electrostatic forces of attraction.

Isotope A different atomic form of the same element, which has the same number of protons, but a different number of neutrons.

J

Joule The standard unit of energy.

L

Lattice A closely-packed regular arrangement of particles.

Le Chatelier's principle The idea that if the conditions of a reaction are changed when a reversible reaction is at equilibrium, the system will try to counteract the change.

Life cycle assessment An assessment of the environmental impact of a product over the course of its life.

Limiting reactant A reactant that gets completely used up in a reaction, so limits the amount of product that's formed.

Litmus A single indicator that's blue in alkalis and red in acids.

M

Macromolecule A large molecule made up of a very large number of atoms held together by covalent bonds (also known as a giant covalent structure).

Mass number The total number of protons and neutrons in an atom.

Mean (average) A measure of average found by adding up all the data and dividing by the number of values there are.

Metal An element that can form positive ions when it reacts.

Metal ore Rocks that are found naturally in the Earth's crust containing enough metal to make the metal profitable to extract.

Metallic bond The attraction between metal ions and delocalised electrons in a metal.

Methyl orange A single indicator that's yellow in alkalis and red in acids.

Mixture A substance made from two or more elements or compounds that aren't chemically bonded to each other.

Mobile phase In chromatography, the mobile phase is a gas or liquid where the molecules are able to move.

Model Something used to describe or display how an object or system behaves in reality.

Mole A unit of amount of substance — the mass of one mole of a substance is equal to the value of the relative formula mass of that substance in grams, and contains 6.02×10^{23} particles of the substance.

Molecular formula A chemical formula showing the actual number of atoms of each element in a compound.

Molecule A particle made up of at least two atoms held together by covalent bonds.

Monomer A small molecule that can be joined together with other small molecules to form a polymer.

Glossary

N

Nanoscience The study of particles and structures on the scale of nanometres.

Natural resource A resource formed without human input.

Neutral substance A substance with a pH of 7.

Neutralisation reaction The reaction between acids and bases that leads to the formation of neutral products — usually a salt and water.

Non-metal An element that doesn't form positive ions when it reacts, with the exception of hydrogen.

NPK Fertiliser A formulation providing nitrogen, phosphorus and potassium as nutrients to help plants to grow.

Nuclear model An accepted theory which describes the atom as having a tiny, positively charged nucleus surrounded by shells which are occupied by negative electrons.

Nucleus The central part of an atom, made up of protons and neutrons.

O

Organic compound A chemical compound that contains carbon atoms.

Oxidation A reaction where electrons are lost or oxygen is gained by a species.

P

Paper chromatography An analytical technique that can be used to separate and analyse coloured substances.

Peer-review The process in which other scientists check the results and explanations of an investigation before they are published.

Percentage yield The amount of product formed in a reaction, given as a percentage of the theoretical amount of product that could be formed.

Period A row in the periodic table.

Periodic table A table of all the known elements, arranged in order of atomic number so that elements with similar chemical properties are in groups.

pH scale A scale from 0 to 14 that is used to measure how acidic or alkaline a solution is.

Phenolphthalein A single indicator that's pink in alkalis and colourless in acids.

Phytomining The process by which a metal is extracted from soil by using plants.

Plum pudding model A disproved theory of the atom as a ball of positive charge with electrons inside it.

Polymer A long chain molecule that is formed by joining lots of smaller molecules (monomers) together.

Polymerisation The process of joining lots of small molecules (monomers) together to form a much longer molecule (a polymer).

Polypeptide A polymer made of amino acid monomers.

Potable water Water that is safe for drinking.

Precipitate A solid that is formed in a solution during a chemical reaction.

Precise result When all the data is close to the mean.

Prediction A statement based on a hypothesis that can be tested.

Product A substance that is formed in a chemical reaction.

Protein A large molecule containing one or more long chains of polypeptides.

Pure substance A substance that only contains one compound or element throughout.

R

Random error A difference in the results of an experiment caused by things like human error in measuring.

Range The difference between the smallest and largest values in a set of data.

Rate of reaction How fast the reactants in a reaction are changed into products.

Reactant A substance that reacts in a chemical reaction.

Reaction profile A graph that shows how the energy in a reaction changes as the reaction progresses (also known as an energy level diagram).

Reactivity series A list of elements arranged in order of their reactivity. The most reactive elements are at the top and the least reactive at the bottom.

Redox reaction A reaction where one substance is reduced and another is oxidised.

Reduction A reaction where electrons are gained or oxygen is lost.

Relative atomic mass (A_r) The average mass of the atoms of an element measured relative to the mass of one atom of carbon-12. The relative atomic mass of an element is the same as its mass number in the periodic table.

Relative formula mass (M_r) All the relative atomic masses (A_r) of the atoms in a compound added together.

Reliable result A result that is repeatable and reproducible.

Renewable resource A resource that can be made at the same or similar rate as it's being used.

Repeatable result A result that will come out the same if the experiment is repeated by the same person using the same method and equipment.

Repeating unit The shortest repeating section of a polymer.

Reproducible result A result that will come out the same if someone different does the experiment, or a slightly different method or piece of equipment is used.

Resolution The smallest change a measuring instrument can detect.

Reversible reaction A reaction where the products of the reaction can themselves react to produce the original reactants.

R_f value In chromatography, the ratio between the distance travelled by a dissolved substance and the distance travelled by a solvent.

Risk The chance that a hazard will cause harm.

Glossary

S

S.I. unit A standard unit of measurement, recognised by scientists all over the world.

Scaling prefix A word or symbol which goes before a unit to indicate a multiplying factor (e.g. 1 km = 1000 m).

Significant figure The first significant figure of a number is the first non-zero digit. The second, third and fourth significant figures follow on immediately after it.

Simple distillation A way of separating a liquid out from a mixture if there are large differences in the boiling points of the substances.

Simple molecule A molecule made up of only a few atoms held together by covalent bonds.

Solute A substance dissolved in a solvent to make a solution.

Solution A mixture made up of one substance (the solute) dissolved in another (the solvent).

Solvent A liquid in which another substance (a solute) can be dissolved.

Solvent front The point the solvent has reached up the filter paper during paper chromatography.

State symbol The letter, or letters, in brackets that are placed after a substance in an equation to show what physical state it's in. E.g. gaseous carbon dioxide is shown as $CO_{2(g)}$.

Stationary phase In chromatography, the stationary phase is a solid or really thick liquid where molecules are unable to move.

Strong acid An acid which fully ionises in an aqueous solution.

Surface area to volume ratio The amount of surface area per unit volume of a particle.

Sustainable development An approach to development that takes into the account the needs of present society while not damaging the lives of those in the future.

Systematic error An error that is consistently made throughout an experiment.

T

Tangent A straight line that touches a curve at a particular point without crossing it.

Theoretical yield The amount of product you would expect to be formed in a reaction.

Theory A hypothesis which has been accepted by the scientific community because there is good evidence to back it up.

Thermal decomposition A reaction where one substance chemically changes into at least two new substances when it's heated.

Thermosetting polymer A polymer that has cross-links between its chains.

Thermosoftening polymer A polymer made of individual tangled polymer chains, with no cross-links between them.

Titration A type of experiment that you can use to find the concentration of a solution.

U

Uncertainty The amount by which a given result may differ from the true value.

Universal indicator A wide range indicator that changes colour depending on the pH of the solution that it's in.

V

Valid result A result that is repeatable, reproducible and answers the original question.

Viscosity How runny or gloopy a substance is.

Y

Yield The amount of product made in a reaction.

Z

Zero error A type of systematic error caused by using a piece of equipment that isn't zeroed properly.

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The Periodic Table

Periods	Group 1	Group 2										Group 3	Group 4	Group 5	Group 6	Group 7	Group 0
1							1 H Hydrogen 1										4 He Helium 2
2	7 Li Lithium 3	9 Be Beryllium 4										11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10
3	23 Na Sodium 11	24 Mg Magnesium 12										27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulfur 16	35.5 Cl Chlorine 17	40 Ar Argon 18
4	39 K Potassium 19	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	63.5 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36
5	85 Rb Rubidium 37	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	[97] Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54
6	133 Cs Caesium 55	137 Ba Barium 56	139 La Lanthanum 57	178 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	[209] Po Polonium 84	[210] At Astatine 85	[222] Rn Radon 86
7	[223] Fr Francium 87	[226] Ra Radium 88	[227] Ac Actinium 89	[267] Rf Rutherfordium 104	[268] Db Dubnium 105	[269] Sg Seaborgium 106	[270] Bh Bohrium 107	[269] Hs Hassium 108	[277] Mt Meitnerium 109	[281] Ds Darmstadtium 110	[282] Rg Roentgenium 111	[285] Cn Copernicium 112	[286] Nh Nihonium 113	[290] Fl Flerovium 114	[290] Mc Moscovium 115	[293] Lv Livermorium 116	[294] Ts Tennessine 117	[294] Og Oganesson 118

Relative atomic mass

Atomic (proton) number

The Lanthanides (atomic numbers 58-71) and the Actinides (atomic numbers 90-103) are not shown in this table.

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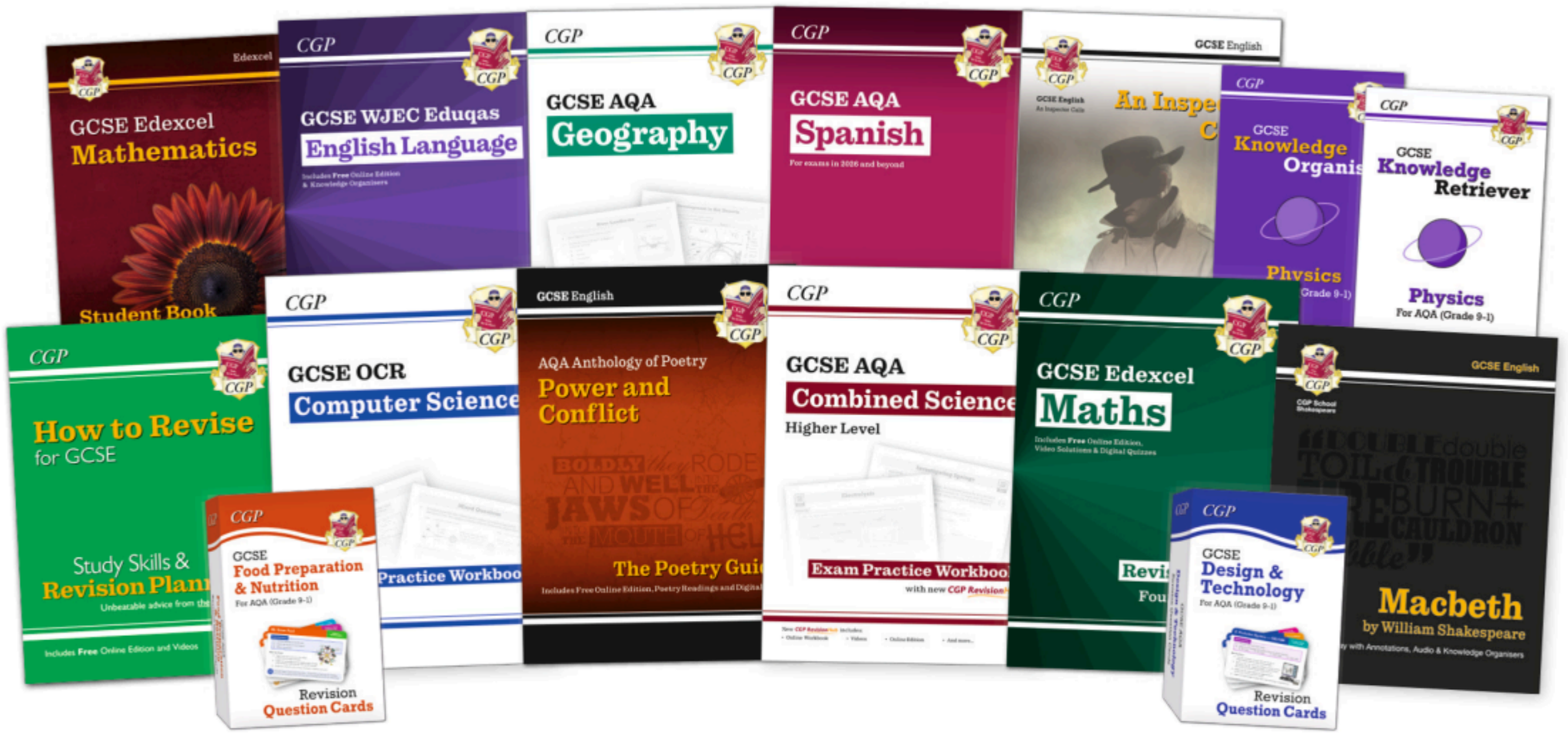

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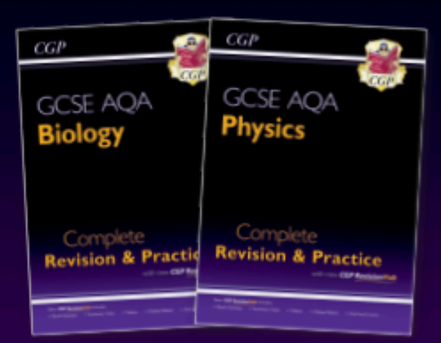

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